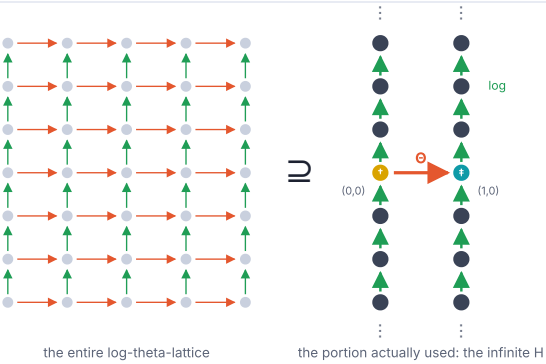
MOTIZUKIPDF · THE IUT TOOLKIT

Inter-universal Teichmüller Theory

data types and processes

A manual and demo: every data type in its form and with its reference to the original papers, and the processes that act on them, walked through step by step.

identifiers	375 data types and notations, 73 processes (`src/iut_ids.c`, `src/iut_proc.c`)
demo Θ-data	abc(1,2400,2401) l=13 — the abc triple 1 + 2400 = 2401, Frey curve $y^2 = x(x-1)(x+2400)$, l = 13
papers	the PDFs in `pdf/`; page numbers are PDF page numbers, which coincide with the printed ones
generated	2026-11-10, from commit `bb46409` by `scripts/manual/build.js`



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0 · How to read this manual

This manual is generated from the toolkit's own tables, so it matches the code exactly: the identifier table `src/iut_ids.c` (also `docs/IDENTIFIERS.md`, `lean/IUT/Identifiers.lean`), the process registry `src/iut_proc.c` (`docs/PROCESSES.md`), the C headers `src/iut.h` and `src/iut_i.h`, the diagrams of `scripts/manual/lib/diagrams.js` and the ten progressions of `scripts/manual/lib/processes.js`. Each entry of Part I shows one data type or notation in three ways:

- **form** — the C struct that implements it, field by field, with the meaning of each field (the Lean port in `lean/IUT/` mirrors the same fields; where a Lean structure of the same name exists it is named);
- **reference** — where the object is introduced or recalled in the papers, resolved to the paper, the section, the statement and the page of the PDF in `pdf/`, e.g. *[IUT I] §3, Definition 3.1 p. 61*. Papers are abbreviated as in the source code: [AbsTopIII], [Alien], [CanLift], [EssLgc], [EtTh], [ExpEst], [FormIUT], [FrdI], [FrdII], [IUT I], [IUT II], [IUT III], [IUT IV], [Pano] (see the table below);
- **identifier** — the English-letters-only name (`^[A-Za-z]+$`, unique case-insensitively) under which it appears in the C code, the JSON, the web app and Lean, obtained from the notation through the symbol dictionary of §1.

Part II shows the processes: the ten illustrated progressions (scripts/manual/lib/processes.js), step by step with a diagram for every step, and then all 73 processes of the registry with the transcript of `iut run <id>` on the demo Θ-data. Part III is a demo session: generation and validation of the Θ-data and a walk through the log-theta-lattice with the text views of the C program. The appendix is an alphabetical index of all identifiers with page numbers.

The program models the *combinatorics and the numbers* of the theory — which labels, which copies are identified, which indeterminacies are incurred, which inequality results — not the anabelian reconstruction algorithms; see `docs/DATA_MODEL.md` §8 for what is and is not computed.

The papers

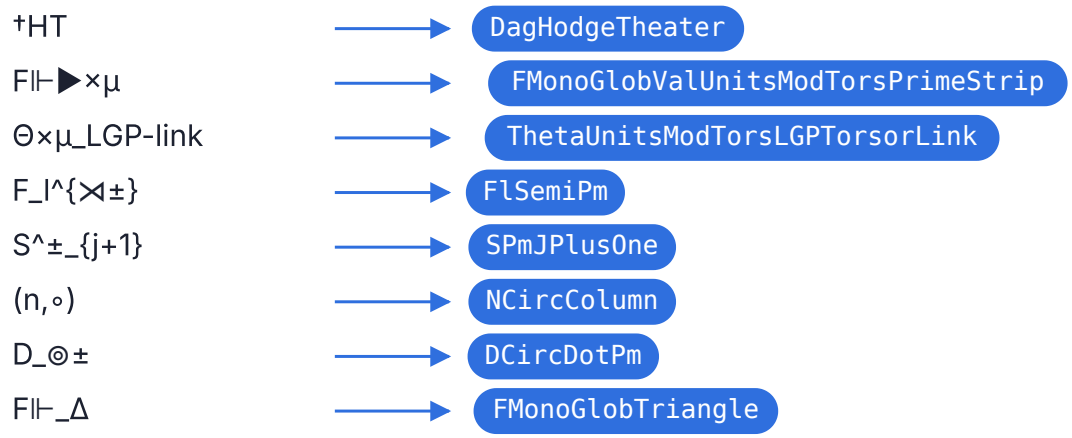
tag	title	file in `pdf/`	pages	indexed statements
[AbsTopIII]	Topics in Absolute Anabelian Geometry III: Global Reconstruction Algorithms	`Topics in Absolute Anabelian Geometry III.pdf`	164	138
[Alien]	The Mathematics of Mutually Alien Copies: from Gaussian Integrals to Inter-universal Teichmüller Theory	`Alien Copies, Gaussians, and Inter-universal Teichmuller Theory.pdf`	170	81
[CanLift]	A Theory of Ordinary p-adic Curves / Canonical Liftings	`Canonical Liftings.pdf`	34	34
[EssLgc]	On the Essential Logical Structure of Inter-universal Teichmüller Theory	`Essential Logical Structure of Inter-universal Teichmuller Theory.pdf`	158	65
[EtTh]	The Étale Theta Function and its Frobenioid-theoretic Manifestations	`The Etale Theta Function and its Frobenioid-theoretic Manifestations.pdf`	112	128
[ExpEst]	Explicit Estimates in Inter-universal Teichmüller Theory	`Explicit estimates in IUTeich.pdf`	58	67
[FormIUT]	On the Formalization of Inter-universal Teichmüller Theory	`Formalization of IUT (2026-04).pdf`	14	4
[FrdI]	The Geometry of Frobenioids I: The General Theory	`The Geometry of Frobenioids I.pdf`	126	90
[FrdII]	The Geometry of Frobenioids II: Poly-Frobenioids	`The Geometry of Frobenioids II.pdf`	72	51
[IUT I]	Inter-universal Teichmüller Theory I: Construction of Hodge Theaters	`Inter-universal Teichmuller Theory I.pdf`	186	158
[IUT II]	Inter-universal Teichmüller Theory II: Hodge–Arakelov-theoretic Evaluation	`Inter-universal Teichmuller Theory II.pdf`	174	151
[IUT III]	Inter-universal Teichmüller Theory III: Canonical Splittings of the Log-theta-lattice	`Inter-universal Teichmuller Theory III.pdf`	199	111
[IUT IV]	Inter-universal Teichmüller Theory IV: Log-volume Computations and Set-theoretic Foundations	`Inter-universal Teichmuller Theory IV.pdf`	87	55
[Pano]	A Panoramic Overview of Inter-universal Teichmüller Theory	`Panoramic Overview of Inter-universal Teichmuller Theory.pdf`	45	21

Statements were located by scanning the text of each PDF for the headings *Definition*, *Proposition*, *Corollary*, *Theorem*, *Remark*, *Example*, *Lemma*, *Fig.* and *Section / §* (`scripts/manual/pageindex.py` → `docs/manual/pageindex.json`). A citation whose statement could not be located is printed without a page number.

1 · The symbol dictionary and the identifier rule

Most of the notation of IUT uses symbols that are not letters: †HT, ‡HT, F⊩×μ-prime-strip, F⊩_Δ, Θ×μ_LGP-link, F_l^{⋊±}, S^±_{j+1}, … Every such object has an identifier matching `^[A-Za-z]+$`, built by replacing each symbol by a word. The 28 symbols:

notation with symbols → English-letters-only identifier (^[A-Za-z]+$, unique case-insensitively)



448 identifiers generated from src/iut_ids.c; full table in explorer tab 7.

English-letters-only identifiers. Every symbol in a notation maps to a word; the identifiers are the keys used in the JSON of the C program and the web model and are checked for uniqueness by `make check`. docs/IDENTIFIERS.md

`Dag` `Ddag` `Mono` `MonoGlob` `UnitsModTors` `BoxPlus` `CircAst` `Triangle`

symbol	word	meaning	introduced / recalled
†	`Dag`	left-hand (domain) copy of a Hodge theater / prime-strip across a Θ-link	**[IUT I]** Cor 3.7 p. 88; **[IUT III]** Introduction p. 2
‡	`Ddag`	right-hand (codomain) copy of a Hodge theater / prime-strip across a Θ-link	**[IUT I]** Cor 3.7 p. 88; **[IUT III]** Introduction p. 2
∗	`Ast`	a further distinct copy (e.g. ∗HT, F∗) or a 'full' set F_l^∗	**[IUT II]** Cor 4.10 p. 158; **[IUT III]** §1 p. 23
∘	`Circ`	'any element of the vertical line': (n,∘) denotes the whole column	**[IUT III]** Thm 3.11 p. 153
•	`Bullet`	vertex of the log-theta-lattice: one Θ±ell NF-Hodge theater	**[IUT III]** Fig I.1
⊢	`Mono`	mono-analytic (forgetting the ring/holomorphic structure)	**[IUT I]** Fig I1.2 p. 6
⊩	`MonoGlob`	mono-analytic with global realified Frobenioid	**[IUT I]** Fig I1.2 p. 6
▶	`Val`	value-group portion O▶ of a local monoid	**[IUT II]** Def 4.9 p. 153
×	`Units`	unit-group portion O× of a local monoid	**[IUT II]** Def 4.9 p. 153
×μ	`UnitsModTors`	units modulo torsion O×μ = O×/Oμ	**[IUT II]** Def 4.9 p. 153
μ	`Tors`	torsion (roots of unity) Oμ	**[IUT II]** Def 4.9 p. 153
⊥	`Perp`	'perp'-superscript: the ±-synchronized / 'perfection-free' variant	**[IUT II]** Def 2.7 p. 80
±	`Pm`	plus-minus (the ±-symmetry, I±, Θ±)	**[IUT I]** §6 p. 7
⋊	`Semi`	semi-direct product, as in F_l^{⋊±}	**[IUT I]** §6 p. 7
⊞	`BoxPlus`	the label of the Θ±-bridge target F_⊞ / D_⊞ (additive side); the paper's own glyph for this label does not survive text extraction, so this toolkit writes it ⊞	**[IUT I]** Def 6.4 p. 162, Def 6.11 p. 172; **[Alien]** §3.6 p. 87
◎	`CircDot`	the label ◎ (D_◎, D_◎±) of the Θell-bridge target	**[IUT I]** §6 p. 7
⊛	`CircAst`	the label ⊛ (D_⊛, F_⊛) of the NF-bridge target	**[IUT I]** §4 p. 95
Δ	`Triangle`	the label Δ obtained by identifying the labels 0 and F_l^* (F⊩_Δ)	**[IUT III]** Thm B p. 21; **[IUT II]** Cor 4.10 p. 158
>	`Gt`	the label > of the Θ-bridge target F_> (multiplicative)	**[IUT I]** §4 p. 95; **[Alien]** §3.6 p. 87
∞	`Inf`	the ∞-subscript: taking all N-th roots (∞Θ, ∞κ, ∞H)	**[IUT II]** §1 p. 20
κ	`Kappa`	κ-coric functions / κ-cuspidal structure	**[IUT I]** §5 p. 123; **[IUT II]** §3 p. 87
_	`Under`	underline, as in X̲, q̲_v, Θ̲_v, ε̲ ('underline' = the reduced/bar-free copy)	**[IUT I]** Def 3.1 p. 61
¯	`Bar`	overline: algebraic closure F̄, or a quotient such as O▷¯	**[IUT I]** Def 3.1 p. 61
^gp	`Gp`	groupification of a monoid	**[Frdl]** §0 p. 10
^pf	`Pf`	perfection of a monoid	**[Frdl]** §0 p. 10
tp	`Tp`	tempered (tempered fundamental group Π^tp, tempered Frobenioid)	**[EtTh]** §1 p. 10
′	`Prime`	prime-superscript variant of an object	**[IUT II]** §4 p. 120
¨	`Ddot`	double-dot: the ±-synchronized / 'doubly-dotted' copy (e.g. Ẍ, D̈)	**[IUT I]** §6 p. 7; **[IUT II]** §2 p. 64

Numbers are spelled out (`Zero`, `One`, `TwoL`, `JSq`, `LStar`, `LPm`); I*, I± and S^±_{j+1} become `LStar`, `LPm`, `SPmJPlusOne`; LGP / lgp and MOD / mod become `LGPTorsor` / `LgpIdeal` and `MODTorsor` / `ModIdeal`. `make check` fails if any identifier is not letters-only or collides with another one case-insensitively.

[p. 31]

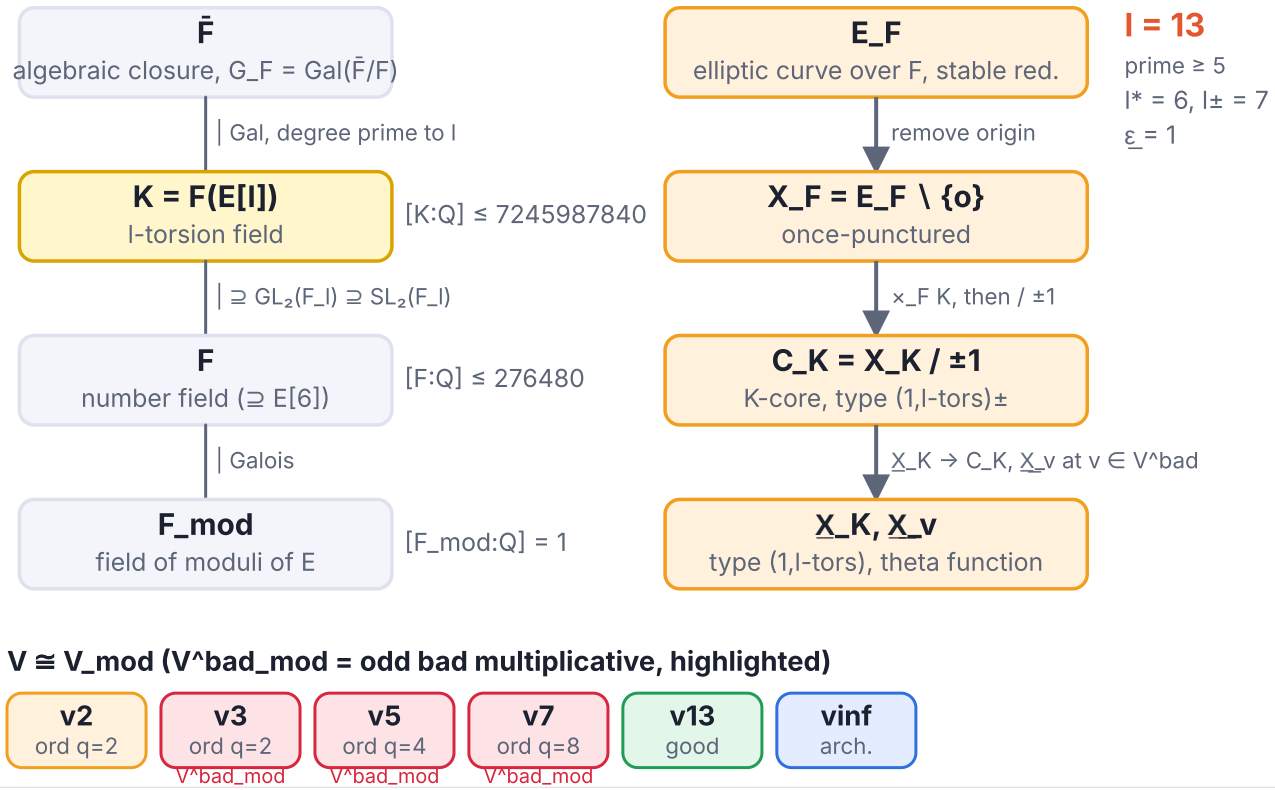
Part I

The data types

347 objects in 9 chapters, grouped by kind and ordered by the paper and statement that introduce them. An entry with a **form** block is implemented by a C struct of its own; the others are fields, kinds or labelled instances of a struct named in the entry.

2 · Initial Θ-data, fields, places and quantities

The input of the whole theory is a collection of initial Θ-data ([IUTchI] Def 3.1): a number field F with an elliptic curve, the tower F_mod ⊆ F ⊆ K = F(E[l]) ⊆ F̄, the prime l, the orbicurve C_K, the set of places V ≅ V_mod and the odd bad places V^bad_mod. The toolkit manufactures it from an abc triple through the Frey curve, records the degrees as upper bounds and the five non-numeric hypotheses as asserted flags, and computes from it the quantities log q, |log q|, log f, log d that the final inequality is about.



Initial Θ-data. The input of the whole theory: a number field F with E_F, the prime l, the orbicurve C_K, the set of valuations V ≅ V_mod with its nonempty set V^bad_mod of odd bad multiplicative places, and ε. Here it is produced from an abc triple via the Frey curve. **[IUT I]** Def 3.1 (via [IUTchII–IV] p. 61, [Pano] §3, [Alien] §3.3 (i))
`InitialThetaData` `FBarOverF` `XF` `CK` `VBadMod` `EpsilonUnder`

54 entries: 14 data, 8 fields, 14 sets of places, 7 constants, 11 quantities.

`InitialThetaData` (F̄/F, X_F, l, C_K, V, V^bad_mod, ε)

data

a collection of initial Θ-data: the input of the whole theory

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

LEAN `structure InitialThetaData — lean/IUT/InitialThetaData.lean`

`typedef struct … iut_initial_theta_data` iut.h

---------- initial Θ-data (species) (F̄/F, X_F, l, C_K, V, Vbad_mod, ε) [IUTchI] Def. 3.1; [Alien] §3.3 (i).

field	type	meaning
`name[96]`	`char`	
number fields Fmod ⊆ F ⊆ K = F(E[l]) ⊆ F̄		
`dmod`	`unsigned`	[Fmod : Q] (Fmod = field of moduli = Q(j_E))
`dF`	`unsigned long long`	upper bound for [F : Q]
`dK`	`unsigned long long`	upper bound for [K : Q]
`logdF`	`double`	log(d^F): normalized degree of different of F/Q
`logdK`	`double`	log(d^K)
the elliptic curve E_F — recorded through its arithmetic invariants		
`curve[96]`	`char`	description, e.g. "Frey(a,b,c)"
`j_log_height`	`double`	h(j_E) (Weil height of j), optional
the prime l ≥ 5 and derived constants		
`l`	`unsigned`	prime l ≥ 5
`lstar`	`unsigned`	l* = (l-1)/2
`lpm`	`unsigned`	l± = (l+1)/2
V ≅ Vmod		
`nV`	`unsigned`	
`V[IUT_MAX_V]`	`iut_valuation`	

field	type	meaning
ε (underlined epsilon) [IUTchI] Def 3.1 (f): a cusp of C_K that arises from a NONZERO element of the quotient Q ≅ Z/lZ, recorded by its label in F_l^* = {1..l}. At v ∈ V^bad, ε_v must be the cus*		
`epsilon`	`int`	
hypotheses of [IUTchI] Def. 3.1 that cannot be machine-verified from the numeric data alone are recorded as asserted flags.		
`h_F_galois_over_Fmod`	`int`	F/Fmod Galois, degree prime to l
`h_F_contains_six_torsion`	`int`	F ⊇ fields of definition of E[2·3]
`h_image_contains_SL2`	`int`	Im(Gal(K/F) → GL2(F_l)) ⊇ SL2(F_l)
`h_stable_reduction`	`int`	E_F has stable reduction everywhere
`h_CK_is_K_core`	`int`	C_K is a K-core
provenance: generated by the toolkit, or read from a JSON file of the schema iut-initial-theta-data/1 (docs/initial-theta-data/)		
`from_json`	`int`	
`source_id[48]`	`char`	e.g. "frey-1-8-9"
`have_expected`	`int`	the JSON's own "derived" numbers
`expected_logq`	`double`	
`expected_abslogq`	`double`	
`expected_logcond`	`double`	
`expected_ht`	`double`	

VSet **V** — sets of places

subset V ⊆ V(K) mapping bijectively to V_mod

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

`typedef struct … iut_valuation` — iut.h

field	type	meaning
`name[IUT_MAX_NAME]`	`char`	e.g. "v7", "vinf"
`kind`	`iut_vkind`	Vnon or Varc
`p`	`unsigned`	residue characteristic p_v (0 if arc)
`e_mod`	`unsigned`	ram. index / res. degree of Fmod_v/Qp
`f_mod`	`unsigned`	ram. index / res. degree of Fmod_v/Qp
`e_K`	`unsigned`	same for K_v / Q_p (v ∈ V ⊆ V(K))
`f_K`	`unsigned`	same for K_v / Q_p (v ∈ V ⊆ V(K))
`d_K`	`double`	exponent of the different of K_v/Q_p in units of ord(p) = 1; valid with IUT_KX_D
`red`	`iut_reduction`	reduction type of E at v
`ordq`	`unsigned`	ord_v(q_v) = -ord_v(j_E) (bad mult.)
the Tate parameter itself, when known (e.g. read from a JSON file): q_v = p^ordq · u with u ≡ q_unit mod p^q_prec (q_prec = 0: unknown)		
`q_unit`	`unsigned long long`	
`q_prec`	`unsigned`	

VBadMod **V^bad_mod** — sets of places

nonempty set of odd nonarchimedean v of F_mod of bad multiplicative reduction, split in K

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

form: `iut_valuation` — see VSet

EpsilonUnder **ε** — data

a cusp of C_K from a nonzero element of Q ≅ Z/lZ; at v ∈ V^bad the cusp of the canonical generator ±1 (label 1)

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 (f) p. 61

form: `iut_initial_theta_data` — see InitialThetaData

LPrime **l** — constants

the prime l ≥ 5 (stays a letter, listed for completeness)

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

`LEAN structure LPrime — lean/IUT/Labels.lean`

LStar **l* = (l−1)/2** — constants

l-star

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

LPm **l± = (l+1)/2** — constants

l-plus-minus

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

LogQ **log(q)** — quantities

normalized degree of the q-divisor (6 × height)

INTRODUCED / RECALLED **[IUT IV]** §1 p. 9

`AbsLogQ` **|log(q)|** — quantities

log-volume of the q-pilot = log(q)/(2l) (q-divisor normalized degree)

INTRODUCED / RECALLED **[IUT III]** §3, Corollary 3.12 p. 173

form: `iut_multiradial_rep` — see MultiradialRepresentation

`CAF` **CAF** — fields

complex archimedean field: a topological field isomorphic to C (K_v for v ∈ V^arc)

INTRODUCED / RECALLED **[IUT I]** §0 p. 33; **[IUT I]** §3, Example 3.4 p. 80

form: `iut_field_kind` — see NF

`JSq` **j²** — constants

the exponent of q in the j-th Θ-value / Gaussian monoid

INTRODUCED / RECALLED **[IUT I]** Introduction p. 2; **[Alien]** §2.5 p. 16

`MLF` **MLF** — fields

mixed-characteristic local field: a finite extension of Q_p (K_v for v ∈ V^non)

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

form: `iut_field_kind` — see NF

`NF` **NF** — fields

number field: a finite extension of Q

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

`typedef enum … iut_field_kind` — iut_i.h

fields [IUTchI] §0: NF number field, MLF mixed-characteristic local field, CAF complex archimedean field.

```
IUT_FIELD_NF = 0
IUT_FIELD_MLF
IUT_FIELD_CAF
```

`PV` **p_v** — constants

the residue characteristic of v ∈ V^non; by convention p_v = e at archimedean v

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

form: `iut_valuation` — see VSet

form: `iut_field_kind` — see NF

`FBarOverF` **F̄/F** — fields

number field F with algebraic closure F̄; G_F = Gal(F̄/F)

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

form: `iut_initial_theta_data` — see InitialThetaData

`Fmod` **F_mod** — fields

field of moduli of E_F (= Q(j_E) for Fmod = Q(j)); V_mod = V(F_mod)

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

form: `iut_initial_theta_data` — see InitialThetaData

`KField` **K = F(E[l])** — fields

the field obtained by adjoining the l-torsion of E_F

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

form: `iut_initial_theta_data` — see InitialThetaData

`VArc` **V^arc** — sets of places

archimedean valuations of V

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

form: `iut_valuation` — see VSet

`VBad` **V^bad** — sets of places

elements of V over V^bad_mod

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

form: `iut_valuation` — see VSet

`VGood` **V^good** — sets of places

elements of V over V^good_mod

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

form: `iut_valuation` — see VSet

`VGoodMod` **V^good_mod** — sets of places

complement V_mod \ V^bad_mod

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

form: `iut_valuation` — see VSet

`VK` **V(K)** — sets of places

all valuations of K; V ⊆ V(K) is a section over V_mod

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 (d) p. 61
form: `iut_labcusp_torsor` — see LabCusp
form: `iut_global_base` — see LabCusp
form: `iut_valuation_orbit` — see LabCusp

`VNon` **V^non** — sets of places

nonarchimedean valuations of V

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61
form: `iut_valuation` — see VSet

`CriticalPoint` **critical point** — data

a point of |C_L| ≅ A¹ over which C_L → |C_L| ramifies: the images of the 2-torsion points

INTRODUCED / RECALLED **[IUT I]** §3, Remark 3.1.7 (i) p. 66
form: `iut_kappa_function` — see CFmod
form: `iut_kappa_verdict` — see CFmod
form: `iut_kappa_sol` — see CFmod

`FMuL` **F(μ_l)** — fields

F with the l-th roots of unity; Gal(K/F(μ_l)) is the unique subgroup ≅ SL_2(F_l)

INTRODUCED / RECALLED **[IUT I]** §3, Remark 3.1.7 (iii) p. 66
form: `iut_kappa_function` — see CFmod
form: `iut_kappa_verdict` — see CFmod
form: `iut_kappa_sol` — see CFmod

`InfKappaCoric` **∞κ-coric function** — data

κ-coric function with all N-th roots adjoined (a divisible, cyclotomic pseudo-monoid)

INTRODUCED / RECALLED **[IUT I]** §3, Remark 3.1.7 (ii) p. 66
form: `iut_kappa_function` — see CFmod
form: `iut_kappa_verdict` — see CFmod
form: `iut_kappa_sol` — see CFmod

`InfKappaUnitsCoric` **∞κ×-coric function** — data

∞κ-coric functions multiplied by constants (F̄×)

INTRODUCED / RECALLED **[IUT I]** §3, Remark 3.1.7 (ii) p. 66
form: `iut_kappa_function` — see CFmod
form: `iut_kappa_verdict` — see CFmod
form: `iut_kappa_sol` — see CFmod

`KappaCoric` **κ-coric function** — data

rational function on |C_L| with exactly one pole, ≥ 2 zeroes, divisor over an NF avoiding the critical points, roots of unity at strictly critical points

INTRODUCED / RECALLED **[IUT I]** §3, Remark 3.1.7 (ii) p. 66
form: `iut_kappa_function` — see CFmod
form: `iut_kappa_verdict` — see CFmod
form: `iut_kappa_sol` — see CFmod

`StrictlyCriticalPoint` **strictly critical point** — data

a critical point at which κ-coric functions must take root-of-unity values

INTRODUCED / RECALLED **[IUT I]** §3, Remark 3.1.7 (i) p. 66
form: `iut_kappa_function` — see CFmod
form: `iut_kappa_verdict` — see CFmod
form: `iut_kappa_sol` — see CFmod

`LogTheta` **log(Θ)** — quantities

the formal symbol of the Θ-versions: N·log_Φ(q_v)·log(Θ), log_mod(p_v)·log(Θ)

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 (v) p. 69; **[IUT I]** §3, Example 3.5 p. 84
form: `iut_local_frobenioid` — see DV

`QUnderV` **q_v** — data

the 2l-th root of the q-parameter q_v at v ∈ V^bad: q_v = q_v^{1/2l}

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 p. 69

`ThetaUnderV` **Θ_v** — data

the (reciprocal of the l-th root of the) theta function at v ∈ V^bad

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 p. 69; **[EtTh]**

`TwoL` **2l** — constants

the root order of q_v and the denominator of |log(q)|

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 p. 69

`LogPV` **log(p_v)** quantities

the generator of the realified divisor monoid at v ∈ V^good: log_mod(p_v) ↦ (1/[K_v:(F_mod)_v])·log_Φ(p_v)

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.3 p. 77; **[IUT I]** §3, Example 3.5 p. 84

form: `iut_local_frobenioid` — see DV

`ABox` **A_□** fields

the CAF (≅ C) attached to an Aut-holomorphic orbispace

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.4 (i) p. 80

form: `iut_local_frobenioid` — see DV

`FrobeniusElement` **Frobenius element of (R≥0)_v** data

the distinguished element ×e_v (nonarch) or ÷2π (arch) of the realified monoid at v

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.5 (iv) p. 84

form: `iut_global_realified` — see CMonoGlobMod

form: `iut_f_monoglob` — see CMonoGlobMod

form: `iut_theta_hodge_theater_def` — see CMonoGlobMod

`LogModPV` **log_mod(p_v)** quantities

the generator of Φ_{C⊩-mod,v}: the realified divisor of p_v on F_mod

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.5 (i) p. 84

form: `iut_global_realified` — see CMonoGlobMod

form: `iut_f_monoglob` — see CMonoGlobMod

form: `iut_theta_hodge_theater_def` — see CMonoGlobMod

`VDCircAst` **V(†D^⊛), V̄(†D^⊛)** sets of places

the valuations of the number field reconstructed from †D^⊛, and their quotient by π_1

INTRODUCED / RECALLED **[IUT I]** §4, Definition 4.1 (v) p. 95

form: `iut_labcusp_torsor` — see LabCusp

form: `iut_global_base` — see LabCusp

form: `iut_valuation_orbit` — see LabCusp

`VBor` **V^Bor** sets of places

Aut(C_K)·V ⊆ V(K): the basepoints of the multiplicative (F_l^*) symmetry

INTRODUCED / RECALLED **[IUT I]** §4, Example 4.3 (i) p. 98

form: `iut_labcusp_torsor` — see LabCusp

form: `iut_global_base` — see LabCusp

form: `iut_valuation_orbit` — see LabCusp

`VPmUn` **V^±un** sets of places

Aut_ε(C_K)·V ⊆ V(K); equals V^± = Aut_csp(D^⊚±)·V

INTRODUCED / RECALLED **[IUT I]** §4, Example 4.3 (i) p. 98; **[IUT I]** §6, Remark 6.1.1 p. 159

form: `iut_labcusp_torsor` — see LabCusp

form: `iut_global_base` — see LabCusp

form: `iut_valuation_orbit` — see LabCusp

`EvaluationPoint` **evaluation point** data

a μ_−-translate of a cusp of X_v, labelled by |F_l|, K_v-rational; Θ_v takes the value q_v^{j²} there (up to μ_2l)

INTRODUCED / RECALLED **[IUT I]** §4, Example 4.4 (i)–(ii) p. 105

form: `iut_eval_points` — see MuMinus

form: `iut_theta_bridge_piece` — see MuMinus

form: `iut_label_transport` — see MuMinus

form: `iut_iso_torsor` — see MuMinus

form: `iut_forgetful_counts` — see MuMinus

`MuMinus` **μ_−** data

the 2-torsion point of X_v by which the cusps are translated to obtain the evaluation points

INTRODUCED / RECALLED **[IUT I]** §4, Example 4.4 (i) p. 105

`typedef struct … iut_eval_points` iut_i.h

Ex 4.4: |F_l| = F_l/{±1} ↔ cusps of C_v; μ_− the 2-torsion point; the evaluation points are the μ_−-translates of the cusps; Θ_v takes the value q_v^{j²} (up to μ_2l) at the points labelled j; the evaluation sections G_v → Π_v are labelled by |F_l|

field	type	meaning
`v_index`	`unsigned`	
`n`	`unsigned`	l± = |{0} ⊔ F_l^*|
`label[IUT_MAX_LPM]`	`int`	0, 1, …, l*
`theta_value[IUT_MAX_LPM]`	`iut_q`	j² (in units of ord q_v), up to μ_2l
`K_v_rational`	`int`	evaluation points of X_v are K_v-rational
`sections_group_theoretic`	`int`	the sections G_v → Π_v are constructible from Π_v
`mu_minus_order`	`unsigned`	2

`typedef struct … iut_theta_bridge_piece` iut_i.h

Ex 4.4 (iii): the poly-morphism φ^Θ_{v_j} : D_{v_j} → D_{>,v} is given by the evaluation section labelled |j| at bad v, and is the full poly-isomorphism at good v.

field	type	meaning
`v_index`	`unsigned`	
`j`	`int`	
`via_evaluation_section`	`int`	
`full_poly_iso`	`int`	
`label_used`	`int`	

`typedef struct … iut_label_transport` iut_i.h

Prop 4.7 (ΘNF), Prop 6.5 (Θ±ell): transport of label classes. ᵗχ : J ≅ F_l^*, ᵗφ^LC_j : LabCusp(ᵗD^⊛) ≅ LabCusp(ᵗD_>), [ᵗε] ↦ j, ᵗζ : LabCusp(ᵗD^⊛) ≅ J ᵗζ^Θell_{v_t}, the global ±-synchronization ᵗξ, ᵗζ_± : LabCusp±(ᵗD^⊚±) ≅ T

field	type	meaning
`signed_labels`	`int`	0: ΘNF (F_l^*), 1: Θ±ell (F_l)
`n`	`unsigned`	
`chi[2*IUT_MAX_LSTAR+1]`	`int`	index in the capsule ↦ label (χ : J ≅ F_l^* or T ≅ F_l)
`zeta[2*IUT_MAX_LSTAR+1]`	`int`	LabCusp(ᵗD^⊛) ↦ J (resp. LabCusp±(ᵗD^⊚±) ↦ T)
`epsilon_class_image`	`int`	[ᵗε] ↦ 1 under φ^LC_1 (resp. the ±-synchronized ξ ↦ ±1)
`synchronized`	`int`	Rem 6.12.4: ζ (NF side) / ζ±, ξ (± side) agree across v ∈ V

`typedef struct … iut_iso_torsor` iut_i.h

Prop 4.8 (ΘNF) and Prop 6.6 (Θ±ell): the isomorphism torsors of bridges and Hodge theaters

field	type	meaning
`object`	`const char *`	
`group`	`const char *`	
`order`	`unsigned long long`	
`ref`	`const char *`	

`typedef struct … iut_forgetful_counts` iut_i.h

Prop 4.9 (ΘNF) and Prop 6.8 (Θ±ell): forgetting structure (i) F_l^* acts simply transitively on J (F_l^{⋊±} doubly transitively on T) (ii) holomorphic capsule: S_{l*} (resp. S_{l±}) symmetry, (l*)^{l*} (resp. (l±)^{l±}) label assignments (iii) mono-analytic capsule: the same after D ↦ D⊩ Prop 4.11 / 6.9: processions cut (l*)^{l*} down to l*! (resp. (l±)^{l±} down to l±!)

field	type	meaning
`signed_labels`	`int`	
`n`	`unsigned`	l* or l±
`symmetric_group_order`	`unsigned long long`	n!
`label_assignments`	`double`	n^n
`procession_assignments`	`unsigned long long`	n!
`reduction_factor`	`double`	n^n / n!
`transitivity`	`const char *`	"simply transitive" / "doubly transitive"

`DistributionVJ` V_j, V_J, V_⟨J⟩

sets of places

V_j = {v_j}_{v∈V}; V_J = ∏_j V_j; the diagonal V_⟨J⟩ ↪ V_J (constant versus arbitrary distribution)

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (vii) p. 123; **[IUT I]** §5, Remark 5.1.1 p. 131

form: `iut_global_frobenioid` — see FCircAstGlobal

form: `iut_distribution` — see FCircAstGlobal

form: `iut_strip_functor` — see FCircAstGlobal

`PrimeCMonoGlob` Prime(C⊩)

sets of places

set of primes of C⊩, in bijection with V

INTRODUCED / RECALLED **[IUT I]** §5, Definition 5.2 p. 134

`DeltaValuation` δ-valuation

data

a valuation of the field determined by ᵗD^⊛ lying over a label δ ∈ LabCusp(ᵗD^⊛); gives the localization ᵗF^⊛|_δ

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.4 (iv) p. 147

form: `iut_global_frobenioid` — see FCircAstGlobal

form: `iut_distribution` — see FCircAstGlobal

form: `iut_strip_functor` — see FCircAstGlobal

`SignVector` α ∈ {±1}^V

data

a choice of sign at every v ∈ V; the D-Θ±-bridge isomorphisms form a torsor over {±1} × {±1}^V

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (iv) p. 155; **[IUT I]** §6, Example 6.2 (iii) p. 159; **[IUT I]** §6, Proposition 6.6 (i) p. 165

form: `iut_fl_semi_pm_set` — see FlSemiPmGroup

form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup

form: `iut_sign_vector` — see FlSemiPmGroup

form: `iut_prop_six_seven` — see FlSemiPmGroup

`VPm` V^±

sets of places

Aut_±·V = Aut_csp·V ⊆ V(K); equals V^±un: the basepoints of the additive (F_l^{⋊⊲±}) symmetry

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (v) p. 155; **[IUT I]** §6, Remark 6.1.1 p. 159

form: `iut_fl_semi_pm_set` — see FlSemiPmGroup

form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup

form: `iut_sign_vector` — see FISemiPmGroup
form: `iut_prop_six_seven` — see FISemiPmGroup

PStar **p*** — constants

p* = p for p odd, p* = p² for p = 2

INTRODUCED / RECALLED **[IUT III]** §1, Definition 1.1 (i) p. 23
form: `iut_log_shell` — see LogShell

LogVolume **μ_log** — quantities

normalized log-volume on log-shells / tensor packets

INTRODUCED / RECALLED **[IUT III]** §3, Proposition 3.9 p. 115; **[IUT IV]** §1, Proposition 1.2 p. 10

AbsLogTheta **|log(Θ)|** — quantities

log-volume of the Ind1-3-hull of the Θ-pilot

INTRODUCED / RECALLED **[IUT III]** §3, Corollary 3.12 p. 173
form: `iut_multiradial_rep` — see MultiradialRepresentation

CTheta **C_Θ** — quantities

the constant with −|log(Θ)| ≥ −|log(q)| ⇔ C_Θ ≥ −1

INTRODUCED / RECALLED **[IUT III]** §3, Corollary 3.12 p. 173; **[IUT IV]** §1, Theorem 1.10 p. 22
form: `iut_multiradial_rep` — see MultiradialRepresentation

LogDK **log(d^K)** — quantities

normalized log-different of K

INTRODUCED / RECALLED **[IUT IV]** §1 p. 9

LogFK **log(f^K)** — quantities

normalized log-conductor of E over K

INTRODUCED / RECALLED **[IUT IV]** §1 p. 9

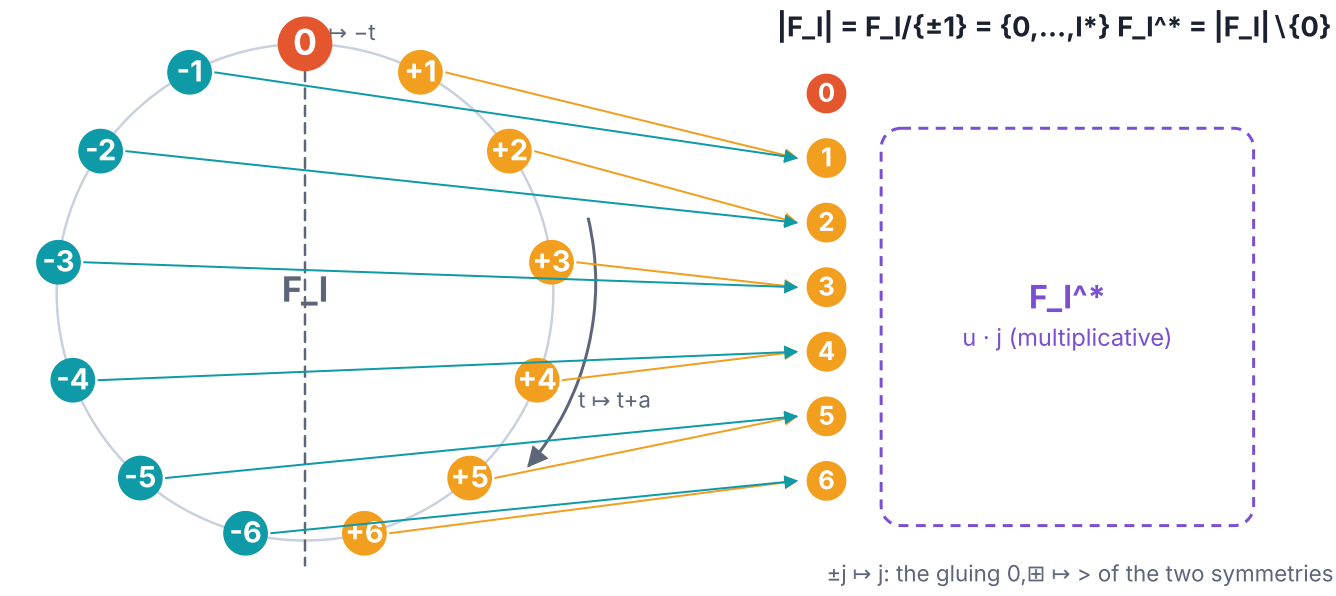
HtE **ht_E = (1/6)log(q)** — quantities

the height of E_F

INTRODUCED / RECALLED **[IUT IV]** §1, Theorem 1.10 p. 22; **[GenEll]**

3 · Label sets, labels, symmetries and capsules

Every Hodge theater is indexed by labels: F_l = {−l*,...,l*} for the additive F_l^{⋊±}-symmetry of the Θ±ell-side, F_l^* = {1,...,l*} for the multiplicative F_l^*-symmetry of the ΘNF-side, |F_l| = F_l/{±1} in between. Capsules are finite collections of prime-strips indexed by these sets; LabCusp and LabCusp± are the torsors of cusp labels that the anabelian side reconstructs.



Label sets and their symmetries. F_l carries the additive symmetry F_l^{⋊±}, its quotient |F_l| = F_l/±1 and F_l^* = |F_l| \{0} carry the multiplicative symmetry F_l^*. The Θ±ell side uses t ∈ F_l, the ΘNF side uses j ∈ F_l^*; the two are glued by ±j ↦ j. **[IUT I]** §4 p. 95, §6 p. 7; **[Alien]** §3.6 (ii) p. 87

`Fl` `FlAbs` `FlStar` `FlSemiPm` `FlStarSym` `LabelT` `LabelJ`

26 entries: 8 label sets, 7 labels, 6 symmetries, 5 capsules.

`Fl` **F_l** — label sets

the field with l elements, {−l*,...,0,...,l*}, labels t of the Θ±ell-HT capsule

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

`FlAbs` **|F_l|** — label sets

F_l/{±1} = {0,1,...,l*}

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

`FlStar` **F_l^*** — label sets

F_l^×/{±1} = {1,...,l*}, labels j of the ΘNF-HT capsule

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

`FlUnits` **F_l^×** — label sets

multiplicative group of F_l

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

`FlSemiPm` **F_l^{⋊±}** — symmetries

F_l ⋊ {±1}: additive / geometric symmetry (conjugate synchronization)

INTRODUCED / RECALLED **[IUT I]** §6 p. 7; **[Alien]** §3.3 (ii) p. 58

LEAN `structure FlSemiPm — lean/IUT/Labels.lean`

`FlStarSym` **F_l^* ↷** — symmetries

multiplicative / arithmetic symmetry on labels j

INTRODUCED / RECALLED **[IUT I]** §4 p. 95; **[Alien]** §3.3 (ii) p. 58

`Capsule` **capsule {A_j}_{j∈J}** — capsules

finite collection of objects indexed by a finite set J (a |J|-capsule); morphism = injection of index sets + morphisms of members

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

```
typedef struct … iut_capsule_member                                                                    iut.h
---------------------------------------------- Hodge theaters (species) Θ±ell NF-Hodge theater = Θ±ell-Hodge theater glued to ΘNF-Hodge theater along F_l ⊇ F_l^× ↠ F_l^* ("0, ⊞ ↦ >").
[IUTchI] §I1, Def. 6.13; [Pano] §3 Fig. 3.3; [Alien] §3.3 (v) Fig. 3.8; [IUTchII] Rem. 3.8.2. one member of a capsule, labeled by t or j
```

field	type	meaning
label	int	t ∈ F_l or j ∈ F_l^*
basepoint	unsigned	distinct copy id (conjugate synchronization)

form: iut_poly_morphism — see PolyMorphism

FT **F_T** capsules

capsule of F-prime-strips indexed by T = F_l

INTRODUCED / RECALLED **[IUT I]** §6 p. 7
form: iut_capsule_member — see Capsule

FJ **F_J** capsules

capsule of F-prime-strips indexed by J = F_l^*

INTRODUCED / RECALLED **[IUT I]** §4 p. 95
form: iut_capsule_member — see Capsule

LabCusp **LabCusp(†D_v)** label sets

the F_l^*-torsor of label classes of cusps of †D_v, with canonical element †η_v; compatible for all v

INTRODUCED / RECALLED **[IUT I]** §4, Definition 4.1 (ii) p. 95; **[IUT I]** §4, Proposition 4.2 p. 98

typedef struct … iut_labcusp_torsor iut_i.h

========== §4 ======= Def 4.1 (ii), Prop 4.2: LabCusp(†D_v) is an F_l^*-torsor with canonical element †η_v; the bijections LabCusp(†D_v) ≅ F_l^* for the various v are compatible.

field	type	meaning
nV	unsigned	
torsor_size	unsigned	l*
eta[IUT_MAX_V]	int	†η_v ∈ F_l^*: the canonical element (label 1)
compatible	int	Prop 4.2: one F_l^*-torsor LabCusp(†D)

typedef struct … iut_global_base iut_i.h

field	type	meaning
kind	iut_global_base_kind	
id	const char *	B(C_K)^0, Π_{C_K} / B(X_K)^0, Π_{X_K}
notation	const char *	B(C_K)^0, Π_{C_K} / B(X_K)^0, Π_{X_K}
curve	const char *	B(C_K)^0, Π_{C_K} / B(X_K)^0, Π_{X_K}
group	const char *	B(C_K)^0, Π_{C_K} / B(X_K)^0, Π_{X_K}
labcusp_size	unsigned	|LabCusp(D^⊛)| = l*; |LabCusp±(D^⊚±)| = l
nV_section	unsigned	|V| ⊆ V(D^⊛): the section over V_mod
aut_image_order	unsigned long long	|Borel/{±1}| = l(l−1)²/2
aut_eps_image_order	unsigned long long	|semi-unipotent/{±1}| = l(l−1)
quotient_order	unsigned long long	|Aut/Aut_ε| = l* (D^⊛); |Aut_±/Aut_csp| = 2l (D^⊚±)
quotient_group	const char *	"F_l^*" / "F_l^{⋊±}"
double_cover	int	D^⊚± → D^⊛ is a double covering

typedef struct … iut_valuation_orbit iut_i.h

Ex 4.3 (i), Def 6.1 (v), Rem 6.1.1: V ⊆ V^±un = Aut_ε·V = V^± = Aut_csp·V ⊆ V^Bor = Aut·V ⊆ V(K)

field	type	meaning
id	const char *	
notation	const char *	
orbit_of	const char *	
lower_bound	unsigned	
upper_bound	unsigned long long	

LabCuspPm **LabCusp±(†D_v)** label sets

the cusps of †D^±_v with the F_l^×-action; ≅ F_l up to ±1 (an F_l^{⋊±}-group); (LabCusp± \ {η^o})/{±1} ≅ LabCusp

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (ii)–(iii) p. 155
form: iut_fl_semi_pm_set — see FlSemiPmGroup
form: iut_labcusp_pm_torsor — see FlSemiPmGroup
form: iut_sign_vector — see FlSemiPmGroup
form: iut_prop_six_seven — see FlSemiPmGroup

IotaInvolution **ι** symmetries

the involution of X over C: swaps l_ε′ and l_ε″, acts as −1 on Δ_E ⊗ Z/l

INTRODUCED / RECALLED **[IUT I]** §1 p. 37
form: iut_cusps_section_one — see CuspEpsilon

`LabelJ` **j ∈ F_l^*** — labels

capsule label of the ΘNF-Hodge theater

INTRODUCED / RECALLED **[IUT I]** §4 p. 95
form: `iut_capsule_member` — see Capsule

`EtaV` **ťη_v** — labels

the canonical element of LabCusp(ťD_v) (image of ε_v)

INTRODUCED / RECALLED **[IUT I]** §4, Definition 4.1 (ii) p. 95
form: `iut_labcusp_torsor` — see LabCusp
form: `iut_global_base` — see LabCusp
form: `iut_valuation_orbit` — see LabCusp

`SJPlusOne` **S_{j+1}** — label sets

procession container {1,...,j+1} (multiplicative version)

INTRODUCED / RECALLED **[IUT I]** §4, Proposition 4.11 p. 120
form: `iut_procession_step` — see Procession

`LabelT` **t ∈ F_l** — labels

capsule label of the Θ±ell-Hodge theater

INTRODUCED / RECALLED **[IUT I]** §6 p. 7
form: `iut_capsule_member` — see Capsule

`LabelZero` **0 ∈ F_l** — labels

the zero label, identified with ⊞ and then with > under the gluing

INTRODUCED / RECALLED **[IUT I]** §6 p. 7

`EtaPmV` **ťη^±_v** — labels

the ±-canonical element of LabCusp±(ťD_v), well-defined up to ±1

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (iii) p. 155
form: `iut_fl_semi_pm_set` — see FlSemiPmGroup
form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup
form: `iut_sign_vector` — see FlSemiPmGroup
form: `iut_prop_six_seven` — see FlSemiPmGroup

`EtaZeroV` **ťη°_v** — labels

the zero element of LabCusp±(ťD_v)

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (ii) p. 155
form: `iut_fl_semi_pm_set` — see FlSemiPmGroup
form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup
form: `iut_sign_vector` — see FlSemiPmGroup
form: `iut_prop_six_seven` — see FlSemiPmGroup

`PositiveElement` **positive / negative element** — labels

elements of an F_l^{⋊±}-group mapping to {1..l*} (positive) or {−l*..−1} (negative) under the bijections

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (i) p. 155
form: `iut_fl_semi_pm_set` — see FlSemiPmGroup
form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup
form: `iut_sign_vector` — see FlSemiPmGroup
form: `iut_prop_six_seven` — see FlSemiPmGroup

`AlphaThetaPm` **α^Θ±** — symmetries

the poly-automorphism of order ≤ 2 of the model base-Θ±-bridge determined by α ∈ {±1}^V

INTRODUCED / RECALLED **[IUT I]** §6, Example 6.2 (iii) p. 159
form: `iut_fl_semi_pm_set` — see FlSemiPmGroup
form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup
form: `iut_sign_vector` — see FlSemiPmGroup
form: `iut_prop_six_seven` — see FlSemiPmGroup

`MinusOneFl` **−1_{F_l}** — symmetries

the poly-automorphism of order two of the model base-Θ±-bridge: t ↦ −t, D_t ≅ D_{−t}, negative sign at every v

INTRODUCED / RECALLED **[IUT I]** §6, Example 6.2 (ii) p. 159
form: `iut_fl_semi_pm_set` — see FlSemiPmGroup
form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup
form: `iut_sign_vector` — see FlSemiPmGroup
form: `iut_prop_six_seven` — see FlSemiPmGroup

`GammaPm` **γ_±** — symmetries

the poly-action of F_l^{⋊±} on the model base-Θell-bridge

INTRODUCED / RECALLED **[IUT I]** §6, Example 6.3 (iii) p. 160
form: `iut_fl_semi_pm_set` — see FlSemiPmGroup
form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup
form: `iut_sign_vector` — see FlSemiPmGroup
form: `iut_prop_six_seven` — see FlSemiPmGroup

`TAbs` **†D_{|T|}** capsules

the l±-capsule obtained from †D_T by the quotient |T| = T/{±1}

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.4 (i) p. 162

form: `iut_fl_semi_pm_set` — see FISemiPmGroup

form: `iut_labcusp_pm_torsor` — see FISemiPmGroup

form: `iut_sign_vector` — see FISemiPmGroup

form: `iut_prop_six_seven` — see FISemiPmGroup

`TStar` **†D_{T*}** capsules

the l*-capsule over T* = |T| \ {0}

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.4 (i) p. 162

form: `iut_fl_semi_pm_set` — see FISemiPmGroup

form: `iut_labcusp_pm_torsor` — see FISemiPmGroup

form: `iut_sign_vector` — see FISemiPmGroup

form: `iut_prop_six_seven` — see FISemiPmGroup

`SPmJPlusOne` **S^±_{j+1}** label sets

procession container {0,1,...,j}

INTRODUCED / RECALLED **[IUT I]** §6, Proposition 6.9 p. 169; **[IUT III]** §3 p. 92

form: `iut_procession_step` — see Procession

4 · Orbicurves, cusps, groups and anabelioids

The geometric side of [IUTchI] §1–§2: the once-punctured elliptic curve X_F, its quotient C_K, the coverings X_K, C_K, X→, C→ and their local versions X_v, C_v at v ∈ V^bad, the cusps ε, ε′, ε″, 2ε used to pin down labels, the (tempered) fundamental groups Π_v ↠ G_v, Π_{X_K}, Π_{C_K} and the anabelioids B(Π), B(X)^0, B^temp(Π) through which the étale-like data is expressed.

59 entries: 12 orbicurves, 5 cusps, 37 groups, 5 anabelioids.

XF X_F

orbicurves

once-punctured elliptic curve E_F \ {origin}

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

`typedef struct … iut_orbicurve` — iut_i.h

field	type	meaning
kind	iut_orbicurve_kind	
id	const char *	base: "k" / "K" / "K_v"
notation	const char *	base: "k" / "K" / "K_v"
type	const char *	base: "k" / "K" / "K_v"
base	const char *	base: "k" / "K" / "K_v"
covers	const char *	the orbicurve it is a covering of
degree	unsigned	degree of that covering
covering_group	const char *	"Z/l", "Z/2", "Z/2l" (Def 1.1), "—"
covering_group_order	unsigned	
ncusps	int	number of cusps; -1 if not recorded
aut_k_order	unsigned	|Aut_k| when stated (Rem 1.2.1), else 0
ref	const char *	

CK C_K

orbicurves

hyperbolic orbicurve X_K/±1 of type (1,l-tors)_±; must be a K-core

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

form: `iut_orbicurve` — see XF

XUnderK X_K

orbicurves

hyperbolic orbicurve of type (1,l-tors) (the Θ-data orbicurve)

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

form: `iut_orbicurve` — see XF

PiV Π_v

groups

(tempered) arithmetic fundamental group at v (holomorphic, étale-like)

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

form: `iut_local_datum` — see BasicPrimeStrip

GV G_v

groups

absolute Galois group of K_v (mono-analytic, étale-like)

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

form: `iut_local_datum` — see BasicPrimeStrip

BPi B(Π)

anabelioids

the connected anabelioid (Galois category) of finite sets with continuous Π-action

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

`typedef struct … iut_anabelioid` — iut_i.h

field	type	meaning
kind	iut_anabelioid_kind	
id	const char *	
notation	const char *	
of	const char *	
connected	int	
tempered	int	

BTemp B^temp(Π)

anabelioids

the connected temperoid of a tempered group Π; D_v = B^temp(X_v)^0 at v ∈ V^bad

INTRODUCED / RECALLED **[IUT I]** §0 p. 33; **[IUT I]** §3, Example 3.2 p. 69

form: `iut_anabelioid` — see BPi

BX B(X)

anabelioids

the anabelioid of finite étale coverings of a connected scheme / orbicurve X

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

form: `iut_anabelioid` — see BPi

BXZero **B(X)^0** anabelioids

the connected component of B(X); D_v = B(X→_v)^0, D^⊛ = B(C_K)^0, D^⊚± = B(X_K)^0

INTRODUCED / RECALLED **[IUT I]** §0 p. 33; **[IUT I]** §3, Example 3.2 –3.3 p. 69; **[IUT I]** §4, Definition 4.1 (v) p. 95

form: `iut_anabelioid` — see BPi

DeltaV **Δ_v** groups

geometric fundamental group, kernel of Π_v ↠ G_v

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

Temperoid **connected temperoid** anabelioids

the category of tempered coverings; the tempered fundamental group is recovered from it

INTRODUCED / RECALLED **[IUT I]** §0 p. 33; **[SemiAnbd]**

form: `iut_anabelioid` — see BPi

COrbicurve **C = X/{±1}** orbicurves

hyperbolic orbicurve of type (1,1)±: the quotient of the once-punctured elliptic curve by the inversion

INTRODUCED / RECALLED **[IUT I]** §1 p. 37

form: `iut_orbicurve` — see XF

CuspEpsilon **ε** cusps

a nonzero cusp of C (its label in F_l^*); ε of Def 3.1 (f) is the corresponding cusp of C_K

INTRODUCED / RECALLED **[IUT I]** §1 p. 37

`typedef struct … iut_cusps_section_one` iut_i.h

the cusps of §1: ε a nonzero cusp of C, ε′, ε″ the cusps of X over it (swapped by ι), 2ε the cusp used to define the section σ : G_k → J_C (needs 2 ≢ ±1 mod l)

field	type	meaning
`epsilon`	`int`	label of ε in F_l^* (we take 1)
`epsilon_prime`	`int`	±ε ∈ F_l: the two cusps of X over ε
`epsilon_dprime`	`int`	±ε ∈ F_l: the two cusps of X over ε
`two_epsilon`	`int`	2ε ∈ F_l, reduced to {−l*..l*}
`two_epsilon_ok`	`int`	2 ≢ ±1 (mod l), automatic for l ≥ 5
`zero_cusp`	`int`	the cusp 0 of X (fixed by ι)
`inertia_product_rank`	`unsigned`	l_ε′ × l_ε″ ≅ Ẑ² : rank 2
`delta_E_mod_l_order`	`unsigned`	|Δ_E ⊗ Z/l| = l²
`iota_swaps_inertia`	`int`	ι: l_ε′ ↔ l_ε″, −1 on Δ_E ⊗ Z/l
`iota_acts_minus_one`	`int`	ι: l_ε′ ↔ l_ε″, −1 on Δ_E ⊗ Z/l
`delta_eps_splits`	`int`	Δ_ε ≅ Δ_ε^+ × Δ_ε^−

CuspEpsilonDoublePrime **ε″** cusps

the other cusp of X over ε (label −ε); ι swaps ε′ and ε″

INTRODUCED / RECALLED **[IUT I]** §1 p. 37

form: `iut_cusps_section_one` — see CuspEpsilon

CuspEpsilonPrime **ε′** cusps

one of the two cusps of X over ε (label +ε)

INTRODUCED / RECALLED **[IUT I]** §1 p. 37

form: `iut_cusps_section_one` — see CuspEpsilon

CuspTwoEpsilon **2ε** cusps

the cusp 2ε (2 ≢ ±1 mod l) whose decomposition group gives the normal section σ : G_k → J_C

INTRODUCED / RECALLED **[IUT I]** §1 p. 37

form: `iut_cusps_section_one` — see CuspEpsilon

CuspZero **0 (cusp)** cusps

the cusp of X fixed by ι: the zero element ᵀη°_v of LabCusp±

INTRODUCED / RECALLED **[IUT I]** §1 p. 37; **[IUT I]** §6, Definition 6.1 (ii) p. 155

form: `iut_cusps_section_one` — see CuspEpsilon

DeltaE **Δ_E ⊗ Z/l** groups

the l-torsion of the geometric fundamental group of the elliptic curve (order l²); ι acts as −1

INTRODUCED / RECALLED **[IUT I]** §1 p. 37

form: `iut_cusps_section_one` — see CuspEpsilon

DeltaEpsilon **Δ_ε** groups

the geometric fundamental group of X minus the cusps ε′, ε″; Δ_ε ≅ Δ_ε^+ × Δ_ε^−

INTRODUCED / RECALLED **[IUT I]** §1 p. 37

form: `iut_cusps_section_one` — see CuspEpsilon

InertiaEpsilonPrime **I_ε′ × I_ε″** — groups

inertia groups of the cusps ε′, ε″ in Δ_ε: 0 → I_ε′ × I_ε″ → Δ_ε → Δ_E ⊗ Z/I → 0

INTRODUCED / RECALLED **[IUT I]** §1 p. 37

form: `iut_cusps_section_one` — see CuspEpsilon

JC **J_C** — groups

the quotient Π_C ↠ J_C; J_X ⊆ J_C of index 2

INTRODUCED / RECALLED **[IUT I]** §1 p. 37

form: `iut_cusps_section_one` — see CuspEpsilon

JX **J_X** — groups

the quotient Π_X ↠ J_X by Δ_ε → Δ_E ⊗ Z/I, an extension of G_k by Δ_E ⊗ Z/I

INTRODUCED / RECALLED **[IUT I]** §1 p. 37

form: `iut_cusps_section_one` — see CuspEpsilon

PiCUnder **Π_C** — groups

arithmetic fundamental group of C (Π_{C_K} over K)

INTRODUCED / RECALLED **[IUT I]** §1 p. 37; **[IUT I]** §1, Corollary 1.2 p. 39

form: `iut_cor_one_two` — see CorOneTwo

PiXUnder **Π_X** — groups

arithmetic fundamental group of X (Π_{X_K} over K)

INTRODUCED / RECALLED **[IUT I]** §1 p. 37; **[IUT I]** §1, Corollary 1.2 p. 39

form: `iut_cor_one_two` — see CorOneTwo

CArrow **C→** — orbicurves

hyperbolic orbicurve of type (1,l-tors→)±: cyclic covering of C of degree l; Aut_k(C→) ≅ Z/l; D_v = B(X→_v)^0 at good v

INTRODUCED / RECALLED **[IUT I]** §1, Definition 1.1 p. 38; **[IUT I]** §1, Remark 1.2.1 p. 40

form: `iut_orbicurve` — see XF

XArrow **X→** — orbicurves

hyperbolic orbicurve of type (1,l-tors→): cyclic covering of X of degree l determined by the section σ; Aut_k(X→) ≅ Z/2l

INTRODUCED / RECALLED **[IUT I]** §1, Definition 1.1 p. 38; **[IUT I]** §1, Remark 1.2.1 p. 40

form: `iut_orbicurve` — see XF

HSubgroup **H = Ker(Δ_X ↠ Δ_X^ab ⊗ Z/l)** — groups

the kernel used in the reconstruction Π_X = Π_X→ · H

INTRODUCED / RECALLED **[IUT I]** §1, Corollary 1.2 p. 39

form: `iut_cor_one_two` — see CorOneTwo

PiCArrow **Π_C→** — groups

arithmetic fundamental group of C→: Gal(X→/C→) ≅ Z/2 is the unique maximal subgroup of odd index of Gal(X→/C) ≅ Z/2l

INTRODUCED / RECALLED **[IUT I]** §1, Corollary 1.2 p. 39

form: `iut_cor_one_two` — see CorOneTwo

PiXArrow **Π_X→** — groups

arithmetic fundamental group of X→; from it Π_X, Π_C→, Π_C are reconstructed group-theoretically

INTRODUCED / RECALLED **[IUT I]** §1, Corollary 1.2 p. 39

form: `iut_cor_one_two` — see CorOneTwo

PiGProfinite **Π_G** — groups

profinite completion of Π^tp_G

INTRODUCED / RECALLED **[IUT I]** §2 p. 16

form: `iut_tempered_fact` — see PiTpG

form: `iut_tempered_at_v` — see PiTpG

PiTpG **Π^tp_G** — groups

tempered fundamental group of a semi-graph of anabelioids G (pro-Σ), dense in the profinite Π_G

INTRODUCED / RECALLED **[IUT I]** §2 p. 16

`typedef struct … iut_tempered_fact` — iut_i.h

========== §2 ======= tempered versus profinite fundamental groups at v ∈ V^bad [IUTchI] §2

field	type	meaning
`id`	`const char *`	
`statement`	`const char *`	
`ref`	`const char *`	
`holds`	`int`	1 in this toolkit's model (the statements are theorems)

`typedef struct … iut_tempered_at_v` — iut_i.h

Cor 2.5 applied to v ∈ V: a decomposition / inertia group of Π_{X} contained in Π^tp_{X} is a tempered decomposition / inertia group — so the conjugacy indeterminacies of tempered and profinite data are synchronized at bad v.

field	type	meaning
v_index	unsigned	
tempered_available	int	v ∈ V^bad: Π_v = Π^tp_{X_v}
profinite_conjugates_stay_tempered	int	Prop 2.1 (compact subgroups)
commensurably_terminal	int	Prop 2.2: Π^tp ⊆ Π is commensurably terminal
decomposition_groups_synchronized	int	Cor 2.5
galois_countable	int	Rem 2.5.3 (T1)–(T6)

`TemperedSubcover` **Δ^tp_{X,H}** groups

the tempered group of the covering of X determined by an open H ⊆ Π_X

INTRODUCED / RECALLED **[IUT I]** §2, Corollary 2.3 p. 47
form: `iut_tempered_fact` — see PiTpG
form: `iut_tempered_at_v` — see PiTpG

`TemperedVerticialEdge` **verticial / edge-like subgroups** groups

the subgroups of Π^tp associated to the vertices / edges of the dual graph, characterized profinitely

INTRODUCED / RECALLED **[IUT I]** §2, Proposition 2.4 p. 50
form: `iut_tempered_fact` — see PiTpG
form: `iut_tempered_at_v` — see PiTpG

`CUnderUnderV` **C_v** orbicurves

hyperbolic orbicurve of type (1,(Z/l)^Θ)± over K_v: model of C_v for the theta function

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 (b) p. 61
form: `iut_orbicurve` — see XF

`CUnderV` **C_v** orbicurves

hyperbolic orbicurve of type (1,Z/l)± over K_v, v ∈ V^bad; its cusps are labelled by |F_l|

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 (b) p. 61; **[IUT I]** §4, Example 4.4 p. 105
form: `iut_orbicurve` — see XF

`XUnderUnderK` **X_v** orbicurves

double-underline: the 'l-th power' covering at v ∈ Vbad used for the theta function

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61
form: `iut_orbicurve` — see XF

`LDeltaTheta` **l·Δ_Θ** groups

the index-l subgroup of the theta cyclotome Δ_Θ identified with M_X

INTRODUCED / RECALLED **[IUT I]** §3, Remark 3.1.2 p. 64; **[EtTh]**
form: `iut_theta_approach` — see ThetaApproach

`MXCyclotome` **M_X** groups

the cyclotome of the once-punctured elliptic curve X; M_X ≅ l·Δ_Θ

INTRODUCED / RECALLED **[IUT I]** §3, Remark 3.1.2 p. 64
form: `iut_theta_approach` — see ThetaApproach

`AutKappaSol` **Aut^κ-sol, Out^κ-sol** groups

automorphisms compatible with the κ-sol structure; the κ-sol-open subgroups are center-free

INTRODUCED / RECALLED **[IUT I]** §3, Remark 3.1.7 (iii) p. 66
form: `iut_kappa_function` — see CFmod
form: `iut_kappa_verdict` — see CFmod
form: `iut_kappa_sol` — see CFmod

`CFmod` **C_{F_mod}** orbicurves

the orbicurve C over F_mod; |C_L| ≅ A¹ has 4 critical points (images of the 2-torsion), all strictly critical

INTRODUCED / RECALLED **[IUT I]** §3, Remark 3.1.7 p. 66

`typedef struct … iut_kappa_function` iut_i.h

Rem 3.1.7: κ-coric functions on C_{F_mod}. |C_L| ≅ A¹ has 4 critical points (the images of the 2-torsion points), all strictly critical. A rational function on |C_L| is κ-coric if it has exactly one pole, at least two zeroes, its divisor is defined over an NF and avoids the critical points, and its value at every strictly critical point is a root of unity.

field	type	meaning
npoles	unsigned	
nzeros	unsigned	
divisor_over_NF	int	
divisor_avoids_critical	int	
roots_of_unity_at_strictly_critical	int	
all_roots_adjoined	int	∞κ: all N-th roots taken
constants_adjoined	int	∞κ×: multiplied by constants

`typedef struct … iut_kappa_verdict` iut_i.h

field	type	meaning
ncritical	unsigned	

field	type	meaning
nstrictly_critical	unsigned	
kappa_coric	int	
inf_kappa_coric	int	
inf_kappa_units_coric	int	
failed	const char *	

typedef struct … iut_kappa_sol — iut_i.h

Rem 3.1.7 (iii): Gal(K/F(μ_l)) ≅ SL_2(F_l) is the unique such subgroup; L_C(C_K) and L_C(κ-sol) linearly disjoint

field	type	meaning
sl2_order	unsigned long long	
k_over_F_mu_l	const char *	
linearly_disjoint	int	
center_free	int	

MuTwoL **μ_2l** — groups

the 2l-th roots of unity: the indeterminacy of q_v = q_v^{1/2l}, of Θ_v and of the splittings τ⊢_v

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 p. 69

form: iut_local_frobenioid — see DV

YDdotV **Ÿ_v** — orbicurves

the tempered covering of X_v on which the theta function lives: Θ_v ∈ O×(T÷_{Ÿ_v})

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 (ii) p. 69; **[EtTh]**

form: iut_local_frobenioid — see DV

PiCK **Π_{C_K}** — groups

arithmetic fundamental group of C_K (D_⊛)

INTRODUCED / RECALLED **[IUT I]** §4 p. 95

AutEpsilon **Aut_ε(C_K)** — groups

automorphisms of C_K fixing the cusp ε: image = semi-unipotent (∗ ∗; 0 ±1)/{±1} ⊆ GL_2(F_l)/{±1}, of order l(l−1); Aut/Aut_ε ≅ F_l^*

INTRODUCED / RECALLED **[IUT I]** §4, Example 4.3 (i) p. 98

form: iut_labcusp_torsor — see LabCusp

form: iut_global_base — see LabCusp

form: iut_valuation_orbit — see LabCusp

AutSL **Aut^SL, Aut^SL_ε** — groups

the automorphisms arising from SL_2(F_l); Aut^SL_ε ⊆ Aut^SL ⊆ Aut(†D^⊛) with quotient ≅ F_l^*

INTRODUCED / RECALLED **[IUT I]** §4, Example 4.3 p. 98; **[IUT I]** §5, Example 5.1 (i) p. 123

form: iut_labcusp_torsor — see LabCusp

form: iut_global_base — see LabCusp

form: iut_valuation_orbit — see LabCusp

BorelSubgroup **Borel (∗ ∗; 0 ∗)** — groups

the image of Aut(C_K) in GL_2(F_l)/{±1}: order l(l−1)²/2

INTRODUCED / RECALLED **[IUT I]** §4, Example 4.3 (i) p. 98

form: iut_labcusp_torsor — see LabCusp

form: iut_global_base — see LabCusp

form: iut_valuation_orbit — see LabCusp

SemiUnipotentSubgroup **semi-unipotent (∗ ∗; 0 ±1)** — groups

the image of Aut_ε(C_K) in GL_2(F_l)/{±1}: order l(l−1)

INTRODUCED / RECALLED **[IUT I]** §4, Example 4.3 (i) p. 98

form: iut_labcusp_torsor — see LabCusp

form: iut_global_base — see LabCusp

form: iut_valuation_orbit — see LabCusp

FmodUnits **F_mod^×** — groups

copies (F_mod^×)_j of the multiplicative group of F_mod

INTRODUCED / RECALLED **[IUT I]** §5 p. 123; **[IUT III]** §3, Theorem 3.11 (c) p. 153

PiRat **π_1^rat(†D^⊛), π_1^κ-sol, π_1^{rat/κ-sol}** — groups

the fundamental groups of the generic point / κ-sol versions from which the pseudo-monoids are reconstructed

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (i) p. 123; **[IUT I]** §5, Definition 5.2 (v) p. 134

form: iut_global_frobenioid — see FCircAstGlobal

form: iut_distribution — see FCircAstGlobal

form: iut_strip_functor — see FCircAstGlobal

PiXK **Π_{X_K}** — groups

arithmetic fundamental group of X_K (D_⊚)

INTRODUCED / RECALLED **[IUT I]** §6 p. 7

`XDdotK` **Ẍ_K** orbicurves

the ±-synchronized (doubly dotted) orbicurve of type (1,l-tors^Θ)_±

INTRODUCED / RECALLED **[IUT I]** §6 p. 7

`AutAlpha` **Aut_α(†D)** groups

for α ∈ {±1}^V: automorphisms with sign α_v at each v; α-signed +-full poly-isomorphisms

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (iv) p. 155

form: `iut_fl_semi_pm_set` — see FlSemiPmGroup

form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup

form: `iut_sign_vector` — see FlSemiPmGroup

form: `iut_prop_six_seven` — see FlSemiPmGroup

`AutCsp` **Aut_csp(D^⊙±)** groups

the automorphisms fixing the cusps; Aut_csp ⊆ Aut_±; V^± = Aut_csp·V

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (v) p. 155

form: `iut_fl_semi_pm_set` — see FlSemiPmGroup

form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup

form: `iut_sign_vector` — see FlSemiPmGroup

form: `iut_prop_six_seven` — see FlSemiPmGroup

`AutMinus` **Aut_−(†D_v)** groups

the complement Aut(†D_v) ∖ Aut_+(†D_v): the sign-reversing automorphisms

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (iii) p. 155

form: `iut_fl_semi_pm_set` — see FlSemiPmGroup

form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup

form: `iut_sign_vector` — see FlSemiPmGroup

form: `iut_prop_six_seven` — see FlSemiPmGroup

`AutPlus` **Aut_+(†D_v), Aut_+(†D)** groups

the kernel of Aut(†D_v) ↠ {±1} (automorphisms preserving the sign of LabCusp±); Aut_+(†D) = ∏_v Aut_+

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (iii)–(iv) p. 155

form: `iut_fl_semi_pm_set` — see FlSemiPmGroup

form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup

form: `iut_sign_vector` — see FlSemiPmGroup

form: `iut_prop_six_seven` — see FlSemiPmGroup

`AutPm` **Aut_±(D^⊙±)** groups

the kernel of Aut(D^⊙±) ↠ F_l^*; Aut_±/Aut_csp ≅ Aut_K(X_K) ≅ F_l^{⋊±}

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (v) p. 155

form: `iut_fl_semi_pm_set` — see FlSemiPmGroup

form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup

form: `iut_sign_vector` — see FlSemiPmGroup

form: `iut_prop_six_seven` — see FlSemiPmGroup

`Ism` **Ism** groups

isometries of O×μ (the Ind2 indeterminacy group)

INTRODUCED / RECALLED **[IUT II]** §1, Example 1.8 (iv) p. 35

`ZHatUnits` **Ẑ×** groups

profinite units acting on Kummer classes (Ind2, ×-Kummer)

INTRODUCED / RECALLED **[IUT II]** §1, Remark 1.11.1 p. 49

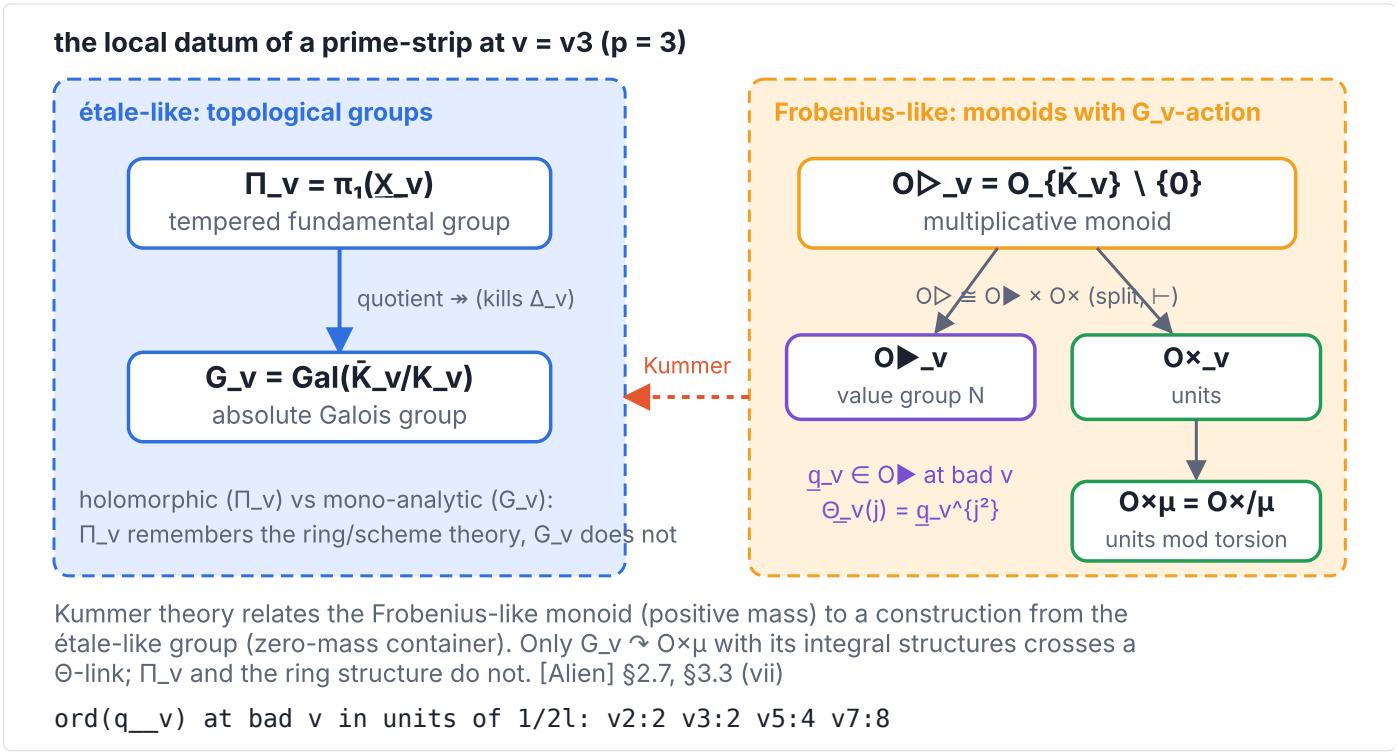
`PiTpV` **Π^tp_v** groups

tempered fundamental group at v ∈ V^bad

INTRODUCED / RECALLED **[EtTh]** §1 p. 10

5 · Monoids, morphisms, categories, Frobenioids and maps

The Frobenius-like side: the local monoids O▷ = O× × O▶, O×μ = O×/Oμ at each v, the pseudo-monoids M^⊛ of global rational functions with the Kummer-ready / Kummer-blind distinction, the poly-morphisms (collections of morphisms, full when every isomorphism is included), the base categories D_v, D^⊛, D^⊛± and the Frobenioids F_v, C_v, C⊢_v, C^Θ_v, C⊩_mod, F^⊛ with the maps ρ_v, φ, χ, ζ between them.



The local datum at v. Inside each prime-strip, at each v: the étale-like groups Π_v ↠ G_v (zero mass) and the Frobenius-like monoid O▷ ≅ O▶ × O× (positive mass), related by Kummer theory. q_v sits in the value group O▶; the units mod torsion O×μ are what survives a Θ-link. **[IUT I]** Def 3.1 p. 61, Def 4.1 p. 95; **[IUT II]** Def 4.9 p. 153; **[Alien]** §2.7 p. 21, §3.3 (vii) p. 58
`PiV` `GV` `OValV` `OUnitsV` `OUnitsModTorsV` `OTorsV` `QUnderV` `ThetaUnderV`

74 entries: 26 monoids, 10 morphisms, 8 categories, 17 Frobenioids, 13 maps.

`OValV` **O▷_{F̄_v}** — monoids

multiplicative monoid of nonzero integers of F̄_v

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 p. 69
form: `iut_local_datum` — see BasicPrimeStrip

`OUnitsV` **O×** — monoids

unit group

INTRODUCED / RECALLED **[IUT II]** §4, Definition 4.9 p. 153
form: `iut_local_datum` — see BasicPrimeStrip

`OUnitsModTorsV` **O×μ** — monoids

units modulo torsion

INTRODUCED / RECALLED **[IUT II]** §4, Definition 4.9 p. 153
form: `iut_local_datum` — see BasicPrimeStrip

`OTorsV` **Oμ** — monoids

roots of unity

INTRODUCED / RECALLED **[IUT II]** §4, Definition 4.9 p. 153
form: `iut_local_datum` — see BasicPrimeStrip

`PseudoMonoid` **pseudo-monoid** — monoids

topological space with a partially defined multiplication; divisible if every element has N-th roots; cyclotomic if it contains μ_∞ ≅ Q/Z

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

`typedef struct … iut_pseudo_monoid` — iut_i.h

field	type	meaning
kind	iut_pseudo_monoid_kind	
id	const char *	
notation	const char *	
ref	const char *	
topological	int	

field	type	meaning
divisible	int	
cyclotomic	int	
is_group	int	
kummer_ready	int	Rem 5.1.3: Kummer-ready (1) or Kummer-blind (0)
etale_like	int	reconstructed from †D^⊛ / ‡D_v (1) or Frobenius-like

`PolyMorphism` **poly-morphism** morphisms

a set of morphisms A → B; composites are sets of composites, multiplicities ignored

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

`typedef struct … iut_poly_morphism` iut_i.h

field	type	meaning
kind	iut_poly_kind	
count	unsigned long long	number of constituent morphisms (0 = "all", i.e. full)
nV	unsigned	+-full only: signs α ∈ {±1}^V
sign[IUT_MAX_V]	int	
index_bijective	int	capsule kinds: index map bijective
ref	const char *	

`FullPolyIsomorphism` **full poly-isomorphism** morphisms

the set of ALL isomorphisms A ≅ B (the Θ-link, the Θ-bridge at good v, the D-links are of this kind)

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

form: `iut_poly_morphism` — see PolyMorphism

`CapsuleFullPolyIsomorphism` **capsule-full poly-isomorphism** morphisms

capsule-full poly-morphism with bijective index map

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

form: `iut_poly_morphism` — see PolyMorphism

`CapsuleFullPolyMorphism` **capsule-full poly-morphism** morphisms

morphism of capsules whose constituent poly-morphisms are full poly-isomorphisms

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

form: `iut_poly_morphism` — see PolyMorphism

`PolyAutomorphism` **poly-automorphism** morphisms

a poly-isomorphism of an object with itself, e.g. −1_{F_l} of Ex 6.2

INTRODUCED / RECALLED **[IUT I]** §0 p. 33; **[IUT I]** §6, Example 6.2 (ii) p. 159

form: `iut_poly_morphism` — see PolyMorphism

`PolyIsomorphism` **poly-isomorphism** morphisms

a poly-morphism all of whose constituents are isomorphisms

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

form: `iut_poly_morphism` — see PolyMorphism

`SectionSigma` **σ : G_k → J_C** maps

the normal section determined by the cusp 2ε; Im(σ) ⊆ J_X defines X→ and C→

INTRODUCED / RECALLED **[IUT I]** §1 p. 37

form: `iut_cusps_section_one` — see CuspEpsilon

`CMonoV` **C⊢_v** Frobenioids

the mono-analytic Frobenioid with divisor monoid Φ_{C⊢v} = N·log_Φ(q_v) / N·log(p_v) / R≥0·log(p_v)

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 (iv) p. 69; **[IUT I]** §3, Example 3.3 (ii) p. 77; **[IUT I]** §3, Example 3.4 (iii) p. 80

form: `iut_local_frobenioid` — see DV

`CThetaV` **C^Θ_v** Frobenioids

the Θ-version of C⊢_v (divisor monoid N·log_Φ(q_v)·log(Θ))

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 (v) p. 69

form: `iut_local_frobenioid` — see DV

`CV` **C_v** Frobenioids

the base-field-theoretic hull of F_v (bad); the p_v-adic Frobenioid (good); the archimedean Frobenioid with O▷(C_v) ≅ O▷_{K̃_v}

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 (iii) p. 69; **[IUT I]** §3, Example 3.3 p. 77; **[IUT I]** §3, Example 3.4 p. 80

form: `iut_local_frobenioid` — see DV

`DMonoV` **D⊢_v** categories

the mono-analytic base: B(K_v)^0 ⊆ D_v (nonarch), an object of TM (arch)

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 –3.4 p. 69

form: `iut_local_frobenioid` — see DV

DThetaV **D^Θ_v** — categories

the Θ-version of D⊢_v at v ∈ V^bad

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 (v) p. 69

form: `iut_local_frobenioid` — see DV

DV **D_v** — categories

the local base category: B^temp(X_v)^0 (bad), B(X→_v)^0 (good nonarch), the Aut-holomorphic orbispace X→_v (arch)

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 –3.4 p. 69

`typedef struct … iut_local_frobenioid` — iut_i.h

field	type	meaning
`kind`	`iut_local_frobenioid_kind`	
`v_index`	`unsigned`	
`ref`	`const char *`	
base categories		
`D_v`	`const char *`	B^temp(X_v)^0 / B(X→_v)^0 / the Aut-holomorphic orbispace X→_v
`D_mono_v`	`const char *`	D⊢_v = B(K_v)^0 / B(K_v)^0 / an object of TM
`D_theta_v`	`const char *`	D^Θ_v (bad v only)
Frobenioids		
`F_v`	`const char *`	F_v tempered / C_v = F_v / (C_v, D_v, κ_v)
`F_birat_v`	`const char *`	F÷_v birationalization (bad)
`C_v`	`const char *`	base-field-theoretic hull
`C_mono_v`	`const char *`	C⊢_v
`C_theta_v`	`const char *`	C^Θ_v (bad)
`divisor_monoid`	`const char *`	Φ_{C⊢v}: N·log_Φ(q_v) / ord(Z_p) = N·log(p_v) / R≥0·log(p_v)
`divisor_monoid_theta`	`const char *`	Θ-version: … · log(Θ)
`generator_value`	`iut_q`	value of the generator in q_v-units: q_v = 1; p_v: e_K (ord of p in q units: 2l·e_K/ord q)…
`generator`	`const char *`	"q_v", "p_v", "p_v (R≥0)"
`splitting`	`const char *`	τ⊢_v: μ_2l-orbit (bad) / canonical from p_v (good) / C× = S^1 × R>0 (arch)
`splitting_indeterminacy`	`unsigned`	|μ_2l| = 2l at bad v, 1 otherwise
`theta_function`	`const char *`	Θ_v ∈ O×(T÷_v̈v) up to μ_2l and l·Z (bad)
`theta_root_indeterminacy`	`unsigned`	2l (μ_2l)
`theta_translation_indeterminacy`	`int`	l·Z (bad)
`is_tempered`	`int`	
`is_split`	`int`	
`has_kummer_structure`	`int`	
`is_birational_available`	`int`	
`formal_symbol`	`const char *`	"log(Θ)" / "log(p_v)·log(Θ)"

FBiratV **F÷_v** — Frobenioids

the birationalization of F_v

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 (ii) p. 69

form: `iut_local_frobenioid` — see DV

FMonoV **F⊢_v = (C⊢_v, τ⊢_v)** — Frobenioids

the split Frobenioid at v: the local datum of an F⊢-prime-strip

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 (v) p. 69; **[IUT I]** §5, Definition 5.2 (ii) p. 134

form: `iut_local_frobenioid` — see DV

FThetaV **F^Θ_v = (C^Θ_v, τ^Θ_v)** — Frobenioids

the Θ-version split Frobenioid; F⊢_v ≅ F^Θ_v

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 (v) p. 69

form: `iut_local_frobenioid` — see DV

FUnderV **F_v** — Frobenioids

the tempered Frobenioid over D_v at v ∈ V^bad (at other v: F_v = C_v, resp. the triple (C_v, D_v, κ_v))

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 p. 69; **[IUT I]** §3, Definition 3.6 p. 87

form: `iut_local_frobenioid` — see DV

LogPhi **log_Φ** — maps

the formal logarithm turning a generator of a divisor monoid into a real/formal number: Φ_{C⊢v} = N·log_Φ(q_v)

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 (iv) p. 69

form: `iut_local_frobenioid` — see DV

PhiCMonoV **Φ_{C⊢v}** — monoids

the divisor monoid of C⊢_v: N·log_Φ(q_v) at bad v, ord(Z_{p_v}) at good v, R≥0·log(p_v) at arch v

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 (iv) p. 69

form: `iut_local_frobenioid` — see DV

`TMCategory` **TM** categories

the category of split topological monoids C ≅ O▷_C with C× ≅ S¹, C→ ≅ R≥0; D⊩_v ∈ TM at v ∈ V^arc

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.4 (ii) p. 80

form: `iut_local_frobenioid` — see DV

`CMonoGlobMod` **C⊩_mod** Frobenioids

the realification of the Frobenioid of arithmetic divisors of F_mod; Prime(C⊩_mod) ≅ V_mod

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.5 (i) p. 84

`typedef struct … iut_global_realified` iut_i.h

field	type	meaning
`kind`	`iut_global_realified_kind`	
`id`	`const char *`	
`notation`	`const char *`	
`ref`	`const char *`	
`nprimes`	`unsigned`	Prime(C⊩_mod) ≅ V_mod (= |V|)
`rho[IUT_MAX_V]`	`double`	ρ_v: log_mod(p_v) ↦ (1/[K_v : (F_mod)_v])·log_Φ(p_v)
`local_degree[IUT_MAX_V]`	`unsigned`	[K_v : (F_mod)_v] = e_K f_K / (e_mod f_mod)
`frobenius_element[IUT_MAX_V]`	`double`	D-version: ×e_v (nonarch) or ÷2π (arch) on (R≥0)_v
`with_theta`	`int`	ρ^Θ_v at bad v: log(Θ) ↦ log_Φ(Θ_v)/log_Φ(q_v)

`typedef struct … iut_f_monoglob` iut_i.h

Ex 3.5 (iii): F⊩_mod = (C⊩_mod, Prime ≅ V, {F⊩_v}, {ρ_v}), F⊩_tht ≅ F⊩_mod; (iv) F⊩_D from {D⊩_v}

field	type	meaning
`id`	`const char *`	
`notation`	`const char *`	
`global`	`iut_global_realified`	
`nV`	`unsigned`	
`loc[IUT_MAX_V]`	`iut_local_frobenioid`	
`theta_version`	`int`	

`typedef struct … iut_theta_hodge_theater_def` iut_i.h

Def 3.6: a Θ-Hodge theater ᵀHT^Θ = ({ᵀF_v}_{v∈V}, ᵀF⊩_mod) with (a) F_v ≅ model, (b) F⊩_mod ≅ model, (c) compatibility

field	type	meaning
`nV`	`unsigned`	
`loc[IUT_MAX_V]`	`iut_local_frobenioid`	
`mod`	`iut_f_monoglob`	
`tht`	`iut_f_monoglob`	
`cond_a`	`int`	
`cond_b`	`int`	
`cond_c`	`int`	

`CMonoGlobTht` **C⊩_tht** Frobenioids

the Θ-version of C⊩_mod with generators log_mod(p_v)·log(Θ)

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.5 (ii) p. 84

form: `iut_global_realified` — see CMonoGlobMod

form: `iut_f_monoglob` — see CMonoGlobMod

form: `iut_theta_hodge_theater_def` — see CMonoGlobMod

`DMonoGlobMod` **D⊩_mod** Frobenioids

the étale-like (D-)version reconstructed from {D⊩_v}: (R≥0)_v with Frobenius element ×e_v (nonarch) or ÷2π (arch)

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.5 (iv) p. 84

form: `iut_global_realified` — see CMonoGlobMod

form: `iut_f_monoglob` — see CMonoGlobMod

form: `iut_theta_hodge_theater_def` — see CMonoGlobMod

`RhoDV` **ρ^D_v** maps

the D-version gluing Φ_{D⊩mod,v} ≅ (R≥0)_v

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.5 (iv) p. 84

form: `iut_global_realified` — see CMonoGlobMod

form: `iut_f_monoglob` — see CMonoGlobMod

form: `iut_theta_hodge_theater_def` — see CMonoGlobMod

`RhoThetaV` **ρ^Θ_v** maps

log_mod(p_v)·log(Θ) ↦ (log_Φ(p_v)/[K_v:(F_mod)_v])·(log_Φ(Θ_v)/log_Φ(q_v)) at v ∈ V^bad

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.5 (ii) p. 84

form: `iut_global_realified` — see CMonoGlobMod

form: `iut_f_monoglob` — see CMonoGlobMod

form: `iut_theta_hodge_theater_def` — see CMonoGlobMod

DCircAstGlobal **D^⊛ = B(C̲_K)^0** — categories

the global base category of the NF-bridge target; Aut(D^⊛) ≅ Aut(C̲_K) ≅ Out(Π_{C̲_K})

INTRODUCED / RECALLED **[IUT I]** §4, Definition 4.1 (v) p. 95
form: `iut_labcusp_torsor` — see LabCusp
form: `iut_global_base` — see LabCusp
form: `iut_valuation_orbit` — see LabCusp

PhiNFBulletV **φ^NF_{•,v}, φ^NF_v** — morphisms

the model NF-bridge constituents D_v → D^⊛, φ^NF_v = β ∘ φ_{•,v} ∘ α with α ∈ Aut(D_v), β ∈ Aut_ε(D^⊛)

INTRODUCED / RECALLED **[IUT I]** §4, Example 4.3 (ii)–(iii) p. 98
form: `iut_labcusp_torsor` — see LabCusp
form: `iut_global_base` — see LabCusp
form: `iut_valuation_orbit` — see LabCusp

EvaluationSection **evaluation section G_v → Π_v** — maps

the decomposition-group section labelled by j ∈ |F_l|, group-theoretically constructible from Π_v

INTRODUCED / RECALLED **[IUT I]** §4, Example 4.4 (ii) p. 105
form: `iut_eval_points` — see MuMinus
form: `iut_theta_bridge_piece` — see MuMinus
form: `iut_label_transport` — see MuMinus
form: `iut_iso_torsor` — see MuMinus
form: `iut_forgetful_counts` — see MuMinus

PhiThetaVJ **φ^Θ_{v_j}** — morphisms

the Θ-bridge constituent D_{v_j} → D_{>,v}: evaluation section labelled |j| at bad v, full poly-iso at good v

INTRODUCED / RECALLED **[IUT I]** §4, Example 4.4 (iii) p. 105
form: `iut_eval_points` — see MuMinus
form: `iut_theta_bridge_piece` — see MuMinus
form: `iut_label_transport` — see MuMinus
form: `iut_iso_torsor` — see MuMinus
form: `iut_forgetful_counts` — see MuMinus

PhiLC **φ^LC_j** — maps

LabCusp(D^⊛) ≅ LabCusp(D_>) transporting [ε] ↦ j; φ^LC_j = φ^LC_1 ∘ j

INTRODUCED / RECALLED **[IUT I]** §4, Example 4.5 p. 107; **[IUT I]** §4, Proposition 4.7 (ii) p. 112
form: `iut_eval_points` — see MuMinus
form: `iut_theta_bridge_piece` — see MuMinus
form: `iut_label_transport` — see MuMinus
form: `iut_iso_torsor` — see MuMinus
form: `iut_forgetful_counts` — see MuMinus

ChiJ **†χ : J ≅ F_l^*** — maps

the bijection of the capsule index set with F_l^* determined by a D-ΘNF-Hodge theater

INTRODUCED / RECALLED **[IUT I]** §4, Proposition 4.7 (i) p. 112
form: `iut_eval_points` — see MuMinus
form: `iut_theta_bridge_piece` — see MuMinus
form: `iut_label_transport` — see MuMinus
form: `iut_iso_torsor` — see MuMinus
form: `iut_forgetful_counts` — see MuMinus

ZetaLabel **†ζ : LabCusp(†D^⊛) ≅ J** — maps

the label synchronization of the ΘNF side

INTRODUCED / RECALLED **[IUT I]** §4, Proposition 4.7 (iii) p. 112; **[IUT I]** §6, Remark 6.12.4 p. 177
form: `iut_eval_points` — see MuMinus
form: `iut_theta_bridge_piece` — see MuMinus
form: `iut_label_transport` — see MuMinus
form: `iut_iso_torsor` — see MuMinus
form: `iut_forgetful_counts` — see MuMinus

DCircAstMod **†D^⊛_mod** — categories

the C_{F_mod}-version of †D^⊛ (†D^⊛ → †D^⊛_mod, π_1 open)

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (i) p. 123
form: `iut_global_frobenioid` — see FCircAstGlobal
form: `iut_distribution` — see FCircAstGlobal
form: `iut_strip_functor` — see FCircAstGlobal

FCircAstAngleJ **†F^⊛_⟨J⟩** — Frobenioids

the diagonal copy over V_⟨J⟩ (constant distribution); log-volumes are averages (÷ l*)

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (vii) p. 123; **[IUT I]** §5, Remark 5.4.2 p. 151
form: `iut_global_frobenioid` — see FCircAstGlobal
form: `iut_distribution` — see FCircAstGlobal
form: `iut_strip_functor` — see FCircAstGlobal

FCircAstGlobal **F^⊛(†D^⊛), †F^⊛** — Frobenioids

the global Frobenioid over †D^⊛ (model / Frobenius-like); Kummer-ready

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (ii)–(iii) p. 123

`typedef struct … iut_global_frobenioid` — iut_i.h

field	type	meaning
`kind`	`iut_global_frobenioid_kind`	
`id`	`const char *`	
`notation`	`const char *`	
`base`	`const char *`	
`ref`	`const char *`	
`kummer_ready`	`int`	Rem 5.1.3
`realified_available`	`int`	^R
`ncopies`	`unsigned`	1, l* (F_J), …
`log_volume_weight`	`double`	Rem 5.4.2: F_(J) averages (1/l*), F_J sums (1)

`typedef struct … iut_distribution` — iut_i.h

Ex 5.1 (vii): V_j = {v_j}, V_J = ∏_j V_j, the diagonal V_(J) ↪ V_J

field	type	meaning
`nj`	`unsigned`	
`nV`	`unsigned`	
`total`	`unsigned long long`	
`diagonal`	`int`	
`notation`	`const char *`	

`typedef struct … iut_strip_functor` — iut_i.h

Def 5.2, Rem 5.2.1: the functorial algorithms F ↦ D, F ↦ F⊩ (mono-analyticization), F ↦ F⊩, and Cor 5.3: Isom(F-strips) → Isom(D-strips) bijective, F⊩ → D⊩ surjective (not injective)

field	type	meaning
`from`	`const char *`	
`to`	`const char *`	
`name`	`const char *`	
`bijective_on_isoms`	`int`	
`surjective_on_isoms`	`int`	
`ref`	`const char *`	

FCircAstJ **†F^⊛_j** — Frobenioids

copy of †F^⊛_mod for j ∈ F_l^*, living over V_j

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (vii) p. 123

form: `iut_global_frobenioid` — see FCircAstGlobal

form: `iut_distribution` — see FCircAstGlobal

form: `iut_strip_functor` — see FCircAstGlobal

FCircAstJCapsule **†F^⊛_J = ∏_j †F^⊛_j** — Frobenioids

the product over F_l^* (arbitrary distribution); †F^⊛_(J) ↪ †F^⊛_J

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (vii) p. 123

form: `iut_global_frobenioid` — see FCircAstGlobal

form: `iut_distribution` — see FCircAstGlobal

form: `iut_strip_functor` — see FCircAstGlobal

FCircAstMod **†F^⊛_mod** — Frobenioids

restriction of †F^⊛′ to the terminal objects: the Frobenioid of F_mod; Kummer-blind; Prime(†F^⊛_mod) ≅ V_mod

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (iii) p. 123

form: `iut_global_frobenioid` — see FCircAstGlobal

form: `iut_distribution` — see FCircAstGlobal

form: `iut_strip_functor` — see FCircAstGlobal

FCircAstPrime **†F^⊛′** — Frobenioids

the Frobenioid over †D^⊛_mod with †F^⊛ = †F^⊛′|_{†D^⊛}; Kummer-ready

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (iii) p. 123

form: `iut_global_frobenioid` — see FCircAstGlobal

form: `iut_distribution` — see FCircAstGlobal

form: `iut_strip_functor` — see FCircAstGlobal

MCircAst **M^⊛(†D^⊛)** — monoids

isomorph of F̄× with its field structure, reconstructed from †D^⊛; Kummer-ready

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (i) p. 123

form: `iut_pseudo_monoid` — see PseudoMonoid

`MCircAstInfKappa` **M^⊛_∞κ** — monoids

pseudo-monoid of ∞κ-coric functions (divisible, cyclotomic); Kummer-ready

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (i) p. 123

form: `iut_pseudo_monoid` — see PseudoMonoid

`MCircAstInfKappaUnits` **M^⊛_∞κ×** — monoids

pseudo-monoid of ∞κ×-coric functions; Kummer-ready; its ∞κ×-coric structure on †F^⊛ is unique (Ex 5.1 (v))

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (i) p. 123

form: `iut_pseudo_monoid` — see PseudoMonoid

`MCircAstKappa` **M^⊛_κ** — monoids

pseudo-monoid of κ-coric rational functions; Kummer-blind

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (i) p. 123

form: `iut_pseudo_monoid` — see PseudoMonoid

`MCircAstKappaSol` **M^⊛_κ-sol** — monoids

the κ-sol version; Kummer-ready; used for κ-sol-conjugate synchronization

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (i) p. 123

form: `iut_pseudo_monoid` — see PseudoMonoid

`MCircAstMod` **M^⊛_mod** — monoids

isomorph of F_mod^× inside M^⊛; Kummer-blind

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (i) p. 123

form: `iut_pseudo_monoid` — see PseudoMonoid

`MCircAstSol` **M^⊛_sol** — monoids

isomorph of F_sol^× (maximal solvable extension) inside M^⊛; Kummer-ready

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (i) p. 123

form: `iut_pseudo_monoid` — see PseudoMonoid

`PhiCircAst` **Φ^⊛(†D^⊛)** — monoids

the monoid of arithmetic divisors reconstructed from †D^⊛; defines the model Frobenioid F^⊛(†D^⊛)

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.1 (ii) p. 123

form: `iut_global_frobenioid` — see FCircAstGlobal

form: `iut_distribution` — see FCircAstGlobal

form: `iut_strip_functor` — see FCircAstGlobal

`CMonoGlob` **C⊩** — Frobenioids

global realified Frobenioid

INTRODUCED / RECALLED **[IUT I]** §5, Definition 5.2 p. 134

`MInfKappaUnitsV` **M_∞κ×v** — monoids

the local ∞κ×-coric pseudo-monoid; the ∞κ-coric structures on ‡F_v are unique

INTRODUCED / RECALLED **[IUT I]** §5, Definition 5.2 (vi) p. 134

form: `iut_pseudo_monoid` — see PseudoMonoid

`MInfKappaV` **M_∞κv** — monoids

the local ∞κ-coric pseudo-monoid (divisible, cyclotomic); Kummer-ready

INTRODUCED / RECALLED **[IUT I]** §5, Definition 5.2 (vi) p. 134

form: `iut_pseudo_monoid` — see PseudoMonoid

`MKappaV` **M_κv** — monoids

the local κ-coric pseudo-monoid; Kummer-blind

INTRODUCED / RECALLED **[IUT I]** §5, Definition 5.2 (vi) p. 134

form: `iut_pseudo_monoid` — see PseudoMonoid

`MV` **M_v(‡D_v)** — monoids

the local pseudo-monoid ≅ O▷_{F̃_v} reconstructed from ‡D_v (via π_1^rat(‡D_v)); Kummer-ready

INTRODUCED / RECALLED **[IUT I]** §5, Definition 5.2 (v) p. 134

form: `iut_pseudo_monoid` — see PseudoMonoid

`RhoV` **ρ_v** — maps

isomorphisms of topological monoids gluing local and global portions

INTRODUCED / RECALLED **[IUT I]** §5, Definition 5.2 p. 134

`FDCircAst` **F^⊛(†D^⊛) ↦ †D^⊛** — maps

Isom(†F^⊛, ‡F^⊛) → Isom(†D^⊛, ‡D^⊛) is bijective

INTRODUCED / RECALLED **[IUT I]** §5, Corollary 5.3 (i) p. 143

form: `iut_global_frobenioid` — see FCircAstGlobal

form: `iut_distribution` — see FCircAstGlobal

form: `iut_strip_functor` — see FCircAstGlobal

`StripLifting` **F ↦ D lifting** — maps

Isom(F-strips) → Isom(D-strips) bijective; Isom(F⊢) → Isom(D⊢) surjective; Aut(F_v) → Aut(D_v) bijective at bad v

INTRODUCED / RECALLED **[IUT I]** §5, Corollary 5.3 (ii)–(iv) p. 143; **[IUT I]** §5, Remark 5.3.1 p. 145

form: `iut_global_frobenioid` — see FCircAstGlobal

form: `iut_distribution` — see FCircAstGlobal

form: `iut_strip_functor` — see FCircAstGlobal

`DCircDotPmGlobal` **D^⊙± = B(X_K)^0** — categories

the global base category of the Θell-bridge target, a double cover of D^⊛; Aut(D^⊙±) ↠ F_l^* with kernel Aut_±

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (v) p. 155

form: `iut_fl_semi_pm_set` — see FISemiPmGroup

form: `iut_labcusp_pm_torsor` — see FISemiPmGroup

form: `iut_sign_vector` — see FISemiPmGroup

form: `iut_prop_six_seven` — see FISemiPmGroup

`DPmV` **†D^±_v** — categories

the X_v-version of †D_v (double cover of the C_v-version), from which LabCusp± is defined

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (ii) p. 155

form: `iut_fl_semi_pm_set` — see FISemiPmGroup

form: `iut_labcusp_pm_torsor` — see FISemiPmGroup

form: `iut_sign_vector` — see FISemiPmGroup

form: `iut_prop_six_seven` — see FISemiPmGroup

`PlusFullPolyIsomorphism` **+-full poly-isomorphism** — morphisms

between D-prime-strips: at each v the Aut_+-orbit (positive) or Aut_−-orbit (negative) of an isomorphism; α-signed for α ∈ {±1}^V

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (iv) p. 155

form: `iut_poly_morphism` — see PolyMorphism

`PhiThetaEllBulletV` **φ^Θell_{•,v}, φ^Θell_{v_t}** — morphisms

the model Θell-bridge constituents D_v → D^⊙±; φ^Θell_{v_0} = β ∘ φ ∘ α with α ∈ Aut_+(D_{v_0}), β ∈ Aut_csp; φ^Θell_t via the poly-action of t

INTRODUCED / RECALLED **[IUT I]** §6, Example 6.3 p. 160

form: `iut_fl_semi_pm_set` — see FISemiPmGroup

form: `iut_labcusp_pm_torsor` — see FISemiPmGroup

form: `iut_sign_vector` — see FISemiPmGroup

form: `iut_prop_six_seven` — see FISemiPmGroup

`XiSync` **†ξ** — maps

the global ±-synchronization of the signs of LabCusp±(†D_t) across t and v

INTRODUCED / RECALLED **[IUT I]** §6, Proposition 6.5 (ii) p. 163; **[IUT I]** §6, Remark 6.12.4 p. 177

form: `iut_fl_semi_pm_set` — see FISemiPmGroup

form: `iut_labcusp_pm_torsor` — see FISemiPmGroup

form: `iut_sign_vector` — see FISemiPmGroup

form: `iut_prop_six_seven` — see FISemiPmGroup

`ZetaPm` **†ζ_±, †ζ^Θell_{v_t}** — maps

the ±-label synchronizations: LabCusp±(†D^⊙±) ≅ T and the local versions at v_t

INTRODUCED / RECALLED **[IUT I]** §6, Proposition 6.5 p. 163

form: `iut_fl_semi_pm_set` — see FISemiPmGroup

form: `iut_labcusp_pm_torsor` — see FISemiPmGroup

form: `iut_sign_vector` — see FISemiPmGroup

form: `iut_prop_six_seven` — see FISemiPmGroup

`PsiTheta` **Ψ_Θ** — monoids

theta monoid (mono-theta environment)

INTRODUCED / RECALLED **[IUT II]** §2, Corollary 2.8 p. 81

`PsiGau` **Ψ_gau** — monoids

Gaussian monoid {q^{j²}}

INTRODUCED / RECALLED **[IUT II]** §3, Corollary 3.6 p. 99

`OValgroupV` **O▶** — monoids

value group portion O▷/O×

INTRODUCED / RECALLED **[IUT II]** §4, Definition 4.9 p. 153

form: `iut_local_datum` — see BasicPrimeStrip

`PsiPerpEnvInf` **Ψ^⊥_{∞env}** — monoids

'perp'-infinite theta-environment monoid

INTRODUCED / RECALLED **[IUT II]** §4, Corollary 4.10 p. 158

`PsiPerpGauInf` **Ψ^⊥_{∞gau}** — monoids

'perp'-infinite Gaussian monoid

INTRODUCED / RECALLED **[IUT II]** §4, Corollary 4.10 p. 158

`PsiLGP` **Ψ_LGP** monoids

log-Gaussian procession monoid

INTRODUCED / RECALLED **[IUT III]** §3, Proposition 3.4 p. 101

`ModIdeal` **F⊩_mod / M^mod** monoids

global Frobenioid via local fractional ideals ('mod', lower-case)

INTRODUCED / RECALLED **[IUT III]** §3, Definition 3.8 p. 112

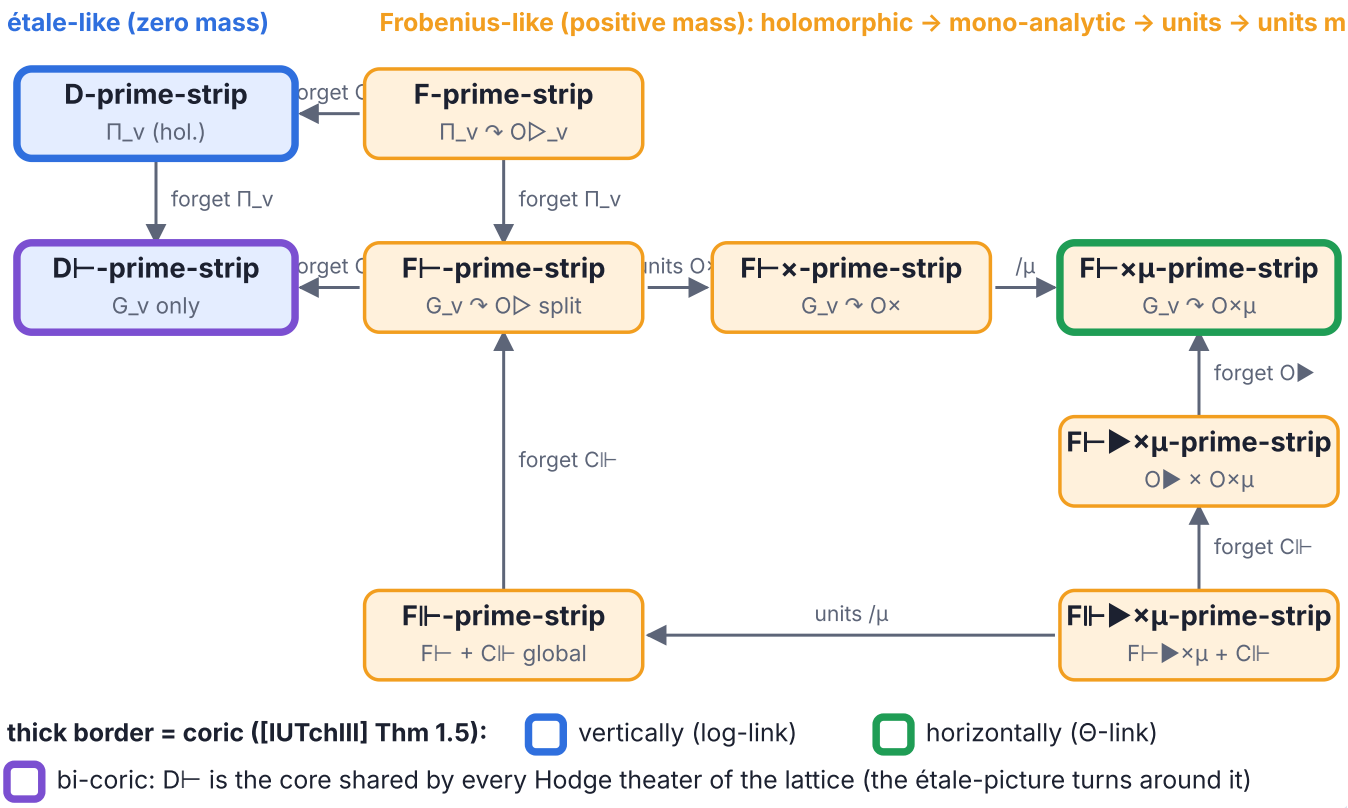
`MODTorsor` **F⊩_MOD / M^MOD** monoids

global Frobenioid of F_mod^×-torsors of arithmetic line bundles (multiplicative, 'MOD')

INTRODUCED / RECALLED **[IUT III]** §3, Definition 3.8 p. 112

6 · Prime-strips

A prime-strip is a collection of local data indexed by v ∈ V ([IUTchI] Fig I1.2). The nine kinds form a forgetful ladder from the holomorphic Frobenius-like F-prime-strip down to the mono-analytic étale-like D⊢-prime-strip; which rungs are preserved by the log-link (vertical coricity) and by the Θ-link (horizontal coricity) is the content of [IUTchIII] Thm 1.5. The labelled strips F_⊞, D_◎±, F_>, F_⊛, F⊩⊢_Δ, F⊩⊢_LGP, … are the targets of the bridges and the homes of the pilots.



Prime-strips: the forgetful ladder. Collections of local data indexed by V. Arrows forget structure. The horizontally coric F⊢×μ-strip is what a Θ-link glues; the vertically coric D-strip is what a log-link shares; the bi-coric D⊢-strip is shared by everything. **[IUT I]** Fig I1.2 p. 6, Def 4.1 p. 95, Def 5.2 p. 134; **[IUT II]** Def 4.9 p. 153; **[IUT III]** Thm 1.5 p. 48

`DPrimeStrip` `DMonoPrimeStrip` `FPrimeStrip` `FMonoPrimeStrip` `FMonoUnitsModTorsPrimeStrip` `FMonoGlobValUnitsModTorsPrimeStrip`

25 entries: 25 prime-strips.

`BasicPrimeStrip` **BPS = F⊢×μ** prime-strips

'basic prime-strip': the F⊢×μ-prime-strip (horizontally coric gluing datum)

INTRODUCED / RECALLED **[FormIUT]** §1 p. 1

`typedef struct … iut_prime_strip` iut.h

field	type	meaning
`kind`	`iut_ps_kind`	
`label[IUT_MAX_NAME]`	`char`	e.g. "t=-3", "j=2", "BoxPlus", "Gt", "Triangle"
`nV`	`unsigned`	
`loc[IUT_MAX_V]`	`iut_local_datum`	
global portion for ⊩⊢ kinds: realified Frobenioid C⊩⊢ with Prime(C⊩⊢)→V		
`has_global_realified`	`int`	
`global_degree`	`double`	arithmetic degree of the pilot object (≤0)

`typedef struct … iut_local_datum` iut.h

Local datum at v of a prime-strip. Numbers are valuations measured in units of ord_v(q_v) where q_v = q_v^{1/2I} (so q_v itself has value 2I).

field	type	meaning
étale-like portion		
`Pi_copy`	`unsigned`	copy/basepoint id of Π_v (0 = absent)
`G_copy`	`unsigned`	copy id of G_v (quotient Π_v ↠ G_v)
Frobenius-like portion		
`units_copy`	`unsigned`	copy id of O× / O×μ (0 = absent)
`units_mod_tors`	`int`	O×μ (1) rather than O× (0)
`has_valgroup`	`int`	O▶ value-group portion present
`has_splitting`	`int`	split Frobenioid: O▷ = O× × O▶ splitting

field	type	meaning
`nval`	`unsigned`	generators of the value-group monoid(s)
`val[IUT_MAX_LSTAR+1]`	`iut_q`	valuations of generators (q̲-units)
Kummer structure on units: Ẑ×-orbit (×) or Ism-orbit (×μ)		
`kummer_kind`	`int`	0 none, 1 ×-Kummer (unique), 2 ×μ-Kummer

`DPrimeStrip` **D-prime-strip** prime-strips

{Π_v}: holomorphic, étale-like

INTRODUCED / RECALLED **[IUT I]** Introduction, Figure I1.2 p. 6
form: `iut_prime_strip` — see BasicPrimeStrip

`DMonoPrimeStrip` **D⊢-prime-strip** prime-strips

{G_v}: mono-analytic, étale-like, bi-coric

INTRODUCED / RECALLED **[IUT I]** Introduction, Figure I1.2 p. 6
form: `iut_prime_strip` — see BasicPrimeStrip

`FPrimeStrip` **F-prime-strip** prime-strips

{Π_v ↷ O▷}: holomorphic, Frobenius-like

INTRODUCED / RECALLED **[IUT I]** Introduction, Figure I1.2 p. 6
form: `iut_prime_strip` — see BasicPrimeStrip

`FMonoPrimeStrip` **F⊢-prime-strip** prime-strips

{G_v ↷ O▷ split}: mono-analytic, Frobenius-like

INTRODUCED / RECALLED **[IUT I]** Introduction, Figure I1.2 p. 6
form: `iut_prime_strip` — see BasicPrimeStrip

`FMonoUnitsPrimeStrip` **F⊢×-prime-strip** prime-strips

{G_v ↷ O×}: unit portion

INTRODUCED / RECALLED **[IUT II]** §4, Definition 4.9 p. 153
form: `iut_prime_strip` — see BasicPrimeStrip

`FMonoUnitsModTorsPrimeStrip` **F⊢×μ-prime-strip** prime-strips

{G_v ↷ O×μ} with ×μ-Kummer structure; horizontally coric

INTRODUCED / RECALLED **[IUT II]** §4, Definition 4.9 p. 153; **[IUT III]** §1, Theorem 1.5 p. 48
form: `iut_prime_strip` — see BasicPrimeStrip

`FMonoValUnitsModTorsPrimeStrip` **F⊢▶×μ-prime-strip** prime-strips

{G_v ↷ O▶ × O×μ}

INTRODUCED / RECALLED **[IUT II]** §4, Definition 4.9 p. 153
form: `iut_prime_strip` — see BasicPrimeStrip

`FMonoGlobPrimeStrip` **F⊩-prime-strip** prime-strips

F⊢ plus global realified Frobenioid C⊩ with Prime(C⊩) → V

INTRODUCED / RECALLED **[IUT I]** Introduction, Figure I1.2 p. 6
form: `iut_prime_strip` — see BasicPrimeStrip

`FMonoGlobValUnitsModTorsPrimeStrip` **F⊩▶×μ-prime-strip** prime-strips

F⊢▶×μ plus global realified Frobenioid; carries the pilot objects

INTRODUCED / RECALLED **[IUT II]** §4, Definition 4.9 p. 153
form: `iut_prime_strip` — see BasicPrimeStrip

`FMonoGlobD` **F⊩_D** prime-strips

the D-version (D⊩_mod, Prime ≅ V, {D⊢_v}, {ρ^D_v}) of F⊩_mod

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.5 (iv) p. 84
form: `iut_global_realified` — see CMonoGlobMod
form: `iut_f_monoglob` — see CMonoGlobMod
form: `iut_theta_hodge_theater_def` — see CMonoGlobMod

`FMonoGlobMod` **F⊩_mod** prime-strips

F⊩-prime-strip of the Θ-Hodge theater carrying q̲_v

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.6 p. 87
form: `iut_prime_strip` — see BasicPrimeStrip

`FMonoGlobTht` **F⊩_tht** prime-strips

F⊩-prime-strip of the Θ-Hodge theater carrying Θ̲_v

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.6 p. 87
form: `iut_prime_strip` — see BasicPrimeStrip

`DGt` **D_>** prime-strips

D-prime-strip of the label >

INTRODUCED / RECALLED **[IUT I]** §4 p. 95
form: `iut_prime_strip` — see BasicPrimeStrip

`FGt` **F_>** prime-strips

target F-prime-strip of the Θ-bridge

INTRODUCED / RECALLED **[IUT I]** §4 p. 95

form: `iut_prime_strip` — see BasicPrimeStrip

`FCircAst` **F_⊛** prime-strips

target of the NF-bridge (Π_{C_K} ↷ F_mod^×)

INTRODUCED / RECALLED **[IUT I]** §5 p. 123

form: `iut_prime_strip` — see BasicPrimeStrip

`DCircDotPm` **D_⊙±** prime-strips

target D-prime-strip of the Θell-bridge (Π_{X_K})

INTRODUCED / RECALLED **[IUT I]** §6 p. 7

form: `iut_prime_strip` — see BasicPrimeStrip

`FBoxPlus` **F_⊞** prime-strips

target F-prime-strip of the Θ±-bridge

INTRODUCED / RECALLED **[IUT I]** §6 p. 7

form: `iut_prime_strip` — see BasicPrimeStrip

`DMonoGt` **D⊢_>** prime-strips

D⊢-prime-strip of the label >

INTRODUCED / RECALLED **[IUT II]** §4, Corollary 4.10 p. 158

form: `iut_prime_strip` — see BasicPrimeStrip

`FMonoGlobEnv` **F⊩_env** prime-strips

theta-monoid version

INTRODUCED / RECALLED **[IUT II]** §4, Corollary 4.10 (ii) p. 158

form: `iut_prime_strip` — see BasicPrimeStrip

`FMonoGlobGau` **F⊩_gau** prime-strips

Gaussian-monoid version

INTRODUCED / RECALLED **[IUT II]** §4, Corollary 4.10 (ii) p. 158

form: `iut_prime_strip` — see BasicPrimeStrip

`FMonoGlobTriangle` **F⊩_Δ** prime-strips

F⊩-prime-strip for the label Δ (0 and F_l^* identified); home of the q-pilot

INTRODUCED / RECALLED **[IUT II]** §4, Corollary 4.10 (i) p. 158; **[IUT III]** Introduction, Theorem B p. 21

form: `iut_prime_strip` — see BasicPrimeStrip

`DMonoTriangle` **D⊢_Δ** prime-strips

D⊢-prime-strip of the label Δ (bi-coric)

INTRODUCED / RECALLED **[IUT III]** §1, Theorem 1.5 p. 48

form: `iut_prime_strip` — see BasicPrimeStrip

`FMonoGlobLgpIdeal` **F⊩_lgp** prime-strips

log-Gaussian-procession version (via mod, local fractional ideals)

INTRODUCED / RECALLED **[IUT III]** §3, Definition 3.8 (ii) p. 112

form: `iut_prime_strip` — see BasicPrimeStrip

`FMonoGlobLGPTorsor` **F⊩_LGP** prime-strips

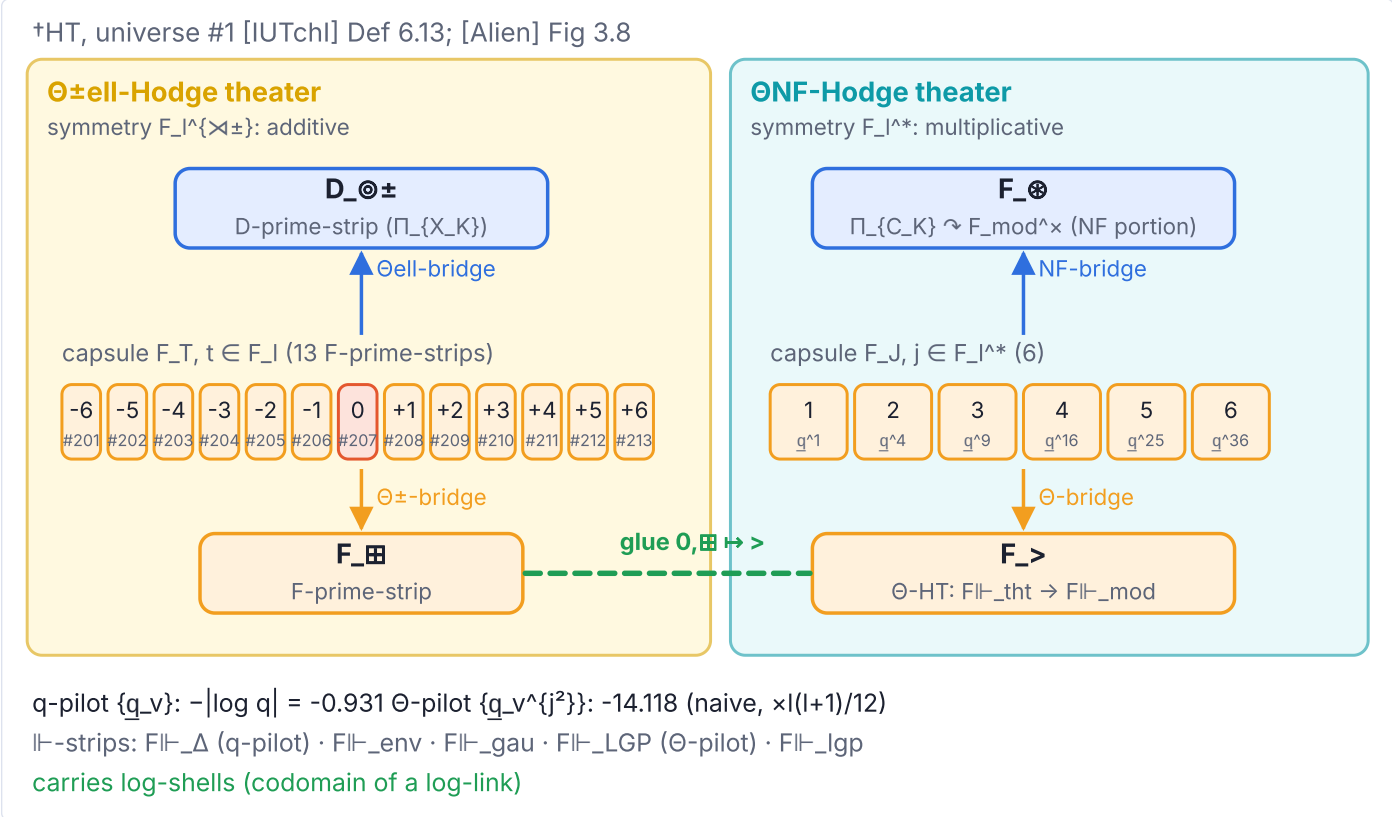
log-Gaussian-procession version (via MOD, F_mod^×-torsors)

INTRODUCED / RECALLED **[IUT III]** §3, Definition 3.8 (ii) p. 112

form: `iut_prime_strip` — see BasicPrimeStrip

7 · Hodge theaters, bridges and the gluing

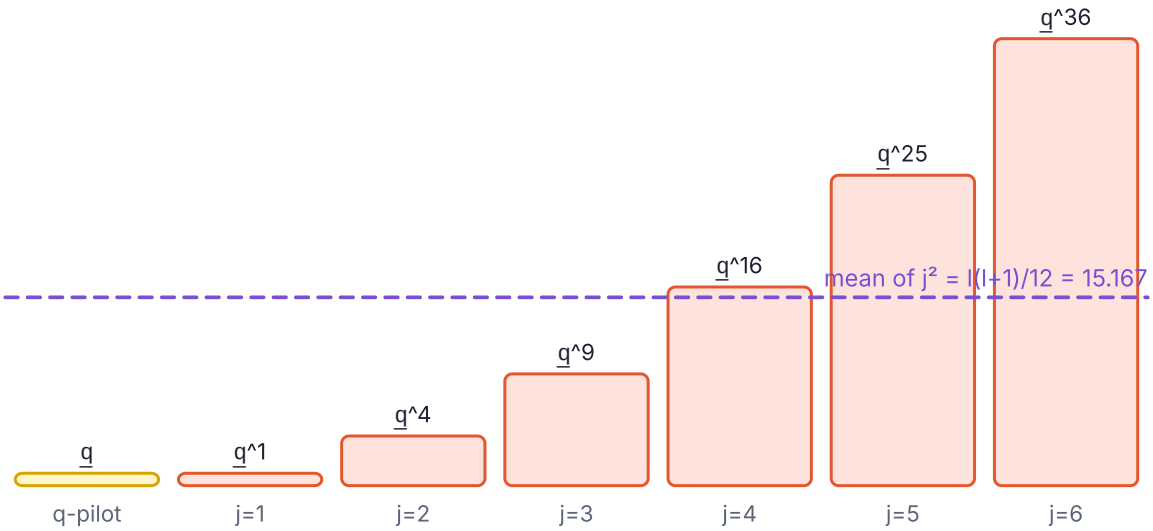
A Θ±ell NF-Hodge theater ([IUTchI] Def 6.13) is a Θ±ell-Hodge theater (capsule F_T, Θ±-bridge to F_⊞, Θell-bridge to D_◎±) glued along 0, ⊞ ↦ > to a ΘNF-Hodge theater (capsule F_J, Θ-bridge to F_>, NF-bridge to F_⊛), which contains the Θ-Hodge theater of Def 3.6 carrying q_v and Θ_v. Every vertex • of the log-theta-lattice is one such theater in its own universe.



Θ±ell NF-Hodge theater. One vertex • of the lattice: a Θ±ell-HT (capsule F_T, Θ±-bridge to F_⊞, Θell-bridge to D_◎±) and a ΘNF-HT (capsule F_J, Θ-bridge to F_>, NF-bridge to F_⊛), glued along 0,⊞ ↦ >. It is a single, complete copy of conventional scheme theory around E_F — a "universe". **[IUT I]** Def 6.13 p. 182; **[Alien]** §3.3 (v) p. 58, Fig 3.8 p. 67

`ThetaPmEllNFHodgeTheater` `ThetaPmEllHodgeTheater` `ThetaNFHodgeTheater` `FT` `FJ` `FBoxPlus` `DCircDotPm` `FGt` `FCircAst` `GluingZeroBoxPlusToGt`

Θ-Hodge theater: Θ_v(j) = q_v^{j²}; the pilots as arithmetic line bundles [IUTchIII] Def 3.8 (i)



q-pilot degree −|log q| = −log(q)/2I = -0.931 Θ-pilot {q^{j²}}_{j=1..I*}: naive degree -14.118

The Θ-link glues the Θ-pilot of †HT to the q-pilot of ‡HT ({q^{j²}} ↦ {q}), so the two "equal" bundles differ by the factor I(I+1)/12: the discrepancy the multiradial representation measures.

```
log(q) = 24.202 (Σ ord(q_v)·f_v·log p over bad v, over [F_mod:Q])
```

Θ-Hodge theater, q-pilot and Θ-pilot. The theta values Θ_v(j) = q_v^{j²} (j = 1..I*) form the Θ-pilot, the single value q_v the q-pilot; the Θ-link identifies them. Averaged over j the Θ-pilot is I(I+1)/12 times the q-pilot: this is the "q ↦ q^N" of [Alien] §2.4 that the theory wants to justify. **[IUT I]** Ex 3.2 p. 69; **[IUT III]** Def 3.8 (i) p. 112; **[Alien]** §3.3 (vii) p. 58

`ThetaHodgeTheater` `QPilot` `ThetaPilot` `JSq` `FMonoGlobTriangle` `FMonoGlobLGPTorsor`

23 entries: 12 Hodge theaters, 10 bridges, 1 gluings.

`ThetaPmEllNFHodgeTheater` **Θ±ell NF-Hodge theater**

Hodge theaters

Θ±ell-HT glued to ΘNF-HT along 0,⊞ ↦ >; a vertex • of the log-theta-lattice

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.13 (iii) p. 182

LEAN `structure ThetaPmEllNFHodgeTheater — lean/IUT/HodgeTheater.lean`

`typedef struct … iut_hodge_theater` iut.h

field	type	meaning
position in the log-theta-lattice and "alien copy" labels		
`n`	`int`	coordinates (n = column, m = row)
`m`	`int`	coordinates (n = column, m = row)
`copy_label[32]`	`char`	"Dag", "Ddag", "NMnm" (letters-only)
`universe`	`unsigned`	distinct label/universe id, [IUTchIV] §3
`data`	`const iut_initial_theta_data *`	
`pmell`	`iut_theta_pm_ell_hodge_theater`	
`nf`	`iut_theta_nf_hodge_theater`	
`glued`	`int`	gluing 0,⊞ ↦ > performed
data constructed in [IUTchII] Cor. 4.10 / [IUTchIII] Def. 3.8		
`FMonoGlobTriangle`	`iut_prime_strip`	F⊩_Δ : labels 0 and F_l^* identified (constant)
`FMonoGlobEnv`	`iut_prime_strip`	F⊩_env (theta monoids)
`FMonoGlobGau`	`iut_prime_strip`	F⊩_gau (Gaussian monoids)
`FMonoGlobLGP`	`iut_prime_strip`	F⊩_LGP (log-Gaussian procession, MOD)
`FMonoGlobLgp`	`iut_prime_strip`	F⊩_lgp (local fractional ideal, mod)
`q_pilot`	`iut_pilot`	in F⊩_Δ
`theta_pilot`	`iut_pilot`	in F⊩_gau / F⊩_LGP
log-shells at this lattice point (constructed by the log-link INTO it)		
`nshell`	`unsigned`	
`shell[IUT_MAX_V]`	`iut_log_shell`	
`has_incoming_log`	`int`	this • is the codomain of a log-link
Frobenius-like "state": how many log-links applied along the column		
`log_iterate`	`int`	

`ThetaPmEllHodgeTheater` **Θ±ell-Hodge theater** Hodge theaters

additive / geometric: Θ±-bridge, Θell-bridge, capsule indexed by F_l

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.13 p. 182

LEAN `structure ThetaPmEllHodgeTheater — lean/IUT/HodgeTheater.lean`

`typedef struct … iut_theta_pm_ell_hodge_theater` iut.h

Θ±ell-Hodge theater [IUTchI] §6

field	type	meaning
`nT`	`unsigned`	2l*+1 = l
`T[2*IUT_MAX_LSTAR+1]`	`iut_capsule_member`	capsule F_T, t ∈ F_l
`FBoxPlus`	`iut_prime_strip`	F_⊞ : target of the Θ±-bridge
`DCircDotPm`	`iut_prime_strip`	D_⊙± : target of the Θell-bridge (Π_{X_K})
`theta_pm_bridge_ok`	`int`	
`theta_ell_bridge_ok`	`int`	

`ThetaNFHodgeTheater` **ΘNF-Hodge theater** Hodge theaters

multiplicative / arithmetic: Θ-bridge, NF-bridge, capsule indexed by F_l^*

INTRODUCED / RECALLED **[IUT I]** §4, Definition 4.6 p. 111, §5, Definition 5.5 p. 151

LEAN `structure ThetaNFHodgeTheater — lean/IUT/HodgeTheater.lean`

`typedef struct … iut_theta_nf_hodge_theater` iut.h

ΘNF-Hodge theater [IUTchI] §4, §5

field	type	meaning
`nJ`	`unsigned`	l*
`J[IUT_MAX_LSTAR+1]`	`iut_capsule_member`	capsule F_J, j ∈ F_l^*
`FGt`	`iut_prime_strip`	F_> : target of the Θ-bridge
`FCircAst`	`iut_prime_strip`	F_⊛ : target of the NF-bridge (Π_{C_K}, F_mod^×)
`tht`	`iut_theta_hodge_theater`	
`theta_bridge_ok`	`int`	
`nf_bridge_ok`	`int`	

`ThetaHodgeTheater` **Θ-Hodge theater** Hodge theaters

(Fl⊢_mod, Fl⊢_tht) modelling the q-parameters and theta function

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.6 p. 87

LEAN `structure ThetaHodgeTheater — lean/IUT/HodgeTheater.lean`

`typedef struct … iut_theta_hodge_theater` iut.h

Θ-Hodge theater [IUTchI] Def. 3.6

field	type	meaning
`FMonoGlobMod`	`iut_prime_strip`	Fl⊢_mod (q-parameters q_v)
`FMonoGlobTht`	`iut_prime_strip`	Fl⊢_tht (theta function Θ_v)
`theta_value[IUT_MAX_LSTAR+1]`	`iut_q`	Θ_v evaluated at label j: j²·ord(q)

`DThetaPmEllNFHodgeTheater` **D-Θ±ell NF-Hodge theater** Hodge theaters

base-category (étale-like) version; vertically coric

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.13 p. 182

LEAN `structure DThetaPmEllNFHodgeTheater — lean/IUT/HodgeTheater.lean`

`GluingZeroBoxPlusToGt` **0,⊞ ↦ >** gluings

identification of the label 0 ∈ F_l and ⊞ with > (F_l ⊇ F_l^× ↠ F_l^*)

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.13 (iii) p. 182; **[Alien]** §3.6 p. 87

`DagHodgeTheater` **†HT** Hodge theaters

the dagger (domain) copy of a Hodge theater

INTRODUCED / RECALLED **[IUT I]** §3, Corollary 3.7 p. 88

form: `iut_hodge_theater` — see ThetaPmEllNFHodgeTheater

`DdagHodgeTheater` **‡HT** Hodge theaters

the double-dagger (codomain) copy

INTRODUCED / RECALLED **[IUT I]** §3, Corollary 3.7 p. 88

form: `iut_hodge_theater` — see ThetaPmEllNFHodgeTheater

`NFBridge` **NF-bridge** bridges

F_J → F_⊛ : capsule to Π_{C_K} and F_mod^× (NF = number field)

INTRODUCED / RECALLED **[IUT I]** §4, Definition 4.1 / 5.5 p. 95

`ThetaBridge` **Θ-bridge** bridges

F_J → F_> : capsule of F-prime-strips to the >-strip

INTRODUCED / RECALLED **[IUT I]** §4, Definition 4.1 / 5.5 p. 95

`DNFBridge` **D-NF-bridge †φ^NF_⊛** bridges

poly-morphism †D_J → †D^⊛ (capsule-full + Aut_ε-orbit); isomorphisms form an F_l^*-torsor (Prop 4.8 (i))

INTRODUCED / RECALLED **[IUT I]** §4, Definition 4.6 (i) p. 111

form: `iut_eval_points` — see MuMinus

form: `iut_theta_bridge_piece` — see MuMinus

form: `iut_label_transport` — see MuMinus

form: `iut_iso_torsor` — see MuMinus

form: `iut_forgetful_counts` — see MuMinus

`DThetaBridge` **D-Θ-bridge †φ^Θ_⊛** bridges

poly-morphism †D_J → †D_> (capsule-full + full poly-iso); isomorphisms: trivial (Prop 4.8 (ii))

INTRODUCED / RECALLED **[IUT I]** §4, Definition 4.6 (ii) p. 111

form: `iut_eval_points` — see MuMinus

form: `iut_theta_bridge_piece` — see MuMinus

form: `iut_label_transport` — see MuMinus

form: `iut_iso_torsor` — see MuMinus

form: `iut_forgetful_counts` — see MuMinus

`DThetaNFHodgeTheater` **D-ΘNF-Hodge theater** Hodge theaters

†D^⊛ ← †D_J → †D_>; isomorphisms trivial (Prop 4.8 (iii)); gluing of the bridges an F_l^*-torsor (Prop 4.8 (iv))

INTRODUCED / RECALLED **[IUT I]** §4, Definition 4.6 (iii) p. 111

form: `iut_eval_points` — see MuMinus

form: `iut_theta_bridge_piece` — see MuMinus

form: `iut_label_transport` — see MuMinus

form: `iut_iso_torsor` — see MuMinus

form: `iut_forgetful_counts` — see MuMinus

`PsiNFBridge` **ψ^NF_⊛** bridges

the NF-bridge ‡F_J → ‡F^⊛ ⇝ ‡F^⊛′ lifting φ^NF_⊛ (unique over it)

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.4 (v) p. 147; **[IUT I]** §5, Definition 5.5 (i) p. 151

form: `iut_global_frobenioid` — see FCircAstGlobal

form: `iut_distribution` — see FCircAstGlobal

form: `iut_strip_functor` — see FCircAstGlobal

`PsiThetaBridge` **ψ^Θ_j, ψ^Θ_⊛** bridges

the Θ-bridge ‡F_J → ‡F_> ⇝ ‡HT^Θ lifting φ^Θ_⊛

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.4 (ii) p. 147; **[IUT I]** §5, Definition 5.5 (ii) p. 151

form: `iut_global_frobenioid` — see FCircAstGlobal

form: `iut_distribution` — see FCircAstGlobal

form: `iut_strip_functor` — see FCircAstGlobal

`DThetaEllBridge` **D-Θell-bridge †φ^Θell_±** bridges

poly-morphism †D_T → †D^⊚± with T an F_l^{⋊±}-torsor (Aut_csp-orbit); isomorphisms an F_l^{⋊±}-torsor (Prop 6.6 (ii))

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.4 (ii) p. 162

form: `iut_fl_semi_pm_set` — see FlSemiPmGroup

form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup

form: `iut_sign_vector` — see FlSemiPmGroup

form: `iut_prop_six_seven` — see FlSemiPmGroup

`DThetaPmBridge` **D-Θ±-bridge †φ^Θ±_±** bridges

poly-morphism †D_T → †D_⊞ with T an F_l^{⋊±}-group; isomorphisms a torsor over {±1} × {±1}^V (Prop 6.6 (i))

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.4 (i) p. 162

form: `iut_fl_semi_pm_set` — see FlSemiPmGroup

form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup

form: `iut_sign_vector` — see FlSemiPmGroup

form: `iut_prop_six_seven` — see FlSemiPmGroup

`DThetaPmEllHodgeTheater` **D-Θ±ell-Hodge theater** Hodge theaters

†D_⊞ ← †D_T → †D^⊚±; isomorphisms a {±1}-torsor (Prop 6.6 (iii)); gluing a torsor over F_l^{⋊±} × {±1}^V (Prop 6.6 (iv))

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.4 (iii) p. 162

form: `iut_fl_semi_pm_set` — see FlSemiPmGroup

form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup

form: `iut_sign_vector` — see FlSemiPmGroup

form: `iut_prop_six_seven` — see FlSemiPmGroup

`ThetaEllBridge` **Θell-bridge** bridges

D_T → D_⊚± : the capsule to Π_{X_K} (ell = elliptic)

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.4 p. 162

`ThetaPmBridge` **Θ±-bridge** — bridges

F_T → F_⊞ : the capsule of F-prime-strips to the ⊞-strip

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.4 p. 162

`DagHTDThetaPmEllNF` **†HT^{D-Θ±ell NF}** — Hodge theaters

the D-version of a Θ±ell NF-Hodge theater: (D-Θ±ell-HT, D-ΘNF-HT, gluing via Prop 6.7)

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.13 (ii) p. 182

form: `iut_fl_semi_pm_set` — see FlSemiPmGroup

form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup

form: `iut_sign_vector` — see FlSemiPmGroup

form: `iut_prop_six_seven` — see FlSemiPmGroup

`NMHodgeTheater` **{n,m}HT** — Hodge theaters

the Hodge theater at coordinate (n,m) of the log-theta-lattice

INTRODUCED / RECALLED **[IUT III]** §1, Definition 1.4 p. 45

form: `iut_hodge_theater` — see ThetaPmEllNFHodgeTheater

`EtaleHodgeTheater` **etHT** — Hodge theaters

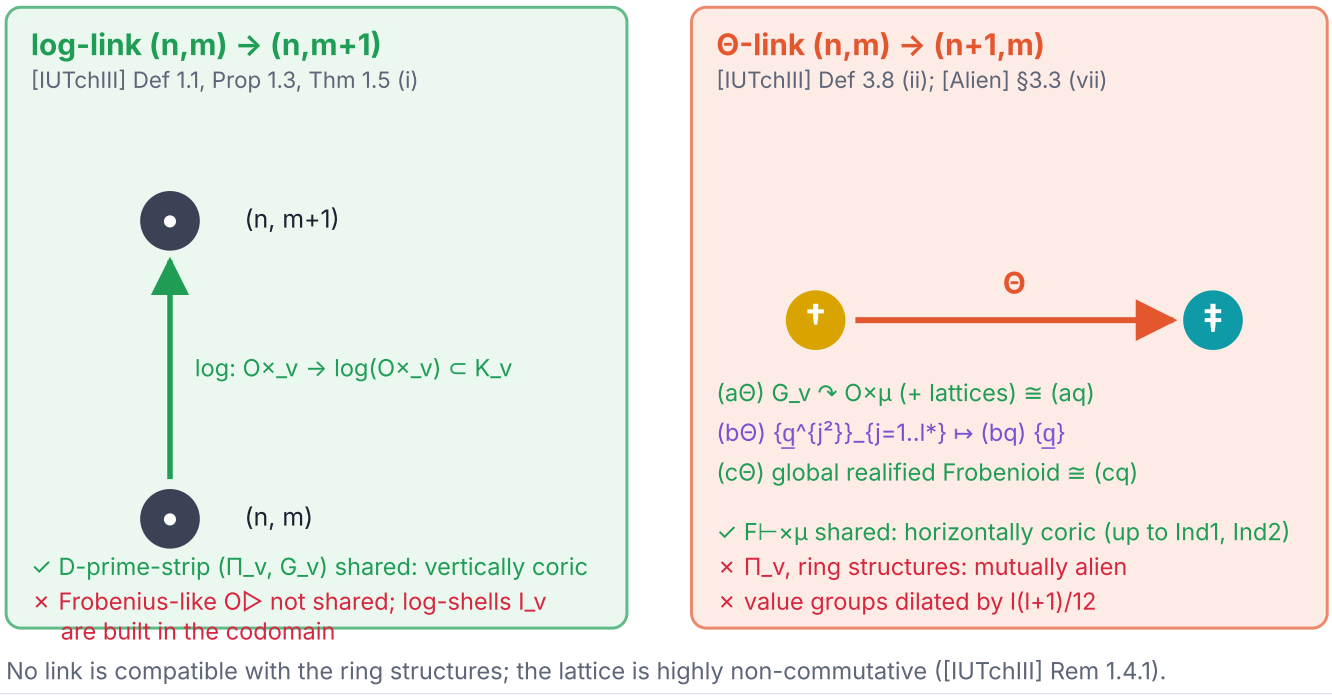
étale-like Hodge theater (D-Θ±ell NF-HT) in the Formalization paper's notation

INTRODUCED / RECALLED **[FormIUT]** §1 p. 1

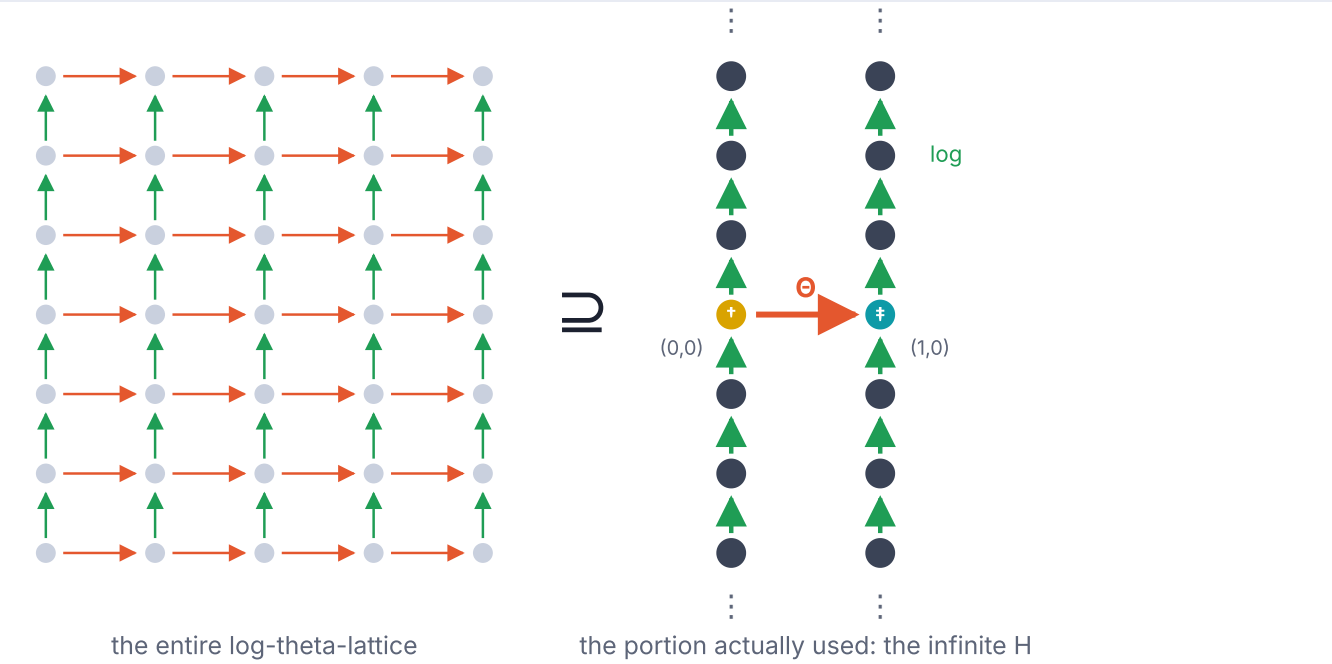
form: `iut_hodge_theater` — see ThetaPmEllNFHodgeTheater

8 · Links, the log-theta-lattice, log-shells and pilots

The vertical arrows of the lattice are log-links (fresh Frobenius-like monoids, shared étale-like data, log-shells $\mathfrak{l}_v = (p^*)^{-1}\log(\mathcal{O}^\times)$ in the codomain); the horizontal arrows are Θ-links in their Θ, Θ×μ, Θ×μ_gau, Θ×μ_LGP, Θ×μ_lgp flavours, gluing the Θ-pilot $\{\underline{q}^{\{j^2\}}\}$ of † to the q-pilot $\{\underline{q}\}$ of ‡ while identifying only G_v and $\mathcal{O}^\times\mu$. The two sides of a Θ-link are mutually alien copies; the proof uses only the infinite H of columns 0 and 1.

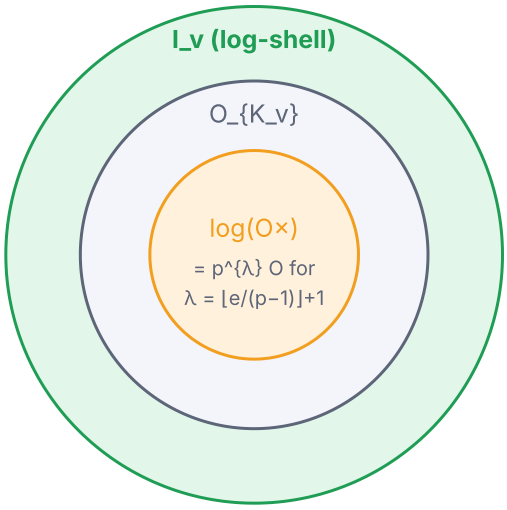


log-link and Θ-link. The two kinds of arrow of the lattice. The log-link uses both ring structures (the p-adic logarithm) and shares the étale-like D-strip; the Θ-link uses only multiplicative structure and shares $G_v \curvearrowright \mathcal{O}^\times\mu$ while mapping $\{q^{\{j^2\}}\}$ to $\{q\}$. **[IUT III]** Def 1.1 p. 23, Def 3.8 (ii) p. 112, Thm 1.5 p. 48; **[Alien]** §3.3 (vi) p. 58, (vii)
`LogLink` `ThetaLink` `ThetaUnitsModTorsLGPTorsorLink` `VerticallyCoric` `HorizontallyCoric` `BiCoric` `AlienCopy`



The log-theta-lattice and the infinite H. Hodge theaters at $(n,m) \in \mathbb{Z}^2$, log-links vertical, Θ-links horizontal. Only the two columns $(0,\circ)$, $(1,\circ)$ and the single Θ-link (0,0) → (1,0) are actually used: the "infinite H" ([EssLgc] §3.10 (Stp7) works entirely inside it). **[IUT III]** Def 1.4 p. 45, Def 3.8 (iii) p. 112; **[Alien]** Fig 3.6 p. 60; **[EssLgc]** §3.3 (InfH) p. 86
`LogThetaLattice` `InfiniteH` `NMHodgeTheater` `NCircColumn` `Dag` `Ddag`

log-shell $l_v = (p^*)^{-1} \cdot \log(O^\times_{K_v}) \subseteq K_v$, $p^* = p$ (p odd), 4 ($p = 2$)



additive container for the multiplicative monoids

log-shells at (0,1), the codomain of the log-link from (0,0):

v	p*	l_v = p^{r} O	μ_log(l_v)
v2	4	r = -25/26	1.3063
v3	3	r = -12/26	1.0141
v5	5	r = -38/52	1.5475
v7	7	r = -162/195	1.9359
v13	13	r = -10/12	2.1375
vinf	0	r = 0	0.0000

μ_log(log_p O×) = −(1/e + m/(ef)) log p (m = 1 iff p = 2): an unramified odd p gives l_v = O_v with log-volume 0 ([IUTchIV] Prop 1.4 (ii)); the log-link preserves log-volumes (Stp8). Étale-like, vertically coric shells hold the Frobenius-like shells of every (n,m) ([Alien] Fig 3.15).

Log-shells. $l_v = (p^*)^{-1} \log(O^\times_{K_v})$: a compact additive module that contains the image of the units. Holomorphic / mono-analytic and Frobenius-like / étale-like versions all appear; the mono-analytic ones are the multiradial containers of (Stp3)–(Stp6). **[IUT III]** Def 1.1 p. 23, Prop 1.2 p. 30; **[IUT IV]** Prop 1.2 p. 10, Prop 1.4 p. 13; **[Alien]** Ex 2.12.3 p. 38, Fig 3.15 p. 99
`LogShell` `HolLogShell` `MonoLogShell` `FrobLogShell` `EtLogShell` `PStar` `LogVolume`

21 entries: 9 links, 5 lattices, 5 log-shells, 2 pilot objects.

`LogThetaLattice` **log-theta-lattice** — lattices

Z^2-indexed diagram of Θ±ell NF-HTs: vertical log-links, horizontal Θ×μ_LGP-links

INTRODUCED / RECALLED **[IUT III]** §1, Definition 1.4 p. 45, §3, Definition 3.8 (iii) p. 112

LEAN `structure LogThetaLattice — lean/IUT/LogThetaLattice.lean`

`typedef struct … iut_log_theta_lattice` — iut.h

---------- log-theta-lattice [IUTchIII] Def. 1.4, Def. 3.8 (iii); Fig. I.1. Vertices • = Θ±ell NF-HTs, vertical arrows = log-links, horizontal arrows = Θ(×μ,gau,LGP,lgp)-links.

field	type	meaning
`data`	`iut_initial_theta_data`	
`horizontal`	`iut_link_kind`	which Θ-link version
`ht[IUT_LAT_N][IUT_LAT_M]`	`iut_hodge_theater *`	lazily created
`next_universe`	`unsigned`	
`nlinks`	`unsigned`	
`links[IUT_LAT_N*IUT_LAT_M*2 + 8]`	`iut_link`	
the "infinite H" actually used: columns 0 and 1, single Θ-link (0,0)→(1,0)		
`focus_n`	`int`	
`focus_m`	`int`	

`InfiniteH` **∞H** — lattices

the two columns (0,∘),(1,∘) joined by a single Θ-link (0,0)→(1,0)

INTRODUCED / RECALLED **[EssLgc]** §3.10 p. 136; **[Alien]** §3.11 Fig 3.14 p. 134

form: `iut_log_theta_lattice` — see LogThetaLattice

`LogLink` **log-link** — links

{n,m}HT → {n,m+1}HT via log: O× → K_v (ring structures not identified)

INTRODUCED / RECALLED **[IUT III]** §1, Definition 1.1 p. 23, §1, Proposition 1.3 p. 41

form: `iut_link` — see ThetaLink

`ThetaLink` **Θ-link** — links

†FⅡ-_tht ≅ ‡FⅡ-_mod : glues Θ_v to q_v, full poly-isomorphism of FⅡ-prime-strips

INTRODUCED / RECALLED **[IUT I]** §3, Corollary 3.7 p. 88

`typedef struct … iut_link` — iut.h

field	type	meaning
`kind`	`iut_link_kind`	

field	type	meaning
dom_n	int	
dom_m	int	
cod_n	int	
cod_m	int	
glued_kind	iut_ps_kind	kind of prime-strip glued (FI⊢, FI⊢▶×μ, ...)
full_poly_isomorphism	int	gluing is a FULL poly-isomorphism
what is preserved ("coric") and what is dilated ("mutated")		
preserves_Pi	int	Π_v identified (log-link: yes; Θ: no)
preserves_G	int	G_v identified up to indeterminate iso
preserves_units_mod_tors	int	O×μ identified (Θ×μ..: yes)
preserves_ring_structure	int	never
dilation	double	value-group dilation factor (Θ: avg j²)

ThetaUnitsModTorsLGPTorsorLink **Θ×μ_LGP-link** links

log-Gaussian-procession version (MOD)

INTRODUCED / RECALLED **[IUT III]** §3, Definition 3.8 (ii) p. 112

form: iut_link — see ThetaLink

LogShell **I = log-shell** log-shells

I_v = (p*)^{-1}·log(O×_{K_v}) ⊆ K_v, a compact container

INTRODUCED / RECALLED **[IUT III]** §1, Definition 1.1 (i) p. 23; **[AbsTopIII]** §5, Definition 5.4 (iii) p. 124

LEAN structure LogShell — lean/IUT/HodgeTheater.lean

typedef struct … iut_log_shell iut.h

---------- log-shells [IUTchIII] Def. 1.1 (i): I_v = (p*_v)^{-1} · log(O×_{K_v}) ⊆ K_v, p* = p (p odd), p* = p² (p = 2). [AbsTopIII] Def. 5.4 (iii).

field	type	meaning
v_index	unsigned	index into V
pstar	unsigned	p or p² (0 for archimedean: I = π-disc)
radius	iut_q	p-adic valuation of the generator of I_v
log_volume	double	normalized log-volume of I_v ([IUTchIV] Prop 1.2/1.4)
holomorphic	int	1: from F-prime-strip, 0: mono-analytic (FI⊢×μ / DI⊢)
etale_like	int	1: constructed from D/DI⊢, 0: Frobenius-like
lattice_n	int	which (n,m) the units came from
lattice_m	int	which (n,m) the units came from

QPilot **q-pilot** pilot objects

object of FI⊢_Δ determined by {q̲_v}_{v ∈ V^bad}

INTRODUCED / RECALLED **[IUT III]** §3, Definition 3.8 (i) p. 112

typedef struct … iut_pilot iut.h

field	type	meaning
kind	iut_pilot_kind	
nj	unsigned	1 for q-pilot, I* for Θ-pilot
exps[IUT_MAX_LSTAR+1]	unsigned	exponents of q̲: {1} or {1,4,9,...}
log_volume	double	−|log(q)| or (upper bound for) −|log(Θ)|

ThetaPilot **Θ-pilot** pilot objects

object of FI⊢_gau/LGP determined by {q̲_v^{j²}}_{j=1..I*}

INTRODUCED / RECALLED **[IUT III]** §3, Definition 3.8 (i) p. 112

form: iut_pilot — see QPilot

DThetaNFLink **D-ΘNF-link** links

the full poly-isomorphism †DI⊢_> ≅ ‡DI⊢_> between D-ΘNF-Hodge theaters; étale picture with core DI⊢_>

INTRODUCED / RECALLED **[IUT I]** §4, Corollary 4.12 p. 121

form: iut_eval_points — see MuMinus

form: iut_theta_bridge_piece — see MuMinus

form: iut_label_transport — see MuMinus

form: iut_iso_torsor — see MuMinus

form: iut_forgetful_counts — see MuMinus

DThetaPmEllLink **D-Θ±ell-link** links

the full poly-isomorphism †DI⊢_> ≅ ‡DI⊢_> between D-Θ±ell-Hodge theaters; étale picture with core DI⊢_>

INTRODUCED / RECALLED **[IUT I]** §6, Corollary 6.10 p. 170

form: iut_fl_semi_pm_set — see FISemiPmGroup

form: iut_labcusp_pm_torsor — see FISemiPmGroup

form: iut_sign_vector — see FISemiPmGroup

form: iut_prop_six_seven — see FISemiPmGroup

`DThetaPmEllNFLink` **D-Θ±ell NF-link** links

link between D-Θ±ell NF-Hodge theaters (the étale picture)

INTRODUCED / RECALLED **[IUT I]** §6, Corollary 6.10 p. 170
form: `iut_link` — see ThetaLink

`ThetaUnitsModTorsGauLink` **Θ×μ_gau-link** links

Gaussian version: {q_v^{j²}} ↦ {q_v}

INTRODUCED / RECALLED **[IUT II]** §4, Corollary 4.10 (iii) p. 158
form: `iut_link` — see ThetaLink

`ThetaUnitsModTorsLink` **Θ×μ-link** links

Θ-link on Fl⊢▶×μ-prime-strips, units modulo torsion

INTRODUCED / RECALLED **[IUT II]** §4, Corollary 4.10 (iii) p. 158
form: `iut_link` — see ThetaLink

`EtLogShell` **I(D) (étale-like)** log-shells

étale-like log-shell, vertically coric

INTRODUCED / RECALLED **[IUT III]** §1, Proposition 1.2 (iii) p. 30
form: `iut_log_shell` — see LogShell

`FrobLogShell` **I (Frobenius-like)** log-shells

Frobenius-like log-shell

INTRODUCED / RECALLED **[IUT III]** §1, Proposition 1.2 (ii) p. 30
form: `iut_log_shell` — see LogShell

`HolLogShell` **I (holomorphic)** log-shells

log-shell constructed from an F-prime-strip

INTRODUCED / RECALLED **[IUT III]** §1, Proposition 1.2 p. 30
form: `iut_log_shell` — see LogShell

`MonoLogShell` **Il⊢ (mono-analytic)** log-shells

log-shell constructed from an Fl⊢×μ- or Dl⊢-prime-strip (bi-coric)

INTRODUCED / RECALLED **[IUT III]** §1, Proposition 1.2 p. 30
form: `iut_log_shell` — see LogShell

`LgpIdealGaussianLogThetaLattice` **lgp-Gaussian log-theta-lattice** lattices

the version with Θ×μ_lgp-links

INTRODUCED / RECALLED **[IUT III]** §3, Definition 3.8 (iii) p. 112
form: `iut_log_theta_lattice` — see LogThetaLattice

`LGPTorsorGaussianLogThetaLattice` **LGP-Gaussian log-theta-lattice** lattices

the version with Θ×μ_LGP-links (used in Thm 3.11)

INTRODUCED / RECALLED **[IUT III]** §3, Definition 3.8 (iii) p. 112
form: `iut_log_theta_lattice` — see LogThetaLattice

`ThetaUnitsModTorsLgpIdealLink` **Θ×μ_lgp-link** links

log-Gaussian-procession version (mod)

INTRODUCED / RECALLED **[IUT III]** §3, Definition 3.8 (ii) p. 112
form: `iut_link` — see ThetaLink

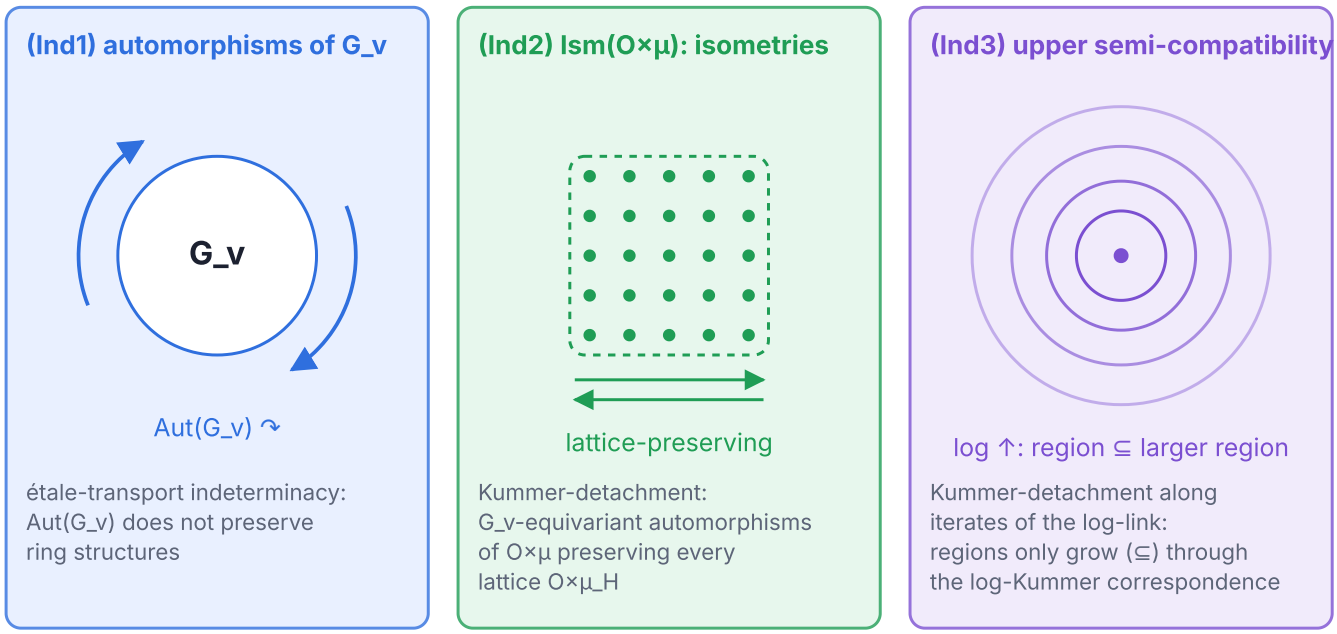
`NCircColumn` **(n,∘)** lattices

the whole vertical line n of the log-theta-lattice, i.e. its vertically coric (étale-like) data

INTRODUCED / RECALLED **[IUT III]** §3, Theorem 3.11 p. 153; **[EssLgc]** §3.6 p. 104
form: `iut_log_theta_lattice` — see LogThetaLattice

9 · Indeterminacies, the multiradial representation, structures and properties

What survives the passage across the Θ-link: processions S^±_{j+1} of tensor packets of log-shells, the splitting monoids of the LGP-monoids and copies of F_mod^×, all up to the indeterminacies Ind1 (automorphisms of processions of D⊢-prime-strips), Ind2 (Ism on O×μ) and Ind3 (upper semi-compatibility of the log-Kummer correspondence) — the multiradial representation of [IUTchIII] Thm 3.11, whose log-volume gives Cor 3.12.



The three indeterminacies. (Ind1) Aut(G_v) when identifying G_v's, (Ind2) isometries of O×μ when confusing Frobenius-like and étale-like log-shells, (Ind3) the upper semi-compatibility of the log-Kummer correspondence. They are the v̈'s of the ∧(v̈)-chain. **[IUT III]** Thm 3.11 p. 153; **[Alien]** §3.7 (i) p. 103; **[EssLgc]** §3.10 (Stp1)–(Stp3) p. 136, §3.11 p. 152
`IndOne` `IndTwo` `IndThree` `Ism` `UpperSemiCompatibility`

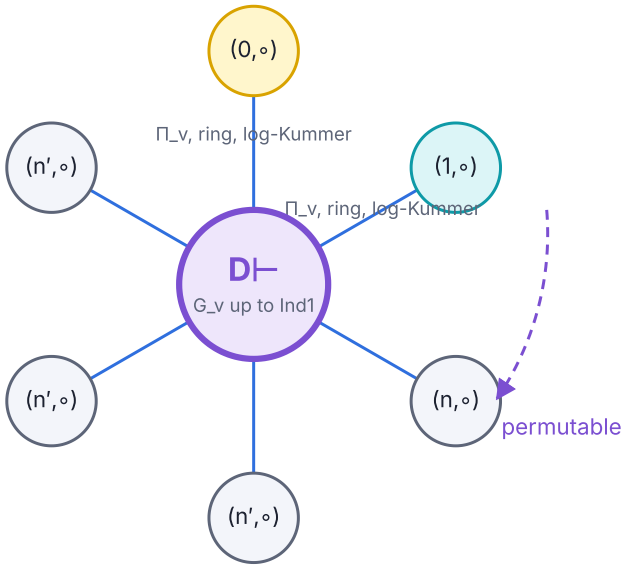
Θ-values acting on tensor packets of log-shells over processions [Alien] Fig 3.16–3.18; [IUTchIII] Thm 3.11



Each row is a tensor packet of mono-analytic log-shells (zero-mass containers) acted on by the theta value q^{j^2} and a copy of F_mod^×, up to (Ind1)–(Ind3). The input is only an abstract F⊢×μ-prime-strip (∧-Input, [EssLgc] §3.6), so the algorithm applies to ᵗHT and ‡HT alike.

Multiradial representation. The Θ-pilot of (0,0), descended to data that makes sense from (1,0): processions S^±_{j+1} of tensor packets of log-shells, the splitting monoids $\{q^{j^2}\}$ and copies of F_mod^×, up to Ind1–3. Its log-volume is bounded by the hull of the shells. **[IUT III]** Thm 3.11 p. 153; **[Alien]** §3.7 (i) p. 103, Fig 3.16 –3.19 p. 103
`MultiradialRepresentation` `SPmJPlusOne` `TensorPacket` `HolomorphicHull` `PsiLGP`

étale-picture: every column (n,∘) is radial data over the one bi-coric core D⊢ [Alien] Fig 3.5, 3.12



Multiradial = the radial algorithm is compatible with arbitrary permutations of the spokes (parallel transport over the coric data); uniradial data, e.g. holomorphic structures, is not ([Alien] Fig 3.1).

Étale-picture and multiradiality. Pass to étale-like data: every column of the lattice becomes a spoke around one core D⊢-prime-strip. A multiradial algorithm is one whose output may be transported along any spoke (parallel transport up to the monodromy Ind1–3). **[IUT I]** Fig I1.6 p. 13; **[Alien]** §3.1 p. 49, Fig 3.1 p. 52, Fig 3.5 p. 57, Fig 3.12 p. 90
`EtalePicture` `Multiradial` `Uniradial` `DMonoPrimeStrip`

39 entries: 5 indeterminacies, 21 structures, 13 properties.

`MultiradialRepresentation` **R^LGP_{n,∘}** structures

tensor packets + splitting monoids + copies of F_mod^×, up to Ind1-3

INTRODUCED / RECALLED **[IUT III]** §3, Theorem 3.11 (i) p. 153

`typedef struct … iut_multiradial_rep` iut.h

---------- multiradial representation R^LGP_{n,∘} = (a) tensor packets of log-shells, (b) splitting monoids of LGP-monoids, (c) copies (F_mod^×)_j — up to (Ind1), (Ind2),(Ind3).

field	type	meaning
`n`	`int`	column (n,∘)
`nsteps`	`unsigned`	
`step[IUT_MAX_LPM]`	`iut_procession_step`	
`ind`	`iut_indeterminacies`	
`logq`	`double`	log(q) normalized degree of q-divisor
`abs_log_q`	`double`	|log(q)| = log(q)/(2l)
`theta_upper_bound`	`double`	−|log(Θ)| upper bound, leading-term version
`leading_term`	`double`	(l²/48)(6 log d^K − log q) (×constant)
`C_theta`	`double`	C_Θ of [IUTchIV] Thm 1.10
`inequality_holds`	`int`	−|log Θ| ≥ −|log q| i.e. C_Θ ≥ −1
`hull_enlargement`	`double`	log-volume added by the holomorphic hull
the upper bound of [IUTchIV] Thm 1.10, proof Steps (v)–(vii), summed over v_Q (iut_hull_bound_compute)		
`theta_hull_bound`	`double`	B with −|log(Θ)| ≤ B: every term of the proof, not only the leading ones
`C_theta_local`	`double`	B/|log q|: an admissible C_Θ computed from the places of this curve

`IndOne` **(Ind1)** indeterminacies

automorphisms of processions of D⊢-prime-strips (perms of S^±_{j+1}, Aut(G_v))

INTRODUCED / RECALLED **[IUT III]** §3, Theorem 3.11 (i) p. 153

`typedef struct … iut_indeterminacies` iut.h

---------- indeterminacies [IUTchIII] Thm. 3.11; [Alien] §3.7 (i); [EssLgc] §3.10 (Stp1)-(Stp3)

field	type	meaning
`ind1`	`int`	automorphisms of processions of D⊢-prime-strips (perms of S±_{j+1}, Aut(G_v))
`ind2`	`int`	Ism / Ẑ×-indeterminacies on O×μ (Kummer-detachment)
`ind3`	`int`	upper semi-compatibility of the log-Kummer correspondence

`IndTwo` **(Ind2)** indeterminacies

Ism-indeterminacies of the ×μ-Kummer structures

INTRODUCED / RECALLED **[IUT III]** §3, Theorem 3.11 (i) p. 153

form: `iut_indeterminacies` — see IndOne

`IndThree` **(Ind3)** indeterminacies

upper semi-compatibility of the log-Kummer correspondence

INTRODUCED / RECALLED **[IUT III]** §3, Theorem 3.11 (ii) p. 153

form: `iut_indeterminacies` — see IndOne

`TensorPacket` **tensor packet** structures

tensor product of log-shells over the labels of S^±_{j+1}

INTRODUCED / RECALLED **[IUT III]** §3, Proposition 3.1 p. 92, §3, Proposition 3.2 p. 97

`Procession` **procession** structures

chain of capsules S_1 ⊂ S_2 ⊂ … of (D⊢-)prime-strips

INTRODUCED / RECALLED **[IUT I]** §4, Proposition 4.11 p. 120, §6, Proposition 6.9 p. 169

`typedef struct … iut_procession_step` iut.h

---------- processions and tensor packets S±_{j+1} = {0,1,…,j} for j = 0..l*; [IUTchI] Prop. 6.9; [IUTchIII] §3

field	type	meaning
`j`	`unsigned`	container S±_{j+1} holds labels 0..j
`size`	`unsigned`	j+1
`perms`	`unsigned`	(j+1)! possibilities for the element "j"
`log_volume_leading`	`double`	(j+1)·log(d^K) leading term per [Alien] §3.7 (iv)
`theta_shift`	`double`	−(j²/2l)·log(q)

`HolomorphicHull` **holomorphic hull** structures

smallest holomorphic region containing the Ind1-3 orbit; adds the log(d^K) terms

INTRODUCED / RECALLED **[IUT III]** §3, Remark 3.9.5 p. 126; **[Alien]** §3.7 (iv) p. 103

form: `iut_multiradial_rep` — see MultiradialRepresentation

`Isomorph` **isomorph** — structures

an object isomorphic to a given one, with the isomorphism forgotten (e.g. M^⊛ is an isomorph of F̄×)

INTRODUCED / RECALLED **[IUT I]** §0 p. 33

form: `iut_poly_morphism` — see PolyMorphism

`OrbicurveType` **(1,l-tors), (1,l-tors)±, (1,l-tors→), (1,(Z/l)^Θ)…** — structures

the types of hyperbolic orbicurves of genus 1 with l-torsion level structure used in IUT

INTRODUCED / RECALLED **[IUT I]** §1 p. 37; **[IUT I]** §1, Definition 1.1 p. 38; **[IUT I]** §3, Definition 3.1 p. 61

form: `iut_orbicurve` — see XF

`CommensurablyTerminal` **commensurably terminal** — properties

Π^tp_X ⊆ Π_X is commensurably terminal

INTRODUCED / RECALLED **[IUT I]** §2, Proposition 2.2 p. 45

form: `iut_tempered_fact` — see PiTpG

form: `iut_tempered_at_v` — see PiTpG

`GaloisCountable` **Galois-countable** — properties

conditions (T1)–(T6) on a topological group under which the tempered theory works; effects (E1)–(E9)

INTRODUCED / RECALLED **[IUT I]** §2, Remark 2.5.3 p. 52

form: `iut_tempered_fact` — see PiTpG

form: `iut_tempered_at_v` — see PiTpG

`ThetaApproach` **Θ-approach** — structures

apply [AbsTopIII] Thm 1.9 to Π_{X_K} ⊆ Π_{C_K}: the cyclotome M_X of the once-punctured curve, not of C_K

INTRODUCED / RECALLED **[IUT I]** §3, Remark 3.1.2 p. 64

`typedef struct … iut_theta_approach` — iut_i.h

Rem 3.1.2: the Θ-approach — [AbsTopIII] Thm 1.9 applied to Π_{X_K} ⊆ Π_{C_K}; the cyclotome M_X of the once-punctured curve is identified with l·Δ_Θ (index l).

field	type	meaning
`cyclotome_index`	`unsigned`	
`cyclotome`	`const char *`	
`theta_cyclotome`	`const char *`	
`applied_to`	`const char *`	
`ref`	`const char *`	

`SMod` **S_mod** — structures

the stack Spec(O_K) // Gal(K/F_mod); its points correspond to V_mod

INTRODUCED / RECALLED **[IUT I]** §3, Remark 3.1.5 p. 65

`typedef struct … iut_s_mod` — iut_i.h

Rem 3.1.5: the stack S_mod = Spec(O_K) // Gal(K/F_mod)

field	type	meaning
`notation`	`const char *`	
`gal_order_bound`	`unsigned long long`	
`npoints`	`unsigned`	

`KappaSolvable` **κ-solvable, κ-sol** — properties

the solvable extensions generated by κ-coric data; L_C(C_K) and L_C(κ-sol) are linearly disjoint

INTRODUCED / RECALLED **[IUT I]** §3, Remark 3.1.7 (iii) p. 66

form: `iut_kappa_function` — see CFmod

form: `iut_kappa_verdict` — see CFmod

form: `iut_kappa_sol` — see CFmod

`TauMonoV` **τ⊢_v** — structures

the characteristic splitting O▷ = O× × Φ of C⊢_v; a μ_2l-orbit at bad v, canonical at good and arch v

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 (iv) p. 69

form: `iut_local_frobenioid` — see DV

`TauThetaV` **τ^Θ_v** — structures

the splitting of the Θ-version C^Θ_v

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.2 (v) p. 69

form: `iut_local_frobenioid` — see DV

`AutHolOrbispace` **Aut-holomorphic orbispace** — structures

the archimedean analogue of a fundamental group: X→_v with its Aut-holomorphic structure

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.4 (i) p. 80; **[AbsTopIII]** §2 p. 50

form: `iut_local_frobenioid` — see DV

`KappaKummerV` **κ_v : O▷(C_v) ↪ A_{D_v}** structures

the archimedean Kummer structure of the triple F_v = (C_v, D_v, κ_v)

INTRODUCED / RECALLED **[IUT I]** §3, Example 3.4 (ii) p. 80

form: `iut_local_frobenioid` — see DV

`ThetaDilation` **Θ-dilation of value groups** properties

across the Θ-link the units are fixed while the value groups are dilated (Θ_v ↦ q_v)

INTRODUCED / RECALLED **[IUT I]** §3, Remark 3.9.3 p. 93

form: `iut_link` — see ThetaLink

form: `iut_global_realified` — see CMonoGlobMod

form: `iut_f_monoglob` — see CMonoGlobMod

form: `iut_theta_hodge_theater_def` — see CMonoGlobMod

`ArithLocalAnalyticSection` **arithmetic local-analytic section** structures

a prime-strip viewed as a section of Spec K → Spec F; requirements (a) local independence, (b) globalizability, (c) étale-like (Rem 4.3.2)

INTRODUCED / RECALLED **[IUT I]** §4, Remark 4.3.1 p. 101

form: `iut_labcusp_torsor` — see LabCusp

form: `iut_global_base` — see LabCusp

form: `iut_valuation_orbit` — see LabCusp

`SynchronizedLabels` **synchronized labels** properties

labels of the capsule members tied together across v ∈ V (via the global data); versus non-synchronized labels

INTRODUCED / RECALLED **[IUT I]** §4, Remark 4.9.2 p. 117

form: `iut_eval_points` — see MuMinus

form: `iut_theta_bridge_piece` — see MuMinus

form: `iut_label_transport` — see MuMinus

form: `iut_iso_torsor` — see MuMinus

form: `iut_forgetful_counts` — see MuMinus

`KummerBlind` **Kummer-blind** properties

data without such a Kummer isomorphism: F^⊛_mod, M_mod, M_κ, F_j, F_⟨J⟩, F_J, F⊢-prime-strips

INTRODUCED / RECALLED **[IUT I]** §5, Remark 5.1.3 p. 131; **[IUT I]** §5, Remark 5.2.2 p. 143

form: `iut_global_frobenioid` — see FCircAstGlobal

form: `iut_distribution` — see FCircAstGlobal

form: `iut_strip_functor` — see FCircAstGlobal

`KummerReady` **Kummer-ready** properties

data admitting a Kummer isomorphism with its étale-like counterpart: F^⊛, F^⊛′, M^⊛, M_sol, M_∞κ, M_∞κ×, M_κ-sol, F-prime-strips

INTRODUCED / RECALLED **[IUT I]** §5, Remark 5.1.3 p. 131; **[IUT I]** §5, Remark 5.2.2 p. 143

form: `iut_global_frobenioid` — see FCircAstGlobal

form: `iut_distribution` — see FCircAstGlobal

form: `iut_strip_functor` — see FCircAstGlobal

`PivotalDistribution` **pivotal distribution** structures

the distribution singled out by the label j = 1

INTRODUCED / RECALLED **[IUT I]** §5, Example 5.4 (vii) p. 147

form: `iut_global_frobenioid` — see FCircAstGlobal

form: `iut_distribution` — see FCircAstGlobal

form: `iut_strip_functor` — see FCircAstGlobal

`FlSemiPmGroup` **F_l^{⋊±}-group** structures

a set with a {±1}-orbit of bijections to F_l (T of a Θ±-bridge); Aut_+(T) ⊆ Aut_±(T) of index 2

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (i) p. 155

`typedef struct … iut_fl_semi_pm_set` iut_i.h

========== §6 ======= Def 6.1 (i): F_l^{⋊±} = F_l ⋊ {±1}; an F_l^{⋊±}-group is a set with a {±1}-orbit of bijections to F_l; an F_l^{⋊±}-torsor a set with an F_l^{⋊±}-orbit of bijections.

field	type	meaning
`n`	`unsigned`	l
`label[2*IUT_MAX_LSTAR+1]`	`int`	one bijection T ≅ F_l
`is_group`	`int`	1: F_l^{⋊±}-group ({±1}-orbit), 0: torsor (F_l^{⋊±}-orbit)
`orbit_size`	`unsigned long long`	2 or 2l
`aut_plus_order`	`unsigned long long`	Aut_+(T) ⊆ Aut_±(T) of index 2 (groups)
`aut_pm_order`	`unsigned long long`	Aut_+(T) ⊆ Aut_±(T) of index 2 (groups)

`typedef struct … iut_labcusp_pm_torsor` iut_i.h

Def 6.1 (ii)–(iii): LabCusp±(†D_v) with the F_l^×-action, the zero element †η°_v, the ±-canonical element †η^±_v (up to ±1); (LabCusp± \ {η°})/{±1} ≅ LabCusp; Aut(†D_v) ↠ {±1} with Aut_+ the kernel, Aut_− the other coset

field	type	meaning
`nV`	`unsigned`	
`size`	`unsigned`	l
`eta_zero[IUT_MAX_V]`	`int`	†η°_v = 0

field	type	meaning
eta_pm[IUT_MAX_V]	int	†η^±_v = ±1 (we store +1; defined up to sign)
aut_plus_index	unsigned long long	[Aut : Aut_+] = 2

typedef struct … iut_sign_vector — iut_i.h

Def 6.1 (iv): Aut_+(†D) and Aut_α(†D) for α ∈ {±1}^V; signed +-full poly-isomorphisms

field	type	meaning
nV	unsigned	
alpha[IUT_MAX_V]	int	
all_plus	int	
all_minus	int	

typedef struct … iut_prop_six_seven — iut_i.h

Prop 6.7: from a D-Θ±-bridge to a D-Θ-bridge: †D_T|_{T\{0}} → †D_{T*} (two labels ±t ↦ one |t|), and the two strips †D_0, †D_⊞ ↦ the single †D_>; +-full replaced by full (good v) / Ex 4.4 (bad v)

field	type	meaning
n_from	unsigned	I → I*
n_to	unsigned	I → I*
from[2*IUT_MAX_LSTAR+1]	int	t ↦ |t| (t ≠ 0)
to[2*IUT_MAX_LSTAR+1]	int	t ↦ |t| (t ≠ 0)
strips_identified	const char *	"D_0, D_⊞ ↦ D_>"
poly_replaced	int	+-full → full (good v) / evaluation-section (bad v)

FlSemiPmTorsor **F_I^{⋊±}-torsor** — structures

a set with an F_I^{⋊±}-orbit of bijections to F_I (T of a Θell-bridge)

INTRODUCED / RECALLED **[IUT I]** §6, Definition 6.1 (i) p. 155
form: iut_fl_semi_pm_set — see FlSemiPmGroup
form: iut_labcusp_pm_torsor — see FlSemiPmGroup
form: iut_sign_vector — see FlSemiPmGroup
form: iut_prop_six_seven — see FlSemiPmGroup

FigSixSix **Frobenioids of a Θ±ell NF-HT** — structures

F-prime-strips Π_v ↷ O▷_{F̃_v}; F_v tempered at bad v; F^⊛_mod; F^⊛; F^⊛′

INTRODUCED / RECALLED **[IUT I]** §6, Figure 6.6 p. 183
form: iut_fl_semi_pm_set — see FlSemiPmGroup
form: iut_labcusp_pm_torsor — see FlSemiPmGroup
form: iut_sign_vector — see FlSemiPmGroup
form: iut_prop_six_seven — see FlSemiPmGroup

HalfPlaneDictionary **upper half-plane dictionary** — structures

F_I^{⋊±} ↔ z ↦ ±z+a (cusp, q = e^{2πiz}, basepoints V^±); F_I^* ↔ toral w = (z−i)/(z+i) (node, basepoints V^Bor); coric {±1}

INTRODUCED / RECALLED **[IUT I]** §6, Remark 6.12.3 p. 175; **[IUT I]** §6, Figure 6.4 p. 178
form: iut_fl_semi_pm_set — see FlSemiPmGroup
form: iut_labcusp_pm_torsor — see FlSemiPmGroup
form: iut_sign_vector — see FlSemiPmGroup
form: iut_prop_six_seven — see FlSemiPmGroup

GeometricBasepoint **geometric vs arithmetic basepoints** — properties

F_I^{⋊±} uses V^± (geometric, semi-unipotent is not normally terminal); F_I^* uses V^Bor (arithmetic, Borel is normally terminal)

INTRODUCED / RECALLED **[IUT I]** §6, Remark 6.12.6 p. 180
form: iut_fl_semi_pm_set — see FlSemiPmGroup
form: iut_labcusp_pm_torsor — see FlSemiPmGroup
form: iut_sign_vector — see FlSemiPmGroup
form: iut_prop_six_seven — see FlSemiPmGroup

Multiradial **multiradial** — properties

a radial algorithm compatible with arbitrary permutations of the radial data over the coric data (parallel transport)

INTRODUCED / RECALLED **[IUT II]** §1, Example 1.7 p. 32; **[Alien]** §3.1 p. 49, §3, Figure 3.1 p. 52

Uniradial **uniradial** — properties

a radial algorithm that is not multiradial (e.g. holomorphic structures)

INTRODUCED / RECALLED **[IUT II]** §1, Example 1.7 p. 32; **[Alien]** §3.1 p. 49, §3, Figure 3.1 p. 52

UnitsKummer **×-Kummer structure** — structures

unique Kummer isomorphism on O×

INTRODUCED / RECALLED **[IUT II]** §4, Definition 4.9 (i) p. 153
form: iut_local_datum — see BasicPrimeStrip

UnitsModTorsKummer **×μ-Kummer structure** — structures

Ism-orbit of isomorphisms O×μ ≅ étale-like O×μ

INTRODUCED / RECALLED **[IUT II]** §4, Definition 4.9 (i) p. 153

form: `iut_local_datum` — see BasicPrimeStrip

`BiCoric` **bi-coric** — properties

both: D⊢-prime-strips and mono-analytic log-shells

INTRODUCED / RECALLED **[IUT III]** §1, Theorem 1.5 (iii) p. 48

`HorizontallyCoric` **horizontally coric** — properties

invariant under Θ×μ-links (F⊢×μ-prime-strips, D⊢)

INTRODUCED / RECALLED **[IUT III]** §1, Theorem 1.5 (ii) p. 48

`VerticallyCoric` **vertically coric** — properties

invariant under log-links (étale-like data D, D⊢, étale log-shells)

INTRODUCED / RECALLED **[IUT III]** §1, Theorem 1.5 (i) p. 48

`KummerDetachment` **Kummer-detachment** — indeterminacies

detaching Frobenius-like data from étale-like data via Kummer theory

INTRODUCED / RECALLED **[IUT III]** §1, Remark 1.5.4 p. 56

`LogKummerCorrespondence` **log-Kummer correspondence** — structures

compatible Kummer isomorphisms along a vertical line, upper semi-compatible

INTRODUCED / RECALLED **[IUT III]** §3, Proposition 3.5 p. 103; **[IUT III]** §3, Theorem 3.11 (ii) p. 153

`UpperSemiCompatibility` **upper semi-compatibility** — indeterminacies

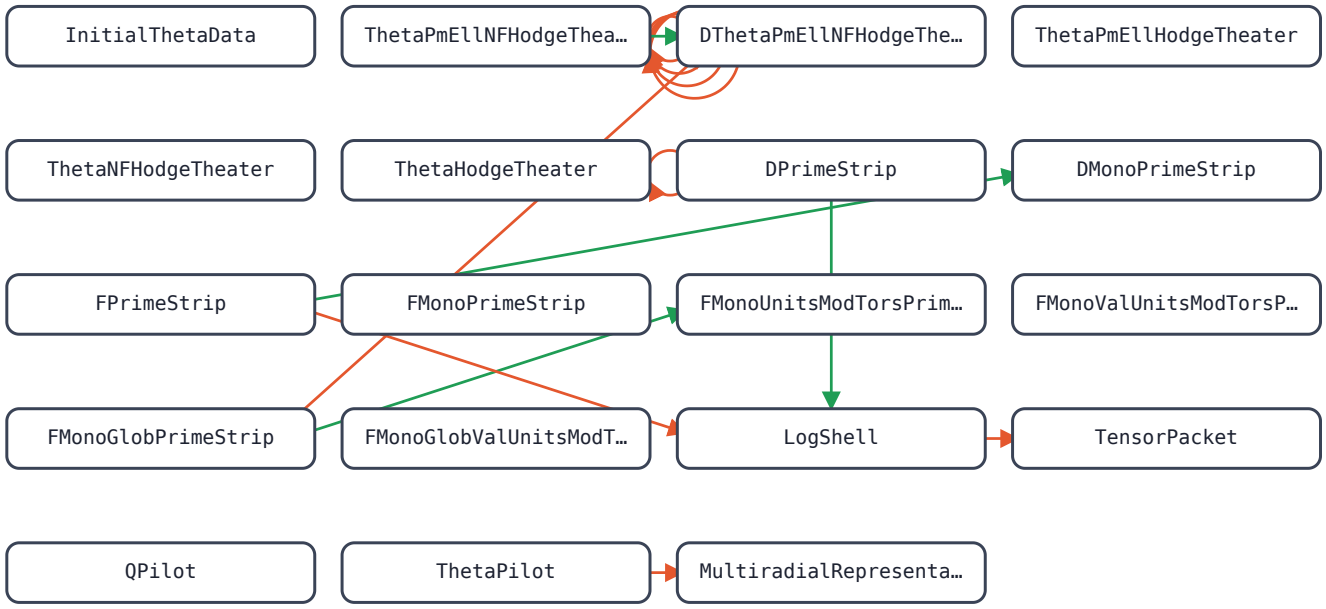
the log-shells contain the images at all stages (the source of Ind3)

INTRODUCED / RECALLED **[IUT III]** §3, Theorem 3.11 (ii) p. 153

10 · Species, mutations, theorems and diagrams

[IUTchIV] §3 makes the bookkeeping explicit: each data type is a species (a set-theoretic formula), each construction a mutation, a chain of them a mutation-history; the log-theta-lattice is the mutation-history that matters, and a core is an observable all of whose arrows are isomorphisms (the étale-picture around D⊢). The named theorems and the two pictures (Frobenius-picture and étale-picture) are listed here as objects too.

species and mutations ([IUTchIV] §3): green = core (isomorphisms), red = non-isomorphic mutation



Species and mutations. The set-theoretic bookkeeping: each data type is a species (a formula defining a collection of sets), each construction a mutation; a chain of mutations is a mutation-history. Different universes (Hodge theaters) are related only through the species. **[IUT IV]** §3 Def 3.1 p. 67, 3.3

`ZeroSpecies` `Mutation` `MutationHistory` `Core`

notation with symbols → English-letters-only identifier (^[A-Za-z]+$, unique case-insensitively)

ᵀHT	→	DagHodgeTheater
Fl⊢▶×μ	→	FMonoGlobValUnitsModTorsPrimeStrip
Θ×μ_LGP-link	→	ThetaUnitsModTorsLGPTorsorLink
F_l^{⋊±}	→	FlSemiPm
S^±_{j+1}	→	SPmJPlusOne
(n,∘)	→	NCircColumn
D_◎±	→	DCircDotPm
Fl⊢_Δ	→	FMonoGlobTriangle

448 identifiers generated from src/iut_ids.c; full table in explorer tab 7.

English-letters-only identifiers. Every symbol in a notation maps to a word; the identifiers are the keys used in the JSON of the C program and the web model and are checked for uniqueness by `make check`. docs/IDENTIFIERS.md

`Dag` `Ddag` `Mono` `MonoGlob` `UnitsModTors` `BoxPlus` `CircAst` `Triangle`

26 entries: 5 species, 6 mutations, 13 theorems and definitions as objects, 2 diagrams.

`ZeroSpecies` **0-species** species

set-theoretic formula Φ0(E) defining the 'objects'

INTRODUCED / RECALLED **[IUT IV]** §3, Definition 3.1 (i) p. 68

`typedef struct … iut_species` iut.h

---------- species / mutations (IUTchIV)

field	type	meaning
`id`	`const char *`	letters-only
`name`	`const char *`	
`ref`	`const char *`	
`species_data`	`const char *`	informal description of the set-theoretic formula Φ0

`Mutation` **mutation** mutations

species-theoretic functor between species

INTRODUCED / RECALLED **[IUT IV]** §3, Definition 3.3 (i) p. 72

LEAN `structure Mutation — lean/IUT/Species.lean`

`typedef struct … iut_mutation` iut.h

field	type	meaning
`id`	`const char *`	
`name`	`const char *`	
`dom`	`const char *`	species ids
`cod`	`const char *`	species ids
`ref`	`const char *`	
`is_core`	`int`	observable all of whose Ve are isomorphisms

`MutationHistory` **mutation-history** mutations

oriented graph Γ with species at vertices and mutations at edges (e.g. the log-theta-lattice)

INTRODUCED / RECALLED **[IUT IV]** §3, Definition 3.3 (iii) p. 72

LEAN `structure MutationHistory — lean/IUT/Species.lean`

form: `iut_mutation` — see Mutation

`Core` **core** mutations

observable all of whose mutations are isomorphisms (coric data)

INTRODUCED / RECALLED **[IUT IV]** §3, Definition 3.3 (v) p. 72

form: `iut_mutation` — see Mutation

`CorThreeTwelve` **[IUTchIII] Cor 3.12** theorems and definitions as objects

$-|\log(\Theta)| \geq -|\log(q)|$ (the log-volume estimate)

INTRODUCED / RECALLED **[IUT III]** §3, Corollary 3.12 p. 173

`ThmThreeEleven` **[IUTchIII] Thm 3.11** theorems and definitions as objects

multiradial algorithms for the Θ-pilot (the 'main theorem')

INTRODUCED / RECALLED **[IUT III]** §3, Theorem 3.11 p. 153

LEAN `structure ThmThreeEleven — lean/IUT/StageTwo.lean`

`ThmOneTen` **[IUTchIV] Thm 1.10** theorems and definitions as objects

the Diophantine inequality derived from Cor 3.12

INTRODUCED / RECALLED **[IUT IV]** §1, Theorem 1.10 p. 22

`CorOneTwo` **[IUTchI] Cor 1.2** theorems and definitions as objects

characteristic nature of coverings: functorial group-theoretic reconstruction of $\Pi_{\underline{X}}$, $\Pi_{C\to}$, $\Pi_{\underline{C}}$ from $\Pi_{X\to}$; l from $l^2 = [\Delta_X : \Delta_{X\to}]$

INTRODUCED / RECALLED **[IUT I]** §1, Corollary 1.2 p. 39

LEAN `structure CorOneTwo — lean/IUT/Construction.lean`

`typedef struct … iut_cor_one_two` iut_i.h

Cor 1.2 / Rem 1.2.1: what is reconstructed group-theoretically from $\Pi_{X\to}$ (resp. $\Pi_{C\to}$)

field	type	meaning
`l_squared`	`unsigned`	$l^2 = [\Delta_X : \Delta_{\underline{X}}]\cdot[\Delta_{\underline{X}} : \Delta_{X\to}] = [\Delta_X : \Delta_{X\to}] = [\Delta_C : \Delta_{X\to}]/2$
`idx_X_XUnder`	`unsigned`	
`idx_XUnder_XArrow`	`unsigned`	
`idx_C_XArrow`	`unsigned`	
`gal_XUnder_over_CUnder`	`unsigned`	2 $\mathrm{Gal}(\underline{X}/\underline{C}) \cong Z/2$
`gal_CArrow_over_CUnder`	`unsigned`	l $\mathrm{Gal}(C\to/\underline{C}) \cong Z/l$
`gal_XArrow_over_CUnder`	`unsigned`	$2l$ $\mathrm{Gal}(X\to/\underline{C}) \cong Z/2 \times Z/l$
`gal_XArrow_over_CArrow`	`unsigned`	2 the unique maximal subgroup of odd index
`aut_k_XArrow`	`unsigned`	$Z/2l$, Z/l (Rem 1.2.1)
`aut_k_CArrow`	`unsigned`	$Z/2l$, Z/l (Rem 1.2.1)
`steps[8]`	`const char *`	the algorithm, step by step

field	type	meaning
`stabilized_cusps[3]`	`int`	0, ε', ε'': the splitting Gal(X/C) is the one stabilizing them

TemperedConjugates **[IUTchI] Prop 2.1** theorems and definitions as objects

a Π_G-conjugate of a nontrivial compact subgroup of Π^tp_G that lies in Π^tp_G is a Π^tp_G-conjugate

INTRODUCED / RECALLED **[IUT I]** §2, Proposition 2.1 p. 45
form: `iut_tempered_fact` — see PiTpG
form: `iut_tempered_at_v` — see PiTpG

DecompositionSynchronized **[IUTchI] Cor 2.5** theorems and definitions as objects

a decomposition / inertia group of Π_X contained in Π^tp_X is a tempered one: the conjugacy indeterminacies of tempered and profinite cusps agree

INTRODUCED / RECALLED **[IUT I]** §2, Corollary 2.5 p. 51
form: `iut_tempered_fact` — see PiTpG
form: `iut_tempered_at_v` — see PiTpG

DiscreteAnalogue **[IUTchI] Thm 2.6** theorems and definitions as objects

the discrete (topological fundamental group) analogues of Props 2.1–2.5

INTRODUCED / RECALLED **[IUT I]** §2, Theorem 2.6 p. 56; **[IUT I]** §2, Lemma 2.7 p. 57; **[IUT I]** §2, Corollary 2.8 p. 59
form: `iut_tempered_fact` — see PiTpG
form: `iut_tempered_at_v` — see PiTpG

DefThreeOne **[IUTchI] Def 3.1 (a)-(f)** theorems and definitions as objects

the six conditions on initial Θ-data; (c), (d), (e), (f) are machine-checked here, (a), (b) are asserted flags

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.1 p. 61

`typedef struct … iut_def_three_one_item` iut_i.h

`============================================================` §3 `=======` Def 3.1 (a)–(f): the conditions on initial Θ-data, each checked or asserted

field	type	meaning
`item`	`const char *`	
`condition`	`const char *`	
`ok`	`int`	1 computed, 0 asserted, 2 partly computed and partly asserted
`machine_checked`	`int`	1 computed, 0 asserted, 2 partly computed and partly asserted
`detail`	`const char *`	

DefThreeSix **[IUTchI] Def 3.6** theorems and definitions as objects

a Θ-Hodge theater ᵀHT^Θ = ({ᵀF_v}_{v∈V}, ᵀFℍ_mod) with (a) local models, (b) global model, (c) compatibility

INTRODUCED / RECALLED **[IUT I]** §3, Definition 3.6 p. 87
form: `iut_global_realified` — see CMonoGlobMod
form: `iut_f_monoglob` — see CMonoGlobMod
form: `iut_theta_hodge_theater_def` — see CMonoGlobMod

CorThreeSeven **[IUTchI] Cor 3.7** theorems and definitions as objects

the Θ-link ᵀFℍ_tht ≅ ǂFℍ_mod (full poly-iso); (ii) preserves D⊢_v; (iii) preserves O×_{C⊢v}

INTRODUCED / RECALLED **[IUT I]** §3, Corollary 3.7 p. 88
form: `iut_global_realified` — see CMonoGlobMod
form: `iut_f_monoglob` — see CMonoGlobMod
form: `iut_theta_hodge_theater_def` — see CMonoGlobMod

PropSixSeven **[IUTchI] Prop 6.7** theorems and definitions as objects

from a D-Θ±-bridge to a D-Θ-bridge: ᵀD_T|_{T\{0}} → ᵀD_{T*} (±t ↦ |t|), ᵀD_0 and ᵀD_⊞ ↦ ᵀD_>

INTRODUCED / RECALLED **[IUT I]** §6, Proposition 6.7 p. 167
form: `iut_fl_semi_pm_set` — see FlSemiPmGroup
form: `iut_labcusp_pm_torsor` — see FlSemiPmGroup
form: `iut_sign_vector` — see FlSemiPmGroup
form: `iut_prop_six_seven` — see FlSemiPmGroup

FrobeniusPicture **Frobenius-picture** diagrams

the log-theta-lattice as a diagram of Frobenius-like objects

INTRODUCED / RECALLED **[IUT III]** §1, Definition 1.4 p. 45

EtalePicture **étale-picture** diagrams

D⊢-prime-strips (coric) radiating to the Hodge theaters: the multiradial picture

INTRODUCED / RECALLED **[IUT III]** §3, Corollary 3.12 p. 173; **[IUT I]** §6, Corollary 6.10 p. 170

ZFCG **ZFCG** species

ZFC + Grothendieck's axiom on universes

INTRODUCED / RECALLED **[IUT IV]** §3 p. 67

OneSpecies **1-species** species

set-theoretic formula Φ1(E,F,M) defining the 'morphisms'

INTRODUCED / RECALLED **[IUT IV]** §3, Definition 3.1 (ii) p. 68

form: `iut_species` — see ZeroSpecies

`SpeciesMorphism` **species-morphism** species

a set M satisfying Φ1

INTRODUCED / RECALLED **[IUT IV]** §3, Definition 3.1 (iii) p. 68
form: `iut_species` — see ZeroSpecies

`SpeciesObject` **species-object** species

a set E satisfying Φ0 in a model of ZFCG

INTRODUCED / RECALLED **[IUT IV]** §3, Definition 3.1 (iii) p. 68
form: `iut_species` — see ZeroSpecies

`MorphismOfMutations` **morphism of mutations** mutations

species-theoretic natural transformation

INTRODUCED / RECALLED **[IUT IV]** §3, Definition 3.3 (ii) p. 72
form: `iut_mutation` — see Mutation

`Observable` **observable** mutations

compatible collection of mutations S_v → S_obs over a mutation-history

INTRODUCED / RECALLED **[IUT IV]** §3, Definition 3.3 (iv) p. 72
form: `iut_mutation` — see Mutation

`AlienCopy` **alien copy** mutations

a copy of a ring/scheme theory related to another only via a non-ring-theoretic link

INTRODUCED / RECALLED **[Alien]** §1 p. 6, §3 p. 49
form: `iut_link` — see ThetaLink

`ThmA` **[ExpEst] Thm A** theorems and definitions as objects

effective ABC inequality (|abc| ≤ 2^4 max{exp(1.7·10^30 ε^{−166/81}), rad(abc)^{3(1+ε)}})

INTRODUCED / RECALLED **[ExpEst]** Introduction, Theorem A p. 4

`ThmB` **[ExpEst] Thm B** theorems and definitions as objects

FLT holds for primes p > 1.615·10^14

INTRODUCED / RECALLED **[ExpEst]** Introduction, Theorem B p. 4

[p.381]

Part II

The processes

The verbs of the theory. Chapter 11 walks through the ten progressions step by step, with the diagram of every step; chapter 12 lists all 73 processes of the registry, each with its citation and the transcript of `iut run` on the demo Θ-data (abc(1,2400,2401) l=13).

11 · Ten progressions, step by step

Each progression is a sequence of states of the data model; the web app (`htdocs/index.html`, "Worked examples") walks through the same constructions on the live C engine; the diagrams below are rendered for the demo data.

11.1 From an abc triple to initial Θ-data

SOURCES **[IUT I]** Def 3.1 (via [IUTchIV] §2, [Pano] §3) p. 61; **[IUT IV]** Cor 2.2 p. 41

How the program manufactures the input of the theory: the Frey curve of a + b = c gives E_F over F_mod = Q with multiplicative reduction at the primes of abc; l is chosen; V, V^bad_mod and the fields follow.

STEP 1 / 6 **the triple**

$$1 + 2400 = 2401$$

a = 1	b = 2400	c = 2401
= 1	= $2^5 \cdot 3 \cdot 5^2$	= 7^4

coprime positive integers; rad(abc) = 2·3·5·7

`log(abc) = 15.567 log rad(abc) = 5.347`

The abc conjecture says log(c) ≤ (1+ε) log rad(abc) + C_ε. IUT reaches it through the Szpiro inequality for the Frey curve: (1/6) log q ≤ (1+ε) log(cond) + C, where log q is the sum of the valuations of the q-parameters ([IUTchIV] Thm 1.10, Cor 2.2).

Pick coprime positive integers with a + b = c (the generator takes them from the bench above). The height of the triple is log c; the conductor-like quantity is log rad(abc).

STEP 2 / 6 **the Frey curve**

$E_F : y^2 = x(x - a)(x + b)$ over $F = F_{mod} = Q$

$\Delta = 16(abc)^2$, full rational 2-torsion, so $F_{tpd} = Q$ and $\log d^{Ftpd} = 0$

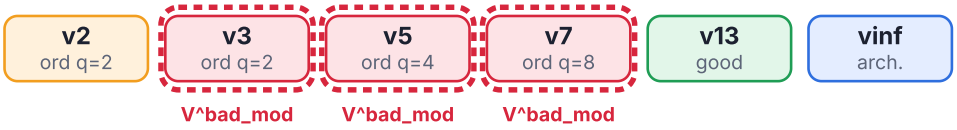
p = 2	p = 3	p = 5	p = 7
$ord_p(abc) = 5$	$ord_p(abc) = 1$	$ord_p(abc) = 2$	$ord_p(abc) = 4$
multiplicative	**multiplicative**	**multiplicative**	**multiplicative**
$ord_p(q) = 2$	$ord_p(q) = 2$	$ord_p(q) = 4$	$ord_p(q) = 8$

At an odd prime $p \mid abc$ the curve has split or non-split multiplicative reduction with Tate parameter q of valuation $ord_p(q) = 2 \cdot ord_p(abc)$. At $p = 2$ the minimal model has $ord_2(q) = 2 \cdot ord_2(abc) - 8$ when $2^5 \mid abc$, good reduction when $2^4 \parallel abc$, and additive reduction otherwise (stable only after an extension — flagged as a warning by the validator).

The Frey curve turns the additive relation $a + b = c$ into an elliptic curve whose bad primes are exactly the primes of abc. The program records the reduction type and $ord_p(q)$ at each of them.

STEP 3 / 6 **V ≅ V_mod and V^bad_mod**

V = V(F) ≅ V_mod = V(F_mod): one valuation per prime of F_mod, plus the archimedean place



V^bad_mod = {v3, v5, v7} must be nonempty: odd residue characteristic, bad multiplicative reduction [IUTchI] D

V^good = good reduction (3, 5 and I are added when absent, since E[6] and E[I] matter); $v_\infty \in$ V^arc

legend:

- odd bad multiplicative (V^bad_mod)
- 2 bad multiplicative (bad but not in V^bad_mod)
- additive
- good
- archimedean

The prime-strips of every Hodge theater are indexed by this V. Each $v \in$ V^bad_mod carries the theta function and the q-parameter q_v whose valuations enter $\log q$; the Θ-link is defined in terms of the value groups at these v.

The valuations become the index set of every prime-strip. The odd bad multiplicative primes form V^bad_mod, which must be nonempty — the Θ-link lives there.

STEP 4 / 6 **l, the fields and the labels**

l = 13 prime ≥ 5, [F : F_mod] prime to l, Im(Gal(K/F) → $GL_2(F_l)$) ⊇ $SL_2(F_l)$

l* = (l−1)/2 = 6 l± = (l+1)/2 = 7 F_l = {−l*,...,l*}, $|F_l|$ = {0,...,l*}, $F_l^{\times}$ = {1,...,l*}



log d^K is estimated as log f + 2 log l (ramification of the l-torsion field over the conductor primes and l); the validator warns when l | ord(q_v) ([IUTchIV] Prop 1.8 (vii)) and rejects l = 5 for this abc family ([F:F_mod] divisible by l).

The prime l fixes the size of all the label sets (capsules of l and l* prime-strips) and the field K = F(E[l]) whose Galois group must be big. Everything in the data model that is "not computable from integers" (K-core, SL_2 image ...) is recorded as an asserted hypothesis.

STEP 5 / 6 **the invariants log q, log f, log d**

log q = Σ_{v bad} ord(q_v)·f_v·log p / [F_mod:Q]	24.202
|log q| = log q / 2l (value group of q_v in units of 2l-th roots)	0.931
ht_E = (1/6) log q (height of the Frey curve)	4.034
log f = Σ_{v bad} f_v·log p / [F_mod:Q] (conductor)	5.347
log d^Ftpd (different of F_tpd = Q)	0.000
log d^K (estimate: log d^Ftpd + log f + 2 log l)	10.477
j-invariant log height of the Frey curve (for reference)	21.112

These are the numbers the final estimate compares: the left side −|log Θ| of Cor 3.12 is controlled by log d^K (through the hull of the log-shells), the right side by log q. The explicit constant of [IUTchIV] Thm 1.10 also involves e_mod and the constant η_prm of [ExpEst].

From the valuations the program computes the quantities that appear in the inequality of [IUTchIII] Cor 3.12 and in the explicit [IUTchIV] Thm 1.10.

STEP 6 / 6 **validate**

validation report for abc(1,2400,2401) l=13: 0 errors, 0 warnings, 8 checks

LPrime	l = 13 prime, l* = 6, l± = 7
Fmod	F_mod ⊆ F ⊆ K with [F_mod:Q]=1, [F:Q]≤276480, [K:Q]≤7245987840
VBadMod	|V^bad_mod| = 3 odd prime(s) of bad multiplicative reduction (2 is ba…
FBarOverF	asserted (not machine-checkable): F Galois over F_mod of degree prime…
FBarOverF	asserted (not machine-checkable): F contains the fields of definition…
KField	asserted (not machine-checkable): Im(Gal(K/F) → GL2(F_l)) ⊇ SL2(F_l)
XF	asserted (not machine-checkable): E_F has stable reduction everywhere
CK	asserted (not machine-checkable): C_K is a K-core

Every condition of [IUTchI] Def 3.1 that can be checked on numbers is checked; the rest is listed as asserted. The same checks run in the C program (`iut validate`) and in `node scripts/manual/lib/test.js`.

11.2 Forgetting: down the prime-strip ladder

SOURCES **[IUT I]** Fig I1.2 p. 6, Def 4.1 p. 95, Def 5.2 p. 134; **[IUT II]** Def 4.9 p. 153; **[IUT III]** Thm 1.5 p. 48

A prime-strip is built from the Θ-data and then progressively stripped: holomorphic → mono-analytic → units → units mod torsion → only G_v. What survives each step is what the corresponding link can share.

STEP 1 / 6 **F-prime-strip (holomorphic)**

F-prime-strip Frobenius-like, holomorphic

FPrimeStrip

	v2	v3	v5	v7	v13	vinf
Π_v	✓	✓	✓	✓	✓	✓
G_v	✓	✓	✓	✓	✓	✓
O× / O▷ (monoid)	✓	✓	✓	✓	✓	✓
mod torsion ×μ	·	·	·	·	·	·
value group O▶	✓	✓	✓	✓	✓	✓
splitting O▶ × O×	·	·	·	·	·	·
Kummer structure	UnitsKum...	UnitsKum...	UnitsKum...	UnitsKum...	UnitsKum...	UnitsKum...
q-values at bad v	·	{q^1}	{q^1}	{q^1}	·	·
global CI⊩	·					

coric: not coric — a fresh copy at every (n,m)

An F-prime-strip remembers at each v the tempered group Π_v (hence the ring/scheme theory: "holomorphic") and the Frobenius-like monoid O▷_v with its Kummer structure. The log-link is defined on these.

```
validator: 0 errors — F-prime-strip demo well-formed
```

Start: the holomorphic Frobenius-like strip, the local data of a Hodge theater. This is the species the log-link needs, since the logarithm uses both ring operations.

STEP 2 / 6 **→ F⊢ (forget Π_v)**

F⊢-prime-strip Frobenius-like, mono-analytic

FMonoPrimeStrip

	v2	v3	v5	v7	v13	vinf
Π_v	·	·	·	·	·	·
G_v	✓	✓	✓	✓	✓	✓
O× / O▷ (monoid)	✓	✓	✓	✓	✓	✓
mod torsion ×μ	✓	✓	✓	✓	✓	✓
value group O▶	✓	✓	✓	✓	✓	✓
splitting O▶ × O×	✓	✓	✓	✓	✓	✓
Kummer structure	UnitsMod...	UnitsMod...	UnitsMod...	UnitsMod...	UnitsMod...	UnitsMod...
q-values at bad v	·	{q^1,q...	{q^1,q...	{q^1,q...	·	·
global CI⊩	·					

coric: not coric — a fresh copy at every (n,m)

Replacing Π_v by its quotient G_v forgets the holomorphic structure; the monoid is now split as O▶ × O× ("mono-analytic"). This is still not something a Θ-link can share, because the value group O▶ is dilated.

```
validator: 0 errors — F⊢-prime-strip demo well-formed
```

Forget Π_v, keep G_v: mono-analytic. The splitting O▷ ≅ O▶ × O× becomes part of the data ([IUTchI] Def 5.2 (ii)).

STEP 3 / 6 → F⊢× (units)

F⊢×-prime-strip Frobenius-like, mono-analytic

FMonoUnitsPrimeStrip

	v2	v3	v5	v7	v13	vinf
Π_v	·	·	·	·	·	·
G_v	✓	✓	✓	✓	✓	✓
O× / O▷ (monoid)	✓	✓	✓	✓	✓	✓
mod torsion ×μ	·	·	·	·	·	·
value group O▶	·	·	·	·	·	·
splitting O▶ × O×	·	·	·	·	·	·
Kummer structure	UnitsKum...	UnitsKum...	UnitsKum...	UnitsKum...	UnitsKum...	UnitsKum...
q-values at bad v	·	·	·	·	·	·
global CI⊢	·					

coric: not coric — a fresh copy at every (n,m)

Keep only the unit group O× with its Kummer structure. The q-values (value group) are gone.

```
validator: 0 errors — F⊢×-prime-strip demo well-formed
```

Drop the value groups: only the units O×_v with G_v-action remain.

STEP 4 / 6 → F⊢×μ (units mod torsion)

F⊢×μ-prime-strip Frobenius-like, mono-analytic

FMonoUnitsModTorsPrimeStrip

	v2	v3	v5	v7	v13	vinf
Π_v	·	·	·	·	·	·
G_v	✓	✓	✓	✓	✓	✓
O× / O▷ (monoid)	✓	✓	✓	✓	✓	✓
mod torsion ×μ	✓	✓	✓	✓	✓	✓
value group O▶	·	·	·	·	·	·
splitting O▶ × O×	·	·	·	·	·	·
Kummer structure	UnitsMod...	UnitsMod...	UnitsMod...	UnitsMod...	UnitsMod...	UnitsMod...
q-values at bad v	·	·	·	·	·	·
global CI⊢	·					

coric: horizontally (across Θ-links)

O×μ = O×/μ with the ×μ-Kummer structure and the integral structures {O×μ_H}. This is exactly the data (aΘ) ≅ (aq) of the Θ-link: horizontally coric ([IUTchIII] Thm 1.5 (ii)). Add back value groups (▶) and CI⊢ to get the FI⊢▶×μ-strip that the Θ×μ-links glue.

```
validator: 0 errors — F⊢×μ-prime-strip demo well-formed
```

Kill the roots of unity: F⊢×μ. This is the strip a Θ-link transports unchanged — the ∧ of the whole theory lives here.

STEP 5 / 6 → D⊩ (only G_v)

D⊩-prime-strip étale-like, mono-analytic

DMonoPrimeStrip

	v2	v3	v5	v7	v13	vinf
Π_v	·	·	·	·	·	·
G_v	✓	✓	✓	✓	✓	✓
O× / O▷ (monoid)	·	·	·	·	·	·
mod torsion ×μ	·	·	·	·	·	·
value group O▶	·	·	·	·	·	·
splitting O▶ × O×	·	·	·	·	·	·
Kummer structure	·	·	·	·	·	·
q-values at bad v	·	·	·	·	·	·
global C⊩	·					

coric: bi-coric (log- and Θ-links)

Only the topological groups G_v remain: étale-like, mono-analytic, hence shared by log-links and Θ-links alike (bi-coric, [IUTchIII] Thm 1.5 (iii)). This is the core of the étale-picture.

```
validator: 0 errors — D⊩-prime-strip demo well-formed
```

Forget the monoid as well: D⊩, the bi-coric core. Every Hodge theater in the lattice sees the same D⊩ up to Ind1.

STEP 6 / 6 F⊩▶×μ: what the Θ×μ-links glue

F⊩▶×μ-prime-strip Frobenius-like, mono-analytic, with globa

FMonoGlobValUnitsModTorsPrimeStrip

	v2	v3	v5	v7	v13	vinf
Π_v	·	·	·	·	·	·
G_v	✓	✓	✓	✓	✓	✓
O× / O▷ (monoid)	✓	✓	✓	✓	✓	✓
mod torsion ×μ	✓	✓	✓	✓	✓	✓
value group O▶	✓	✓	✓	✓	✓	✓
splitting O▶ × O×	✓	✓	✓	✓	✓	✓
Kummer structure	UnitsMod...	UnitsMod...	UnitsMod...	UnitsMod...	UnitsMod...	UnitsMod...
q-values at bad v	·	{q^1,q...	{q^1,q...	{q^1,q...	·	·
global C⊩	arithmetic degree -4.344					

coric: not coric — a fresh copy at every (n,m)

The F⊩▶×μ-prime-strip packs O▶ × O×μ at each v together with a global realified Frobenioid C⊩ (arithmetic degrees). The Θ×μ_LGP-link glues F⊩_LGP of †HT (Θ-pilot, values q^{j²}) to F⊩_Δ of ‡HT (q-pilot, value q).

```
validator: 0 errors — F⊩▶×μ-prime-strip demo well-formed
```

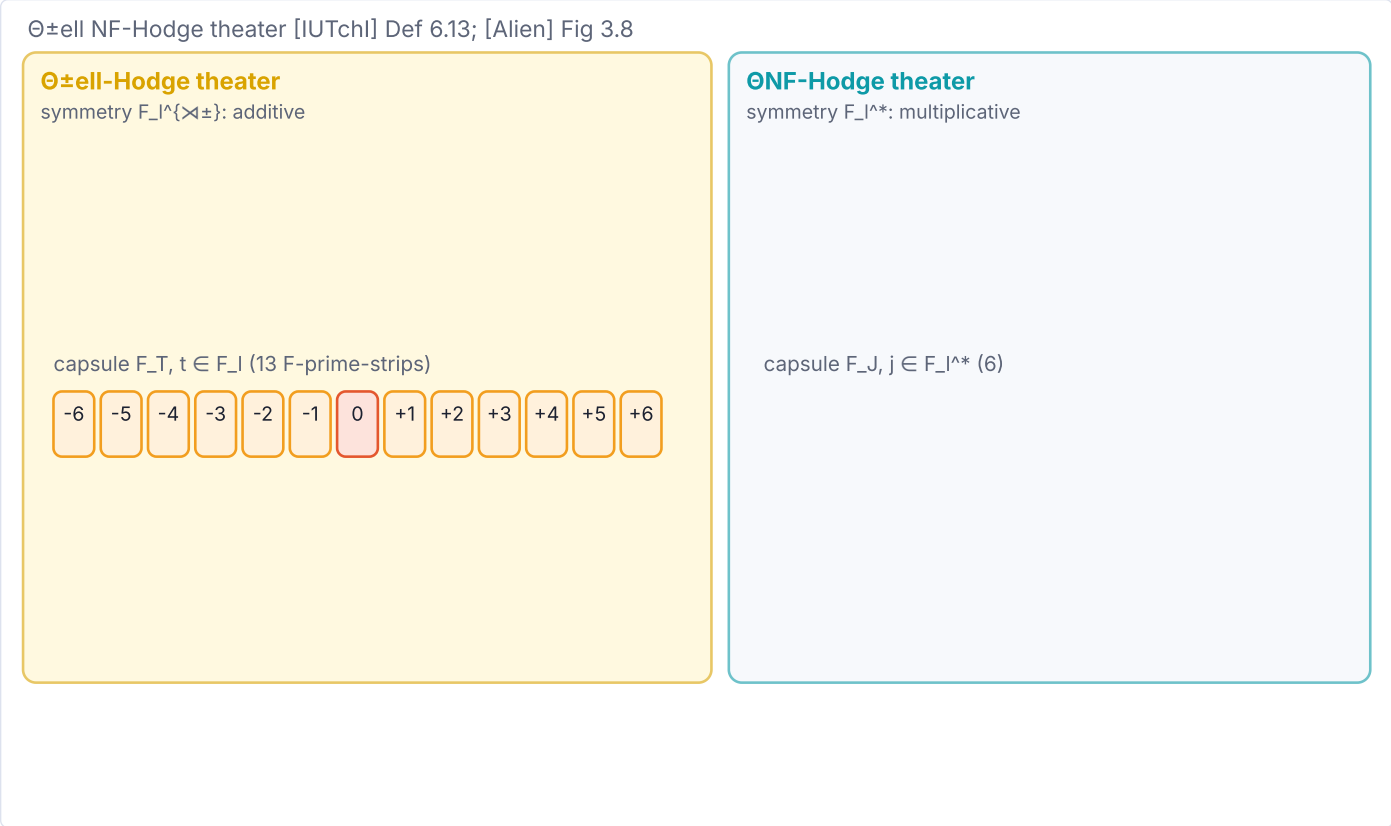
Going back up with global data: the F⊩▶×μ-strips F⊩_Δ, F⊩_env, F⊩_gau, F⊩_LGP, F⊩_lgp are the pilots; the explorer builds all of them at every vertex.

11.3 Assembling a Θ±ell NF-Hodge theater

SOURCES **[IUT I]** Def 4.6 p. 111, Def 5.5 p. 151, Def 6.4 p. 162, Def 6.13 p. 182; **[Alien]** §3.3 (v) p. 58, §3.6 (ii) p. 87, Fig 3.8 p. 67

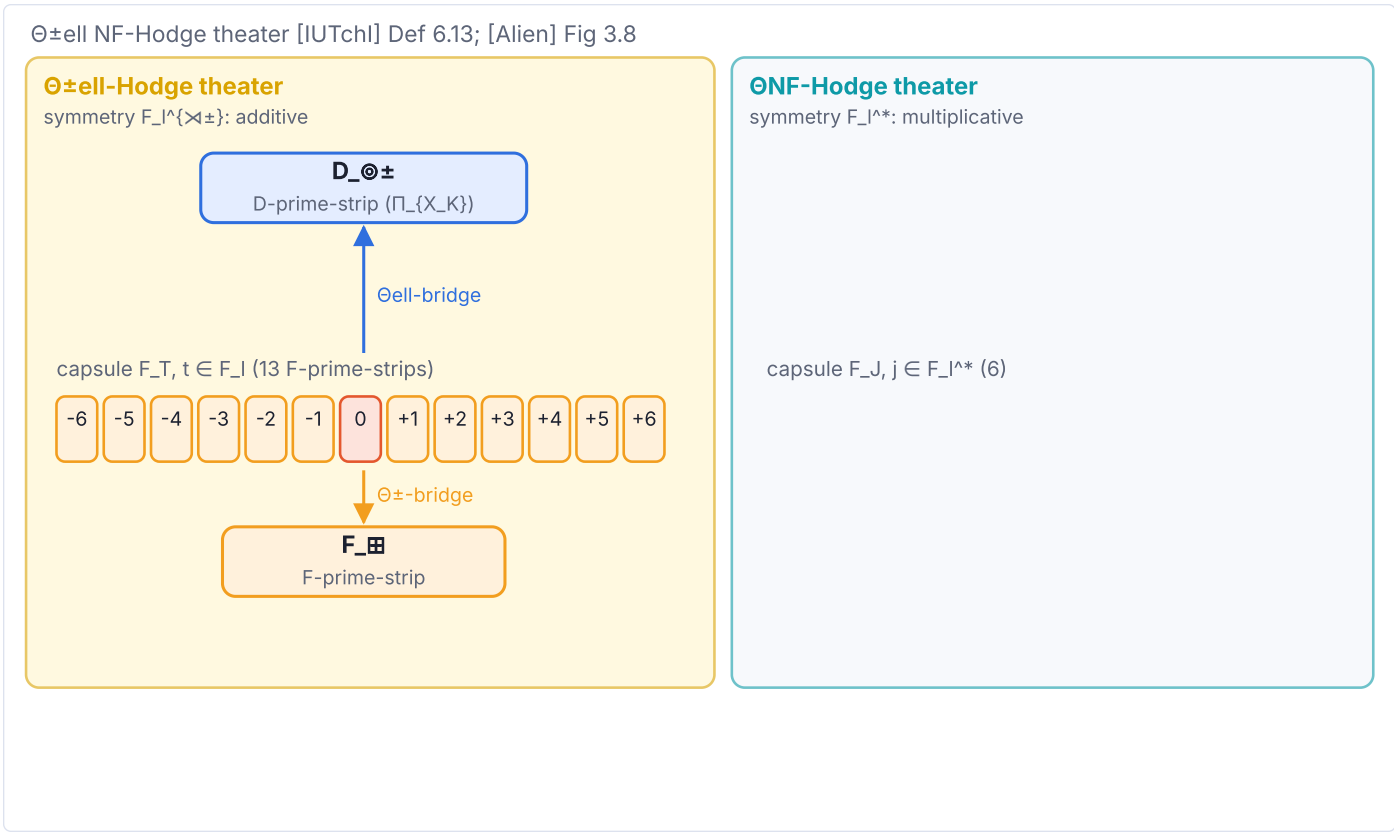
One vertex of the lattice is built in five moves: two capsules of prime-strips, four bridges, one gluing. The additive symmetry acts on the left, the multiplicative one on the right.

STEP 1 / 5 **the capsule F_T**



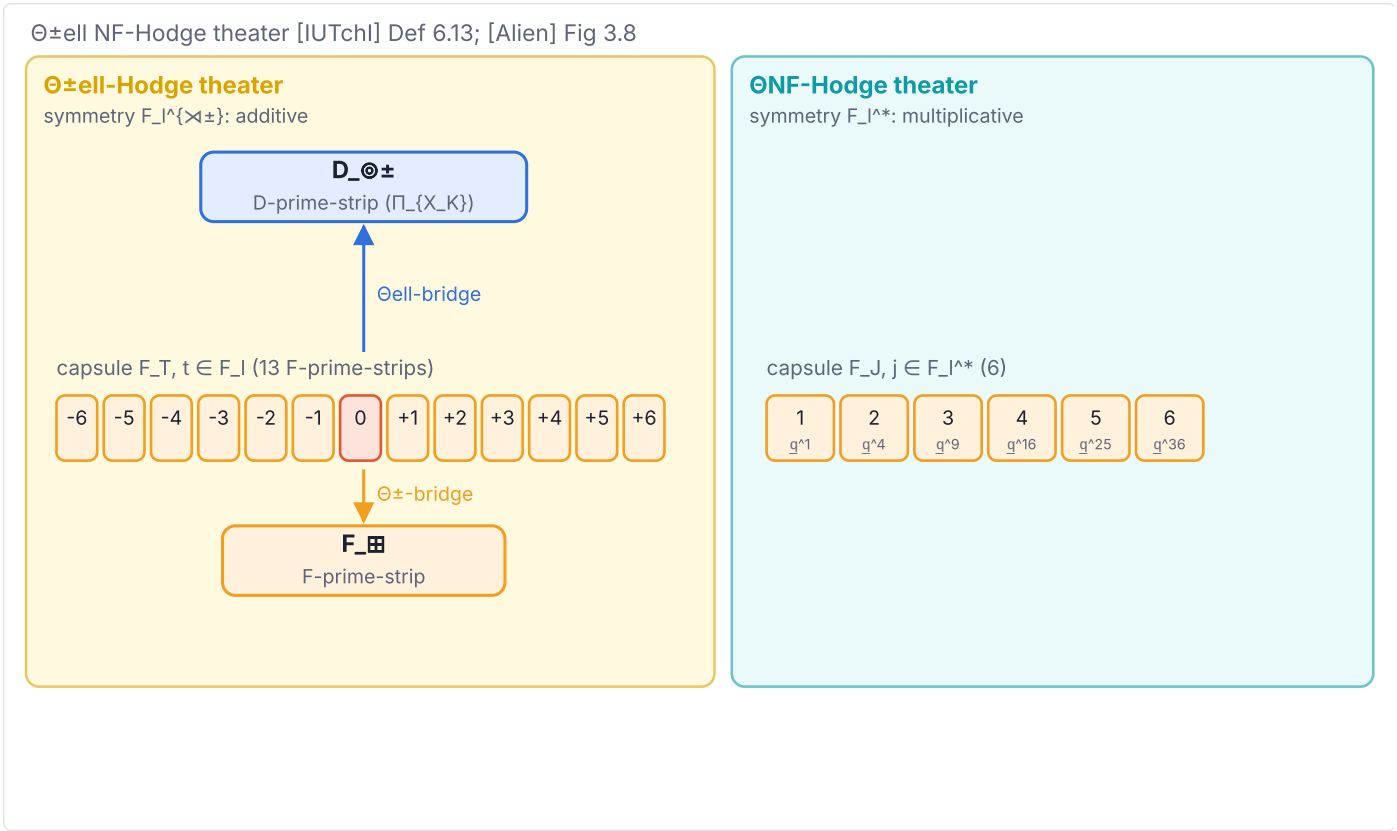
A capsule of l F-prime-strips indexed by t ∈ F_l, with distinct basepoints, on which F_l^{⋈±} acts by conjugate synchronization (the ±-symmetry is geometric: it comes from the automorphisms of the elliptic curve).

STEP 2 / 5 **Θ±-bridge and Θell-bridge**

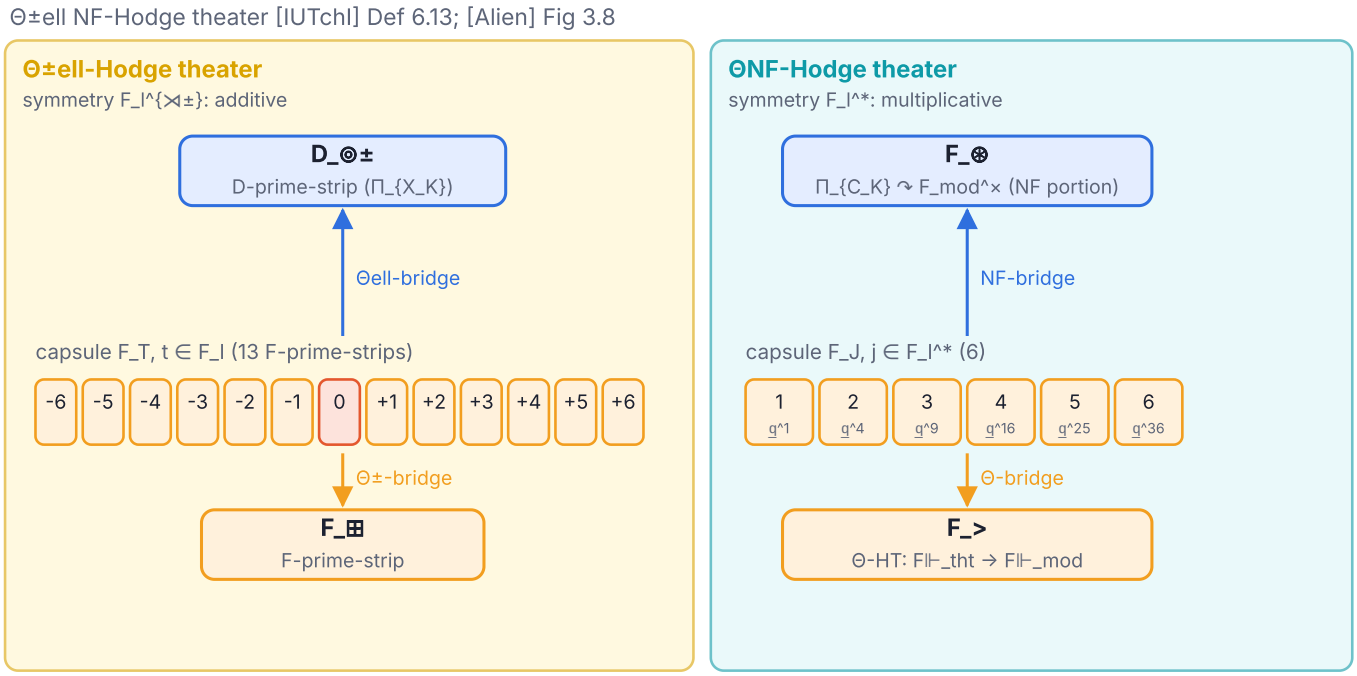


The Θ±-bridge collapses the capsule onto a single F-prime-strip F_⊞ (the additive "⊞" label); the Θell-bridge relates the capsule to the D-prime-strip D_◎± that carries Π_{X_K} and the I-torsion. Together: the Θ±ell-Hodge theater.

STEP 3 / 5 **the capsule F_J**

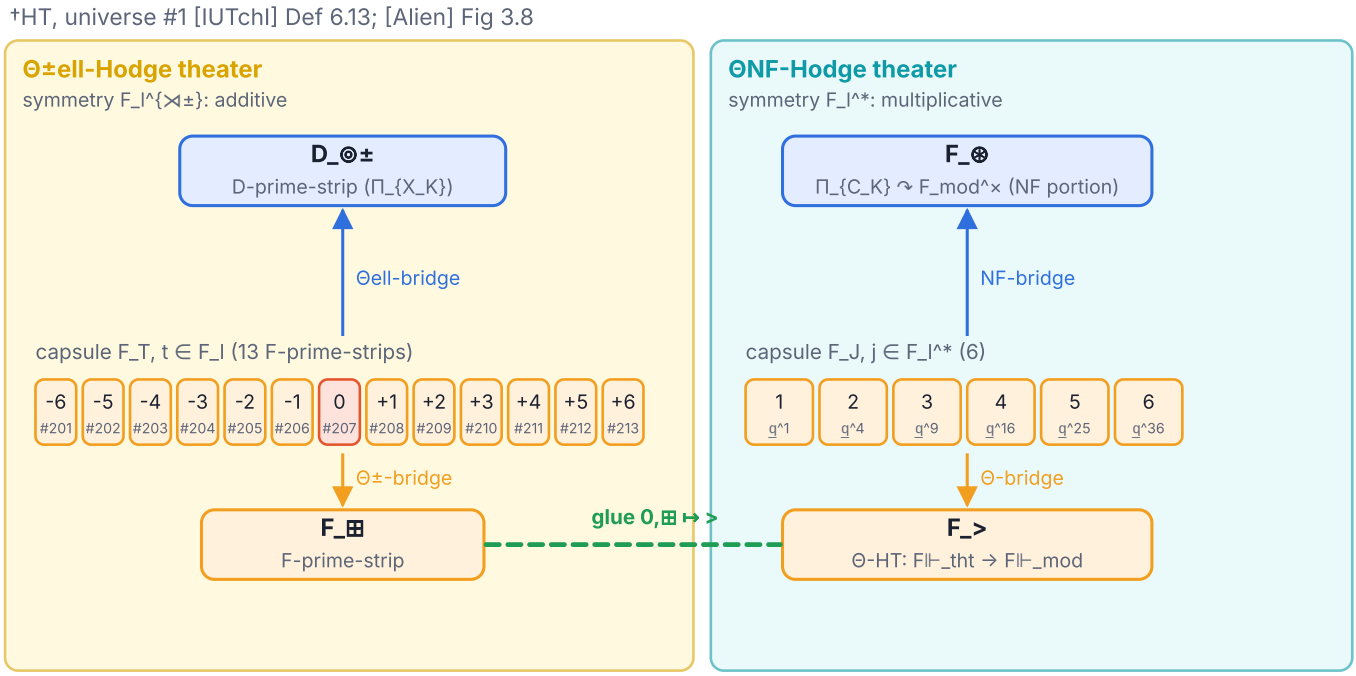


A capsule of l* F-prime-strips indexed by j ∈ F_l^* (the labels |F_l| \ {0}), with the multiplicative symmetry F_l^* acting through the number-field side.



The Θ-bridge sends the capsule to F_> (the ">" label) inside a Θ-Hodge theater that carries the theta values Θ_v(j) = q^{j²} and the strips FIH-_tht → FIH-_mod; the NF-bridge relates it to F_⊛ = (Π_{C_K} ↷ F_mod^×), the number-field portion. Together: the ΘNF-Hodge theater.

STEP 5 / 5 **the gluing 0,⊞ ↦ >**



q-pilot {q_v}: −|log q| = -0.931 Θ-pilot {q_v^{j²}}: -14.118 (naive, ×l(l+1)/12)

IIℍ-strips: FIIℍ_Δ (q-pilot) · FIIℍ_env · FIIℍ_gau · FIIℍ_LGP (Θ-pilot) · FIIℍ_lgp

carries log-shells (codomain of a log-link)

Identify the label 0 of F_T with the "⊞" strip and glue the result to F_> via ±j ↦ j: the Θ±ell NF-Hodge theater. Its q-pilot and Θ-pilot are now defined; the capsule F_J will later become the processions S^±_{j+1} of the multiradial representation.

11.4 Building the log-theta-lattice (the infinite H)

SOURCES **[IUT III]** Def 1.1 p. 23, Def 1.4 p. 45, Def 3.8 (iii) p. 112, Thm 1.5 p. 48; **[Alien]** §3.3 (ii) p. 58, (vi), Fig 3.6 p. 60, Fig 3.15 p. 99; **[EssLgc]** §3.3 p. 86

The explorer does exactly this when you press l, t and H. Each vertex is a fresh universe; links identify only the parts that are coric.

STEP 1 / 6 **one Hodge theater**

the infinite H: columns (0,∘) and (1,∘), one Θ-link [IUTchIII] Def 1.4; [EssLgc] §3.3 (InfH); [Alien] Fig 3.6

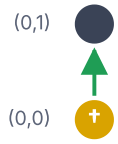


A Θ±ell NF-Hodge theater at (0,0): universe #1. Everything in it — Π_v, G_v, the monoids, the ring structures — is a copy that belongs to this vertex only.

Start with one Hodge theater, labelled (0,0) and written †HT. It is a complete copy of the conventional arithmetic geometry of E_F.

STEP 2 / 6 **apply a log-link**

the infinite H: columns (0,∘) and (1,∘), one Θ-link [IUTchIII] Def 1.4; [EssLgc] §3.3 (InfH); [Alien] Fig 3.6



shared: D-prime-strips Π_v #270002 = #270002 (vertically coric)

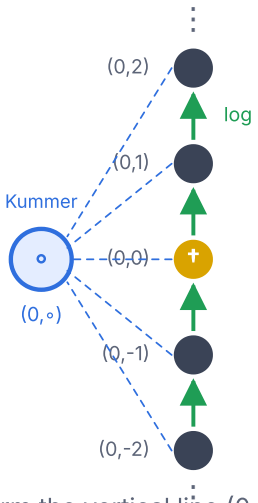
fresh: Frobenius-like units #183003 ≠ #383003; log-shells I_v built at (0,1)

The log-link (0,0) → (0,1) is the p-adic logarithm O×_v → K_v applied at each v: it needs both ring structures, so it is only meaningful inside one column.

The log-link uses the full ring structure, maps units to the additive log-shell of the next vertex, and identifies the étale-like D-prime-strips of domain and codomain ([IUTchIII] Thm 1.5 (i)).

STEP 3 / 6 **iterate: the 0-column**

the infinite H: columns (0,∘) and (1,∘), one Θ-link [IUTchIII] Def 1.4; [EssLgc] §3.3 (InfH); [Alien] Fig 3.6

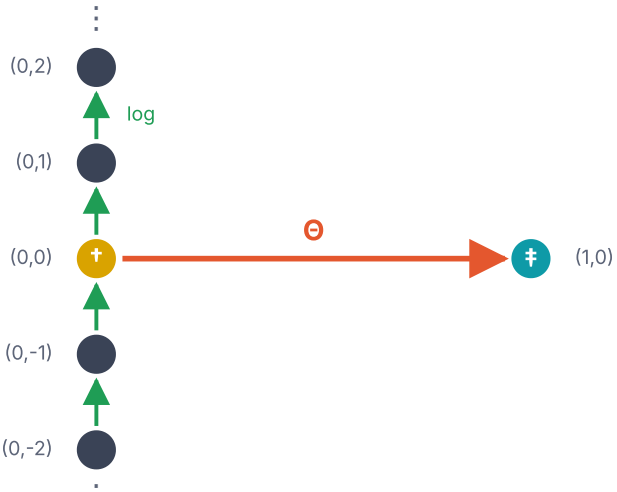


All log-link iterates form the vertical line (0,∘). The Frobenius-like objects • at the different (0,m) are related through Kummer isomorphisms to one vertically coric étale-like ∘ ([Alien] Fig 3.15): this is the log-Kummer correspondence, and its upper semi-compatibility is (Ind3).

The infinite column is symmetric under vertical shifts; only the whole column, not a finite piece, has this symmetry — which is why the theory needs the "infinite" H.

STEP 4 / 6 **apply the Θ-link**

the infinite H: columns (0,∘) and (1,∘), one Θ-link [IUTchIII] Def 1.4; [EssLgc] §3.3 (InfH); [Alien] Fig 3.6



shared: G_v ↷ O×μ #183003 = #183003 (horizontally coric, up to Ind1, Ind2)

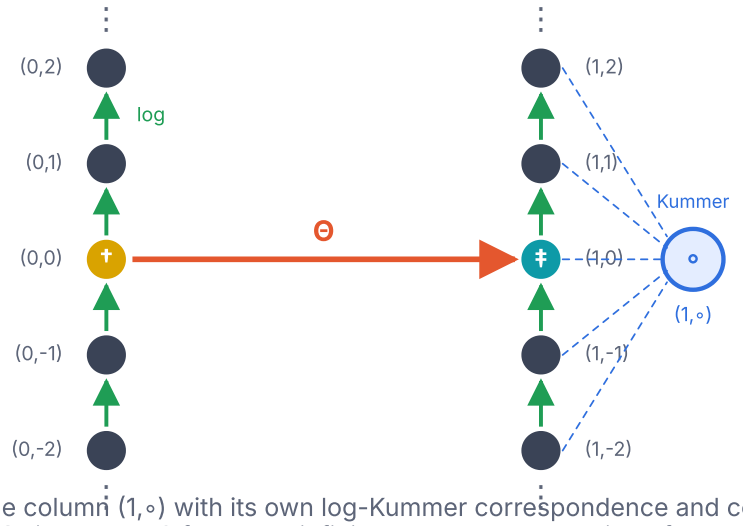
alien: Π_v #270002 vs #470002, value groups {q^{j²}} ↦ {q} dilated by l(l+1)/12 = 15.167

The Θ-link (0,0) → (1,0) glues the Θ-pilot FΙ⊢_LGP of †HT to the q-pilot FΙ⊢_Δ of ‡HT. Only multiplicative data crosses; ‡HT is a mutually alien copy.

The Θ-link creates the second column. From the point of view of (1,0) the log-links of the 0-column are meaningless: the ring structure that defines them is not transported.

STEP 5 / 6 **the 1-column and the H**

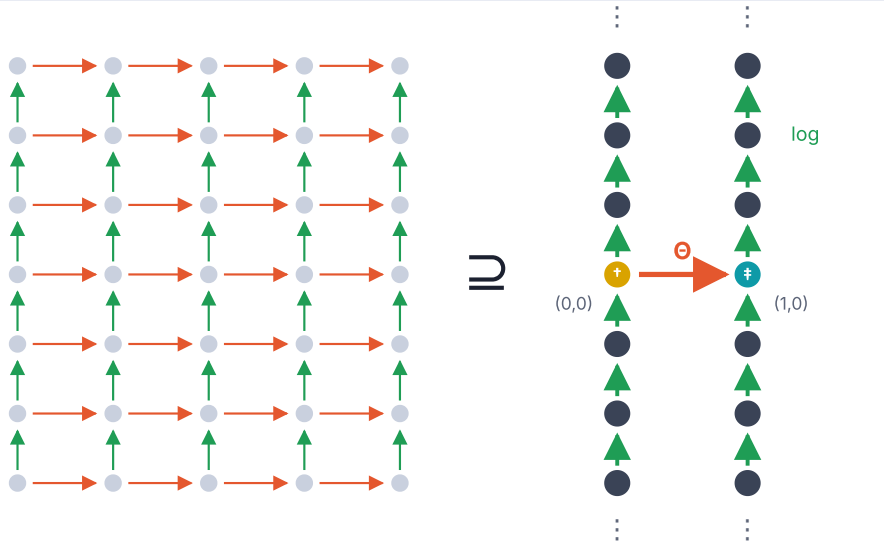
the infinite H: columns (0,∘) and (1,∘), one Θ-link [IUTchIII] Def 1.4; [EssLgc] §3.3 (InfH); [Alien] Fig 3.6



Log-links on the right give the column (1,∘) with its own log-Kummer correspondence and coric ∘. The two columns plus the single Θ-link at m = 0 form the "infinite H" — the only portion of the lattice the proof uses ([EssLgc] §3.10 (Stp7) works on the 1-column).

The explorer's "build infinite H" button produces exactly this configuration; `iut walk` prints it in text.

STEP 6 / 6 **the whole lattice ⊇ the H**



the entire log-theta-lattice

the portion actually used: the infinite H

Formally Hodge theaters sit at all (n,m) ∈ Z^2 with log-links up and Θ-links right, and the diagram is highly non-commutative ([IUTchIII] Rem 1.4.1). Vertically coric = invariant under vertical shifts; horizontally coric = under horizontal shifts; bi-coric = both ([IUTchIII] Thm 1.5).

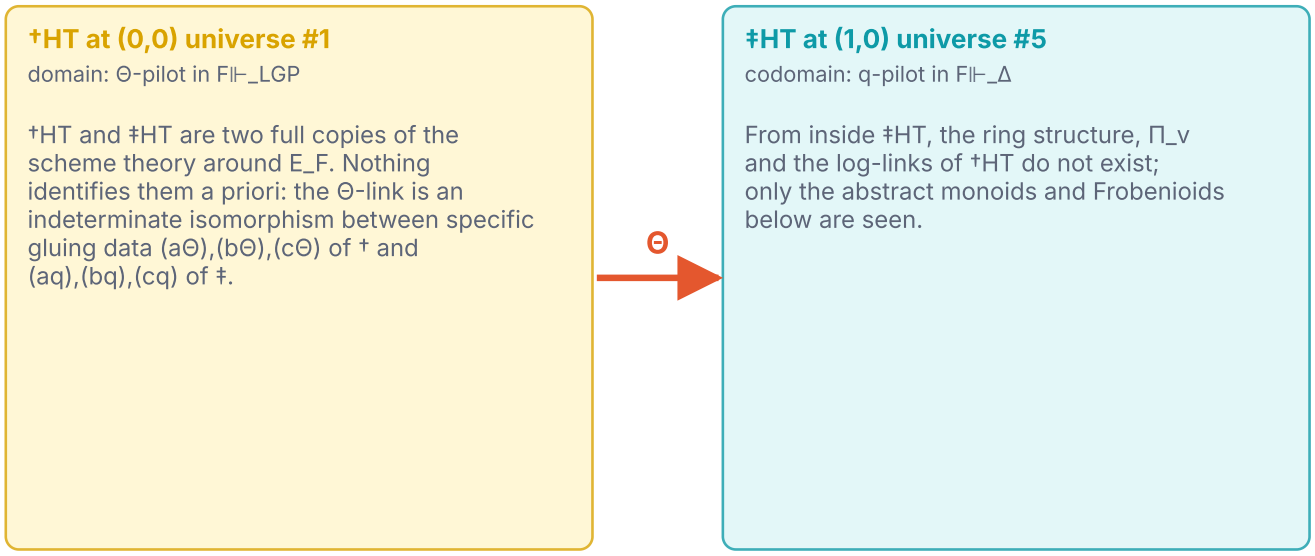
The full lattice is only the ambient bookkeeping. D⊢ (bi-coric), F⊢×μ (horizontally coric) and D (vertically coric) are the three kinds of invariant data ([IUTchIII] Thm 1.5).

11.5 The Θ-link: gluing data and alien copies

SOURCES **[IUT III]** Def 3.8 (ii) p. 112; **[IUT II]** Cor 4.10 p. 158; **[Alien]** §3.3 (vii) p. 58, §3.6 (iv) p. 87, Fig 3.19 p. 106

What exactly is identified by the Θ-link (0,0) → (1,0), what is deliberately not, and why the two Hodge theaters are "mutually alien".

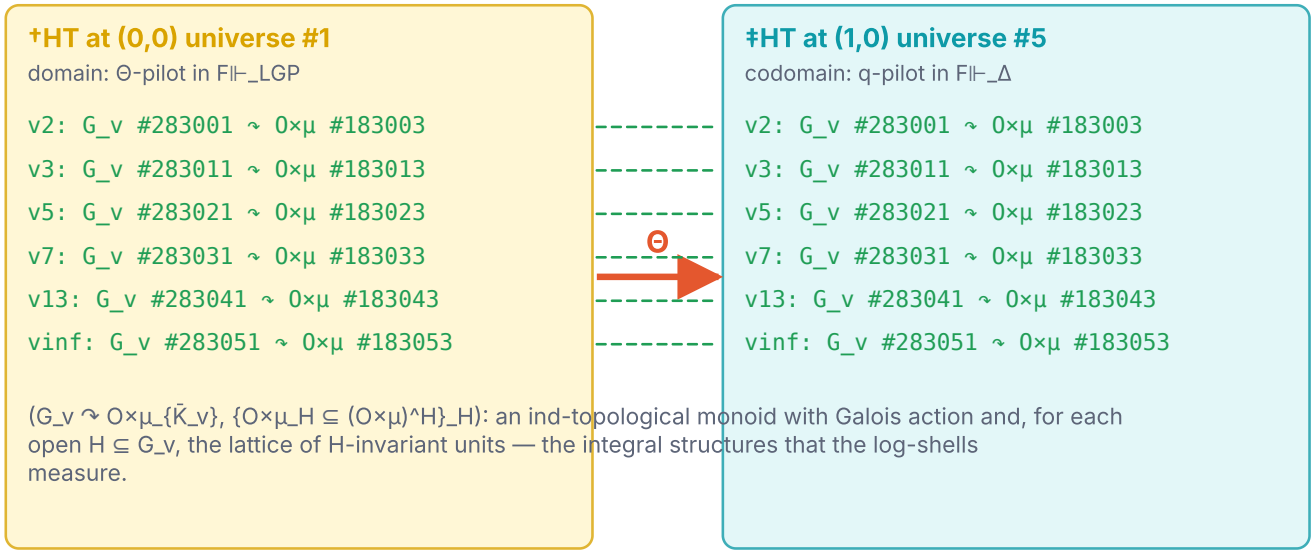
STEP 1 / 6 **two Hodge theaters**



Compare [Alien] §1 Step 3: two identical but mutually independent copies of the Gaussian integral.

The Θ-link relates a Θ-pilot to a q-pilot in two independent universes. It is defined only in terms of abstract monoids, which is what makes the gluing possible at all.

STEP 2 / 6 **(aΘ) ≅ (aq): unit group portion**

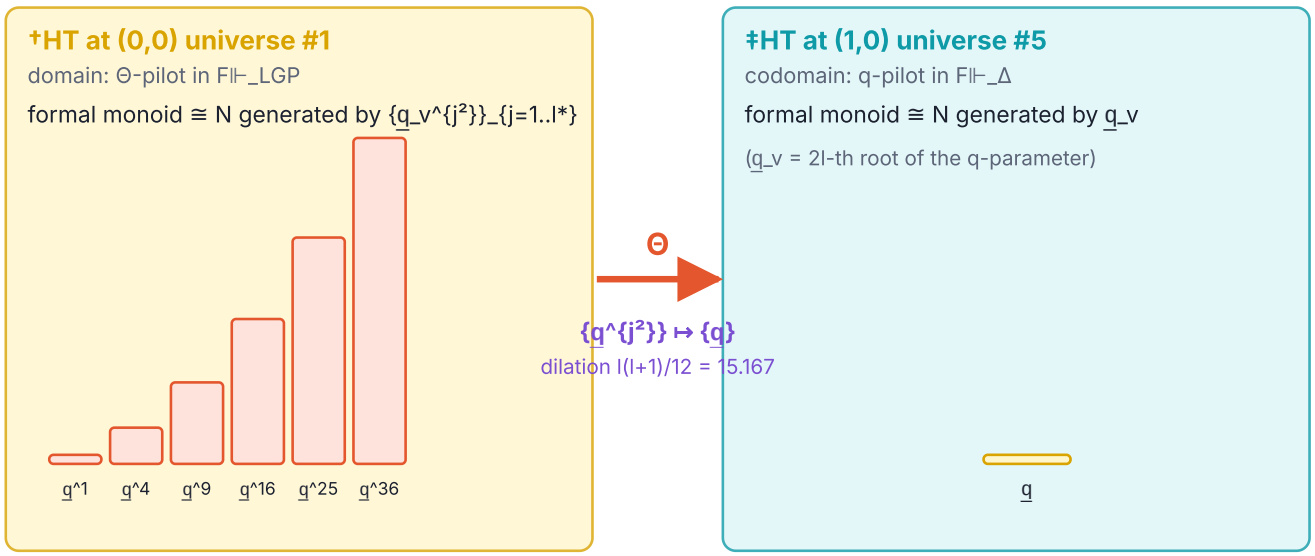


identified up to Aut(G_v) [Ind1] and the isometries Ism(O×μ) [Ind2]

In the explorer the copy numbers show the identification: after a Θ-link the units of †F⊩_LGP and ‡F⊩_Δ carry the same number, Π_v does not.

The unit groups modulo torsion, with their Galois actions and integral structures, are glued. This is the horizontally coric F⊩×μ-prime-strip of [IUTchIII] Thm 1.5 (ii).

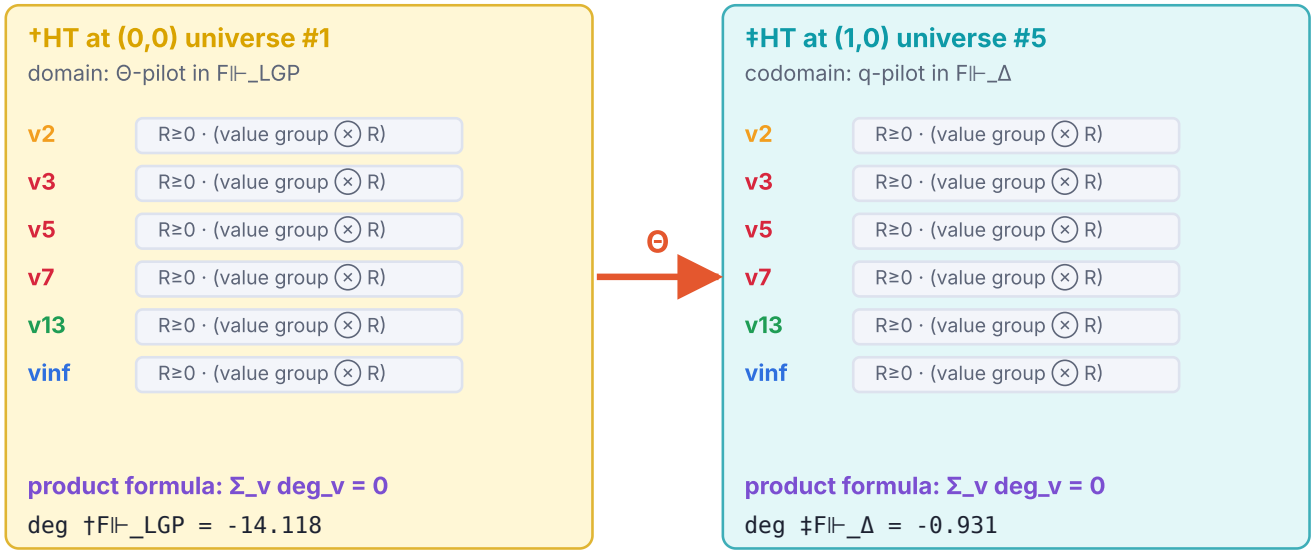
STEP 3 / 6 **(bΘ) ↦ (bq): local value groups**



The identification of these two monoids is an isomorphism of abstract monoids, not of value groups inside a common field: the "arithmetic degrees" differ, $-|\log \Theta|$ vs $-|\log q| = -0.931$. The multiradial representation measures this discrepancy and Cor 3.12 bounds it.

The theta values on the left are sent to the q-parameter on the right: "q ↦ q^{j²}" averaged over j ([Alien] §2.4). This is where the dilation by l(l+1)/12 — the content of the inequality — enters.

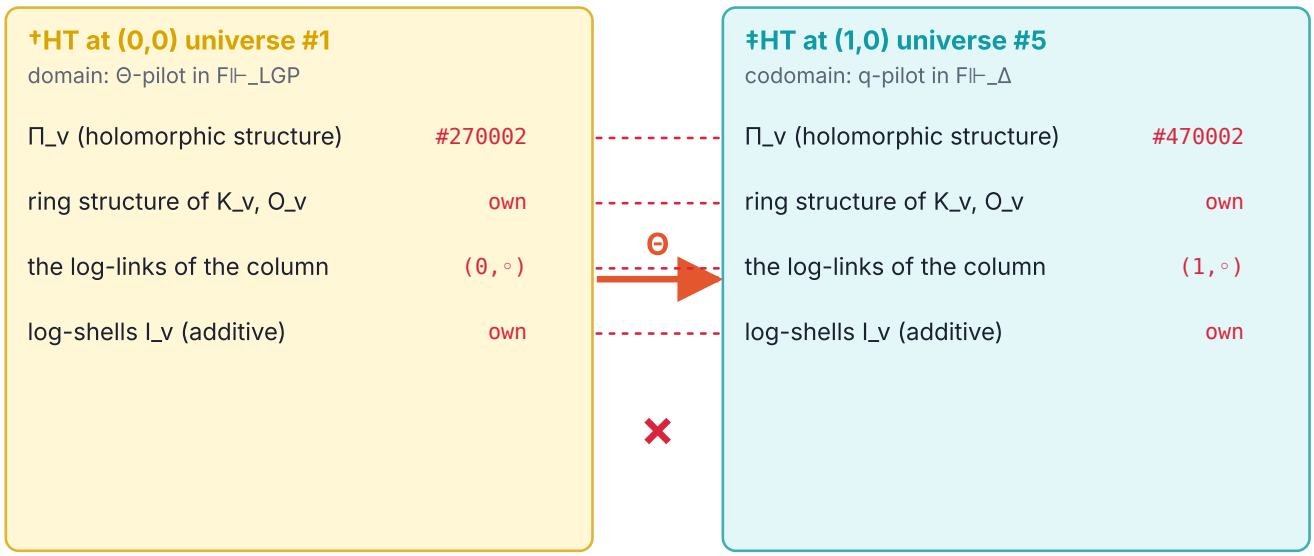
STEP 4 / 6 **(cΘ) ≅ (cq): global realified Frobenioids**



Realify each local monoid (R≥0-multiples) and remember the product-formula relations: a global realified Frobenioid CIH-. It is glued across the Θ-link as an abstract Frobenioid, which is what later allows log-volumes to be compared in a single copy of R ((Stp8)).

The global arithmetic-degree bookkeeping is glued too, so that both pilots have a degree in "the same R" — but only up to the indeterminacies that the chain of ∧'s will track.

STEP 5 / 6 **what is NOT glued**

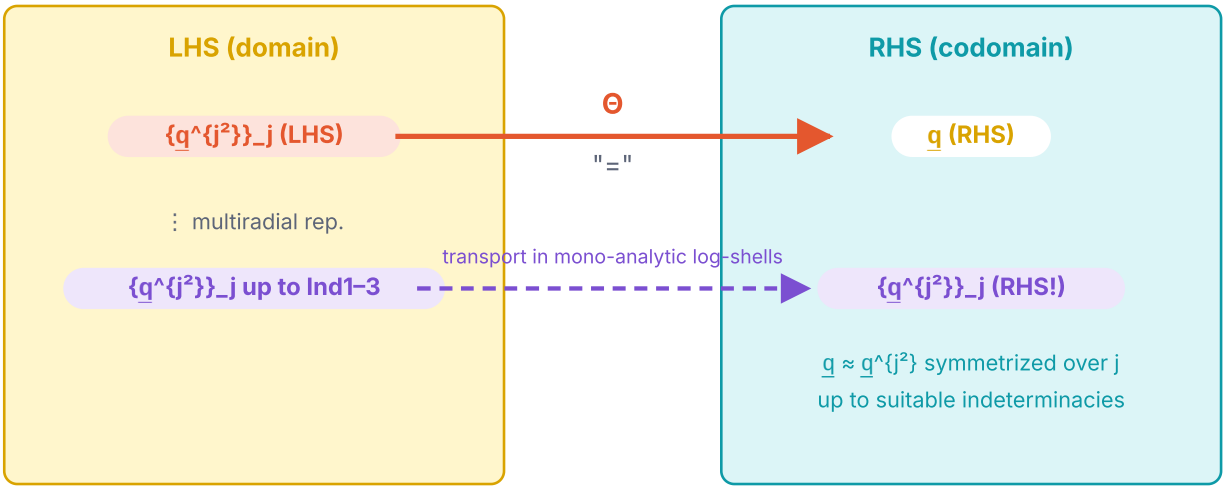


"The log-link, and hence the construction of log-shells at (n,m), are meaningless, at least in any a priori sense, from the point of view of (n+1,m)" ([Alien] §3.6 (iv)). The fundamental dilemma of the theory: the Θ-link needs only multiplication, the log-link needs both operations.

The ring structures, Π_v and everything built with addition stay on their own side. This is the precise meaning of "mutually alien copies": distinct labels that may not be identified ([EssLgc] §3.4, (AOΘ1)–(AOΘ4)).

STEP 6 / 6 **Fig 3.19: the "=" of the Θ-link seen multiradially**

the gluing, or tautological identification "=", of the Θ-link seen from the multiradial representation [Alien] Fig 3.19



Any F⊢×μ-prime-strip isomorphic to the domain data — in particular the codomain data (aq),(bq),(cq) — can be represented via the conventional scheme-like representation of the Θ-pilot. Specialized to the RHS: the line bundle of q may be identified with that of q^N, N = j², up to the indeterminacies ([Alien] §3.7 (i)).

The multiradial representation lets the RHS see the LHS's theta values inside its own log-shells: the two sides of the "=" become comparable, at the price of Ind1–3. This is the ∧ of (FxGl) kept alive ([EssLgc] §3.10).

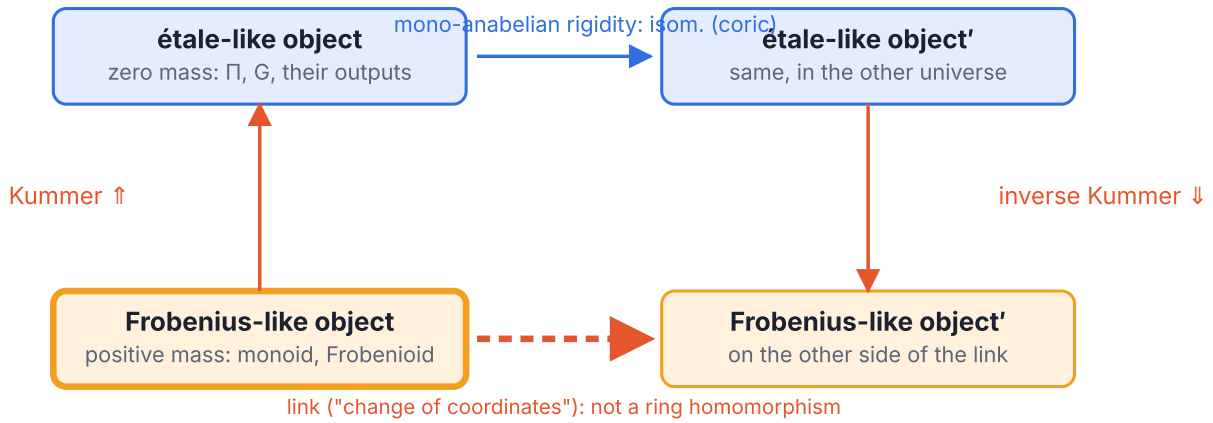
11.6 Mono-anabelian transport

SOURCES **[Alien]** §2.7 p. 21, §2.9 p. 26, Fig 2.1 p. 23; **[IUT III]** Rem 3.11.1 (APT) p. 159; **[AbsTopIII]**

The computational technique behind every step of the theory: carry a Frobenius-like object across a link by lifting it to its étale-like container, which the link does preserve, and coming back down.

STEP 1 / 5 **a Frobenius-like object and a link**

mono-anabelian transport [Alien] §2.7 (v), Fig 2.1

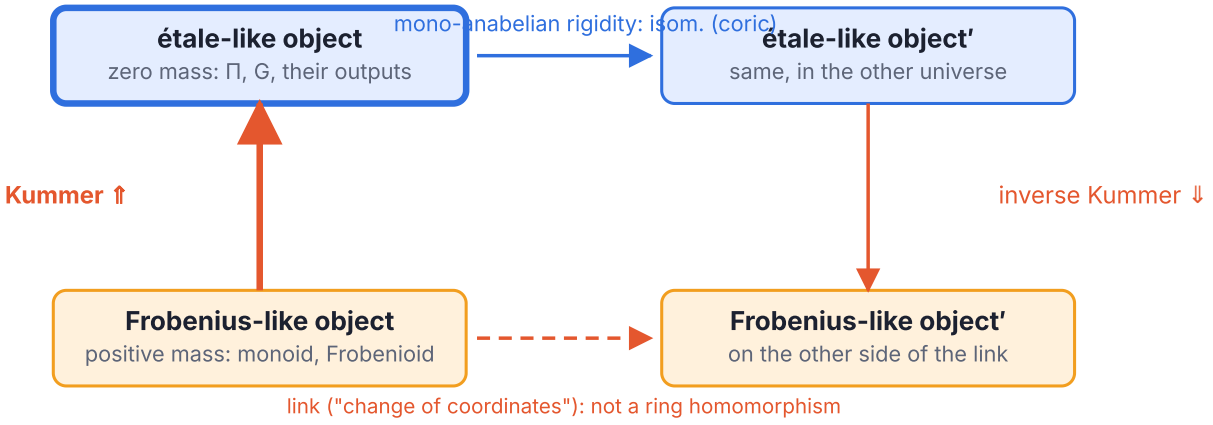


Frobenius-like objects (monoids O▷, Frobenioids, log-shells as Frobenius-like) have "positive mass": they are the things whose log-volumes one wants. A link such as the Θ-link transports some of them but is not a ring homomorphism, so their effect cannot be computed directly.

The question: what does the link do to a Frobenius-like object (e.g. the Θ-pilot)? It cannot be followed through the link as a scheme-theoretic object.

STEP 2 / 5 **Kummer: up to the étale-like container**

mono-anabelian transport [Alien] §2.7 (v), Fig 2.1

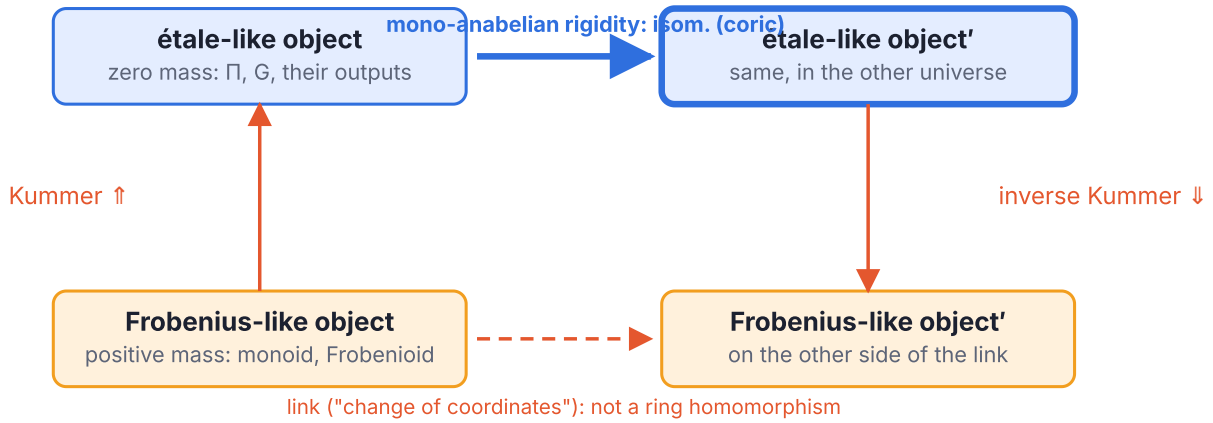


Kummer theory (for units, for theta functions via mono-theta environments, for κ-coric functions) identifies the Frobenius-like object with data constructed from the étale-like group by anabelian algorithms — "zero mass" containers that are rigid.

Kummer isomorphisms lift the object into étale-like data: Galois groups and what can be reconstructed from them by group theory alone.

STEP 3 / 5 **mono-anabelian rigidity across the link**

mono-anabelian transport [Alien] §2.7 (v), Fig 2.1

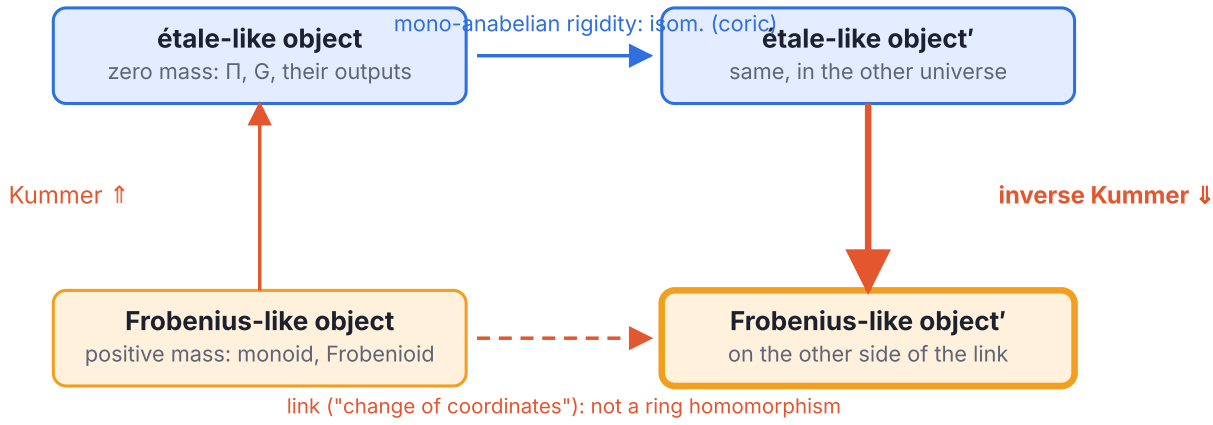


Étale-like objects, regarded up to isomorphism, are coric: the link identifies them (G_v across a Θ-link, Π_v across a log-link). Anabelian reconstruction (mono-anabelian: from the group alone) rebuilds the ring-theoretic data on the other side — up to the automorphisms of the group, i.e. up to (Ind1).

On the étale-like level the link is an isomorphism, and anabelian geometry reconstructs the arithmetic on either side from the group. This is the content of [AbsTopIII] used in IUT.

STEP 4 / 5 **inverse Kummer: back down**

mono-anabelian transport [Alien] §2.7 (v), Fig 2.1

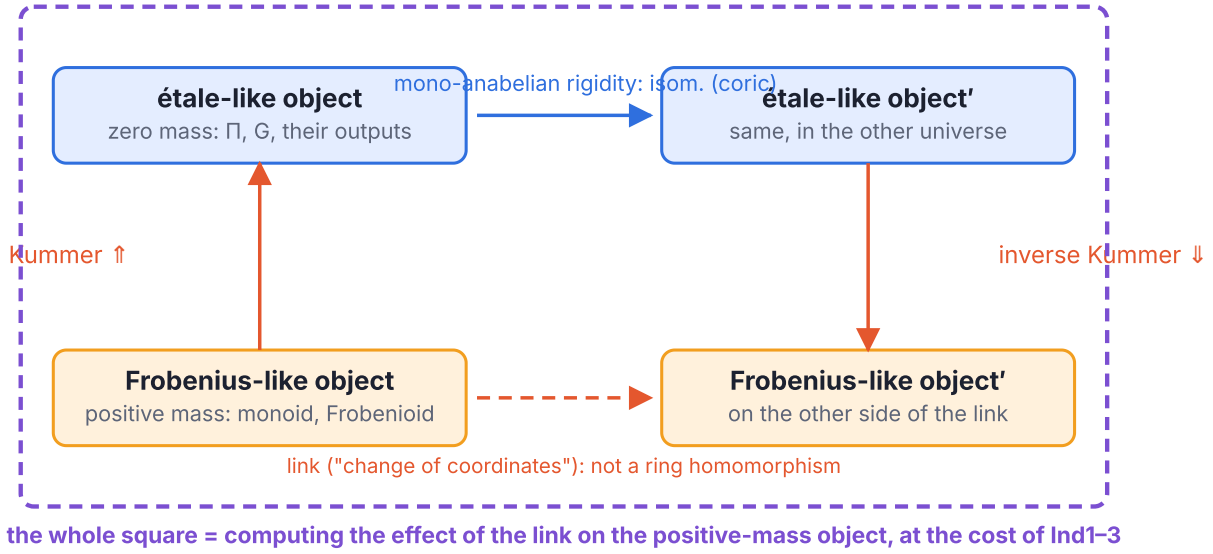


Going back through the Kummer isomorphism on the other side produces the Frobenius-like object as seen in the other universe. The confusion between the Frobenius-like and étale-like versions of the containers costs (Ind2); following iterates of the log-link costs (Ind3) ("Kummer-detachment indeterminacies").

The output is the original object re-expressed in the other universe's terms, with the indeterminacies as the price of the detour.

STEP 5 / 5 **the square: an arithmetic change of coordinates**

mono-anabelian transport [Alien] §2.7 (v), Fig 2.1



the whole square = computing the effect of the link on the positive-mass object, at the cost of Ind1–3

Compare [Alien] §1 Steps 6–7: the effect of a change of coordinates on an integral is computed through the Jacobian, i.e. locally through linear algebra. Mono-anabelian transport plays that role here; Example 2.6.1 (Frobenius in characteristic p) is the prototype, and the Kodaira–Spencer morphism its degenerate classical version (§2.9).

Everything in processes 7 and 10 is an iterate of this square: (Stp1) is the log-Kummer correspondence and Galois evaluation, (Stp2)–(Stp3) are descents to weaker étale-like data.

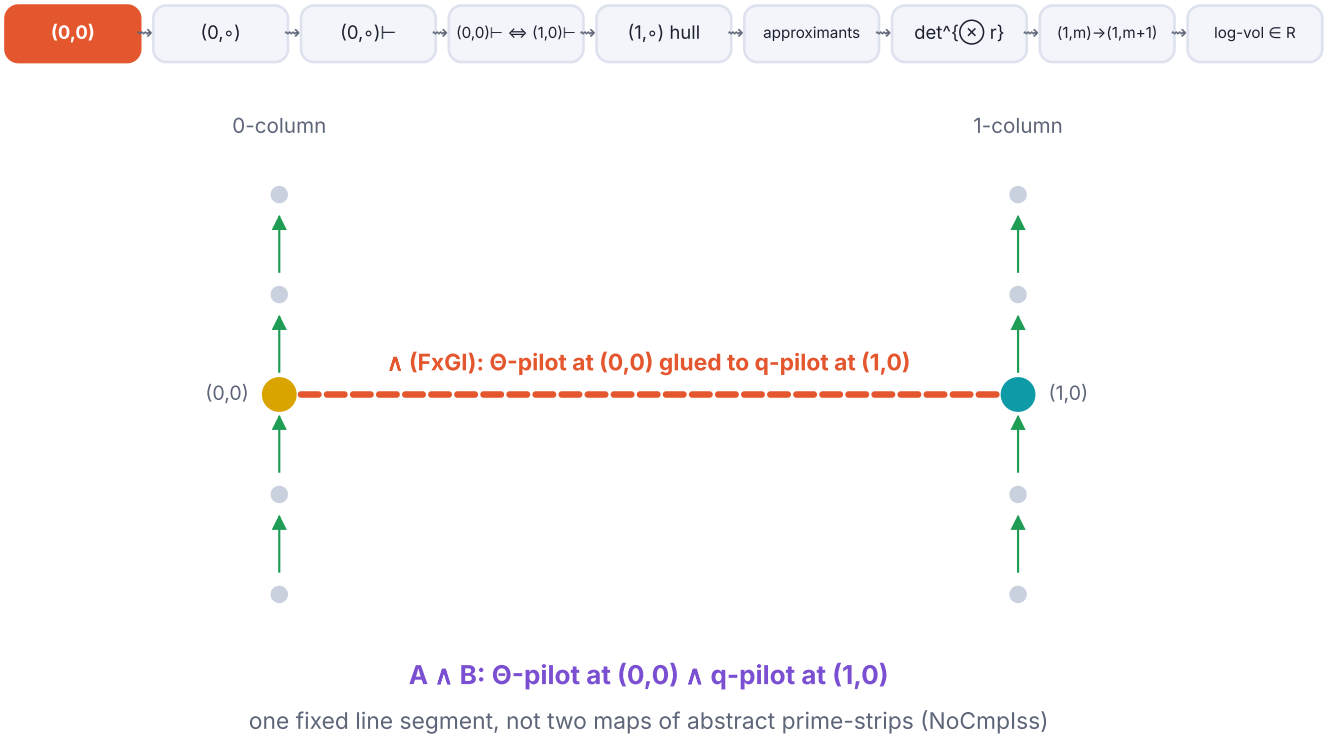
11.7 The chain of logical AND relations: (Stp1)–(Stp8)

SOURCES **[EssLgc]** §3.6 (∧-Chn) p. 104, (ExtInd), §3.10 (Stp1)–(Stp8) p. 136; **[IUT III]** Thm 3.11 p. 153, Cor 3.12 p. 173

Mochizuki's own description of the essential logical structure: A ∧ B ⇒ A ∧ (B_1 ∨̇ B_2 ∨̇ ...). The ∧ comes from the Θ-link (multiplicative structure common to both sides); the ∨̇'s come from the log-Kummer correspondence (additive structures, not common). The proof never concatenates inequalities; it preserves one ∧.

STEP 1 / 10 **(FxGl): the fixed line segment**

fix the gluing •====• (FxGl) and "fuzzify" the left end until it descends to the label of the right end [EssLgc] §3.10



Fix the Frobenius-like q-pilot at (1,0) and the gluing, via the Θ-link, of the Frobenius-like Θ-pilot at (0,0) to it. This is the one and only ∧ (simultaneous execution, gluing). Everything that follows adds possibilities (∨) to the left end; it never considers two distinct mappings of abstract prime-strips (DstMp) — so there is no commutativity issue (NoCmpIss).

REF **[EssLgc]** §3.10 (FxGl) p. 136, (DstMp), (NoCmpIss); **[IUT III]** Rem 3.12.2 (ii) p. 187

STEP 2 / 10 **(Stp1) log-Kummer correspondence and Galois evaluation**

fix the gluing •====• (FxGl) and "fuzzify" the left end until it descends to the label of the right end [EssLgc] §3.10

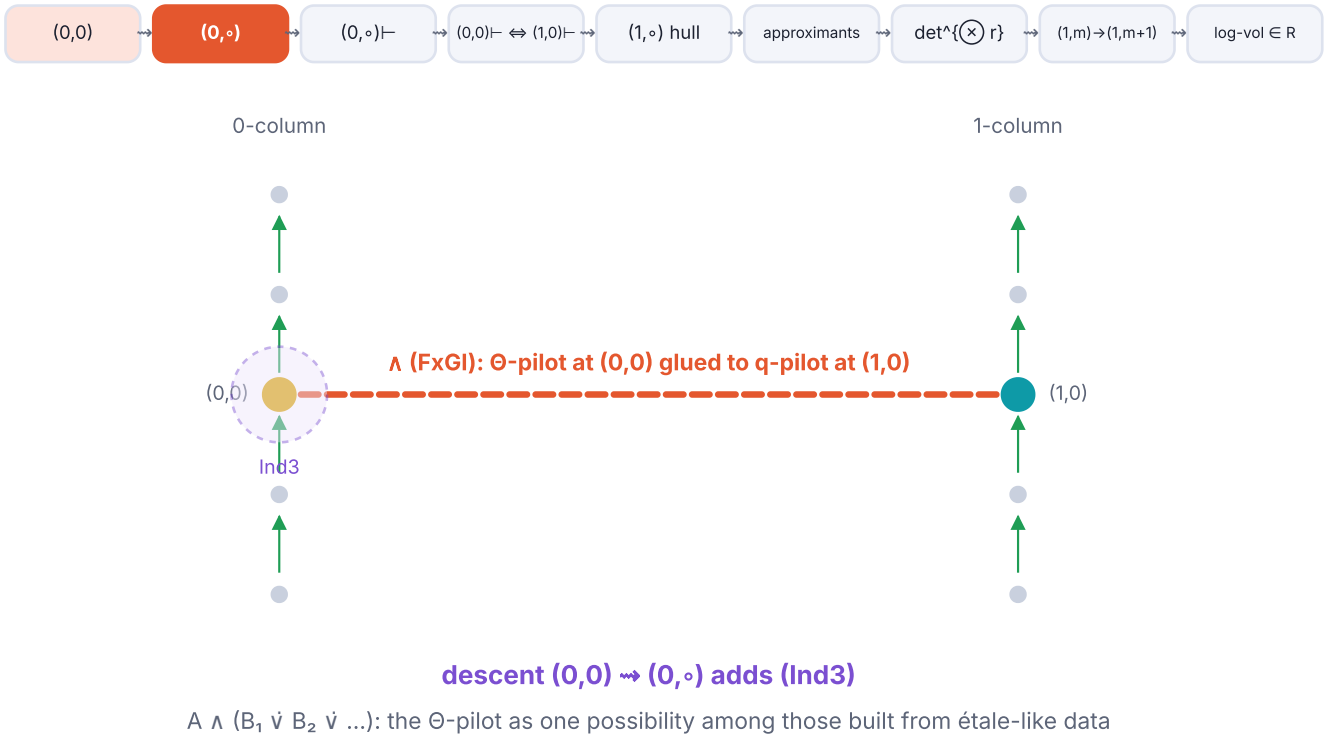
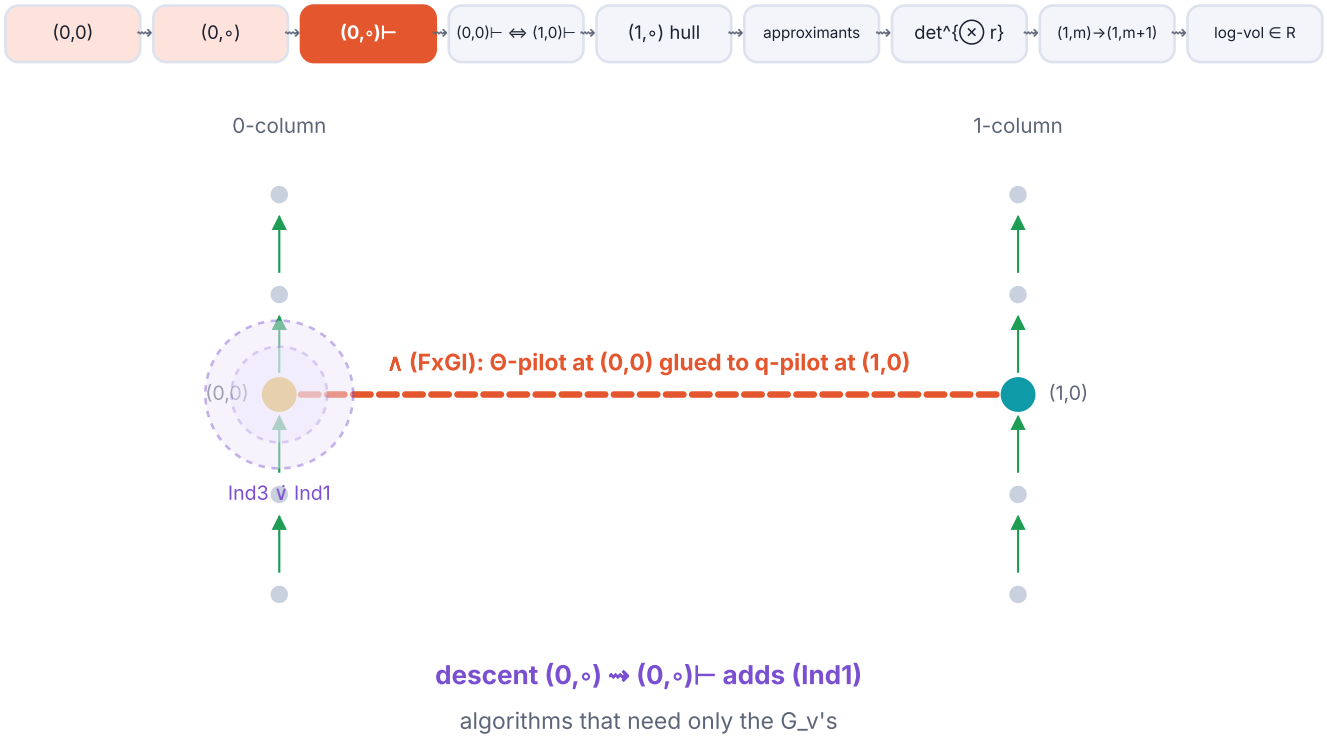


Exhibit the Frobenius-like Θ-pilot at (0,0) — the data giving the Fℍ×μ-prime-strip in the domain of the Θ-link — as one possibility within a collection of possibilities constructed by anabelian algorithms from holomorphic étale-like data of the 0-column. Label descent (0,0) ⇝ (0,∘). Introduces (Ind3), the coarse approximation of the whole log-Kummer apparatus. This is the step that occupies the bulk of [IUTchI–III].

REF **[EssLgc]** §3.10 (Stp1) p. 136, §3.11 p. 152; **[IUT III]** Thm 3.11 (ii) p. 153, (iii), Rem 3.9.5 (viii) (sQ1) p. 126, (sQ2)

STEP 3 / 10 **(Stp2) introduce (Ind1)**

fix the gluing •====• (FxGl) and "fuzzify" the left end until it descends to the label of the right end [EssLgc] §3.10

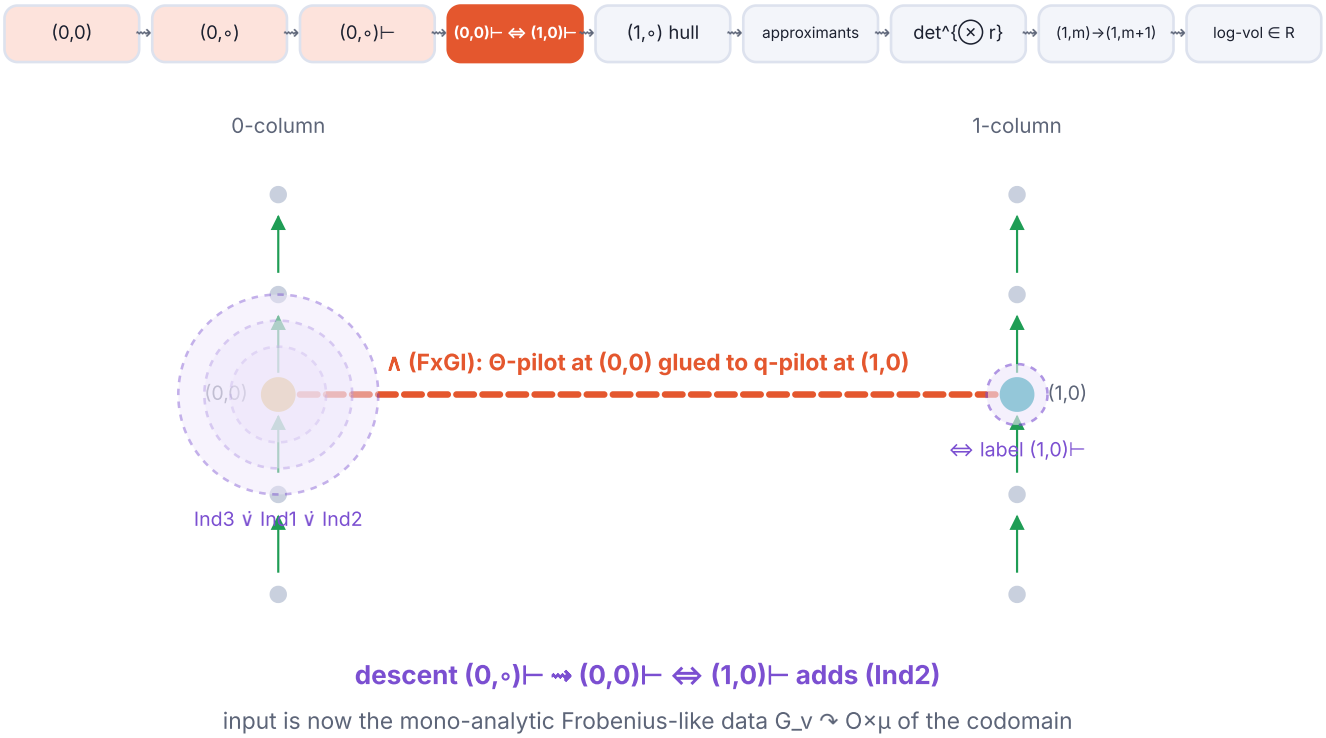


The anabelian algorithms of (Stp1) descend to algorithms needing only the mono-analytic étale-like data, the G_v's: (0,∘) ⇝ (0,∘)⊢. Very mildly enlarges the collection of possibilities by Aut(G_v).

REF **[EssLgc]** §3.10 (Stp2) p. 136; **[IUT III]** Thm 3.11 (i) p. 153

STEP 4 / 10 **(Stp3) introduce (Ind2): cross the Θ-link**

fix the gluing •====• (FxGl) and "fuzzify" the left end until it descends to the label of the right end [EssLgc] §3.10



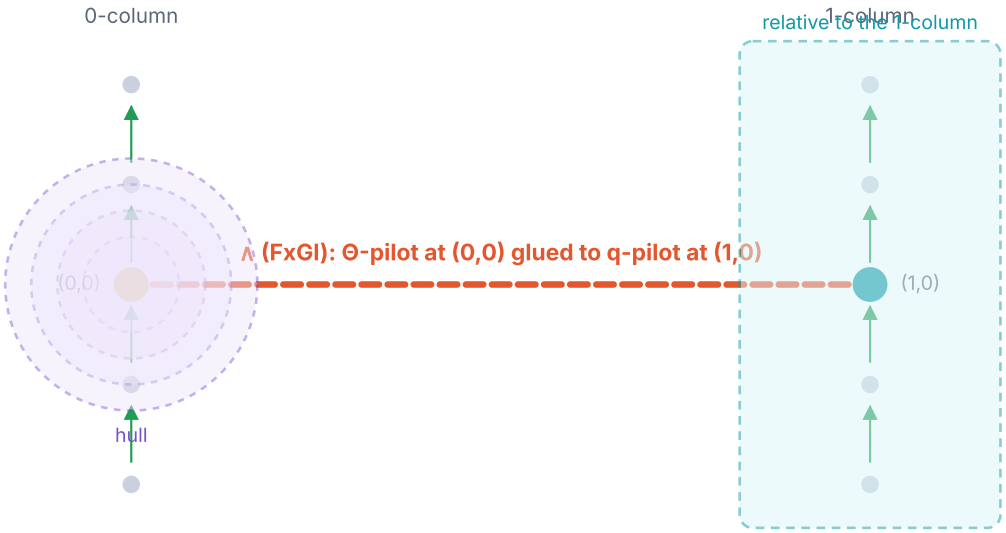
The algorithms descend further to the mono-analytic Frobenius-like data G_v ↷ O×μ (log-shells as submonoids of O×μ rather than subquotients of G_v): (0,∘)⊢ ↠ (0,0)⊢. Since the Θ-link is an equivalence of labels (0,0)⊢ ⇔ (1,0)⊢ (the full poly-isomorphism of F⊢×μ-strips), the algorithm now has input data (1,0)⊢ from the codomain. This isolates the log-link indeterminacies of the two columns from each other.

REF **[EssLgc]** §3.10 (Stp3) p. 136; **[IUT III]** Rem 3.9.5 (vii) (Ob7-2) p. 126; **[Alien]** §3.6 (iv) p. 87

STEP 5 / 10 **(Stp4) holomorphic hull**

fix the gluing •====• (FxGI) and "fuzzify" the left end until it descends to the label of the right end [EssLgc] §3.10

(0,0) ⇝ (0,∘) ⇝ (0,∘)⊢ ⇝ (0,0)⊢ ⇔ (1,0)⊢ ⇝ **(1,∘) hull** ⇝ approximants ⇝ det^{⊗ r} ⇝ (1,m)→(1,m+1) ⇝ log-vol ∈ R



regions ⊆ holomorphic hull relative to the 1-column

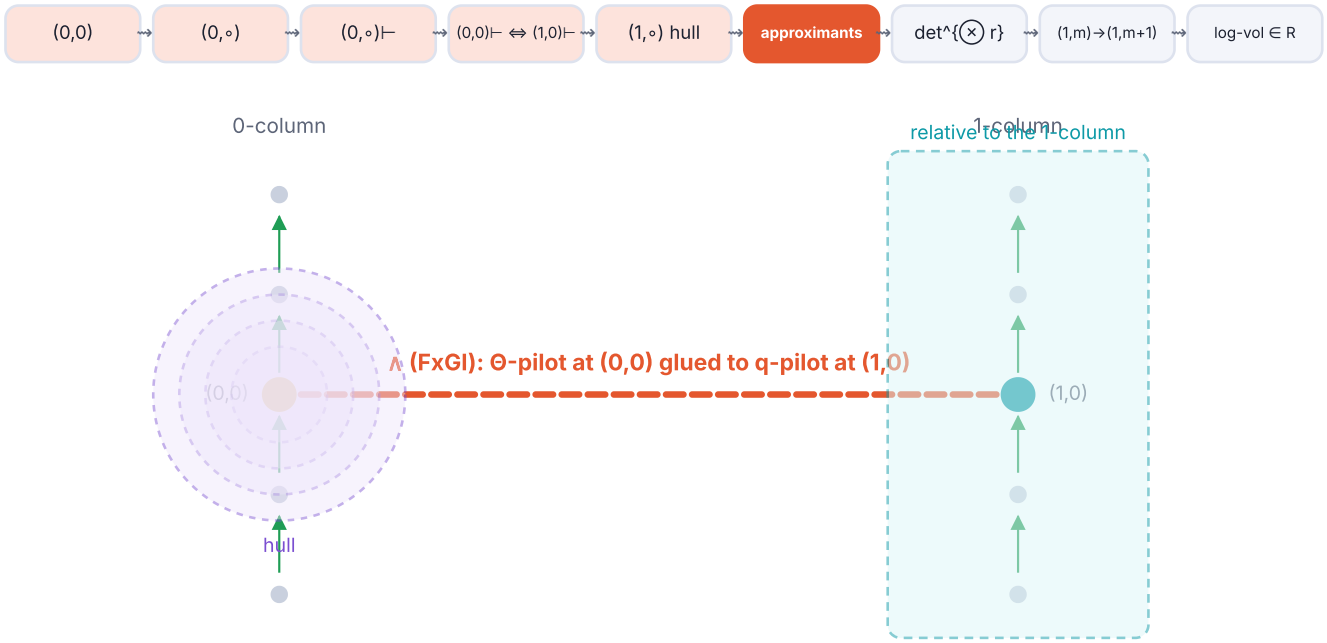
(ExtInd1): more v̇'s, towards arithmetic vector bundles

Pass from the collection of possible regions output by (Stp3) to the regions contained in the holomorphic hull, relative to the 1-column, of their union: a plain application of (ExtInd1), enlarging the set of possibilities (more v̇'s).

REF **[EssLgc]** §3.10 (Stp4) p. 136; **[IUT III]** Rem 3.9.5 (vi) p. 126, (vii) (Ob5)

STEP 6 / 10 **(Stp5) hull-approximants**

fix the gluing •====• (FxGl) and "fuzzify" the left end until it descends to the label of the right end [EssLgc] §3.10



hull-approximants: same log-volume, vector bundles

does not change the possible value-group portions (F⊢▶-strips)

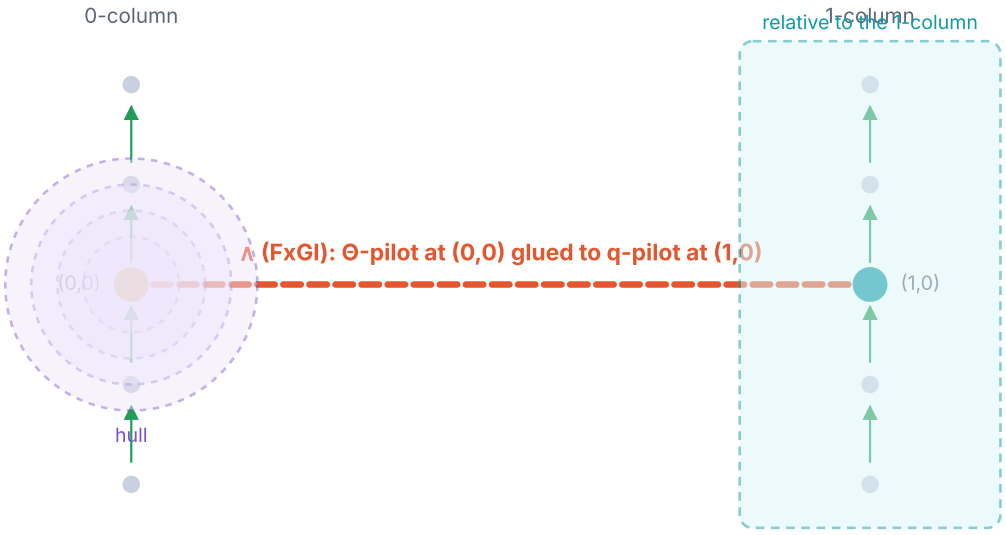
Replace arbitrary regions in the hull by regions with the same global log-volume that correspond to arithmetic vector bundles. Does not change the ∧/∨ structure, because it does not change the possible value-group portions (the F⊢▶-strips) obtained by taking log-volumes.

REF **[EssLgc]** §3.10 (Stp5) p. 136; **[IUT III]** Rem 3.9.5 (vii) (Ob6) p. 126

STEP 7 / 10 **(Stp6) tensor power of the determinant**

fix the gluing •====• (FxGl) and "fuzzify" the left end until it descends to the label of the right end [EssLgc] §3.10

(0,0) ⇝ (0,∘) ⇝ (0,∘)⊢ ⇝ (0,0)⊢ ⇔ (1,0)⊢ ⇝ (1,∘) hull ⇝ approximants ⇝ det^{⊗ r} ⇝ (1,m)→(1,m+1) ⇝ log-vol ∈ R



(det)^{⊗ r}, r ∈ Q_{>0}: an arithmetic line bundle

lives at the holomorphic structure of (1,1), one log-shift above (1,0)

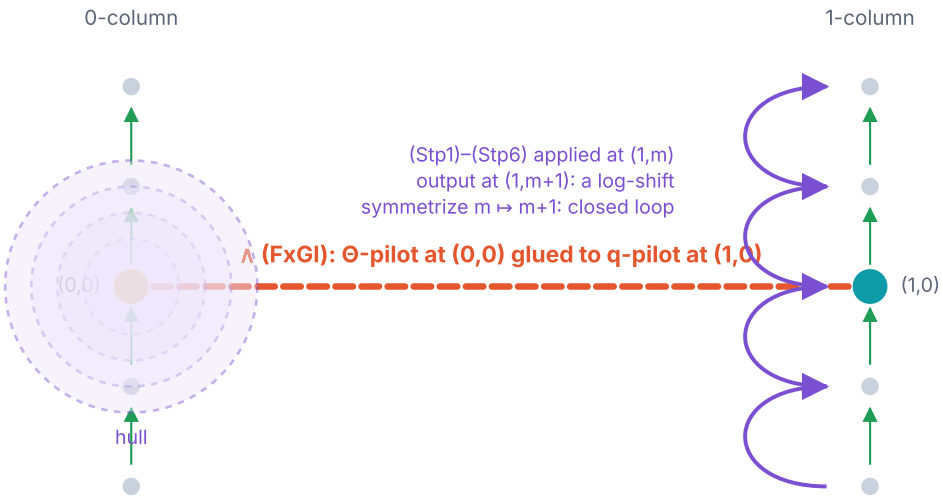
Pass to a suitable positive rational tensor power of the determinant arithmetic line bundle of those vector bundles. Again harmless for the ∧/∨ structure.

REF **[EssLgc]** §3.10 (Stp6) p. 136; **[IUT III]** Rem 3.9.5 (vii) (Ob3) p. 126, (Ob4)

STEP 8 / 10 **(Stp7) log-shift adjustment: the closed loop**

fix the gluing •====• (FxGl) and "fuzzify" the left end until it descends to the label of the right end [EssLgc] §3.10

(0,0) ⇢ (0,∘) ⇢ (0,∘)⊢ ⇢ (0,0)⊢ ⇔ (1,0)⊢ ⇢ (1,∘) hull ⇢ approximants ⇢ det^{⊗ r} ⇢ **(1,m) → (1,m+1)** ⇢ log-vol ∈ R



(1,m) → (1,m+1) for all m: symmetrize m ↦ m+1

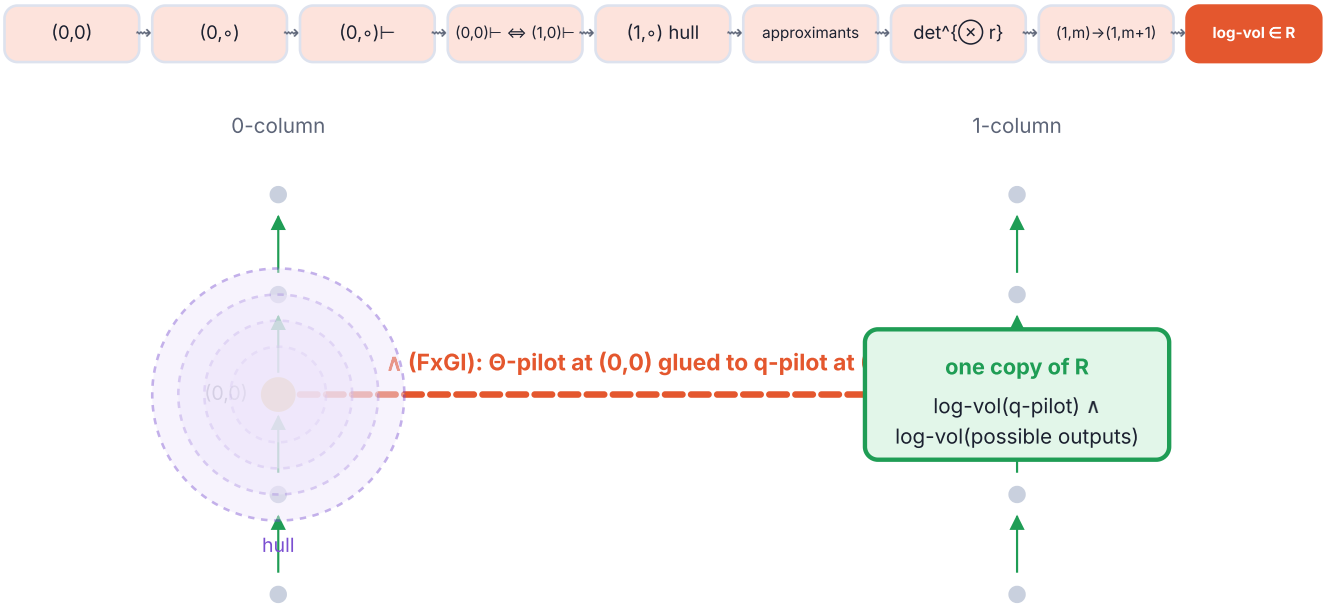
a formal closed loop between distinct labels $(1,m)^{+}$, $(1,m+1)^{-}$; compatible with log-links via log-volume

The line bundles of (Stp6) live at the holomorphic structure of (1,1), one log-shift above the codomain (1,0). Applying (Stp1)–(Stp6) at every (1,m) gives diagonal arrows (1,m) → (1,m+1): an endomorphism of the 1-column. Symmetrizing by m ↦ m+1 gives a "closed loop" — a formal, diagram-theoretic gluing between data with distinct labels $(1,m)^{+}$ and $(1,m+1)^{-}$, never a set-theoretic identification. Compatibility with the log-links of the 1-column is obtained by passing to log-volumes, which the log-link preserves.

REF **[EssLgc]** §3.10 (Stp7) p. 136; **[IUT III]** Rem 3.9.5 (vii) (Ob8) p. 126, (Ob9), (viii) (sQ4)

STEP 9 / 10 **(Stp8) log-volumes and the inequality**

fix the gluing •====• (FxGl) and "fuzzify" the left end until it descends to the label of the right end [EssLgc] §3.10



log-vol(q-pilot) ∧ log-vol(possible outputs) in one R

Cor 3.12 follows from the invariance of log-volume under the log-link; Ind1–3 = monodromy of the loop

Taking log-volumes, the ∧ maintained through (Stp1)–(Stp7) becomes an ∧ between the log-volume of the original q-pilot and the log-volumes of an algorithmically constructed collection of possible outputs, in one copy of R. The inequality of Cor 3.12 is then a formal consequence of the invariance of log-volume under the log-link. The indeterminacies Ind1–3 are the monodromy of the loop, synchronized across all m.

REF **[EssLgc]** §3.10 (Stp8) p. 136; **[IUT III]** Cor 3.12 p. 173, Rem 3.12.2 (iv) p. 187, (v)

STEP 10 / 10 **summary: ∧(v̇)-Chn**

$$A \wedge B$$

$$\Downarrow$$

$$A \wedge (B_1 \dot{\vee} B_2 \dot{\vee} \ldots)$$

$$\Downarrow$$

$$A \wedge (B_1 \dot{\vee} B_2 \dot{\vee} \ldots \dot{\vee} B_1' \dot{\vee} B_2' \dot{\vee} \ldots)$$

$$\vdots$$

∧ ← Θ-link

simultaneous execution, gluing: the multiplicative structures common to both Hodge theaters (cf. ∧ = multiplication of F_2, shared by Z and F_2 via Teichmüller representatives)

v̇ ← log-Kummer correspondence

indeterminacies, poly-morphisms, (sub)quotients: the additive structures (log-shells) are not common to domain and codomain (cf. v̇ = addition of F_2, not shared)

the two Boolean operators ↔ the two combinatorial dimensions of a ring ↔ the two dimensions of the log-theta-lattice

(∧-Dgns): to diagnose a misunderstanding, find the first step at which the preservation of ∧ seems unclear.

The v̇'s may be read as XOR when the possibilities are listed without overlaps. The passage from Thm 3.11 to Cor 3.12 is then "relatively straightforward", as the word corollary suggests ([EssLgc] §3.6, (Syp1)–(Syp3)).

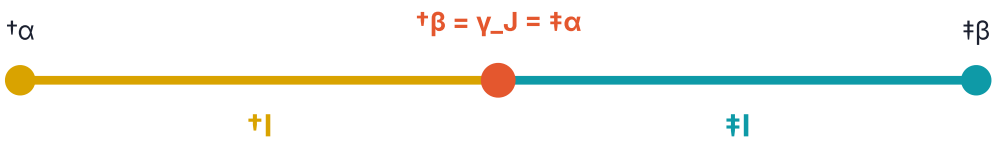
REF **[EssLgc]** §3.10 (∧(v̇)-Chn1)–(Chn4) p. 136, (TrHrc); **[EssLgc]** §3.6 (∧-Dgns) p. 104

11.8 Logical AND versus logical OR

SOURCES **[EssLgc]** §2.3 Ex 2.3.1–2.3.2 p. 43, §2.4 Ex 2.4.1 p. 46, Ex 2.4.5 p. 50, §3.4 (AOΘ1)–(AOΘ4) p. 94, §3.6 (ExtInd) p. 104

The elementary models from [EssLgc] of what goes wrong when "redundant copies" are identified: a glued pair of segments versus a loop, and a three-line numerical model of the whole theory with sliders.

STEP 1 / 5 **two segments glued at a point: ∧**



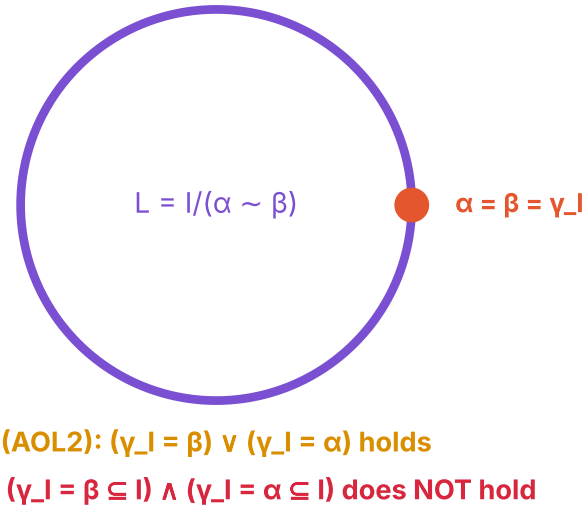
J = †I ∪ ‡I (glued at one point)

(AOL1): (γ_J = †β ⊆ †I) ∧ (γ_J = ‡α ⊆ ‡I) holds

Ex 2.3.2/2.4.1: †I and ‡I are isomorphic but distinct copies of [0,1], glued at †β = ‡α =: γ_J. Both statements about γ_J hold simultaneously. The coordinate functions x on the two copies glue to a function on J.

REF **[EssLgc]** Ex 2.3.2 p. 45, Ex 2.4.1 (i) p. 46, Ex 2.4.2 p. 47

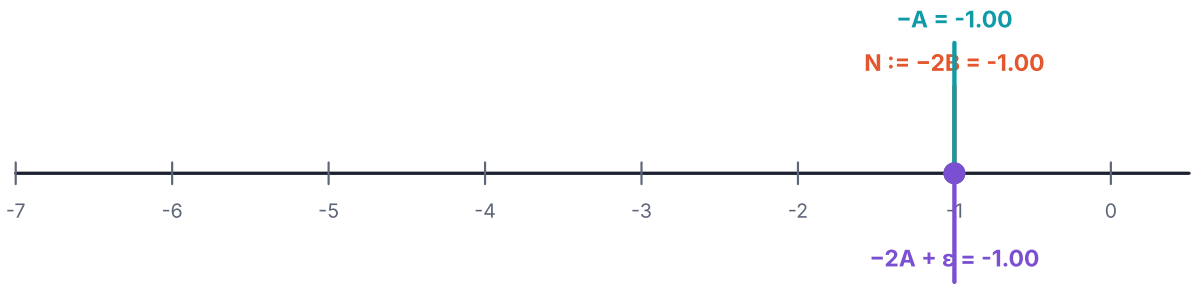
STEP 2 / 5 **identify the copies: a loop, ∨ but not ∧**



Identify †I with ‡I "because they are redundant": the result is a loop, γ_I is either α or β (∨) but cannot be both (∧ fails). The coordinate function no longer glues — exactly the mechanism of the RCS assertions about IUT (§3.2, §3.4).

REF **[EssLgc]** Ex 2.4.1 (ii) p. 46, (iii)

Example 2.4.5 (ii): the ∧ model of the logical flow of IUT



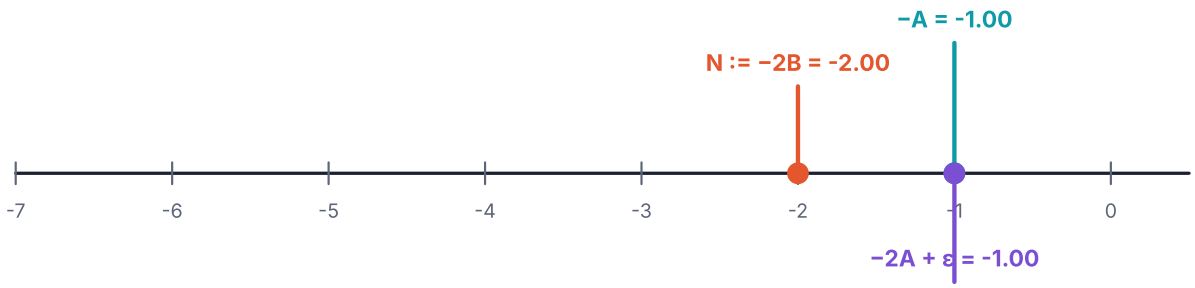
Θ-link: (N = −2B) ∧ (N = −A)	**✓ holds: A = 2B**
multiradial rep.: (N = −2A + ε) ∧ (N = −A)	**✓ holds: ε = A**
final estimate: −2A + ε = −A ⇒ A = ε ≤ 1	**✓ A = 1.00 ≤ 1**

The ∧'s are meaningful precisely because A and B are allowed to be distinct — the two Hodge theaters are different copies. Correspondences: B ↔ ᵗI, −2B ↔ ᵗβ, N ↔ γ_J, A ↔ ‡I, −A ↔ ‡α, −2A ↔ ‡β.

Real numbers A, B > 0 and 0 ≤ ε ≤ 1 model the Θ-link (the definition N := −2B and the gluing N = −A), the multiradial representation (N = −2A + ε) and the final estimate A = ε ≤ 1. Move the sliders: the ∧'s hold only for A = 2B and ε = A; the point is that nothing forces A = B.

REF **[EssLgc]** Ex 2.4.5 (i) p. 50, (ii); **[Alien]** Ex 3.11.4 p. 142

the "∨" version with A = B identified (RCS-IUT)



Θ-link: (N = −2B) ∨ (N = −A)	**∨ fine; ∧ impossible (A = 2A ⇒ A = 0)**
multiradial rep.: (N = −2A + ε) ∨ (N = −A)	**meaningless as ∨**
final estimate: −2A + ε = −A ⇒ A = ε ≤ 1	**abrupt / unjustified**

Identifying A and B "for simplicity" turns the ∧ of the Θ-link into a self-contradiction, so one is forced to read it as ∨ (indeed XOR); then the passage to the estimate looks like an unjustified replacement of ∨ by ∧. RCS-IUT ⇔ XOR-IUT ⇔ OR-IUT.

Replace ∧ by ∨ and the distinct numbers look superfluous, so one sets A = B — which makes the original ∧ version of the Θ-link self-contradictory and the passage to the estimate "abrupt, mysterious, unjustified". AND-IUT (the real one) vs OR-IUT / XOR-IUT / RCS-IUT (equivalent) vs EssOR-IUT (accepts the ∧ formally, never uses it).

REF **[EssLgc]** Ex 2.4.5 (iii)–(vi) p. 50, (XOR/RCS), (EssOR/RCS)

STEP 5 / 5 **(ExtInd): how new ∧'s are made from old**

$$A \wedge B \Rightarrow A \wedge (B \vee C)$$

since $B \Rightarrow B \vee C$

(ExtInd1)

add more possibilities / indeterminacies inside a fixed container: upper semi-continuity of (Ind3), passage to holomorphic hulls. Prepares descent data; establishes symmetry / multiradiality.

(ExtInd2)

crush together objects with distinct labels, at the cost of knowing the object only up to isomorphism: Π_v at (n,m), $(n,m') \rightsquigarrow (n,\circ)$; G_v at any $(n,m) \rightsquigarrow (\circ,\circ)$; the $F\vdash\times\mu$-strips of $\dagger$ and $\ddagger$ up to isomorphism.

This is how the log-Kummer correspondence exhibits the Frobenius-like Θ-pilot as one possibility among many constructed from étale-like data (ExtInd1), while the Kummer isomorphisms pass from labels (n,m) to (n,∘) (ExtInd2) — the "chain of (sub)quotients" aspect of the SHE property ([IUTchIII] Rem 3.11.1 (iii)).

The principle of extension of indeterminacies: the only legitimate way to grow the $\dot{v}$ side without introducing new labels, which is what (sQLTL) — projecting the left column of the H onto the right — requires.

REF **[EssLgc]** §3.6 (ExtInd) p. 104, (ExtInd1), (ExtInd2), (sQLTL)

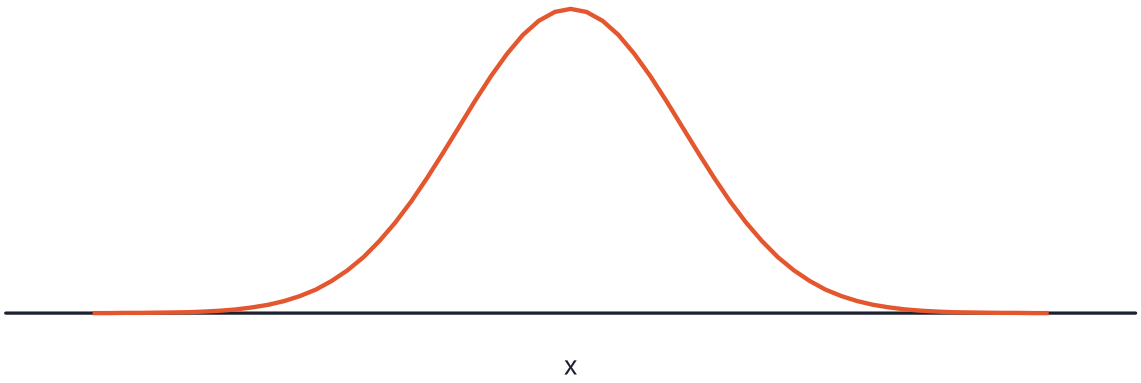
11.9 The Gaussian-integral analogy

SOURCES **[Alien]** §1.1 –§1.7 (Steps 1–12) p. 6, §3.8 p. 112, Fig 3.20.1 –3.20.2 p. 113

The "game plan" of IUT explained to a high-school student: a naive substitution is wrong on one copy of the real line, but becomes correct — up to the factor $\sqrt{\pi}$ — once two mutually independent copies are considered and rotated into each other.

STEP 1 / 7 **the naive substitution fails**

$$\int e^{-x^2}\,dx = ?$$



naive: $e^{-x^2} \rightsquigarrow u$ gives $\int_0^1 du = 1$ — wrong: dx was mistreated (Step 1)

partial integrations lead nowhere (Step 2)

The student tries $e^{-x^2} \rightsquigarrow u$ and gets 1; the infinitesimal dx was not transformed. Partial integrations do not help. This is formally reminiscent of the Θ-link's $\{\underline{q}^{j^2}\} \mapsto \underline{q}$, [IUTchII] Rem 1.12.5.

REF **[Alien]** §1 p. 6, §3.8 (1gau) p. 112

two identical — but mutually independent! — copies (Step 3)

$$\left(\int e^{-x^2}\,dx\right)\cdot\left(\int e^{-y^2}\,dy\right)$$

"utterly ludicrous" to the student: no clever substitution, just a second copy



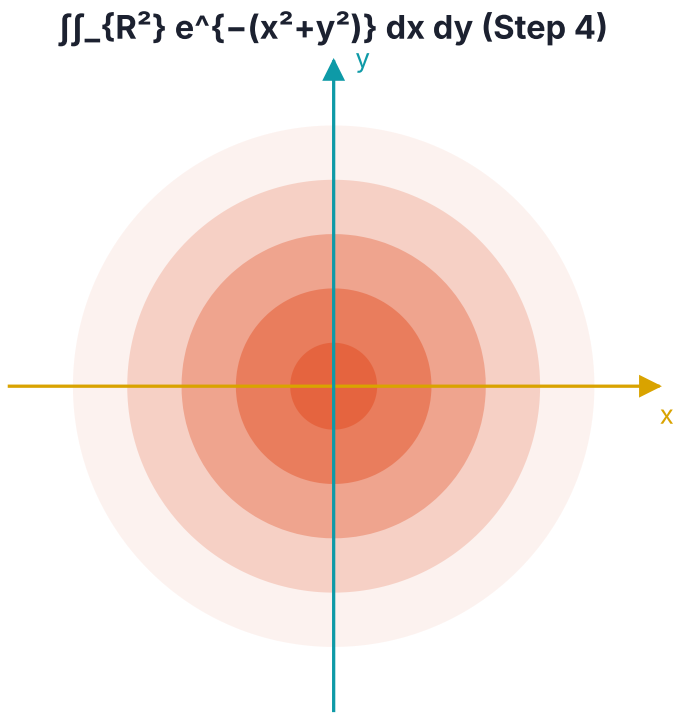
x: $\int e^{-x^2}\,dx$ y: $\int e^{-y^2}\,dy$

↔ the two Θ±ell NF-Hodge theaters • in the domain and codomain of the Θ-link

Square the integral: consider two identical but independent copies. Nothing clever; the student is indignant. In IUT: the two Hodge theaters in the domain and codomain of the Θ-link, i.e. two copies of the scheme theory around E_F.

REF **[Alien]** §1 p. 6, §3.8 (2gau) p. 112

STEP 3 / 7 **one integral over the plane**

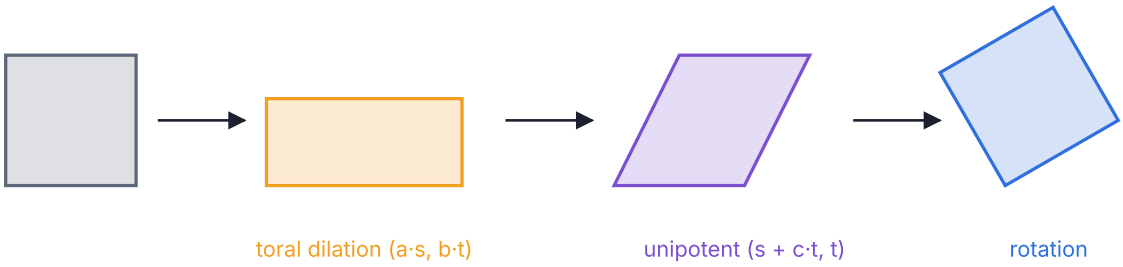


The product becomes a single integral over R^2 — a two-dimensional object, like the lattice, like a ring with its two operations. Integrals are net masses: sums of infinitesimals of zero mass (étale-like containers for positive-mass Frobenius-like objects).

REF **[Alien]** §1 p. 6, §3.8 (3gau) p. 112

∫_V ψ dx dy = ∫_U φ · J ds dt (Step 6)

the Jacobian: locally the map is linear; areas of small parallelograms scale by det (Step 7)

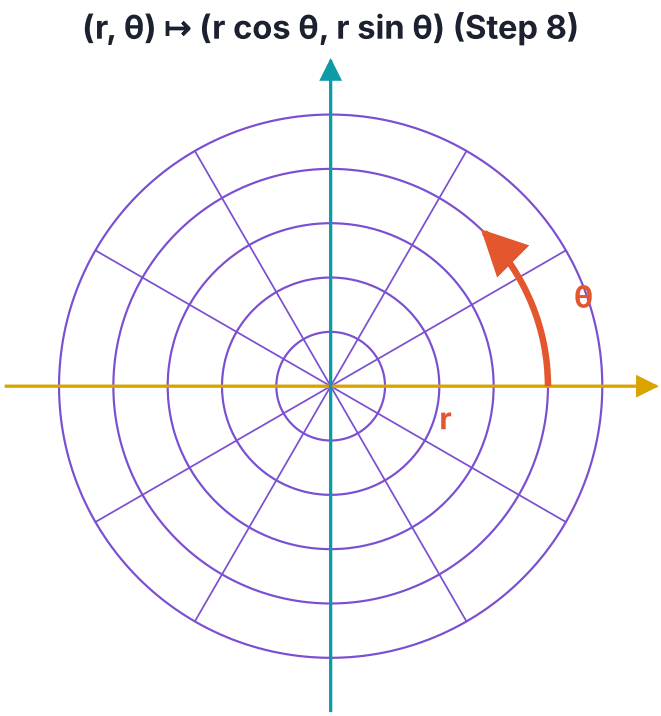


↔ mono-anabelian transport computes the effect of arithmetic changes of coordinates;
dilations / unipotents / rotations ↔ the examples for MLF's of [Alien] §2.12

The Jacobian computes the effect of a change of coordinates locally through linear maps: dilations, unipotents, rotations. Mono-anabelian transport is the arithmetic analogue; the MLF examples of §2.12 the analogues of the three linear maps.

REF **[Alien]** §1 p. 6, §3.8 (4gau) p. 112

STEP 5 / 7 **polar coordinates**



a rotation / continuous deformation between the two alien copies x and y

↔ parallel transport between radial data: the étale-picture and the multiradial representation

Rotating the x-copy into the y-copy — passing to polar coordinates — is the analogue of the étale-picture: the two columns become spokes around one coric core, and the multiradial representation is the parallel transport along the rotation.

REF **[Alien]** §1 p. 6, §3.8 (5gau) p. 112

STEP 6 / 7 **decoupling: 1 × π = √π²**

$$\iint e^{-(x^2+y^2)}\,dx\,dy = \iint e^{-r^2}\,r\,dr\,d\theta = \left(\int_0^{\infty} e^{-r^2}\cdot 2r\,dr\right)\cdot\left(\int_{-\pi}^{\pi}\tfrac{1}{2}\,d\theta\right)$$

radial and angular coordinates decouple (Step 9)

radial integral = 1

by the substitution $e^{-r^2} \rightsquigarrow u$ — the very substitution that was erroneous in Step 1, now justified (Step 10). Works because the exponent is quadratic.

angular integral = π

the imaginary part of $\log(r\,e^{i\theta}) = \log r + i\theta$ (Step 11). The "error factor" by which the naive answer 1 must be corrected.

$$\int e^{-x^2}\,dx = \sqrt{\pi}\ \text{(Step 12)}$$

no "explicit portion" of the one-dimensional integral corresponds to the radial or angular part

↔ unit / value group decoupling and cyclotomic rigidity (§3.4); quadratic exponent ↔ quadratic commutators of theta groups

Radial and angular parts separate: the radial integral is the naive substitution, now valid; the angular integral is the error factor. In IUT the unit-group and value-group portions decouple by cyclotomic rigidity, the quadratic structure of theta groups playing the role of the quadratic exponent.

REF **[Alien]** §1 p. 6, §3.8 (6gau) p. 112

STEP 7 / 7 **the dictionary**

classical computation of the Gaussian integral	inter-universal Teichmüller theory
naive change of coordinates $e^{-x^2} \rightsquigarrow u$	the assignment $\{\underline{q}^{j^2}\}_j \mapsto \underline{q}$ of the Θ-link
two mutually alien copies of the integral	domain and codomain • of the Θ-link
the two dimensions of $\mathbb{R}^2$	the two dimensions of the lattice / rings / NF's / MLF's
net mass = lim Σ zero-mass infinitesimals	prime-strips; heights as log-volumes; Frobenius-like / étale-like
effect of local changes of coordinates (Jacobian)	mono-anabelian transport of arithmetic changes of coordinates
unipotents, toral dilations, rotations	examples for MLF's ([Alien] §2.12)
cartesian → polar coordinates	log-theta-lattice → étale-picture → multiradial representation
decoupling radial × angular	multiradial decouplings / cyclotomic rigidity (§3.4)
quadratic exponent makes the radial integral easy	quadratic commutators of theta groups (mono-theta rigidity)
the error factor √π	the indeterminacies and the constant of the estimate

[Alien] Fig 3.20.1, 3.20.2 (the last row is this app's paraphrase)

Mochizuki's comparison table ([Alien] Fig 3.20.1–3.20.2).

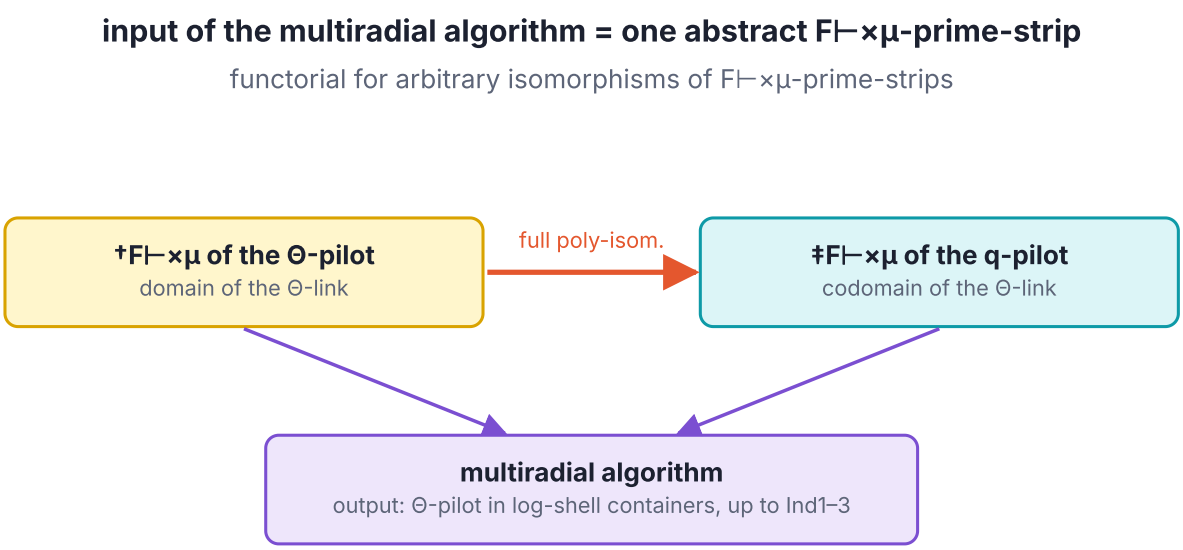
REF **[Alien]** §1 p. 6, §3.8 (7gau) p. 112

11.10 From the multiradial representation to the log-volume estimate and abc

SOURCES **[IUT III]** Thm 3.11 p. 153, Cor 3.12 p. 173; **[IUT IV]** Thm 1.10 p. 22, Cor 2.2 p. 41; **[Alien]** §3.7 (i)–(iv) p. 103; **[ExpEst]**

The last ∧ becomes an inequality of real numbers; with explicit constants it is the Szpiro-type bound that implies abc. The numbers below are computed for the current data (a numerical check of the shape of the estimate, not a proof).

STEP 1 / 6 **(∧-Input): an abstract F⊢×μ-prime-strip**



Because only this strip is needed, the algorithm can be run on the codomain's strip as well: any F⊢×μ-strip may serve as gluing data between the ‡ Hodge theater and the content of the algorithm. This is the starting point of the ∧'s of Thm 3.11.

The multiradial algorithm asks only for an F⊢×μ-prime-strip, exactly what the Θ-link glues; hence it runs equally on either side.

REF **[EssLgc]** §3.6 (∧-Input) p. 104; **[IUT III]** Rem 3.11.1 (ii) p. 159

STEP 2 / 6 **processions of log-shells as containers**

Θ-values acting on tensor packets of log-shells over processions [Alien] Fig 3.16–3.18; [IUTchIII] Thm 3.11



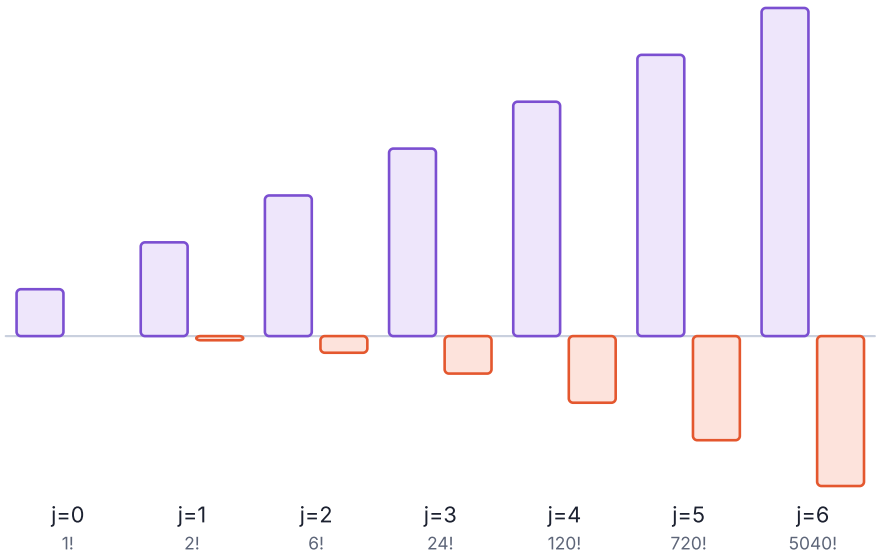
Each row is a tensor packet of mono-analytic log-shells (zero-mass containers) acted on by the theta value q^{j^2} and a copy of F_mod^×, up to (Ind1)–(Ind3). The input is only an abstract F⊢×μ-prime-strip (∧-Input, [EssLgc] §3.6), so the algorithm applies to ᵀHT and ‡HT alike.

The output lives in tensor packets of mono-analytic log-shells indexed by the processions S^±_{j+1} = {0,...,j} (with their (j+1)! permutation symmetries), on which the theta value q^{j^2} and a copy (F_mod^×)_j act.

REF **[Alien]** §3.6 (iv) p. 87, (v), Fig 3.16 –3.18 p. 103

STEP 3 / 6 **hull versus Θ-shift, step by step**

per procession step j: hull enlargement (j+1)·log d^K (up) vs Θ-shift −(j²/2l)·log q (down) [Alien] §3.7 (iv); [IUTchIII] Thm 3.11



Σ_j (hull + Θ-shift)
198.171

(l²/48)(6 log d^K − log q)
136.115

log d^K ≈ 10.477
log q = 24.202

÷ l*: −|log Θ| heuristic
33.029

−|log q| = −log q / 2l
-0.931

log-volume of the Θ-pilot's container ≤ hull of the tensor packets of log-shells; the Θ-values push it down by j²/2l · log q each.

For each j the container can grow at most by the hull of j+1 log-shells, roughly (j+1) log d^K, while the theta value contributes −(j²/2l) log q. Summing over j = 1..l* and dividing by l* gives the heuristic bound for −|log Θ|.

REF **[Alien]** §3.7 (iv) p. 103; **[IUT IV]** Prop 1.4 p. 13

$-|\log \Theta| \geq -|\log q|$

for any C_Θ with $-|\log \Theta| \leq C_\Theta \cdot |\log q|$ this says $C_\Theta \geq -1$

$-\lvert\log q\rvert = -\log q / 2l$	**-0.931**
leading terms of $-\lvert\log \Theta\rvert$ (hull − shifts, ÷ l*): an estimate	**33.029**
bound B of Thm 1.10, Steps (v)–(vii), every term	**1530.593**
$C_\Theta = B/\lvert\log q\rvert \leq C_\Theta$ of the statement	**1644.29 ≤ 1.880e+22** ✓
consistent with Cor 3.12 (an input): $B \geq -\lvert\log q\rvert$	**yes** ✓

The ∧ of (Stp8) in one copy of R: the log-volume of the q-pilot is simultaneously the log-volume of some possible output of the multiradial algorithm; the log-link preserves log-volumes, so the possible outputs are bounded by the hull. The "closed loop" is the symmetrized diagonal of the 1-column, not an equality of two maps.

The statement of [IUTchIII] Cor 3.12 and how the current data fares. For small abc triples the hull term dominates and the inequality is comfortably true; it is the explicit constant that carries the content.

REF **[IUT III]** Cor 3.12 p. 173; **[EssLgc]** §3.10 (Stp8) p. 136

STEP 5 / 6 **Thm 1.10: the explicit constant**

$$C_0 = (l+1)/(4|\log q|) \cdot \{ (1 + 12\, d_{mod}/l)(\log d^{Ftpd} + \log f^{Ftpd}) + 10(e^*_{mod} \cdot l + \eta_{prm}) - (1/6)(1 - 12/l^2) \log q \} - 1$$

$e^*_{mod} = 2^{12} \cdot 3^3 \cdot 5 \cdot e_{mod}$, $\eta_{prm} = 5 \cdot 10^{20}$ ([ExpEst]), $d_{mod} = [F_{mod}:Q] = 1$, $l = 13$

hence, from $C_\Theta \geq -1$:

$$(1/6) \log q \leq (1 + 20\, d_{mod}/l)(\log d^{Ftpd} + \log f^{Ftpd}) + 20(e^*_{mod} \cdot l + \eta_{prm})$$

(1/6) log q (height of E)	**4.034**
log d^Ftpd + log f^Ftpd	**4.654**
right-hand side	**1.000e+22**
leading term $(l^2/48)(6 \log d^K - \log q)$ [Alien] §3.7 (iv)	**136.115**

Szpiro's shape: the height (1/6) log q is bounded by the conductor and the different up to a factor close to 1 plus a huge additive constant. Choosing l of size about (log q)^{1/2} optimizes the factor; the constant η_prm comes from the prime-number estimates of [ExpEst]. For the Frey curve, log f ≈ log rad(abc) and (1/6) log q ≈ (1/3) log(abc) − c, which yields abc.

The explicit form of [IUTchIV] Thm 1.10 with the data's numbers. The leading-term heuristic $(l^2/48)(6 \log d^K - \log q)$ of [Alien] §3.7 (iv) is the "size of the inequality" that the multiradial representation produces before normalizing.

REF **[IUT IV]** Thm 1.10 p. 22, Cor 2.2 p. 41; **[ExpEst]** ; **[Alien]** §3.7 (iv) p. 103

STEP 6 / 6 **summary for this data**

the numbers are consistent with Cor 3.12 for abc(1,2400,2401) l=13



The toolkit implements the data types and the bookkeeping of these steps and checks the arithmetic that can be checked (valuations, label sets, coricity, log-shell volumes, the l(l+1)/12 dilation, the shape of the estimate). It does not implement anabelian reconstruction, Kummer theory of theta functions or the actual multiradial algorithm — those are the content of [IUTchII–III] and are only represented by their types, indeterminacies and citations. Cor 3.12 is an input, not something the toolkit verifies. Every such stand-in is a named placeholder: `iut gaps`, docs/GAPS.md.

The whole progression from the integers a + b = c to the inequality, as implemented. Change the data in the bench above and every diagram on this page recomputes.

REF docs/DATA_MODEL.md

12 · The registry: all 73 processes

Every row is a family of C functions in `src/iut_ops_*.c` (declared in `src/iut_ops.h`), run live in the web app (`htdocs/index.html`, "Processes"); `iut run <id>` demonstrates it on given Θ-data and `iut run --all` runs them all (exit code = number of failures). The process ids are identifiers of the table too (kind *process*).

id	paper · section	process	inputs → outputs
`MapLabels`	**[IUT I]** §4 -§6 p. 95, Def 6.13 p. 182	label sets F_l ⊇ F_l^× ↠ F_l^*, |F_l| and the gluing 0,⊞ ↦ >	InitialThetaData → labels
`ActSymmetries`	**[IUT I]** Prop 4.9 p. 115, Prop 6.6 p. 165	the F_l^{⋊±}- and F_l^*-symmetries and their orbits	InitialThetaData, label → label, orbit size
`LabelCusps`	**[IUT I]** Def 6.1 p. 155, Prop 6.5 p. 163	LabCusp±(D_⊛) ≅ F_l, LabCusp(D_⊛) ≅ F_l^*, forgetting the sign	InitialThetaData → LabCusp, LabCusp±
`ExtractCapsules`	**[IUT I]** §0 p. 33, Def 4.1 p. 95, Def 6.4 p. 162	the capsules F_T, F_J and morphisms of capsules	ThetaPmEllNFHodgeTheater → capsule, capsule morphism
`BuildProcessions`	**[IUT I]** Prop 4.11 p. 120, Prop 6.9 p. 169	processions Prc(D_J), Prc(D_T) with their automorphisms	InitialThetaData → procession
`BuildBridges`	**[IUT I]** Def 4.6 p. 111, Def 5.5 p. 151, Def 6.4 p. 162	Θ-, NF-, Θ±- and Θell-bridges and their symmetry groups	ThetaPmEllNFHodgeTheater → bridge
`ForgetToDHodgeTheater`	**[IUT I]** Def 6.13 p. 182, Rem 6.12.2 p. 174	Θ±ell NF-HT ↦ D-Θ±ell NF-HT, and the Θ±ell/ΘNF/Θ projections	ThetaPmEllNFHodgeTheater → DThetaPmEllNFHodgeTheater
`ChainFrobeniusPicture`	**[IUT I]** Cor 3.8 p. 89	the Frobenius picture: a chain of Θ-linked Hodge theaters	LogThetaLattice → chain of universes
`DrawEtalePicture`	**[IUT I]** Cor 4.12 p. 121, Cor 6.10 p. 170	the étale picture: D-HTs as spokes around the core DI⊢	LogThetaLattice → étale picture
`BuildGlobalNF`	**[IUT I]** Ex 5.1 p. 123, Cor 5.3 p. 143	global number-field data from Π_{C_K} and its Kummer isomorphism	InitialThetaData → F^⊛, F^⊛_mod, F^⊛R
`GlueHodgeTheaters`	**[IUT I]** Prop 6.7 p. 167, Def 6.13 p. 182	gluing Θ±ell-HT to ΘNF-HT along 0,⊞ ↦ >	ThetaPmEllHodgeTheater, ThetaNFHodgeTheater → ThetaPmEllNFHodgeTheater
`BuildPrimeStrip`	**[IUT I]** Def 4.1 p. 95, Def 5.2 p. 134	construct and validate a prime-strip of any of the 9 kinds	InitialThetaData → prime-strip
`GenerateInitialData`	**[IUT I]** Def 3.1 p. 61	initial Θ-data from an abc triple / example and its validation	abc triple → InitialThetaData, report
`ClassifyPseudoMonoids`	**[IUT I]** §0 p. 33, Ex 5.1 (i) p. 123, Def 5.2 (v)-(vi) p. 134, Rem 5.1.3 p. 131	pseudo-monoids M^⊛, M^⊛_mod, M^⊛_sol, M^⊛_κ, M^⊛_∞κ, M^⊛_∞κ×, M^⊛_κ-sol, M_v...: divisible / cyclotomic / Kummer-ready	DCircAstGlobal, DV → pseudo-monoid table
`ComposePolyMorphisms`	**[IUT I]** §0 p. 33, Def 6.1 (iv) p. 155	poly-morphisms, full and +-full poly-isomorphisms, their composites; anabelioids; NF / MLF / CAF	morphisms, α ∈ {±1}^V → poly-morphism
`ClassifyOrbicurves`	**[IUT I]** §1 p. 37, Def 1.1 p. 38, Def 3.1 (b) p. 61	the orbicurves X, C, X̲, C̲, X→, C→, X̲_v, C̲_v, C̲_v: types, degrees, cusps, covering groups	InitialThetaData → orbicurve table
`ReconstructCoverings`	**[IUT I]** §1 p. 37, Cor 1.2 p. 39, Rem 1.2.1 p. 40	the cusps ε, ε′, ε″, 2ε and the group-theoretic reconstruction of Π_X̲, Π_C→, Π_C̲ from Π_X→	PiXArrow → Π_X̲, Π_C→, Π_C̲, I
`CompareTempered`	**[IUT I]** §2 p. 16, Prop 2.1 -Cor 2.8 p. 45, Rem 2.5.3 p. 52	tempered versus profinite fundamental groups at v ∈ V^bad; Cor 2.5 synchronization	InitialThetaData → facts per v
`CheckDefThreeOne`	**[IUT I]** Def 3.1 (a)-(f) p. 61, Rem 3.1.2 p. 64, Rem 3.1.3 p. 65, Rem 3.1.5 p. 65	the six conditions on initial Θ-data; the Θ-approach M_X ≅ I·Δ_Θ; the determining data; S_mod	InitialThetaData → report
`CheckKappaCoric`	**[IUT I]** Rem 3.1.7 p. 66	κ-coric / ∞κ-coric / ∞κ×-coric functions, critical points, Gal(K/F(μ_l)) ≅ SL_2(F_l)	rational function data → verdict
`BuildLocalFrobenioids`	**[IUT I]** Ex 3.2 p. 69, Ex 3.3 p. 77, Ex 3.4 p. 80	F̲_v, C_v, CI⊢_v, τI⊢_v, C^Θ_v at bad / good / archimedean v; q̲_v, Θ̲_v, log(p_v), log(Θ); FI⊢_v ≅ F^Θ_v	InitialThetaData → FUnderV, FMonoV, FThetaV
`BuildGlobalRealified`	**[IUT I]** Ex 3.5 p. 84	CI⊢_mod, CI⊢_tht, DI⊢_mod; ρ_v, ρ^Θ_v, ρ^D_v; FI⊢_mod ≅ FI⊢_tht	InitialThetaData → CMonoGlobMod, FMonoGlobMod
`AssembleThetaHodgeTheater`	**[IUT I]** Def 3.6 p. 87, Cor 3.7 -3.9 p. 88, Rem 3.9.3 p. 93	†HT^Θ = ({F̲_v}, FI⊢_mod) with (a)(b)(c); the Θ-link as full poly-iso FI⊢_tht ≅ FI⊢_mod preserving DI⊢_v and O×	InitialThetaData, Iattice → ThetaHodgeTheater, report
`TorsorLabelCusps`	**[IUT I]** Def 4.1 (ii) p. 95, Prop 4.2 p. 98, Def 6.1 (ii)-(iii) p. 155	LabCusp(†D_v) as F_l^*-torsor with †η_v; LabCusp± (†D_v) with †η°_v, †η^±_v; forgetting the sign	DPrimeStrip → LabCusp, LabCuspPm
`DescribeGlobalBases`	**[IUT I]** Def 4.1 (v) p. 95, Ex 4.3 p. 98, Def 6.1 (v) p. 155, Rem 6.1.1 p. 159	D^⊛ = B(C̲_K)^0, D^⊛± = B(X̲_K)^0; Aut ⊇ Aut_ε (Borel ⊇ semi-unipotent), Aut_± ⊇ Aut_csp; V ⊆ V^±un ⊆ V^Bor ⊆ V(K)	InitialThetaData → global bases, orbits
`EvaluateAtPoints`	**[IUT I]** Ex 4.4 p. 105	evaluation points labelled |F_l|, Θ̲_v ↦ q̲_v^{j²}, evaluation sections G_v → Π_v, the Θ-bridge constituents φ^Θ_{v_j}	InitialThetaData → EvaluationPoint, PhiThetaVJ
`TransportLabelClasses`	**[IUT I]** Prop 4.7 p. 112, Prop 6.5 p. 163, Rem 6.12.4 p. 177	†χ : J ≅ F_l^*, †ζ, †ζ_±, the global ±-synchronization †ξ	DThetaNFHodgeTheater, DThetaPmEllHodgeTheater → label bijections
`CountIsomorphismTorsors`	**[IUT I]** Prop 4.8 p. 115, Prop 6.6 p. 165, Cor 5.6 p. 153, Cor 6.12 p. 173, Rem 6.12.2 p. 174	the isomorphism torsors of the four bridges, the Hodge theaters and their gluings	InitialThetaData → torsor table
`CountForgottenSymmetries`	**[IUT I]** Prop 4.9 p. 115, Prop 4.11 p. 120, Prop 6.8 p. 167, Prop 6.9 p. 169	(l*)^{l*} → l*! and (l±)^{l±} → l±!: what the forgetful functors lose and what processions recover	InitialThetaData → counts
`DistributeGlobalData`	**[IUT I]** Ex 5.1 (ii)-(vii) p. 123, Rem 5.1.1 p. 131, Rem 5.1.3 p. 131, Rem 5.4.2 p. 151	F^⊛, F^⊛′, F^⊛_mod, F^⊛_j, F^⊛_{J}, F^⊛_J; V_j, V_{J}, V_J; Kummer-ready vs Kummer-blind; averaged vs summed log-volumes	DCircAstGlobal → global Frobenioids
`LiftToFrobenioids`	**[IUT I]** Def 5.2 p. 134, Rem 5.2.1 p. 143, Cor 5.3 p. 143, Rem 5.3.1 p. 145	F ↦ D, F ↦ FI⊢, F ↦ FI⊢; Isom(F) ≅ Isom(D); lifting a D-prime-strip to an F-prime-strip	DPrimeStrip → FPrimeStrip
`StructureFlSemiPm`	**[IUT I]** Def 6.1 (i) p. 155, Def 6.4 (i) p. 162, Ex 6.2 p. 159	F_l^{⋊±}-groups and -torsors, positive elements, |T| and T*, sign vectors {±1}^V and −1_{F_l}	InitialThetaData → FlSemiPmGroup, TAbs, TStar

id	paper · section	process	inputs → outputs
`ReduceThetaPmToTheta`	**[IUT I]** Prop 6.7 p. 167, Prop 6.9 (iii) p. 169, Rem 6.12.2 p. 174	D-Θ±-bridge ⇒ D-Θ-bridge (±t ↦ |t|, D_0, D_⊞ ↦ D_>); S*_j ↪ S^±_{j+1}; uniqueness of the gluing	DThetaPmBridge → DThetaBridge
`CompareHalfPlane`	**[IUT I]** Rem 6.12.3 p. 175, Fig 6.4 p. 178, Fig 6.6 p. 183	the two symmetries versus the upper half-plane; the Frobenioids of a Θ±ell NF-Hodge theater	— → tables
`ClassifyRadial`	**[IUT II]** Ex 1.7 -1.9 p. 32	radial environments, multiradiality, multiradialization	radial data, coric data → multiradial / uniradial
`RigidifyCyclotomes`	**[IUT II]** Cor 1.10 -1.12 p. 46	the three cyclotomic rigidity isomorphisms, constant multiple rigidity	Π_v / G_v, M^Θ → rigidity + indeterminacy
`EvaluateTheta`	**[IUT II]** Cor 2.4 p. 69, Cor 2.5 p. 71, Cor 2.8 p. 81	theta evaluation at the l-torsion points: θ ↦ q̲^{t²}, |t|-dependence	InitialThetaData → theta values
`BuildMonoids`	**[IUT II]** Prop 3.1 p. 87, Cor 3.5 -3.7 p. 93, Cor 4.5 -4.6 p. 131	Ψ_env, Ψ_gau, Ψ_cns, Ψ^ss_cns; evaluation and Kummer isomorphisms	prime-strip at v → monoid
`RealifyFromDMono`	**[IUT II]** Cor 4.5 (ii) p. 131	D⊩ ↦ Dǁ⊩(D⊩): the global realified Frobenioid from the core	DMonoPrimeStrip → FMonoGlobPrimeStrip
`Symmetrize`	**[IUT II]** Cor 3.5 (i) p. 93, Cor 4.5 (iii) p. 131, Cor 4.7 (ii) p. 140	symmetrizing isomorphisms, diagonals, conjugate synchronization	capsule → symmetrization
`ForgetLadder`	**[IUT II]** Def 4.9 p. 153	F⊩▶×μ → F⊩▶×μ → F⊩×μ → F⊩× → D⊩ and the pilot object	FMonoGlobValUnitsModTorsPrimeStrip → prime-strips
`LinkThetaUnitsModTors`	**[IUT II]** Cor 4.10 p. 158, Rem 4.11.1 p. 164	F⊩_Δ ≅ F⊩_mod, F⊩_env ≅ F⊩_gau, the Θ×μ(_gau)-link, coric F⊩×μ and Dǁ⊩	two Hodge theaters → link, D-link
`MakeLogShells`	**[IUT III]** Def 1.1 p. 23, Prop 1.2 (v)-(ix) p. 30	log-shells I_v and their properties / isomorphisms	InitialThetaData → LogShell
`ApplyLogLinkStrip`	**[IUT III]** Def 1.1 p. 23, Prop 1.2 p. 30	the log-link on an F-prime-strip; what is coric and what is not	FPrimeStrip → FPrimeStrip, LogShell
`ChainLogLinks`	**[IUT III]** Prop 1.2 (x) p. 30, Prop 1.3 p. 41	the vertical Frobenius picture and the log-link of Hodge theaters	LogThetaLattice → chain, report
`DefineLattice`	**[IUT III]** Def 1.4 p. 45, Rem 1.4.1 p. 46	lattice flavours; non-commutativity; horizontal obligates vertical	LogThetaLattice → report
`CheckBicoric`	**[IUT III]** Thm 1.5 p. 48	vertical / horizontal / bi-coric data of the lattice	LogThetaLattice → bicoric summary
`CoricThetaMonoids`	**[IUT III]** Prop 2.1 p. 58, Thm 2.2 p. 65, Cor 2.3 p. 72	vertically coric theta monoids; the étale-picture radial environment	LogThetaLattice → report, radial items
`BuildTensorPackets`	**[IUT III]** Prop 3.1 -3.3 p. 92	local holomorphic / mono-analytic / global tensor packets	LogShell, InitialThetaData → TensorPacket
`BuildLGPMonoids`	**[IUT III]** Prop 3.4 p. 101, Prop 3.5 p. 103	LGP-monoids in the packets; upper semi-compatibility (Ind3)	TensorPacket → monoid, Δlog-volume
`CompareGlobalVariants`	**[IUT III]** Ex 3.6 p. 106, Prop 3.7 p. 109	F_MOD vs F_mod; F⊩_gau ≅ F⊩_LGP ≅ F⊩_lgp	ThetaPmEllNFHodgeTheater → report
`MakePilots`	**[IUT III]** Def 3.8 p. 112	q-/Θ-pilot objects; the Θ×μ_LGP-link relative to a log-link	InitialThetaData, lattice → QPilot, ThetaPilot
`NormalizeLogVolumes`	**[IUT III]** Prop 3.9 p. 115, Prop 3.10 p. 147	packet-/procession-normalization, arithmetic degree, non-interference	log-volumes → numbers
`RepresentMultiradially`	**[IUT III]** Thm 3.11 p. 153	the multiradial representation (a)(b)(c) with Ind1-Ind3	LogThetaLattice → MultiradialRepresentation
`EstimateLogVolume`	**[IUT III]** Cor 3.12 p. 173	−|log Θ| ≥ −|log q| as numbered steps	LogThetaLattice → numbers
`BoundDifferents`	**[IUT IV]** Prop 1.1 -1.3 p. 9	differents, a_i, b_i, the φ-estimates, bounds for d_i	local fields k_i → exponents
`EstimateNonarchVolume`	**[IUT IV]** Prop 1.4 p. 13	μ_log(log_p R×) and the normalized log-volume bounds	local fields k_i → log-volumes
`EstimateArchMetric`	**[IUT IV]** Prop 1.5 p. 14	archimedean summands 2^{|I|−1}|V|^{|I|} and metric factors	|I|, |V_arc| → counts
`BoundPrimes`	**[IUT IV]** Prop 1.6 p. 16, Prop 2.1 p. 40	prime number theorem and Chebyshev-type bounds	numbers → numbers
`AverageWeighted`	**[IUT IV]** Prop 1.7 p. 16	weighted averages with λ_v = [(F_mod)_v : Q_p]	InitialThetaData → number
`CheckTorsionPoints`	**[IUT IV]** Prop 1.8 p. 18	torsion points, fields of definition, ramification at Vbad	InitialThetaData → report
`DegreeArithDivisors`	**[IUT IV]** Def 1.9 p. 21	arithmetic divisors, deg, normalized degree, deg_E	InitialThetaData → ArithDivisor, numbers
`ComputeThmOneTen`	**[IUT IV]** Thm 1.10 p. 22	(E1)-(E6), the intermediate bounds and the final inequality	InitialThetaData → Thm 1.10 numbers
`ComputeCorTwoTwo`	**[IUT IV]** Cor 2.2 p. 41, Cor 2.3 p. 54	δ, (C1) range of l, ε_E, (C2), the height form	InitialThetaData → Cor 2.2 numbers
`OperateSpecies`	**[IUT IV]** Def 3.1 p. 68, Def 3.3 p. 72	compose/identity mutations, species-isomorphism, mutation-history, cores	species tables, lattice → mutation, history
`EvaluateLogicModel`	**[EssLgc]** Ex 2.4.5 p. 50	the numerical model: ∧ gives A = ε ≤ 1, ∨ gives nothing	A, B, ε → conclusion
`ChainAndSteps`	**[EssLgc]** §3.10 (Stp1)-(Stp8) p. 136, (FxGl)	the chain of logical ∧ relations through the lattice	LogThetaLattice → steps with log-volumes
`Fuzzify`	**[EssLgc]** §3.10 (Englf) p. 136, (ExtInd1/2), §3.11 p. 152	fuzzification, engulfing, extending indeterminacies, ∨ verdicts	value, indeterminacies → interval
`ComputeGaussian`	**[Alien]** §1 Steps 1-12 p. 6	the Gaussian integral and its IUT analogues	— → √π
`TransportMonoAnabelian`	**[Alien]** §2.7 p. 21, Fig 2.1 p. 23	mono-anabelian transport: Kummer-detachment, étale-transport	LogThetaLattice → stages
`SumLabelsFormally`	**[Alien]** §3.6 p. 87	formal sums [0], [|j|], [⊞], [>] and their compatibilities; Ÿ labels	InitialThetaData → formal sums
`StepEstimate`	**[Alien]** §3.7 (1est)-(12est) p. 103, (iv)	the estimate step by step; the leading term	LogThetaLattice → numbers
`TranslateDictionary`	**[Alien]** §3.8 p. 112	the dictionary (1gau)-(11gau)	— → table

12.1 Processes of [IUT I] — Inter-universal Teichmüller Theory I: Construction of Hodge Theaters

`MapLabels` label sets F_l ⊇ F_l^× ↠ F_l^*, \|F_l\| and the gluing 0,⊞ ↦ > ✓ ok

WHERE **[IUT I]** §4 -§6 p. 95, §6, Definition 6.13 p. 182

INPUTS InitialThetaData OUTPUTS labels

```
    F_l = {-6..6}, F_l^× = F_l\{0}, F_l^* = F_l^×/{±1} = {1..6}, |F_l| = {0..6}
    t= -6  |t⊨ 6  F_l^×→F_l^*:  6  glue 0,⊠→>:  6
    t= -5  |t⊨ 5  F_l^×→F_l^*:  5  glue 0,⊠→>:  5
    t= -4  |t⊨ 4  F_l^×→F_l^*:  4  glue 0,⊠→>:  4
    t= -3  |t⊨ 3  F_l^×→F_l^*:  3  glue 0,⊠→>:  3
    t= -2  |t⊨ 2  F_l^×→F_l^*:  2  glue 0,⊠→>:  2
    t= -1  |t⊨ 1  F_l^×→F_l^*:  1  glue 0,⊠→>:  1
    t=  0  |t⊨ 0  F_l^×→F_l^*: -1  glue 0,⊠→>:  0
    t=  1  |t⊨ 1  F_l^×→F_l^*:  1  glue 0,⊠→>:  1
    t=  2  |t⊨ 2  F_l^×→F_l^*:  2  glue 0,⊠→>:  2
    t=  3  |t⊨ 3  F_l^×→F_l^*:  3  glue 0,⊠→>:  3
    t=  4  |t⊨ 4  F_l^×→F_l^*:  4  glue 0,⊠→>:  4
    t=  5  |t⊨ 5  F_l^×→F_l^*:  5  glue 0,⊠→>:  5
    t=  6  |t⊨ 6  F_l^×→F_l^*:  6  glue 0,⊠→>:  6
  ⇒ ok
```

ActSymmetries the F_l^{⋊±}- and F_l^*-symmetries and their orbits

✓ ok

WHERE **[IUT I]** §4, Proposition 4.9 p. 115, §6, Proposition 6.6 p. 165

INPUTS InitialThetaData, label OUTPUTS label, orbit size

```
    F_l^{⋊±} ↷ F_l: orbit of 0 has 13 elements (transitive); F_l^* ↷ F_l^*: orbit of 1 has 6 elements (simply transitive)
    (a=1,+) · t=2 = 3;  (a=0,-) · t=2 = -2;  u=2 · j=3 = 6
  ⇒ ok
```

LabelCusps LabCusp±(D_⊚) ≅ F_l, LabCusp(D_⊛) ≅ F_l^*, forgetting the sign

✓ ok

WHERE **[IUT I]** §6, Definition 6.1 p. 155, §6, Proposition 6.5 p. 163

INPUTS InitialThetaData OUTPUTS LabCusp, LabCusp±

```
    LabCusp±(D_⊚): 13 cusps ≅ F_l (bijection ok);  LabCusp(D_⊛): 6 ≅ F_l^* (ok);  forget sign: 6 labels (ok)
  ⇒ ok
```

ExtractCapsules the capsules F_T, F_J and morphisms of capsules

✓ ok

WHERE **[IUT I]** §0 p. 33, §4, Definition 4.1 p. 95, §6, Definition 6.4 p. 162

INPUTS ThetaPmEllNFHodgeTheater OUTPUTS capsule, capsule morphism

```
    capsule F_T: 13 members (labels -6..6, basepoints 101..113), 1932053504 full poly-isomorphisms
    capsule F_J: 6 members; morphism F_J → F_T (j ↦ +j): valid injection
  ⇒ ok
```

BuildProcessions processions Prc(D_J), Prc(D_T) with their automorphisms

✓ ok

WHERE **[IUT I]** §4, Proposition 4.11 p. 120, §6, Proposition 6.9 p. 169

INPUTS InitialThetaData OUTPUTS procession

```
    Prc(D_J) (Prop 4.11): 6 capsules, |S_k| = 1..6, automorphisms Π|S_k|! = 24883200
      S_1 = {1}  weight 1/1
      S_2 = {1,2}  weight 1/2
      S_3 = {1,2,3}  weight 1/3
      S_4 = {1,2,3,4}  weight 1/4
    Prc(D_T) (Prop 6.9): 7 capsules, |S_k| = 1..7, automorphisms Π|S_k|! = 125411328000
      S±_1 = {0}  weight 1/1
      S±_2 = {0,1}  weight 1/2
      S±_3 = {0,1,2}  weight 1/3
      S±_4 = {0,1,2,3}  weight 1/4
  ⇒ ok
```

BuildBridges Θ-, NF-, Θ±- and Θell-bridges and their symmetry groups

✓ ok

WHERE **[IUT I]** §4, Definition 4.6 p. 111, §5, Definition 5.5 p. 151, §6, Definition 6.4 p. 162

INPUTS ThetaPmEllNFHodgeTheater OUTPUTS bridge

```
    ThetaBridge     FJ → FGt  (6 members)  isomorphisms: trivial ([IUTchI] Prop 4.8 (ii), Cor 5.6 (ii)) (1)  ok
    NFBridge        FJ → FCircAst  (6 members)  isomorphisms: F_l^* (6)  ok
    ThetaPmBridge   FT → FBoxPlus  (13 members)  isomorphisms: {±1} × {±1}^V (128)  ok
    ThetaEllBridge  DT → DCircDotPm  (13 members)  isomorphisms: F_l^{⋊±} (26)  ok
  ⇒ ok
```

ForgetToDHodgeTheater Θ±ell NF-HT ↦ D-Θ±ell NF-HT, and the Θ±ell/ΘNF/Θ projections

✓ ok

WHERE **[IUT I]** §6, Definition 6.13 p. 182, §6, Remark 6.12.2 p. 174

INPUTS ThetaPmEllNFHodgeTheater OUTPUTS DThetaPmEllNFHodgeTheater

```
    HT(universe 1) ↦ D-HT: capsules |T⊨13 |J⊨6, Π copies of F_⊠ 170002.., F_> 170002.., D_⊛± 171002.., F_⊛ 173002..; self-iso: yes
    components: 0±ell-HT (13 strips in T), 0NF-HT (6 in J), 0-HT (F⊩_mod, F⊩_tht with 6 theta values)
  ⇒ ok
```

ChainFrobeniusPicture the Frobenius picture: a chain of Θ-linked Hodge theaters

✓ ok

WHERE **[IUT I]** §3, Corollary 3.8 p. 89

INPUTS LogThetaLattice OUTPUTS chain of universes

```
    chain of 4 0-Hodge theaters (universes 5 1 3 6), 0-link; Z-translation: yes; swap of adjacent: never
    translated by 1: 6 5 1 3
  ⇒ ok
```

DrawEtalePicture the étale picture: D-HTs as spokes around the core D⊩ ✓ ok

WHERE **[IUT I]** §4, Corollary 4.12 p. 121, §6, Corollary 6.10 p. 170

INPUTS LogThetaLattice OUTPUTS étale picture

```
    hub D⊩ (6 copies 180001..), 2 spokes (D-HTs 1 3); reversed: valid permutation
  ⇒ ok
```

BuildGlobalNF global number-field data from Π_{C_K} and its Kummer isomorphism ✓ ok

WHERE **[IUT I]** §5, Example 5.1 p. 123, §5, Corollary 5.3 p. 143

INPUTS InitialThetaData OUTPUTS F^⊛, F^⊛_mod, F^⊛R

```
    Π_CK copy 5001; F_mod^× copy 5002; κ-coric 5003; sol 5004; F^⊛_mod ⊆ F^⊛: yes; realified: yes
    [ ok ] GlobalKummerIso              Kummer isomorphism F^⊛(5002) ≅ F^⊛(D_⊛(6001)) is unique (Q>0 ∩ Ẑ× = {1})   [IUTchI] Cor 5.3; [IUTchII] Cor 4.8
  ⇒ ok
```

GlueHodgeTheaters gluing Θ±ell-HT to ΘNF-HT along 0,⊞ ↦ > ✓ ok

WHERE **[IUT I]** §6, Proposition 6.7 p. 167, §6, Definition 6.13 p. 182

INPUTS ThetaPmEllHodgeTheater, ThetaNFHodgeTheater OUTPUTS ThetaPmEllNFHodgeTheater

```
    0,⊠ ↦ >: F_> now carries the Π copies of F_⊠ (170002 = 170002); label map t ↦ |t|: 6,5,4,3,2,1,0,1,2,3,4,5,6
  ⇒ ok
```

BuildPrimeStrip construct and validate a prime-strip of any of the 9 kinds ✓ ok

WHERE **[IUT I]** §4, Definition 4.1 p. 95, §5, Definition 5.2 p. 134

INPUTS InitialThetaData OUTPUTS prime-strip

```
    built F⊩▶×μ-prime-strip over 6 valuations, pilot degree -14.1180
    [ ok ] FMonoGlobValUnitsModTorsPrimeStrip F⊩▶×μ-prime-strip 'demo' well-formed (Frobenius-like, mono-analytic)   [IUTchI] Fig I1.2
  ⇒ ok
```

GenerateInitialData initial Θ-data from an abc triple / example and its validation ✓ ok

WHERE **[IUT I]** §3, Definition 3.1 p. 61

INPUTS abc triple OUTPUTS InitialThetaData, report

```
    abc(1,2400,2401) l=13: l=13, |V⊨6, log q = 24.2023, log f = 5.3471
    [ ok ] LPrime                       l = 13 prime, l* = 6, l± = 7   [IUTchI] Def 3.1 (c)
    [ ok ] Fmod                         F_mod ⊆ F ⊆ K with [F_mod:Q]=1, [F:Q]≤276480, [K:Q]≤7245987840   [IUTchI] Def 3.1
    [ ok ] VBadMod                      |V^bad_mod| = 3 odd primes of bad multiplicative reduction (2 is bad but excluded from V^bad_mod)   [IUTchI] Def 3.1 (b)
    [info] FBarOverF                    asserted (not machine-checkable here): F Galois over F_mod of degree prime to l   [IUTchI] Def 3.1 (b)
    [info] FBarOverF                    asserted (not machine-checkable here): √-1 ∈ F, and the 2·3-torsion points of E_F are rational over F   [IUTchI] Def 3.1 (a),
(b)
    [info] KField                       asserted (not machine-checkable here): Im(Gal(K/F) → GL2(F_l)) ⊇ SL2(F_l)   [IUTchI] Def 3.1 (c)
    [info] XF                           asserted (not machine-checkable here): E_F has stable reduction at all nonarchimedean valuations of F   [IUTchI] Def 3.1 (b)
    [info] CK                           asserted (not machine-checkable here): C_K is a K-core (no Shimura-curve coverings)   [IUTchI] Def 3.1 (d); [CanLift] Rem 2.1.1
    [ ok ] EpsilonUnder                 ε_ = the cusp ±1 of C_K (nonzero; the canonical generator at every v ∈ V^bad)   [IUTchI] Def 3.1 (f)
  ⇒ ok
```

ClassifyPseudoMonoids pseudo-monoids M^⊛, M^⊛_mod, M^⊛_sol, M^⊛_κ, M^⊛_∞κ, M^⊛_∞κ×, M^⊛_κ-sol, M_v…: divisible / cyclotomic / Kummer-ready ✓ ok

WHERE **[IUT I]** §0 p. 33, §5, Example 5.1 (i) p. 123, §5, Definition 5.2 (v)-(vi) p. 134, §5, Remark 5.1.3 p. 131

INPUTS DCircAstGlobal, DV OUTPUTS pseudo-monoid table

```
    MCircAst              M^⊛(†D^⊛) divisible:yes cyclotomic:yes group:yes Kummer-ready   [IUTchI] Ex 5.1 (i); Rem 5.1.3
    MCircAstMod           M^⊛_mod    divisible:no  cyclotomic:no  group:yes Kummer-blind   [IUTchI] Ex 5.1 (i); Rem 5.1.3
    MCircAstSol           M^⊛_sol    divisible:no  cyclotomic:yes group:yes Kummer-ready   [IUTchI] Ex 5.1 (i); Rem 5.1.3
    MCircAstKappa         M^⊛_κ      divisible:no  cyclotomic:no  group:no  Kummer-blind   [IUTchI] Ex 5.1 (i); Rem 3.1.7; Rem 5.1.3
    MCircAstInfKappa      M^⊛_∞κ  divisible:yes cyclotomic:yes group:no  Kummer-ready   [IUTchI] Ex 5.1 (i); Rem 3.1.7; Rem 5.1.3
    MCircAstInfKappaUnits M^⊛_∞κ× divisible:yes cyclotomic:yes group:no  Kummer-ready   [IUTchI] Ex 5.1 (i); Rem 3.1.7; Rem 5.1.3
    MCircAstKappaSol      M^⊛_κ-sol divisible:no  cyclotomic:no  group:no  Kummer-ready   [IUTchI] Ex 5.1 (i); Rem 3.1.7 (iii); Rem 5.1.3
    MV                    M_v(‡D_v)  divisible:yes cyclotomic:yes group:no  Kummer-ready   [IUTchI] Def 5.2 (v)
    MKappaV               M_κv        divisible:no  cyclotomic:no  group:no  Kummer-blind   [IUTchI] Def 5.2 (vi)
    MInfKappaV            M_∞κv     divisible:yes cyclotomic:yes group:no  Kummer-ready   [IUTchI] Def 5.2 (vi)
    MInfKappaUnitsV       M_∞κ×v   divisible:yes cyclotomic:yes group:no  Kummer-ready   [IUTchI] Def 5.2 (vi)
    M^⊛_mod ⊆ M^⊛: yes;  M^⊛_κ ⊆ M^⊛_∞κ×: yes;  M^⊛_∞κ ⊄ M^⊛_κ: yes
  ⇒ ok
```

ComposePolyMorphisms poly-morphisms, full and +-full poly-isomorphisms, their composites; anabelioids; NF / MLF / CAF ✓ ok

WHERE **[IUT I]** §0 p. 33, §6, Definition 6.1 (iv) p. 155

INPUTS morphisms, α ∈ {±1}^V OUTPUTS poly-morphism

```
    PolyMorphism    , PolyIsomorphism    , FullPolyIsomorphism    , PlusFullPolyIsomorphism    , CapsuleFullPolyMorphism    , CapsuleFullPolyIsomorphism
    (3 isomorphisms) composed with the full poly-isomorphism = FullPolyIsomorphism (count 0 = all)   [IUTchI] §0
    +-full poly-isomorphism with α = (+,-,+,…): positive no, negative no; |{±1}^V| = 64   [IUTchI] Def 6.1 (iv)
    anabelioids: B(Π), B(X), B(X)^0, BTemp = B^temp(Π) of Π^tp_{X__v} (connected yes, tempered yes); fields NF / MLF / CAF: K_v is MLF at v0
  ⇒ ok
```

ClassifyOrbicurves the orbicurves X, C, X̲, C̲, X→, C→, X̲_v, C̲_v: types, degrees, cusps, covering groups ✓ ok

WHERE **[IUT I]** §1 p. 37, §1, Definition 1.1 p. 38, §3, Definition 3.1 (b) p. 61

INPUTS InitialThetaData OUTPUTS orbicurve table

```
    XF            X             type (1,1)           over k    covers —  deg 1   group —                      cusps 1   Aut_k 0   [IUTchI] §1; Def 3.1 (a)
    COrbicurve    C = X/{±1}    type (1,1)±        over k    covers —  deg 1   group —                    cusps 1   Aut_k 0   [IUTchI] §1
```

```
    XUnderK        X_            type (1,l-tors)       over K    covers X    deg 13  group Z/l                          cusps 13  Aut_k 0   [IUTchI] §1; Def 3.1 (b);
Def 6.1 (iii): cusps ↔ F_l
    CK             C_ = X_/{±1} type (1,l-tors)±     over K    covers C    deg 13  group Z/l                        cusps 7   Aut_k 0   [IUTchI] §1; Def 3.1 (b); Ex
4.4 (i): cusps ↔ |F_l|
    XArrow         X→           type (1,l-tors→)     over k    covers X_  deg 13  group Z/l (Gal(X→/C_) ≅ Z/2l) cusps -1  Aut_k 26  [IUTchI] Def 1.1; Rem 1.2.1
    CArrow         C→           type (1,l-tors→)±  over k    covers C_  deg 13  group Z/l                          cusps -1  Aut_k 13  [IUTchI] Def 1.1; Rem 1.2.1
    XUnderUnderK   X__v         type (1,(Z/l)^0)     over K_v  covers X_v deg 13  group Z/l                          cusps -1  Aut_k 0   [IUTchI] Def 3.1 (b); [EtTh]
Def 2.5
    CUnderUnderV   C__v         type (1,(Z/l)^0)±   over K_v  covers C_v deg 13  group Z/l                        cusps -1  Aut_k 0   [IUTchI] Def 3.1 (b)
    CUnderV        C_v           type (1,Z/l)±       over K_v  covers C_v  deg 13  group Z/l                        cusps 7   Aut_k 0   [IUTchI] Def 3.1 (b); Ex 4.4
(i)
  ⇒ ok
```

ReconstructCoverings the cusps ε, ε′, ε″, 2ε and the group-theoretic reconstruction of Π_X, Π_C→, Π_C from Π_X→ ✓ ok

WHERE **[IUT I]** §1 p. 37, §1, Corollary 1.2 p. 39, §1, Remark 1.2.1 p. 40

INPUTS PiXArrow OUTPUTS Π_X, Π_C→, Π_C, I

```
    cusps: ε = 1 (of C_), ε′ = 1, ε″ = -1 (of X_, swapped by ι), 2ε = 2 (2 ≠ ±1 mod l: yes), 0 fixed by ι; |Δ_E ⊗ Z/l| = 169, rank(I_ε′×I_ε″) = 2
    l² = 169 = [Δ_X:Δ_X_]·[Δ_X_:Δ_X→] = 13·13 = [Δ_C:Δ_X→]/2 = 338/2;  Gal(X_/C_) = Z/2, Gal(C→/C_) = Z/13, Gal(X→/C_) = Z/26 ⊇ Gal(X→/C→) = Z/2;  Aut_k(X→) = Z/26,
Aut_k(C→) = Z/13
    step 1: Π_X→, Π_X_, Π_C_ are slim, hence embed in Aut(Π_X→); recover Δ_X→ ⊆ Π_X→ ([AbsTopI] Thm 2.6 (v),(vi))
    step 2: reconstruct Π_C ⊇ Π_X→ and Δ_X ⊆ Δ_C ⊆ Π_C ([AbsTopII] Cor 3.3 (i),(ii))
    step 3: recover l from l² = [Δ_X : Δ_X_]·[Δ_X_ : Δ_X→] = [Δ_X : Δ_X→] = [Δ_C : Δ_X→]/2
    step 4: H = Ker(Δ_X ↠ Δ_X^ab ⊗ Z/l);  Π_X_ = Π_X→ · H
    step 5: decomposition groups of 0, ε′, ε″ in Π_X_ = those of cusps with nontrivial image in Gal(X→/X_) = Π_X_/Π_X→
    step 6: Π_C_ ⊆ Π_C: the unique splitting of Gal(X/C) = Π_C/Π_X ↠ Π_C/Π_X_ = Gal(X_/C_) stabilizing the classes of 0, ε′, ε″ (l prime to 3!)
    step 7: Π_C→ ⊆ Π_C_: Gal(X→/C→) (≅ Z/2) ⊆ Gal(X→/C_) (≅ Z/2l) is the unique maximal subgroup of odd index
    step 8: decomposition groups of ε′, ε″: cusps with nontrivial image in Gal(X→/X_) not fixed up to conjugacy by the outer action of Gal(X_/C_)
  ⇒ ok
```

CompareTempered tempered versus profinite fundamental groups at v ∈ V^bad; Cor 2.5 synchronization ✓ ok

WHERE **[IUT I]** §2 p. 16, §2, Proposition 2.1 -Cor 2.8 p. 45, §2, Remark 2.5.3 p. 52

INPUTS InitialThetaData OUTPUTS facts per v

```
    TemperedConjugates         [IUTchI] Prop 2.1   a profinite conjugate of a nontrivial compact subgroup of Π^tp_G lying in Π^tp_G is a Π^tp_G-conjugate (Σ-tempered
groups of graphs)
    CommensurablyTerminal      [IUTchI] Prop 2.2   Π^tp_X ⊆ Π_X is commensurably terminal
    TemperedSubcover           [IUTchI] Cor 2.3   for H ⊆ Π_X open, Δ^tp_{X,H} := Δ^tp_X ∩ H is the tempered group of the covering determined by H
    TemperedVerticialEdge      [IUTchI] Prop 2.4   verticial / edge-like subgroups of Π^tp are characterized by their profinite conjugacy behaviour
    DecompositionSynchronized  [IUTchI] Cor 2.5   a decomposition (resp. inertia) group of Π_X contained in Π^tp_X is a tempered decomposition (resp. inertia) group
    DiscreteAnalogue           [IUTchI] Thm 2.6, Lem 2.7, Cor 2.8   the discrete analogues: Thm 2.6, Lemma 2.7, Cor 2.8 for the topological fundamental group π_1 and
its tempered completions
    GaloisCountable            [IUTchI] Rem 2.5.3   Galois-countable groups (T1)–(T6) and the effects (E1)–(E9) on the tempered theory
    v2     Π_v tempered: no  commensurably terminal: no  decomposition groups synchronized (Cor 2.5): no  Galois-countable: yes
    v3     Π_v tempered: yes commensurably terminal: yes decomposition groups synchronized (Cor 2.5): yes Galois-countable: yes
    v5     Π_v tempered: yes commensurably terminal: yes decomposition groups synchronized (Cor 2.5): yes Galois-countable: yes
    v7     Π_v tempered: yes commensurably terminal: yes decomposition groups synchronized (Cor 2.5): yes Galois-countable: yes
    v13    Π_v tempered: no  commensurably terminal: no  decomposition groups synchronized (Cor 2.5): no  Galois-countable: yes
    vinf   Π_v tempered: no  commensurably terminal: no  decomposition groups synchronized (Cor 2.5): no  Galois-countable: yes
  ⇒ ok
```

CheckDefThreeOne the six conditions on initial Θ-data; the Θ-approach M_X ≅ I·Δ_Θ; the determining data; S_mod ✓ ok

WHERE **[IUT I]** §3, Definition 3.1 (a)-(f) p. 61, §3, Remark 3.1.2 p. 64, §3, Remark 3.1.3 p. 65, §3, Remark 3.1.5 p. 65

INPUTS InitialThetaData OUTPUTS report

```
    (a) ok   (asserted flag)  F is a number field with √-1 ∈ F; F̄ an algebraic closure of F
    (b) ok   (partly checked) X_F once-punctured E_F with stable reduction; V^bad_mod ≠ ∅, odd, bad multiplicative; F/F_mod Galois of degree prime to l; E[2·3] ⊆ E(F)
    (c) ok   (partly checked) l ≥ 5 prime with Im(G_F → GL_2(F_l)) ⊇ SL_2(F_l); K = F(E[l]); l prime to the residue characteristics of V^bad_mod and to the orders
ord_v(q_v) on V^bad
    (d) ok   (asserted flag)  C__K a hyperbolic orbicurve of type (1, l-tors)± with K-core C_K = C_F ×_F K; it determines X__K of type (1, l-tors)
    (e) ok   (checked)        V ⊆ V(K) is a section of V(K) ↠ V_mod (one v above each v_mod); at v ∈ V^bad, C__v of type (1, Z/lZ)±
    (f) ok   (checked)        ε_ a cusp of C__K from a NONZERO element of Q ≅ Z/lZ; at v ∈ V^bad the cusp of the canonical generator ±1
    Rem 3.1.2 0-approach: Π_{X__K} ⊆ Π_{C__K} (the 0-approach: use the once-punctured curve, not C__K directly); M_X (the cyclotome of the once-punctured elliptic
curve, [AbsTopIII] Thm 1.9) ≅ Δ_0 (the cyclotome of the theta function, [EtTh]) with index 13
    Rem 3.1.3: the data is determined by F̄/F X_F C__K V V^bad_mod
    Rem 3.1.5: S_mod = Spec(O_K) // Gal(K/F_mod), [K:F_mod] ≤ 7245987840, 6 points
  ⇒ ok
```

CheckKappaCoric κ-coric / ∞κ-coric / ∞κ×-coric functions, critical points, Gal(K/F(μ_l)) ≅ SL_2(F_l) ✓ ok

WHERE **[IUT I]** §3, Remark 3.1.7 p. 66

INPUTS rational function data OUTPUTS verdict

```
    |C_L| ≅ A¹ has 4 critical points, 4 strictly critical   [IUTchI] Rem 3.1.7 (i)
    (1 pole, 2 zeroes, divisor over NF avoiding critical points, roots of unity at strictly critical points): κ-coric yes, ∞κ-coric yes, ∞κ×-coric yes
    two poles: exactly one pole required;  non-root-of-unity value at a strictly critical point: values at strictly critical points must be roots of unity
    Rem 3.1.7 (iii): Gal(K/F(μ_l)) is the unique subgroup of Gal(K/F) isomorphic to SL_2(F_l), |SL_2(F_l)| = 2184; linearly disjoint: yes; κ-sol-open subgroups center-
free: yes
  ⇒ ok
```

BuildLocalFrobenioids F_v, C_v, C⊢_v, τ⊢_v, C^Θ_v at bad / good / archimedean v; q_v, Θ_v, log(p_v), log(Θ); F⊢_v ≅ F^Θ_v ✓ ok

WHERE **[IUT I]** §3, Example 3.2 p. 69, §3, Example 3.3 p. 77, §3, Example 3.4 p. 80

INPUTS InitialThetaData OUTPUTS FUnderV, FMonoV, FThetaV

```
    v2     [IUTchI] Ex 3.3
      D_v: D_v = B(X→_v)^0;  D⊢_v: D⊢_v = B(K_v)^0
      F_v: F_v = C_v: the p_v-adic Frobenioid, Φ_{C_v}: Spec L ↦ ord(O_L)^pf
      C⊢_v: C⊢_v ⊆ C_v;  Φ = Φ_{C⊢v} = ord(Z_{p_v}) = N · log(p_v) (generator p_v, value 26);  0-version N · log(p_v) · log(0)
      splitting τ⊢_v: τ⊢_v: the splitting determined by p_v (indeterminacy 1);  tempered no, split yes, Kummer structure no
      reconstruct D⊢_v = B(K_v)^0 from D_v = B(X→_v)^0 (Π_v ↠ G_v)
```

```
      reconstruct C⊢_v ⊆ C_v and the splitting τ⊢_v from p_v
      reconstruct the formal symbol log(p_v)·log(θ)
    v3     [IUTchI] Ex 3.2
      D_v: D_v = B^temp(X___v)^0;  D⊢_v: D⊢_v = B(K_v)^0 ⊆ D_v
      F__v: F__v: the tempered Frobenioid over D_v
      C⊢_v: C⊢_v (with the μ_2l-orbit of characteristic splittings τ⊢_v);  Φ = Φ_{C⊢v} = N · log_Φ(q__v) (generator q__v = q_v^{1/2l} (up to μ_2l), value 1);  θ-version
N · log_Φ(q__v) · log(θ)
      splitting τ⊢_v: τ⊢_v: a μ_2l-orbit of splittings O▷ = O× × Φ (indeterminacy 26);  tempered yes, split yes, Kummer structure no
      θ__v: θ__v ∈ O×(T÷_{Ÿ_v}) up to μ_2l and l·Z (translation of the labels)
      reconstruct (a) the subgroups Π^tp_{X___v} ↠ G_v, the cuspidal decomposition groups and Ÿ_v from the tempered Frobenioid F__v
      reconstruct (b) the birationalization F÷_v and the base-field-theoretic hull C_v
      reconstruct (c) the theta function θ__v ∈ O×(T÷_{Ÿ_v}) up to μ_2l and l·Z  ([EtTh] Thm 1.10)
      reconstruct (d) q_v and q__v = q_v^{1/2l} up to μ_2l; the divisor monoid Φ_{C⊢v} = N·log_Φ(q__v)
      reconstruct (e) the μ_2l-orbit of characteristic splittings τ⊢_v, hence the split Frobenioid F⊢_v = (C⊢_v, τ⊢_v)
      reconstruct (f) the θ-version D^θ_v, C^θ_v, τ^θ_v and the isomorphism F⊢_v ≅ F^θ_v
    v5     [IUTchI] Ex 3.2
      D_v: D_v = B^temp(X___v)^0;  D⊢_v: D⊢_v = B(K_v)^0 ⊆ D_v
      F__v: F__v: the tempered Frobenioid over D_v
      C⊢_v: C⊢_v (with the μ_2l-orbit of characteristic splittings τ⊢_v);  Φ = Φ_{C⊢v} = N · log_Φ(q__v) (generator q__v = q_v^{1/2l} (up to μ_2l), value 1);  θ-version
N · log_Φ(q__v) · log(θ)
      splitting τ⊢_v: τ⊢_v: a μ_2l-orbit of splittings O▷ = O× × Φ (indeterminacy 26);  tempered yes, split yes, Kummer structure no
      θ__v: θ__v ∈ O×(T÷_{Ÿ_v}) up to μ_2l and l·Z (translation of the labels)
      reconstruct (a) the subgroups Π^tp_{X___v} ↠ G_v, the cuspidal decomposition groups and Ÿ_v from the tempered Frobenioid F__v
      reconstruct (b) the birationalization F÷_v and the base-field-theoretic hull C_v
      reconstruct (c) the theta function θ__v ∈ O×(T÷_{Ÿ_v}) up to μ_2l and l·Z  ([EtTh] Thm 1.10)
      reconstruct (d) q_v and q__v = q_v^{1/2l} up to μ_2l; the divisor monoid Φ_{C⊢v} = N·log_Φ(q__v)
      reconstruct (e) the μ_2l-orbit of characteristic splittings τ⊢_v, hence the split Frobenioid F⊢_v = (C⊢_v, τ⊢_v)
      reconstruct (f) the θ-version D^θ_v, C^θ_v, τ^θ_v and the isomorphism F⊢_v ≅ F^θ_v
    v7     [IUTchI] Ex 3.2
      D_v: D_v = B^temp(X___v)^0;  D⊢_v: D⊢_v = B(K_v)^0 ⊆ D_v
      F__v: F__v: the tempered Frobenioid over D_v
      C⊢_v: C⊢_v (with the μ_2l-orbit of characteristic splittings τ⊢_v);  Φ = Φ_{C⊢v} = N · log_Φ(q__v) (generator q__v = q_v^{1/2l} (up to μ_2l), value 1);  θ-version
N · log_Φ(q__v) · log(θ)
      splitting τ⊢_v: τ⊢_v: a μ_2l-orbit of splittings O▷ = O× × Φ (indeterminacy 26);  tempered yes, split yes, Kummer structure no
      θ__v: θ__v ∈ O×(T÷_{Ÿ_v}) up to μ_2l and l·Z (translation of the labels)
      reconstruct (a) the subgroups Π^tp_{X___v} ↠ G_v, the cuspidal decomposition groups and Ÿ_v from the tempered Frobenioid F__v
      reconstruct (b) the birationalization F÷_v and the base-field-theoretic hull C_v
      reconstruct (c) the theta function θ__v ∈ O×(T÷_{Ÿ_v}) up to μ_2l and l·Z  ([EtTh] Thm 1.10)
      reconstruct (d) q_v and q__v = q_v^{1/2l} up to μ_2l; the divisor monoid Φ_{C⊢v} = N·log_Φ(q__v)
      reconstruct (e) the μ_2l-orbit of characteristic splittings τ⊢_v, hence the split Frobenioid F⊢_v = (C⊢_v, τ⊢_v)
      reconstruct (f) the θ-version D^θ_v, C^θ_v, τ^θ_v and the isomorphism F⊢_v ≅ F^θ_v
    v13    [IUTchI] Ex 3.3
      D_v: D_v = B(X→_v)^0;  D⊢_v: D⊢_v = B(K_v)^0
      F__v: F__v = C_v: the p_v-adic Frobenioid, Φ_{C_v}: Spec L ↦ ord(O_L)^pf
      C⊢_v: C⊢_v ⊆ C_v;  Φ = Φ_{C⊢v} = ord(Z_{p_v}) = N · log(p_v) (generator p_v, value 12);  θ-version N · log(p_v) · log(θ)
      splitting τ⊢_v: τ⊢_v: the splitting determined by p_v (indeterminacy 1);  tempered no, split yes, Kummer structure no
      reconstruct D⊢_v = B(K_v)^0 from D_v = B(X→_v)^0 (Π_v ↠ G_v)
      reconstruct C⊢_v ⊆ C_v and the splitting τ⊢_v from p_v
      reconstruct the formal symbol log(p_v)·log(θ)
    vinf   [IUTchI] Ex 3.4
      D_v: X→_v: the Aut-holomorphic orbispace of X→ ×_K K_v (a CAF A_□ = C);  D⊢_v: D⊢_v: a split topological monoid in TM (C ≅ O▷_C, C× ≅ S¹, C→ ≅ R≥0)
      F__v: F__v = (C_v, D_v, κ_v): archimedean Frobenioid C_v with O▷(C_v) ≅ O▷_{K̃_v}, Kummer structure κ_v : O▷(C_v) ↷ A_{D_v}
      C⊢_v: C⊢_v (F⊢_v = (C⊢_v, D⊢_v, τ⊢_v));  Φ = Φ_{C⊢v} = R≥0 · log(p_v) (generator p_v (= e), value 1);  θ-version R≥0 · log(p_v) · log(θ)
      splitting τ⊢_v: τ⊢_v: C× = S¹ × R>0 (the canonical splitting of a CAF) (indeterminacy 1);  tempered no, split yes, Kummer structure yes
      reconstruct the Kummer structure κ_v and O▷(C_v) ≅ O▷_{K̃_v}
      reconstruct D⊢_v ∈ TM: C× ≅ S¹, C→ ≅ R≥0
      reconstruct Φ_{C⊢v} = R≥0·log(p_v) and its θ-version
    F⊢_v ≅ F^θ_v at every v: yes   [IUTchI] Ex 3.2 (v)
  ⇒ ok
```

BuildGlobalRealified C⊩_mod, C⊩_tht, D⊩_mod; ρ_v, ρ^Θ_v, ρ^D_v; F⊩_mod ≅ F⊩_tht ✓ ok

WHERE **[IUT I]** §3, Example 3.5 p. 84

INPUTS InitialThetaData OUTPUTS CMonoGlobMod, FMonoGlobMod

```
    CMonoGlobMod = C⊩_mod, CMonoGlobTht = C⊩_tht, DMonoGlobMod = D⊩_mod;  Prime ≅ V_mod: 6 primes
    v2     [K_v:(F_mod)_v] = 26  ρ_v = 0.0385  ρ^θ_v(j=1) = 0.0267  ρ^θ_v(j=2) = 0.1066  Frobenius element 26.0000
    v3     [K_v:(F_mod)_v] = 26  ρ_v = 0.0385  ρ^θ_v(j=1) = 0.0423  ρ^θ_v(j=2) = 0.1690  Frobenius element 26.0000
    v5     [K_v:(F_mod)_v] = 52  ρ_v = 0.0192  ρ^θ_v(j=1) = 0.0310  ρ^θ_v(j=2) = 0.1238  Frobenius element 52.0000
    v7     [K_v:(F_mod)_v] = 2340  ρ_v = 0.0004  ρ^θ_v(j=1) = 0.0008  ρ^θ_v(j=2) = 0.0033  Frobenius element 195.0000
    v13    [K_v:(F_mod)_v] = 12  ρ_v = 0.0833  ρ^θ_v(j=1) = 0.2137  ρ^θ_v(j=2) = 0.8550  Frobenius element 12.0000
    vinf   [K_v:(F_mod)_v] = 1  ρ_v = 1.0000  ρ^θ_v(j=1) = 1.0000  ρ^θ_v(j=2) = 4.0000  Frobenius element 0.1592
    F⊩_mod = (C⊩_mod, Prime ≅ V, {F⊢_v}, {ρ_v}) over 6 valuations;  F⊩_tht ≅ F⊩_mod: yes   [IUTchI] Ex 3.5 (iii)
  ⇒ ok
```

AssembleThetaHodgeTheater †HT^Θ = ({F̲_v}, F⊩_mod) with (a)(b)(c); the Θ-link as full poly-iso F⊩_tht ≅ F⊩_mod preserving D⊢_v and O× ✓ ok

WHERE **[IUT I]** §3, Definition 3.6 p. 87, §3, Corollary 3.7 -3.9 p. 88, §3, Remark 3.9.3 p. 93

INPUTS InitialThetaData, lattice OUTPUTS ThetaHodgeTheater, report

```
    †HT^θ = ({†F__v}_{v∈V}, †F⊩_mod): 6 local Frobenioids (bad 3, good 2, arch 1); (a) yes (b) yes (c) yes   [IUTchI] Def 3.6
    θ-link (0,0) → (1,0) checked against Cor 3.7:
    [ ok ] ThetaLinkFullPolyIso         θ-link is the full poly-isomorphism F⊩_tht ≅ F⊩_mod   [IUTchI] Cor 3.7 (i)
    [ ok ] ThetaLinkPreservesDMono      D⊢_v = {G_v} preserved, Π_v not   [IUTchI] Cor 3.7 (ii)
    [ ok ] ThetaLinkPreservesUnits      O×_{C⊢v} preserved; value groups dilated by 15.167   [IUTchI] Cor 3.7 (iii); Rem 3.9.3
    [ ok ] ThetaLinkNotRing             the ring structures are not related by the θ-link   [IUTchI] Rem 3.9.3
    Cor 3.8: Frobenius-picture … → †HT^θ → ‡HT^θ → … with ⁿθ__v ↦ ⁿ⁺¹q__v;  Cor 3.9: étale-picture around D⊢_v ↷ O×
  ⇒ ok
```

TorsorLabelCusps LabCusp(†D_v) as F_l^*-torsor with †η_v; LabCusp±(†D_v) with †η°_v, †η^±_v; forgetting the sign ✓ ok

WHERE **[IUT I]** §4, Definition 4.1 (ii) p. 95, §4, Proposition 4.2 p. 98, §6, Definition 6.1 (ii)-(iii) p. 155

INPUTS DPrimeStrip OUTPUTS LabCusp, LabCuspPm

```
    LabCusp(†D_v): F_l^*-torsor of size 6, canonical †n_v = 1 at every v, compatible across V: yes   [IUTchI] Def 4.1 (ii), Prop 4.2
    F_l^* acts: 2 · †n_v = 2, 3 · 2 = 6
    LabCusp±(†D_v): size 13, †n°_v = 0, †n^±_v = ±1 (up to sign), [Aut : Aut_+] = 2   [IUTchI] Def 6.1 (ii)-(iii)
```

```
    (LabCusp± \ {n⁰})/{±1} ≅ LabCusp: -6↦6 -5↦5 -4↦4 -3↦3 -2↦2 -1↦1 0↦-1 1↦1 2↦2 3↦3 4↦4 5↦5 6↦6
  ⇒ ok
```

DescribeGlobalBases D^⊛ = B(C_K)^0, D^⊙± = B(X_K)^0; Aut ⊇ Aut_ε (Borel ⊇ semi-unipotent), Aut_± ⊇ Aut_csp; V ⊆ V^±un ⊆ V^Bor ⊆ V(K) ✓ ok

WHERE **[IUT I]** §4, Definition 4.1 (v) p. 95, §4, Example 4.3 p. 98, §6, Definition 6.1 (v) p. 155, §6, Remark 6.1.1 p. 159

INPUTS InitialThetaData OUTPUTS global bases, orbits

```
    DCircAstGlobal: D^⊛ = B(C__K)^0, group Π_{C__K}, |LabCusp| = 6; Aut image Borel/{±1} of order 936 ⊇ Aut_ε image semi-unipotent/{±1} of order 156; quotient F_l^* =
Aut(C__K)/Aut_ε(C__K) of order 6   [IUTchI] Def 4.1 (v), Ex 4.3
    DCircDotPmGlobal: D^⊙± = B(X__K)^0, group Π_{X__K}, |LabCusp±| = 13, double cover of D^⊛: yes; quotient F_l^{×±} = Aut_±(D^⊙±)/Aut_csp(D^⊙±) of order 26   [IUTchI]
Def 6.1 (v)
    VSet   V             6 ≤ |·| ≤ 6   —
    VPmUn  V^±un = V^±  6 ≤ |·| ≤ 936   Aut_ε(C__K)·V = Aut_csp(D^⊙±)·V  ([IUTchI] Ex 4.3 (i); Def 6.1 (v); Rem 6.1.1)
    VBor   V^Bor         6 ≤ |·| ≤ 5616   Aut(C__K)·V  ([IUTchI] Ex 4.3 (i))
    VK     V(K)          6 ≤ |·| ≤ 43475927040   all valuations of K
  ⇒ ok
```

EvaluateAtPoints evaluation points labelled |F_l|, Θ_v ↦ q_v^{j²}, evaluation sections G_v → Π_v, the Θ-bridge constituents φ^Θ_{v_j} ✓ ok

WHERE **[IUT I]** §4, Example 4.4 p. 105

INPUTS InitialThetaData OUTPUTS EvaluationPoint, PhiThetaVJ

```
    at v3: 7 evaluation points labelled |F_l| = {0,1,2,3,4,5,6}; θ__v values (in units of ord q__v, up to μ_2l): 0 1 4 9 16 25 36
    μ_- of order 2; K_v-rational: yes; sections G_v → Π_v group-theoretic: yes   [IUTchI] Ex 4.4 (i)-(ii)
    φ^Θ_{v_j} at v2, j = -3: full poly-isomorphism (label 3)   [IUTchI] Ex 4.4 (iii)
    φ^Θ_{v_j} at v3, j = -3: evaluation section (label 3)   [IUTchI] Ex 4.4 (iii)
    φ^Θ_{v_j} at v5, j = -3: evaluation section (label 3)   [IUTchI] Ex 4.4 (iii)
    φ^Θ_{v_j} at v7, j = -3: evaluation section (label 3)   [IUTchI] Ex 4.4 (iii)
  ⇒ ok
```

TransportLabelClasses †χ : J ≅ F_l^*, †ζ, †ζ_±, the global ±-synchronization †ξ ✓ ok

WHERE **[IUT I]** §4, Proposition 4.7 p. 112, §6, Proposition 6.5 p. 163, §6, Remark 6.12.4 p. 177

INPUTS DThetaNFHodgeTheater, DThetaPmEllHodgeTheater OUTPUTS label bijections

```
    0NF: †x : J ≅ F_l^* = {1,2,3,4,5,6}, †ζ : LabCusp(†D^⊛) ≅ J, [†ε] ↦ 1, synchronized yes   [IUTchI] Prop 4.7
    0±ell: T ≅ F_l = {-6..6}, †ζ_± : LabCusp±(†D^⊙±) ≅ T, †ξ global ±-synchronization, ε_ ↦ ±1   [IUTchI] Prop 6.5; Rem 6.12.4
    bijections valid: yes
  ⇒ ok
```

CountIsomorphismTorsors the isomorphism torsors of the four bridges, the Hodge theaters and their gluings ✓ ok

WHERE **[IUT I]** §4, Proposition 4.8 p. 115, §6, Proposition 6.6 p. 165, §5, Corollary 5.6 p. 153, §6, Corollary 6.12 p. 173, §6, Remark 6.12.2 p. 174

INPUTS InitialThetaData OUTPUTS torsor table

```
    D-NF-bridge †φ^NF_⊛                  torsor over F_l^*                    order 6        [IUTchI] Prop 4.8 (i)
    D-Θ-bridge †φ^Θ_⊛                   torsor over trivial                  order 1        [IUTchI] Prop 4.8 (ii)
    D-ΘNF-Hodge theater                      torsor over trivial                   order 1        [IUTchI] Prop 4.8 (iii)
    gluing D-NF-bridge U D-Θ-bridge          torsor over F_l^*                     order 6        [IUTchI] Prop 4.8 (iv)
    D-Θ±-bridge †φ^Θ±_±                 torsor over {±1} × {±1}^V          order 128      [IUTchI] Prop 6.6 (i)
    D-Θell-bridge †φ^Θell_±               torsor over F_l^{×±}                order 26       [IUTchI] Prop 6.6 (ii)
    D-Θ±ell-Hodge theater                    torsor over {±1}                      order 2        [IUTchI] Prop 6.6 (iii)
    gluing D-Θ±-bridge U D-Θell-bridge      torsor over F_l^{×±} × {±1}^V     order 1664     [IUTchI] Prop 6.6 (iv)
    NF-/Θ-bridges, ΘNF-HT (F-version)          torsor over same as the D-versions     order 1        [IUTchI] Cor 5.6 (i)-(iii)
    Θ±-/Θell-bridges, Θ±ell-HT (F-version)  torsor over same as the D-versions     order 1        [IUTchI] Cor 6.12 (i)-(iii)
    gluing Θ±ell-HT U ΘNF-HT (Prop 6.7)     torsor over unique                   order 1        [IUTchI] Rem 6.12.2; Def 6.13
  ⇒ ok
```

CountForgottenSymmetries (l*)^{l*} → l*! and (l±)^{l±} → l±!: what the forgetful functors lose and what processions recover ✓ ok

WHERE **[IUT I]** §4, Proposition 4.9 p. 115, §4, Proposition 4.11 p. 120, §6, Proposition 6.8 p. 167, §6, Proposition 6.9 p. 169

INPUTS InitialThetaData OUTPUTS counts

```
    ΘNF (Prop 4.9, 4.11): F_l^* acts simply transitively on J (Prop 4.9 (i)); capsule of 6 strips, S_{l*} of order 720; label assignments (l*)^{l*} = 46656, with
processions l*! = 720 (factor 64.8)
    Θ±ell (Prop 6.8, 6.9): F_l^{×±} acts doubly transitively on T (Prop 6.8 (i)); capsule of 7 strips, S_{l±} of order 5040; (l±)^{l±} = 823543, with processions l±! =
5040 (factor 163.4)
    (ii) holomorphic capsules → (iii) mono-analytic capsules: the same counts after D ↦ D⊢
  ⇒ ok
```

DistributeGlobalData F^⊛, F^⊛′, F^⊛_mod, F^⊛_j, F^⊛_⟨J⟩, F^⊛_J; V_j, V_⟨J⟩, V_J; Kummer-ready vs Kummer-blind; averaged vs summed log-volumes ✓ ok

WHERE **[IUT I]** §5, Example 5.1 (ii)-(vii) p. 123, §5, Remark 5.1.1 p. 131, §5, Remark 5.1.3 p. 131, §5, Remark 5.4.2 p. 151

INPUTS DCircAstGlobal OUTPUTS global Frobenioids

```
    FCircAstGlobal     F^⊛(†D^⊛)        over †D^⊛                      Kummer-ready  copies 1   log-volume weight 1.0000   [IUTchI] Ex 5.1 (ii),(iii); Rem 5.1.3
    FCircAstPrime      F^⊛′               over †D^⊛_mod (the C_{F_mod} version) Kummer-ready  copies 1   log-volume weight 1.0000   [IUTchI] Ex 5.1 (iii); Rem 5.1.3
    FCircAstMod        F^⊛_mod              over terminal objects of †D^⊛_mod Kummer-blind  copies 1   log-volume weight 1.0000   [IUTchI] Ex 5.1 (iii); Rem 5.1.3
    FCircAstJ          F^⊛_j                over V_j = {v_j}                    Kummer-blind  copies 1   log-volume weight 1.0000   [IUTchI] Ex 5.1 (vii); Rem 5.1.3
    FCircAstAngleJ     F^⊛_⟨J⟩          over V_⟨J⟩ (diagonal)             Kummer-blind  copies 1   log-volume weight 0.1667   [IUTchI] Ex 5.1 (vii); Rem 5.1.1; Rem
5.4.2
    FCircAstJCapsule   F^⊛_J = ∏_j F^⊛_j over V_J = ∏_j V_j                 Kummer-blind  copies 6   log-volume weight 1.0000   [IUTchI] Ex 5.1 (vii); Rem 5.1.3
    V_⟨J⟩: 6 valuations (diagonal), V_J: 36 = |V|·l*;  log-volumes of (1,2,…,l*): F_⟨J⟩ average 3.5000, F_J sum 21.0000   [IUTchI] Ex 5.1 (vii), Rem 5.4.2
  ⇒ ok
```

LiftToFrobenioids F ↦ D, F ↦ F⊢, F ↦ F⊩; Isom(F) ≅ Isom(D); lifting a D-prime-strip to an F-prime-strip ✓ ok

WHERE **[IUT I]** §5, Definition 5.2 p. 134, §5, Remark 5.2.1 p. 143, §5, Corollary 5.3 p. 143, §5, Remark 5.3.1 p. 145

INPUTS DPrimeStrip OUTPUTS FPrimeStrip

```
    FPrimeStrip        → DPrimeStrip           F ↦ D (forget the Frobenioids)                bijective on Isom: yes surjective: yes [IUTchI] Rem 5.2.1 (i); Cor 5.3
(ii): Isom(F) → Isom(D) bijective
    FPrimeStrip        → FMonoPrimeStrip       F ↦ F⊢ (mono-analyticization)                 bijective on Isom: no  surjective: yes [IUTchI] Rem 5.2.1 (ii); Rem 5.2.2:
F Kummer-ready, F⊢ Kummer-blind
    FMonoPrimeStrip    → DMonoPrimeStrip       F⊢ ↦ D⊢                                     bijective on Isom: no  surjective: yes [IUTchI] Cor 5.3 (iii): Isom(F⊢) →
Isom(D⊢) surjective, not injective
    FPrimeStrip        → FMonoGlobPrimeStrip   F ↦ F⊩ (adjoin the global realified Frobenioid) bijective on Isom: no  surjective: yes [IUTchI] Rem 5.2.1 (iii); Def
5.2 (iv)
    DPrimeStrip        → DMonoPrimeStrip       D ↦ D⊢ (Π_v ↦ G_v)                          bijective on Isom: no  surjective: yes [IUTchI] Def 4.1 (iv)
    FCircAstGlobal     → DCircAstGlobal        F^⊛ ↦ Base(F^⊛) = D^⊛                       bijective on Isom: yes surjective: yes [IUTchI] Cor 5.3 (i): Isom(F^⊛) →
Isom(D^⊛) bijective
    FUnderV            → DV                    F_v ↦ D_v (v ∈ V^bad)                       bijective on Isom: yes surjective: yes [IUTchI] Cor 5.3 (iv): Aut(F_v) →
Aut(D_v) bijective
    lift of the D-prime-strip (Π copies 21002..) to an F-prime-strip: yes, Π copy preserved: yes   [IUTchI] Cor 5.3 (ii), Rem 5.3.1
  ⇒ ok
```

StructureFlSemiPm F_l^{⋈±}-groups and -torsors, positive elements, |T| and T*, sign vectors {±1}^V and −1_{F_l} ✓ ok

WHERE **[IUT I]** §6, Definition 6.1 (i) p. 155, §6, Definition 6.4 (i) p. 162, §6, Example 6.2 p. 159

INPUTS InitialThetaData OUTPUTS FlSemiPmGroup, TAbs, TStar

```
    F_l^{⋈±}-group T (θ±-bridge index set): 13 elements, {±1}-orbit of 2 bijections, Aut_+(T) ⊆ Aut_±(T) orders 1 ⊆ 2   [IUTchI] Def 6.1 (i)
    F_l^{⋈±}-torsor T (θell-bridge index set): F_l^{⋈±}-orbit of 26 bijections, Aut_+ ⊆ Aut_± orders 13 ⊆ 26
    positive elements: 1 2 3 4 5 6 ;  |T| has 7 elements, T* has 6   [IUTchI] Def 6.4 (i)
    α = (-,…,-) ∈ {±1}^V: all minus yes (the -1_{F_l} of Ex 6.2 (ii)); 64 sign vectors in all
  ⇒ ok
```

ReduceThetaPmToTheta D-Θ±-bridge ⇒ D-Θ-bridge (±t ↦ |t|, D_0, D_⊞ ↦ D_>); S*_j ↪ S^±_{j+1}; uniqueness of the gluing ✓ ok

WHERE **[IUT I]** §6, Proposition 6.7 p. 167, §6, Proposition 6.9 (iii) p. 169, §6, Remark 6.12.2 p. 174

INPUTS DThetaPmBridge OUTPUTS DThetaBridge

```
    D-Θ±-bridge ⇒ D-Θ-bridge: 13 strips ↦ 6, t ↦ |t|: -6↦6 -5↦5 -4↦4 -3↦3 -2↦2 -1↦1 1↦1 2↦2 3↦3 4↦4 5↦5 6↦6
    †D_0, †D_⊠ ↦ †D_>; +-full poly-isomorphisms replaced by full (good v) / evaluation-section (bad v): yes   [IUTchI] Prop 6.7
    [ ok ] ProcessionsCompatible        S*_j ↬ S^±_{j+1} for j = 1..6   [IUTchI] Prop 6.9 (iii)
    Rem 6.12.2: the gluing isomorphism of the Θ±ell-HT to the ΘNF-HT along the Θ±-/Θ-bridges is unique
  ⇒ ok
```

CompareHalfPlane the two symmetries versus the upper half-plane; the Frobenioids of a Θ±ell NF-Hodge theater ✓ ok

WHERE **[IUT I]** §6, Remark 6.12.3 p. 175, §6, Figure 6.4 p. 178, §6, Figure 6.6 p. 183

INPUTS — OUTPUTS tables

```
    additive symmetry F_l^{⋈±} (Θ±ell-HT, geometric) ↔ z ↦ ±z + a, the parabolic/translation symmetries basepoints V^± (basepoints via Aut_csp)            at ∞: the
cusp at infinity; q = e^{2πiz}
    multiplicative symmetry F_l^* (ΘNF-HT, arithmetic)  ↔ toral symmetries w = (z-i)/(z+i), rotations      basepoints V^Bor (basepoints via the Borel subgroup) at ∞:
the node; w ↦ e^{iθ}w
    coric {±1} shared by both                          ↔ z ↦ -z̄ … the common {±1}                 basepoints V^±un = V^±                          at ∞: —
    Fig 6.6, the Frobenioids of a Θ±ell NF-Hodge theater:
      FPrimeStrip      {Π_v ↷ O▷_{F̃_v}}_{v∈V} every F-prime-strip F_t, F_j, F_⊠, F_>            Kummer-ready
      FUnderV          F__v (tempered Frobenioid)  v ∈ V^bad, inside F_>                          Kummer-ready
      FCircAstMod      F^⊛_mod                    NF-bridge target, restricted to terminal objects     Kummer-blind
      FCircAstGlobal   F^⊛                        NF-bridge target over D^⊛                          Kummer-ready
      FCircAstPrime    F^⊛'                     over D^⊛_mod                                       Kummer-ready
      FMonoGlobMod     F⊩_mod (and F⊩_tht)      the Θ-Hodge theater inside the ΘNF-HT              Kummer-blind (mono-analytic)
  ⇒ ok
```

12.2 Processes of [IUT II] — Inter-universal Teichmüller Theory II: Hodge–Arakelov-theoretic Evaluation

ClassifyRadial radial environments, multiradiality, multiradialization ✓ ok

WHERE **[IUT II]** §1, Example 1.7 -1.9 p. 32

INPUTS radial data, coric data OUTPUTS multiradial / uniradial

```
    uniradial    R = holomorphic structure: Π_v ↷ O▷_{F̃_v} (an F-prime-strip) | C = D⊢ = {G_v}   [IUTchII] Ex 1.8 (i): uniradial — the ring structure is not
transportable
    multiradial  R = D⊢ itself | C = D⊢   [IUTchII] Ex 1.8 (ii): the coric data is tautologically multiradial
    multiradial  R = O×μ_v with G_v-action, up to Ism | C = D⊢   [IUTchII] Ex 1.8 (iv), Cor 1.12: units modulo torsion are multiradial (Ind2)
    uniradial    R = O×_v with its Ẑ×-Kummer structure | C = D⊢   [IUTchII] Ex 1.8 (iii): the full Kummer structure on O× is uniradial (needs the cyclotome)
    multiradial  R = the mono-theta environment and its cyclotomic rigidity (l·Δ_Θ)(Π) ≅ Π_μ | C = D⊢   [IUTchII] Cor 1.10, Ex 1.9: multiradial
    multiradial  R = theta monoids Ψ_env(D_>) with their splittings | C = (D⊢, F⊢×μ(D⊢))   [IUTchII] Thm 2.2, Cor 2.3 (étale-picture radial environment)
    multiradial  R = the multiradial representation R^LGP: packets, splitting monoids, (F_mod^×)_j | C = (D⊢, F⊢×μ)   [IUTchIII] Thm 3.11 (i): multiradial up to Ind1,
Ind2, Ind3
    multiradialization (Ex 1.7 (ii)): uni → multi (α = full poly-iso of D⊢ adjoined)
  ⇒ ok
```

RigidifyCyclotomes the three cyclotomic rigidity isomorphisms, constant multiple rigidity ✓ ok

WHERE **[IUT II]** §1, Corollary 1.10 -1.12 p. 46

INPUTS Π_v / G_v, M^Θ OUTPUTS rigidity + indeterminacy

```
    mono-theta: indeterminacy none, multiradial, no ring structure   [IUTchII] Cor 1.10; [IUTchIII] Rem 2.3.3 (i)
    MLF-Galois pair: indeterminacy Ẑ×-orbit (resp. Aut(G_v)-orbit) on the units, uniradial, uses ring structure   [IUTchII] Cor 1.11; [IUTchIII] Rem 2.3.3 (ii)
    number field: indeterminacy none (Q>0 ∩ Ẑ× = {1}), multiradial, no ring structure   [IUTchI] Ex 5.1 (v); [IUTchII] Cor 4.8; [IUTchIII] Rem 2.3.3 (vi)
    constant multiple rigidity (Cor 1.12): ι-orbit of size 2
  ⇒ ok
```

EvaluateTheta theta evaluation at the l-torsion points: Θ ↦ q̲^{t²}, |t|-dependence ✓ ok

WHERE **[IUT II]** §2, Corollary 2.4 p. 69, §2, Corollary 2.5 p. 71, §2, Corollary 2.8 p. 81

INPUTS InitialThetaData OUTPUTS theta values

```
    [Π:Π_Ÿ]=2 [Π±:Π]=13 [Π±:Π_Ÿ]=26; 13 evaluation points; values θ(ν-q^t) in q_-units: |t⊨0→q_^0 |t⊨1→q_^1 |t⊨2→q_^4 |t⊨3→q_^9 |t⊨4→q_^16 |t⊨5→q_^25 |t⊨6→q_^36
(up to μ_26)
  ⇒ ok
```

BuildMonoids Ψ_env, Ψ_gau, Ψ_cns, Ψ^ss_cns; evaluation and Kummer isomorphisms ✓ ok

WHERE **[IUT II]** §3, Proposition 3.1 p. 87, §3, Corollary 3.5 -3.7 p. 93, §4, Corollary 4.5 -4.6 p. 131

INPUTS prime-strip at v OUTPUTS monoid

```
    at v3: Ψ_env = O×·θ^N (split 1), Ψ_gau generators q_^1 q_^4 q_^9 q_^16 q_^25 q_^36; Ψ_cns (1 gen), Ψ^ss_cns = O× × R≥0, log(p_v) = 1.0986
    [ ok ] EvaluationIso               Ψ_env ≅ Ψ_gau at v3: θ ↦ (q_^{j²})_{j=1..6}   [IUTchII] Cor 3.5 (ii); Cor 4.10 (ii)
    [ ok ] KummerIso                   Kummer isomorphism Ψ(F)[4013] ≅ Ψ(D)[3011] at v#1   [IUTchII] Cor 4.6, Cor 4.8
  ⇒ ok
```

RealifyFromDMono D⊢ ↦ D⊩(D⊢): the global realified Frobenioid from the core ✓ ok

WHERE **[IUT II]** §4, Corollary 4.5 (ii) p. 131

INPUTS DMonoPrimeStrip OUTPUTS FMonoGlobPrimeStrip

```
    D⊢ ↦ D⊩(D⊢): kind F⊩, global realified: yes, degree of q-object -0.9309 (defined up to R>0)
  ⇒ ok
```

Symmetrize symmetrizing isomorphisms, diagonals, conjugate synchronization ✓ ok

WHERE **[IUT II]** §3, Corollary 3.5 (i) p. 93, §4, Corollary 4.5 (iii) p. 131, §4, Corollary 4.7 (ii) p. 140

INPUTS capsule OUTPUTS symmetrization

```
    F_l^{×±}: 7 classes (|F_l|), conjugate synchronization: yes (copy 107), diagonal 1079
    F_l^*: 6 classes (F_l^*), conjugate synchronization: no, diagonal 1089
    label subset {1} keeps F_l^{×±}-symmetry: no; full F_l: yes
  ⇒ ok
```

ForgetLadder F⊩▶×μ → F⊢▶×μ → F⊢×μ → F⊢× → D⊢ and the pilot object ✓ ok

WHERE **[IUT II]** §4, Definition 4.9 p. 153

INPUTS FMonoGlobValUnitsModTorsPrimeStrip OUTPUTS prime-strips

```
    F⊩▶×μ global 1, valgroup 1, units O×μ, Kummer Ism-orbit   [IUTchII] Def 4.9 (viii)
  → F⊢▶×μ global 0, valgroup 1, units O×μ, Kummer Ism-orbit   [IUTchII] Def 4.9 (vii)
  → F⊢×μ global 0, valgroup 0, units O×μ, Kummer Ism-orbit   [IUTchII] Def 4.9 (vii)
  → F⊢×   global 0, valgroup 0, units O×, Kummer F̂×   [IUTchII] Def 4.9 (vi)
  → D⊢    global 0, valgroup 0, units —, Kummer —   [IUTchI] Def 4.1 (iii)
  pilot object degree -14.1180 (≤ 0), ×μ-Kummer orbit marker 2
  ⇒ ok
```

LinkThetaUnitsModTors F⊩_Δ ≅ F⊩_mod, F⊩_env ≅ F⊩_gau, the Θ×μ(_gau)-link, coric F⊢×μ and D⊩ ✓ ok

WHERE **[IUT II]** §4, Corollary 4.10 p. 158, §4, Remark 4.11.1 p. 164

INPUTS two Hodge theaters OUTPUTS link, D-link

```
    θ×μ_gau-link (0,0)→(1,0): glued F⊩▶×μ, full poly-iso 1, Π preserved 0, O×μ preserved 1, dilation 15.167; induced D-Θ±ell NF-link on D⊢
    exponents q_ ↦ q_^{1,4,9,16,25,36} (Rem 4.11.1)
    [ ok ] ConstantStripIdentification  F⊩_Δ ≅ F⊩_mod (labels 0 and F_l^* identified; degree -0.9309)   [IUTchII] Cor 4.10 (i)
    [ ok ] EnvelopeIsos                 F⊩_env ≅ F⊩_tht ≅ F⊩_gau (degree -14.1180)   [IUTchII] Cor 4.10 (ii)
    [ ok ] CoricUnitsModTors            †F⊢×μ ≅ ‡F⊢×μ: full poly-isomorphism of F⊢×μ-prime-strips (Ism-orbits)   [IUTchII] Cor 4.10 (iv)
    [ ok ] CoricGlobalRealified         D⊩(†D⊢) ≅ D⊩(‡D⊢) up to R>0 (ratio 1.000)   [IUTchII] Cor 4.10 (v)
  ⇒ ok
```

12.3 Processes of [IUT III] — Inter-universal Teichmüller Theory III: Canonical Splittings of the Log-theta-lattice

MakeLogShells log-shells I_v and their properties / isomorphisms ✓ ok

WHERE **[IUT III]** §1, Definition 1.1 p. 23, §1, Proposition 1.2 (v)-(ix) p. 30

INPUTS InitialThetaData OUTPUTS LogShell

```
    I_v2    p* = 4, generator valuation -25/26, log-volume 1.3063, iso indeterminacy Ism
    I_v3    p* = 3, generator valuation -6/13, log-volume 1.0141, iso indeterminacy Ism
    I_v5    p* = 5, generator valuation -19/26, log-volume 1.5475, iso indeterminacy Ism
    I_v7    p* = 7, generator valuation -54/65, log-volume 1.9359, iso indeterminacy Ism
    I_v13   p* = 13, generator valuation -5/6, log-volume 2.1375, iso indeterminacy Ism
    I_vinf  p* = 0, generator valuation 0, log-volume 0.0000, iso indeterminacy {±1}
    [ ok ] LogShellProperties           I_v2: (a) log(O×) ⊆ I ✓, (b) O ⊆ p^{-?}I ✓, (c) compact Z_p-module ✓   [IUTchIII] Prop 1.2 (v)
    [ ok ] LogShellProperties           I_v3: (a) log(O×) ⊆ I ✓, (b) O ⊆ p^{-?}I ✓, (c) compact Z_p-module ✓   [IUTchIII] Prop 1.2 (v)
    [ ok ] LogShellProperties           I_v5: (a) log(O×) ⊆ I ✓, (b) O ⊆ p^{-?}I ✓, (c) compact Z_p-module ✓   [IUTchIII] Prop 1.2 (v)
    [ ok ] LogShellProperties           I_v7: (a) log(O×) ⊆ I ✓, (b) O ⊆ p^{-?}I ✓, (c) compact Z_p-module ✓   [IUTchIII] Prop 1.2 (v)
    [ ok ] LogShellProperties           I_v13: (a) log(O×) ⊆ I ✓, (b) O ⊆ p^{-?}I ✓, (c) compact Z_p-module ✓   [IUTchIII] Prop 1.2 (v)
    [ ok ] LogShellProperties           I_vinf: (a) log(O×) ⊆ I ✓, (b) O ⊆ p^{-?}I ✓, (c) compact Z_p-module ✓   [IUTchIII] Prop 1.2 (v)
  ⇒ ok
```

ApplyLogLinkStrip the log-link on an F-prime-strip; what is coric and what is not ✓ ok

WHERE **[IUT III]** §1, Definition 1.1 p. 23, §1, Proposition 1.2 p. 30

INPUTS FPrimeStrip OUTPUTS FPrimeStrip, LogShell

```
    †F →log ‡F (full): Π kept (11002 = 11002), units renewed (11003 → 11004), 6 log-shells; coric D 1 D⊢ 1; ring preserved 0; log-volume compatible 1; Kummer compatible
0
    I_F ≅ I_F⊢×μ ≅ I_D⊢ ≅ I_D at v2: yes (hol/étale flags 10 00 01 11)
  ⇒ ok
```

ChainLogLinks the vertical Frobenius picture and the log-link of Hodge theaters ✓ ok

WHERE **[IUT III]** §1, Proposition 1.2 (x) p. 30, §1, Proposition 1.3 p. 41

INPUTS LogThetaLattice OUTPUTS chain, report

```
    vertical chain of 4 HTs in column 0 (universes 5 1 2 6); translation symmetric 1, adjacent swap 0
    [ ok ] LogLinkHodgeTheater          log-link (0,0)→(0,1): D-HTs coric, 6 log-shells built   [IUTchIII] Prop 1.3
  ⇒ ok
```

DefineLattice lattice flavours; non-commutativity; horizontal obligates vertical ✓ ok

WHERE **[IUT III]** §1, Definition 1.4 p. 45, §1, Remark 1.4.1 p. 46

INPUTS LogThetaLattice OUTPUTS report

```
    0-link    → non-Gaussian log-theta-lattice ([IUTchIII] Def 1.4 (i))
    0×μ-link → non-Gaussian log-theta-lattice ([IUTchIII] Def 1.4 (i))
    0×μ_gau-link → Gaussian log-theta-lattice ([IUTchIII] Def 1.4 (ii))
    0×μ_LGP-link → LGP-Gaussian log-theta-lattice ([IUTchIII] Def 3.8 (iii))
    0×μ_lgp-link → lgp-Gaussian log-theta-lattice ([IUTchIII] Def 3.8 (iii))
    squares commute: no (Rem 1.4.1 (i))
    [ ok ] HorizontalObligatesVertical  0 horizontal arrow(s) without an accompanying log-link in their column   [IUTchIII] Rem 1.4.1 (ii)
  ⇒ ok
```

CheckBicoric vertical / horizontal / bi-coric data of the lattice ✓ ok

WHERE **[IUT III]** §1, Theorem 1.5 p. 48

INPUTS LogThetaLattice OUTPUTS bicoric summary

```
    (i) D-HT vertically coric 1 (Π copy 170002)  (ii) F⊢×μ horizontally coric 1 (units copy 183003)  (iii) F⊢×μ(D⊢_Δ) bi-coric 1  (iv) log-shells bi-coric 1  (v)
D⊪(D⊢) bi-coric 1
  ⇒ ok
```

CoricThetaMonoids vertically coric theta monoids; the étale-picture radial environment ✓ ok

WHERE **[IUT III]** §2, Proposition 2.1 p. 58, §2, Theorem 2.2 p. 65, §2, Corollary 2.3 p. 72

INPUTS LogThetaLattice OUTPUTS report, radial items

```
    (aR)   a D-Θ±ell NF-Hodge theater †HT^D (étale-like)
    (bR)   the F⊢×μ-prime-strips F⊢×μ(†D⊢_t) at the labels t ∈ F_l, with their ×μ-Kummer structures
    (cR)   the theta monoids Ψ_env(†D_>), ∞Ψ_env(†D_>) with their canonical splittings (Thm 2.2)
    (dR)   the F_l^{×±}-symmetrizing isomorphisms and the diagonal (conjugate synchronization)
    (eR)   the full poly-isomorphism F⊢×μ(†D⊢_Δ) ≅ (coric F⊢×μ) — the 'α' of the multiradialization
    coric  (D⊢, F⊢×μ(D⊢)): the mono-analytic core and its units modulo torsion
    [ ok ] VerticallyCoricThetaMonoids  Ψ_env(0,∘D_>) is the same at m = 0 and m = 1 (Π copy 170002)   [IUTchIII] Prop 2.1
    [ ok ] ThetaMonoidsMultiradial      theta monoids with mono-theta cyclotomic rigidity: multiradial (indeterminacy: none)   [IUTchIII] Thm 2.2, Cor 2.3
  ⇒ ok
```

BuildTensorPackets local holomorphic / mono-analytic / global tensor packets ✓ ok

WHERE **[IUT III]** §3, Proposition 3.1 -3.3 p. 92

INPUTS LogShell, InitialThetaData OUTPUTS TensorPacket

```
    log(^{S±_1} F_3): 1 factors, |V_p⊨1, 1 summands, shell log-volume 1.0141
    log(^{S±_2} F_3): 2 factors, |V_p⊨1, 1 summands, shell log-volume 2.0282
    log(^{S±_3} F_3): 3 factors, |V_p⊨1, 1 summands, shell log-volume 3.0423
    log(^{S±_4} F_3): 4 factors, |V_p⊨1, 1 summands, shell log-volume 4.0564
    archimedean I(^{S±_2} D⊢_∞): 2 summands (2^{j}·|V_arc|^{j+1})
    global (M_mod)_{S±_2}: 2 factors, 1 summand fields
    [ ok ] GlobalPacketLocalization     (M_mod)_{S±2} localizes into the packet at v_Q = 3   [IUTchIII] Prop 3.3 (ii)
  ⇒ ok
```

BuildLGPMonoids LGP-monoids in the packets; upper semi-compatibility (Ind3) ✓ ok

WHERE **[IUT III]** §3, Proposition 3.4 p. 101, §3, Proposition 3.5 p. 103

INPUTS TensorPacket OUTPUTS monoid, Δlog-volume

```
    Ψ_LGP(S±_2) at v3: splitting monoid generated by q_^1
    Ψ_LGP(S±_3) at v3: splitting monoid generated by q_^4
    Ψ_LGP(S±_4) at v3: splitting monoid generated by q_^9
    v2    ⊆ (log(O×) ⊆ I, inclusion)          Δlog-vol +1.3063   [IUTchIII] Prop 3.5 (ii)(a)
    v3    = up to roots of unity (log(ζ) = 0)     Δlog-vol +0.0000   [IUTchIII] Prop 3.5 (ii)(c)
    v5    = up to roots of unity (log(ζ) = 0)     Δlog-vol +0.0000   [IUTchIII] Prop 3.5 (ii)(c)
    v7    = up to roots of unity (log(ζ) = 0)     Δlog-vol +0.0000   [IUTchIII] Prop 3.5 (ii)(c)
    v13   ⊆ (log(O×) ⊆ I, inclusion)          Δlog-vol +2.1375   [IUTchIII] Prop 3.5 (ii)(a)
    vinf  ↠ (surjection onto the closed disc)    Δlog-vol +0.0000   [IUTchIII] Prop 3.5 (ii)(b)
  ⇒ ok
```

CompareGlobalVariants F_MOD vs F_mod; F⊪_gau ≅ F⊪_LGP ≅ F⊪_lgp ✓ ok

WHERE **[IUT III]** §3, Example 3.6 p. 106, §3, Proposition 3.7 p. 109

INPUTS ThetaPmEllNFHodgeTheater OUTPUTS report

```
    F_MOD: F_mod^×-torsors of rational functions (arithmetic line bundles as global objects): 0-link compatible 1, log-volume compatible 0   [IUTchIII] Ex 3.6 (i), Rem
3.6.2 (i)
    F_mod: collections of local fractional ideals {J_v}_v (local integral structures): 0-link compatible 0, log-volume compatible 1   [IUTchIII] Ex 3.6 (ii), Rem 3.6.2
(ii)
    [ ok ] GauLGPLgpIsos                F⊪_gau ≅ F⊪_LGP ≅ F⊪_lgp (degree -14.1180)   [IUTchIII] Prop 3.7 (ii)
  ⇒ ok
```

MakePilots q-/Θ-pilot objects; the Θ×μ_LGP-link relative to a log-link ✓ ok

WHERE **[IUT III]** §3, Definition 3.8 p. 112

INPUTS InitialThetaData, lattice OUTPUTS QPilot, ThetaPilot

```
q-pilot {q_}: log-volume -0.9309;  θ-pilot {q_^{j²}} (6 exps, last 36): log-volume -14.1180;  arithmetic degree of q_: -0.9309
lattice for θ×μ_LGP: LGP-Gaussian log-theta-lattice ([IUTchIII] Def 3.8 (iii))
[ ok ] LGPLinkRelativeToLog         (0,0) is the codomain of a log-link: the θ×μ_LGP-link out of it is defined   [IUTchII] Def 3.8 (ii)
[warn] LGPLinkRelativeToLog         (1,0) has no incoming log-link: the θ×μ_LGP-link needs the log-shells of (1,θ-1)   [IUTchIII] Def 3.8 (ii)
⇒ ok
```

NormalizeLogVolumes packet-/procession-normalization, arithmetic degree, non-interference ✓ ok

WHERE **[IUT III]** §3, Proposition 3.9 p. 115, §3, Proposition 3.10 p. 147

INPUTS log-volumes OUTPUTS numbers

```
packet-normalization: vol(p·M) = vol(M) - log p: θ ↦ -1.0986 (p=3); arch: e·M ↦ 1.0
procession-normalization: average of 7 capsule volumes = 4.0000; log-link compatibility: preserved
non-interference: F_mod^× ∩ Πθ×_v = μ (order 2), MOD/LGP-link 1, mod/log-volume 1
⇒ ok
```

RepresentMultiradially the multiradial representation (a)(b)(c) with Ind1·Ind3 ✓ ok

WHERE **[IUT III]** §3, Theorem 3.11 p. 153

INPUTS LogThetaLattice OUTPUTS MultiradialRepresentation

```
R^LGP_{θ,∘}: (a) 7 packets, (b) 18 splitting monoids, (c) 6 copies of F_mod^×; Ind1 125411328000, Ind2 on 7 summands, Ind3 1; log-Kummer 1; link compat 1111
-|log q| = -0.9309, -|log θ| ≤ 33.0285, C_θ = 1879989920415953295769 6.000, inequality: holds
⇒ ok
```

EstimateLogVolume −|log Θ| ≥ −|log q| as numbered steps ✓ ok

WHERE **[IUT III]** §3, Corollary 3.12 p. 173

INPUTS LogThetaLattice OUTPUTS numbers

```
(0) |log q| = log q / 2l: log-volume of the q-pilot at (1,0), i.e. -|log q|                  -0.9309   [IUTchIII] Cor 3.12; Def 3.8 (i)
(1) the θ-pilot at (0,0): naive log-volume -avg(j²)·|log q|                              -14.1180   [IUTchIII] Def 3.8 (i)
(2) the θ×μ_LGP-link (0,0)→(1,0) identifies the θ-pilot with the q-pilot (value groups), units up to Ism      -0.9309   [IUTchIII] Def 3.8 (ii), Thm 3.11 (iii)
(3) multiradial representation of the θ-pilot in terms of the mono-analytic data of (1,0): Ind1 count    5040.0000   [IUTchIII] Thm 3.11 (i)
(4) holomorphic hull, leading terms only: ((l+5)/4)·log d^K (an estimate, not a bound)         47.1465   [Alien] §3.7 (iv)
(5) leading terms with the theta shift: ((l+5)/4)·log d^K - ((l+1)/24)·log q (not a bound)      33.0285   [Alien] §3.7 (iv); [IUTchIV] Rem 1.10.1
(6) -|log θ| ≤ B: the bound of the proof of Thm 1.10 with every term, summed over v_Q    1530.5931   [IUTchIV] Thm 1.10, Steps (v)-(vii)
(7) Cor 3.12 (an input, not re-proved here) gives B ≥ -|log q|, i.e. C_θ ≥ -1 for C_θ = B/|log q|    1644.2855   [IUTchIII] Cor 3.12
Cor 3.12 holds: yes
⇒ ok
```

12.4 Processes of [IUT IV] — Inter-universal Teichmüller Theory IV: Log-volume Computations and Set-theoretic Foundations

BoundDifferents differents, a_i, b_i, the φ-estimates, bounds for d_i ✓ ok

WHERE **[IUT IV]** §1, Proposition 1.1 -1.3 p. 9

INPUTS local fields k_i OUTPUTS exponents

```
k_0 (p=7,e=1): a=1.000 b=-1.000  p^a R ⊆ log(R×) ⊆ p^{-b} R (equalities)
k_1 (p=7,e=2): a=0.500 b=-0.500  p^a R ⊆ log(R×) ⊆ p^{-b} R (equalities)
k_2 (p=7,e=6): a=0.333 b=0.833  p^a R ⊆ log(R×) ⊆ p^{-b} R
d_I = 1.500, d_{I*} (*=0) = 1.500; φ(p^λ (R_I)~) ⊆ p^{-3} log(R_I×) ⊆ p^{-3} (R_I)~ for λ=0; (iii) exponent -3; unramified: 1
Prop 1.3: 0.833 ≤ d_2 ≤ 2.000 (actual 1.000)
⇒ ok
```

EstimateNonarchVolume μ_log(log_p R×) and the normalized log-volume bounds ✓ ok

WHERE **[IUT IV]** §1, Proposition 1.4 p. 13

INPUTS local fields k_i OUTPUTS log-volumes

```
μ_log(log_p(R×)) at v3 = -0.0423; Prop 1.4 (iii) bounds for λ=0: log part ≤ 2.5634, tilde part ≤ 7.3567; d_I + a_I ≥ 2; (iv) unramified odd ⇒ μ_log = 0: 1
the same number through iut_log_shell_volume (with p*): 1.0141
⇒ ok
```

EstimateArchMetric archimedean summands 2^{|I|−1}|V|^{|I|} and metric factors ✓ ok

WHERE **[IUT IV]** §1, Proposition 1.5 p. 14

INPUTS |I|, |V_arc| OUTPUTS counts

```
|I⊨1: 1 summands, metric factor 1, ⊗m ∈ λ^{|I|}B with λ=2: 2
|I⊨2: 2 summands, metric factor 2, ⊗m ∈ λ^{|I|}B with λ=2: 4
|I⊨3: 4 summands, metric factor 4, ⊗m ∈ λ^{|I|}B with λ=2: 8
⇒ ok
```

BoundPrimes prime number theorem and Chebyshev-type bounds ✓ ok

WHERE **[IUT IV]** §1, Proposition 1.6 p. 16, §2, Proposition 2.1 p. 40

INPUTS numbers OUTPUTS numbers

```
n ≤ 4p_n/(3 log p_n): p_25 = 97 ⇒ n ≤ 28.27; #{p ≤ n_prm} ≤ 1.399e+19;  log x ≤ x at x=13: 1;  θ(x) ∈ [66.7, 133.3] for x = 100; prime in A with θ_A=50, ξ=5: p ≤
110
⇒ ok
```

AverageWeighted weighted averages with λ_v = [(F_mod)_v : Q_p] ✓ ok

WHERE **[IUT IV]** §1, Proposition 1.7 p. 16

INPUTS InitialThetaData OUTPUTS number

```
    weights λ_v = [(F_mod)_v:Q_3] for 1 valuation(s) above 3; weighted average of β = 1..n: 1.0000
  ⇒ ok
```

CheckTorsionPoints torsion points, fields of definition, ramification at Vbad ✓ ok

WHERE **[IUT IV]** §1, Proposition 1.8 p. 18

INPUTS InitialThetaData OUTPUTS report

```
    [ ok ] SemistableOverK              E[l] rational over K ⇒ semistable reduction over K away from l (asserted: yes)   [IUTchIV] Prop 1.8 (ii),(iii)
    [info] MinimalFieldOfDefinition     k_E ⊆ F_mod = Q(j_E), [F_mod:Q] = 1; F_tpd = F_mod(E[2]) has degree ≤ 6 over F_mod   [IUTchIV] Prop 1.8 (i)
  ⇒ ok
```

DegreeArithDivisors arithmetic divisors, deg, normalized degree, deg_E ✓ ok

WHERE **[IUT IV]** §1, Definition 1.9 p. 21

INPUTS InitialThetaData OUTPUTS ArithDivisor, numbers

```
    q-divisor: deg = 24.2023, normalized deg = 24.2023 (= log q 24.2023), deg over v|3: 2.1972
  ⇒ ok
```

ComputeThmOneTen (E1)-(E6), the intermediate bounds and the final inequality ✓ ok

WHERE **[IUT IV]** §1, Theorem 1.10 p. 22

INPUTS InitialThetaData OUTPUTS Thm 1.10 numbers

```
    (E1) n=6: 3.5 (E2) 15.167 (E3) |GL2(F_13)| = 26208 (E5) 2
    log q = 24.2023, |log q| = 0.9309, log d^Ftpd = 0.0000, log f^Ftpd = 4.6540, log d^K ≤ 30.7839, e_mod = 1, e*_mod = 552960
    C_0 = 187998992041595329576960.000 (Cor 3.12 ⇔ C_0 ≥ -1: yes);  (1/6) log q = 4.0337 ≤ 11.8139 + 1.0000e+22: holds, but only as a comparison with the constant
20(e*_mod·l + n_prm)
  ⇒ ok
```

ComputeCorTwoTwo δ, (C1) range of l, ε_E, (C2), the height form ✓ ok

WHERE **[IUT IV]** §2, Corollary 2.2 p. 41, §2, Corollary 2.3 p. 54

INPUTS InitialThetaData OUTPUTS Cor 2.2 numbers

```
    d = 1, δ = 552960, log qϑ = 25.5886; (C1) 5.06 ≤ l ≤ 4.799e+08: l = 13 in range; ε_E = 3.7337e+15
    (C2) 4.2648 ≤ (1+ε_E)(0.0000 + [4.6540, 7.9121]) + 1.000e+22: holds trivially;  Cor 2.3 slack: 1.000e+22
    this curve is in the exceptional set Exc of Cor 2.2 (ε_E > 1; (log qϑ)^{1/2} < ξ_prm + n_prm), so the corollary asserts nothing about it
  ⇒ ok
```

OperateSpecies compose/identity mutations, species-isomorphism, mutation-history, cores ✓ ok

WHERE **[IUT IV]** §3, Definition 3.1 p. 68, §3, Definition 3.3 p. 72

INPUTS species tables, lattice OUTPUTS mutation, history

```
    compose LogLink then LogLink: ThetaPmEllNFHodgeTheater → ThetaPmEllNFHodgeTheater (core 0)
    identity of InitialThetaData: InitialThetaData → InitialThetaData; species-isomorphic to itself: 1
    mutation-history of the lattice: 4 vertices, 3 edges; cores: D⊢ 1, F⊢×μ 1, D-HT 1, F-HT 0
  ⇒ ok
```

12.5 Processes of [EssLgc] — On the Essential Logical Structure of Inter-universal Teichmüller Theory

EvaluateLogicModel the numerical model: ∧ gives A = ε ≤ 1, ∨ gives nothing ✓ ok

WHERE **[EssLgc]** §2, Example 2.4.5 p. 50

INPUTS A, B, ε OUTPUTS conclusion

```
    AND-IUT (∧)               0-link: N=-1.50 ∧ N=-0.50; multiradial: N=-0.50 ∧ N=-0.50 ⇒ A = ε ≤ 1
    OR-IUT (∨)                0-link: N=-1.50 ∧ N=-0.50; multiradial: N=-0.50 ∧ N=-0.50 ⇒ no conclusion
    XOR-IUT                     0-link: N=-1.50 ∧ N=-0.50; multiradial: N=-0.50 ∧ N=-0.50 ⇒ no conclusion
    RCS-IUT (redundant copies)  0-link: N=-1.50 ∧ N=-0.50; multiradial: N=-0.50 ∧ N=-0.50 ⇒ no conclusion
    EssOR-IUT                   0-link: N=-1.50 ∧ N=-0.50; multiradial: N=-0.50 ∧ N=-0.50 ⇒ no conclusion
  ⇒ ok
```

ChainAndSteps the chain of logical ∧ relations through the lattice ✓ ok

WHERE **[EssLgc]** §3.10 (Stp1)-(Stp8) p. 136, (FxGl)

INPUTS LogThetaLattice OUTPUTS steps with log-volumes

```
    StpOne    (0,0)        → (0,◦)               Ind3         33.0285   [EssLgc] §3.10 (Stp1); [IUTchIII] Thm 3.11 (ii)
    StpTwo    (0,◦)      → (0,◦)⊢             Ind1         33.0285   [EssLgc] §3.10 (Stp2); [IUTchIII] Thm 3.11 (i)
    StpThree  (0,◦)⊢   → (0,0)⊢ ⇔ (1,0)⊢  Ind2         33.0285   [EssLgc] §3.10 (Stp3); [IUTchII] Cor 4.10 (iv)
    StpFour   (1,0)⊢     → (1,◦) hull           none         47.1465   [EssLgc] §3.10 (Stp4); [IUTchIII] Cor 3.12 proof
    StpFive   (1,◦) hull → hull-approximants      none         47.1465   [EssLgc] §3.10 (Stp5)
    StpSix    approximants → determinant^{1/n}      none         47.1465   [EssLgc] §3.10 (Stp6)
    StpSeven  (1,m)↑     → (1,m+1)↓             none         47.1465   [EssLgc] §3.10 (Stp7); [IUTchIII] Prop 1.2 (iii)
    StpEight  log-volumes  → R: -|log 0| ≥ -|log q| monodromy    -0.9309   [EssLgc] §3.10 (Stp8); [IUTchIII] Cor 3.12
    (FxGl) q-pilot fixed at (1,0): -0.9309; glued 0-pilot at (0,0): -14.1180
  ⇒ ok
```

Fuzzify fuzzification, engulfing, extending indeterminacies, ∨ verdicts ✓ ok

WHERE **[EssLgc]** §3.10 (EngIf) p. 136, (ExtInd1/2), §3.11 p. 152

INPUTS value, indeterminacies OUTPUTS interval

```
    0-pilot -14.1180 fuzzified by Ind1-3 to [-14.1180, 33.0285]: engulfs -|log q| = -0.9309: yes;  q-pilot fuzzified symmetrically: [-1.9309, 47.2157]
    (ExtInd1) {1,2} ∪ {2,3} has 3 elements; (ExtInd2) crushed by |·|: 3 classes; (logORInd): harmless — log-volume is invariant under the log-link; (0ORInd): fatal — an
OR across the 0-link asserts nothing
  ⇒ ok
```

12.6 Processes of [Alien] — The Mathematics of Mutually Alien Copies: from Gaussian Integrals to Inter-universal Teichmüller Theory

ComputeGaussian the Gaussian integral and its IUT analogues ✓ ok

WHERE **[Alien]** §1 Steps 1-12 p. 6

INPUTS — OUTPUTS √π

```
    Step  1 the naive approach e^{-x²} ↦ ∪ fails: no elementary antiderivative      —   ⇔ a single ring/scheme theory cannot compute the height
    Step  2 partial integration does not help either                                —   ⇔ no Frobenius morphism in characteristic 0
    Step  3 introduce a SECOND copy of the integral: I² = ∫∫ e^{-x²-y²}  3.14159   ⇔ two mutually alien copies of the ring theory (the two sides of the 0-link)
    Step  4 regard them as coordinates of R²                                3.14159   ⇔ the log-theta-lattice as a plane of copies
    Step  5 infinitesimals dx dy: the measure                                3.14159   ⇔ log-volumes
    Step  6 the Jacobian: how the measure transforms                         1.00000   ⇔ compatibility of log-volumes with the links
    Step  7 linear transformations: toral dilations, unipotent, rotations    1.00000   ⇔ the dilation q ↦ q^{j²} of the 0-link vs the rotation by the log-link
    Step  8 polar coordinates on R² \ (R≤0 × {0})                    1.00000   ⇔ the coordinates (n, m) of the lattice; the slit = the 'Kummer-detachment'
indeterminacy
    Step  9 decoupling of the radial and angular parts                       1.00000   ⇔ multiradiality: the radial data is decoupled from the coric data
    Step 10 radial integral ∫_0^∞ e^{-r²} 2r dr = 1                  1.00000   ⇔ the multiradial representation computes the 0-pilot
    Step 11 angular integral ∫_0^{2π} (1/2) dθ = π                     3.14159   ⇔ the holomorphic hull / the indeterminacies contribute the constant
    Step 12 √π: the result, with its 'error factor' from the slit         1.77245   ⇔ -|log 0| ≥ -|log q| with the indeterminacies as error term
    numerical ∫e^{-x²} = 1.772454, √π = 1.772454
  ⇒ ok
```

TransportMonoAnabelian mono-anabelian transport: Kummer-detachment, étale-transport ✓ ok

WHERE **[Alien]** §2.7 p. 21, §2, Figure 2.1 p. 23

INPUTS LogThetaLattice OUTPUTS stages

```
    Frobenius-like object (input)                                domain   positive mass (rigid, carries the arithmetic)   none
    Kummer-detachment                                            domain   —                                              Kummer-detachment indeterminacies (Ind2,
Ind3)
    étale-like object (reconstructed from Π/G)                  domain   zero mass (an indeterminate isomorphism class)   none
    étale-transport across the link (the coric data)             link     zero mass                                      étale-transport indeterminacies (Ind1:
Aut(G_v), permutations)
    étale-like object                                           codomain zero mass                                      none
    Kummer-detachment (inverse): re-attach to Frobenius-like data  codomain positive mass again                         Kummer-detachment indeterminacies
    Frobenius-like object (output), compared with the input        codomain positive mass                               the comparison holds up to Ind1, Ind2, Ind3
    run: Frobenius-like input at (0,0): 0-pilot log-volume                          -14.1180
    run: Kummer-detach at (0,0): pass to étale-like (0,◦) data (mass 0)              0.0000
    run: étale-transport across the 0-link via D⊢ (Ind1 permutations)       125411328000.0000
    run: re-attach at (1,0): q-pilot log-volume (the mass of the codomain)           -0.9309
  ⇒ ok
```

SumLabelsFormally formal sums [0], [|j|], [⊞], [>] and their compatibilities; Ÿ labels ✓ ok

WHERE **[Alien]** §3.6 p. 87

INPUTS InitialThetaData OUTPUTS formal sums

```
    [0]     1 terms, total mass 1
    [|2|]   2 terms, total mass 1
    [⊠]  13 terms, total mass 1
    [>]    12 terms, total mass 1
    line segment of Ÿ: 13 labels -6..6; evaluation points of t=2: {2,-2} (2)
    [ ok ] FormalSumCompatibility       [0] → [>]: total mass 1, ±-symmetric: compatible   [Alien] §3.6 (i)
    [ ok ] FormalSumCompatibility       [⊠] → [>]: total mass 1, ±-symmetric: compatible   [Alien] §3.6 (i)
    [ ok ] FormalSumCompatibility       (1/(l*+1))([0]+Σ[|j|]) → [>]: total mass 1, ±-symmetric: compatible   [Alien] §3.6 (i)
  ⇒ ok
```

StepEstimate the estimate step by step; the leading term ✓ ok

WHERE **[Alien]** §3.7 (1est)-(12est) p. 103, (iv)

INPUTS LogThetaLattice OUTPUTS numbers

```
    (1est) start with the q-pilot on the RHS (1,0): -|log q|                     -0.9309   [Alien] §3.7 (ii) (1est)
    (2est) the 0-link identifies it with the 0-pilot of the LHS (0,0)            -0.9309   (2est)
    (3est) the multiradial representation expresses the 0-pilot in RHS terms      33.0285   (3est); [IUTchIII] Thm 3.11
    (4est) Ind1-invariance: permutations of the procession                  125411328000.0000   (4est)
    (5est) Ind2-invariance: Ism on the units                                      1.0000   (5est)
    (6est) Ind3 gives an UPPER bound on the 0-pilot's region                     33.0285   (6est)
    (7est) comparison only up to isomorphism class of O_k-modules                 0.0000   (7est)
    (8est) holomorphic hull = ⊗ of j+1 copies of log(O_k^×): log d^K terms      47.1465   (8est)
    (9est) log-volume = arithmetic degree                                        33.0285   (9est); [IUTchIII] Prop 3.9 (iii)
    (10est) log-Kummer on both sides                                              0.0000   (10est)
    (11est) the inequality -|log 0| ≥ -|log q|                                33.9594   (11est); [IUTchIII] Cor 3.12
    (12est) leading term (l²/48)(6 log d^K - log q), l ≈ vheight              136.1146   (12est); [Alien] §3.7 (iv)
    leading term per j (j=1): 20.0232; sum over F_l^*: 198.1713 ≈ (l²/48)(6 log d^K - log q) = 136.1146; l ≈ vht = 2.01
  ⇒ ok
```

TranslateDictionary the dictionary (1gau)-(11gau) see transcript

WHERE **[Alien]** §3.8 p. 112

INPUTS — OUTPUTS table

```
    (1gau) the two copies of ∫ e^{-x²}                                ↔ the two Hodge theaters on either side of the θ-link
    (2gau) R² = the plane of the two coordinates                          ↔ the log-theta-lattice
    (3gau) the measure dx dy                                               ↔ log-volumes
    (4gau) linear coordinate changes / toral dilations                     ↔ the dilation q_ ↦ q_^{j²} of the θ-link
    (5gau) rotations                                                       ↔ the log-link (rotating multiplicative into additive structure)
    (6gau) polar coordinates                                               ↔ the coordinates (n, m) and the multiradial decoupling
    (7gau) the slit R≤0 × {0}                                        ↔ the indeterminacies Ind1-Ind3 (Kummer-detachment / étale-transport)
    (8gau) the radial integral = 1                                         ↔ the multiradial representation of the θ-pilot
    (9gau) the angular integral = π                                      ↔ the holomorphic hull, the constant terms
    (10gau) √π                                                         ↔ -|log θ| ≥ -|log q|
    (11gau) the 'error' from the slit is what makes the computation possible ↔ the indeterminacies are what make the θ-link's gluing consistent with ring theory
  ⇒ ok
73 process(es) run, 0 failure(s)
```

Part III

A demo session

The text views of the C program on the demo Θ-data, and the standalone figures of the web app: the Hodge theater assembled in five stages, the whole lattice and the infinite H, the multiradial representation and the étale-picture.

13 · Generate and validate

`iut validate --example 2` — the initial Θ-data as the program prints it and the validation report: every condition of [IUTchI] Def 3.1 that can be checked on numbers is checked; the rest is listed as asserted.

```
Initial Θ-data  (F̄/F, X_F, l, C_K, V, V^bad_mod, ε_)      [IUTchI] Def 3.1
  name      : abc(1,2400,2401) l=13
  curve     : Frey y^2=x(x-1)(x+2400)
  fields    : F_mod ⊆ F ⊆ K = F(E[l]) ⊆ F̄     [F_mod:Q]=1  [F:Q]≤276480  [K:Q]≤7245987840
  l         : 13   (l* = 6, l± = 7)   ε_ = the cusp ±1
  log(q)    : 24.2023   |log(q)| = log(q)/2l = 0.9309   ht_E = log(q)/6 = 4.0337
  log(f)    : 4.6540 over V^bad_mod (the f of Thm 1.10); 5.3471 over every bad place   log(d^Ftpd) = 0.0000   h(j) ≈ 21.112
  log(d^K)  : ≈ 10.4770 (an estimate; ≤ 30.7839 by Thm 1.10, Step (ii))
  V ≅ V_mod : 6 valuations
    name   kind  p    e/f_m e/f_K reduction  ord(q)  V^bad_mod
    v2     non   2    1/1   ≥26/≥1 bad mult.  2       -
    v3     non   3    1/1   ≥26/≥1 bad mult.  2       yes
    v5     non   5    1/1   ≥52/≥1 bad mult.  4       yes
    v7     non   7    1/1   195/≥12 bad mult.  8       yes
    v13    non   13   1/1   ≥12/≥1 good       0       -
    vinf   arc   -    -     -     -          -       -
    v2: bad multiplicative but over 2, so it stays in V^good (Def 3.1 (b), (c); Thm 1.10 allows bad reduction over 2l)
  asserted  : F/F_mod Galois ✓ | √-1, E[6] ⊆ F ✓ | E[5] ⊆ F ✓ | SL2 ⊆ image ✓ | stable red. ✓ | C_K K-core ✓
  labels    : F_l = {-6..6}  |F_l| = {0..6}  F_l^* = {1..6}
  symmetries: F_l^{×±} (additive, 0±ell-HT)  acts on F_l;  F_l^* (multiplicative, 0NF-HT) acts on F_l^*

Validation report: 0 error(s), 0 warning(s), 19 checks
  [  ok ] LPrime                 l = 13 prime, l* = 6, l± = 7   ([IUTchI] Def 3.1 (c))
  [  ok ] Fmod                   F_mod ⊆ F ⊆ K with [F_mod:Q]=1, [F:Q]≤276480, [K:Q]≤7245987840   ([IUTchI] Def 3.1)
  [  ok ] VBadMod                |V^bad_mod| = 3 odd primes of bad multiplicative reduction (2 is bad but excluded from V^bad_mod)   ([IUTchI] Def 3.1 (b))
  [info ] FBarOverF              asserted (not machine-checkable here): F Galois over F_mod of degree prime to l   ([IUTchI] Def 3.1 (b))
  [info ] FBarOverF              asserted (not machine-checkable here): √-1 ∈ F, and the 2·3-torsion points of E_F are rational over F   ([IUTchI] Def 3.1 (a), (b))
  [info ] KField                 asserted (not machine-checkable here): Im(Gal(K/F) → GL2(F_l)) ⊇ SL2(F_l)   ([IUTchI] Def 3.1 (c))
  [info ] XF                     asserted (not machine-checkable here): E_F has stable reduction at all nonarchimedean valuations of F   ([IUTchI] Def 3.1 (b))
  [info ] CK                     asserted (not machine-checkable here): C_K is a K-core (no Shimura-curve coverings)   ([IUTchI] Def 3.1 (d); [CanLift] Rem 2.1.1)
  [  ok ] EpsilonUnder           ε_ = the cusp ±1 of C_K (nonzero; the canonical generator at every v ∈ V^bad)   ([IUTchI] Def 3.1 (f))
  [  ok ] FPrimeStrip            F-prime-strip 'BoxPlus' well-formed (Frobenius-like, holomorphic)   ([IUTchI] Fig I1.2)
  [  ok ] DPrimeStrip            D-prime-strip 'CircDotPm' well-formed (étale-like, holomorphic)   ([IUTchI] Fig I1.2)
  [  ok ] FPrimeStrip            F-prime-strip 'Gt' well-formed (Frobenius-like, holomorphic)   ([IUTchI] Fig I1.2)
  [  ok ] FPrimeStrip            F-prime-strip 'CircAst' well-formed (Frobenius-like, holomorphic)   ([IUTchI] Fig I1.2)
  [  ok ] FMonoGlobPrimeStrip    FΠ-prime-strip 'MonoGlobMod' well-formed (Frobenius-like, mono-analytic)   ([IUTchI] Fig I1.2)
  [  ok ] FMonoGlobPrimeStrip    FΠ-prime-strip 'MonoGlobTht' well-formed (Frobenius-like, mono-analytic)   ([IUTchI] Fig I1.2)
  [  ok ] FMonoGlobValUnitsModTorsPrimeStrip FΠ→×μ-prime-strip 'Triangle' well-formed (Frobenius-like, mono-analytic)   ([IUTchI] Fig I1.2)
  [  ok ] FMonoGlobValUnitsModTorsPrimeStrip FΠ→×μ-prime-strip 'Gau' well-formed (Frobenius-like, mono-analytic)   ([IUTchI] Fig I1.2)
  [  ok ] FMonoGlobValUnitsModTorsPrimeStrip FΠ→×μ-prime-strip 'LGPTorsor' well-formed (Frobenius-like, mono-analytic)   ([IUTchI] Fig I1.2)
  [  ok ] ThetaPmEllNFHodgeTheater Dag at (0,0) is a well-formed 0±ell NF-Hodge theater (universe #1)   ([IUTchI] Def 6.13)
```

14 · Assemble a Hodge theater

The Θ±ell NF-Hodge theater at (0,0), built in the five stages of [IUTchI] §4–§6: the capsule F_T, the Θ±- and Θell-bridges, the capsule F_J, the Θ- and NF-bridges, the gluing 0, ⊞ ↦ >.

†HT, universe #1 [IUTchI] Def 6.13; [Alien] Fig 3.8

Θ±ell-Hodge theater

symmetry F_l^{⋊±}: additive

capsule F_T, t ∈ F_l (13 F-prime-strips)

-6	-5	-4	-3	-2	-1	0	+1	+2	+3	+4	+5	+6
#201	#202	#203	#204	#205	#206	#207	#208	#209	#210	#211	#212	#213

ΘNF-Hodge theater

symmetry F_l^*: multiplicative

capsule F_J, j ∈ F_l^* (6)

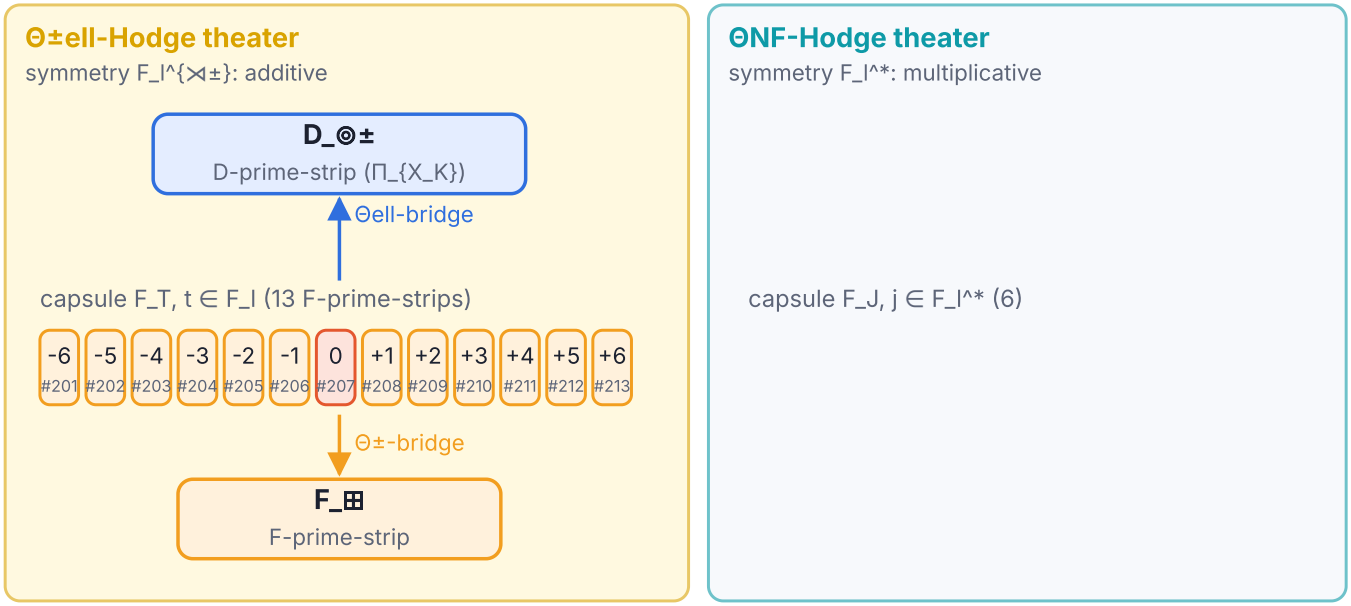
q-pilot {q_v}: −|log q| = -0.931 Θ-pilot {q_v^{j²}}: -14.118 (naive, ×l(l+1)/12)

⊩-strips: F⊩_Δ (q-pilot) · F⊩_env · F⊩_gau · F⊩_LGP (Θ-pilot) · F⊩_lgp

carries log-shells (codomain of a log-link)

stage 1 of 5 — the capsule F_T (t ∈ F_l, distinct basepoints) — **[IUT I]** Def 6.4 p. 162, Def 5.5 p. 151, Def 6.13 p. 182; **[Alien]** Fig 3.8 p. 67

†HT, universe #1 [IUTchI] Def 6.13; [Alien] Fig 3.8



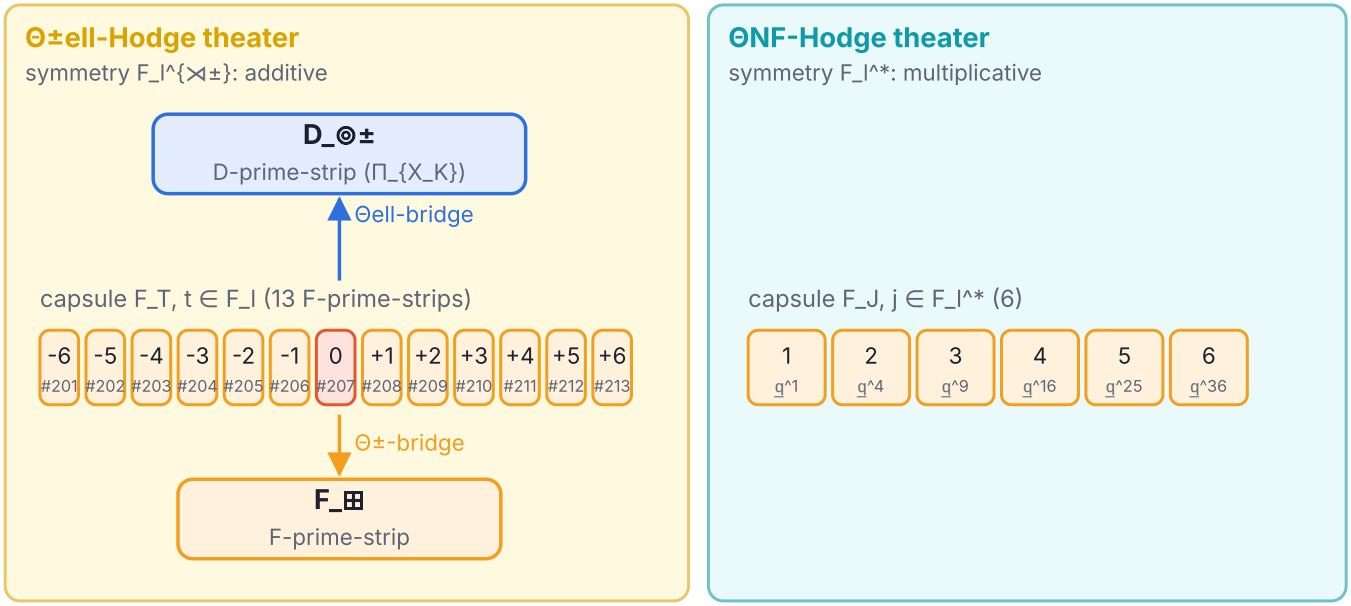
q-pilot {q_v}: −|log q| = -0.931 Θ-pilot {q_v^{j²}}: -14.118 (naive, ×l(l+1)/12)

Ⅱ⊢-strips: FⅡ⊢_Δ (q-pilot) · FⅡ⊢_env · FⅡ⊢_gau · FⅡ⊢_LGP (Θ-pilot) · FⅡ⊢_lgp

carries log-shells (codomain of a log-link)

stage 2 of 5 — the Θ±-bridge F_T → F_⊞ and the Θell-bridge D_T → D_◎± — **[IUT I]** Def 6.4 p. 162, Def 5.5 p. 151, Def 6.13 p. 182; **[Alien]** Fig 3.8 p. 67

†HT, universe #1 [IUTchI] Def 6.13; [Alien] Fig 3.8



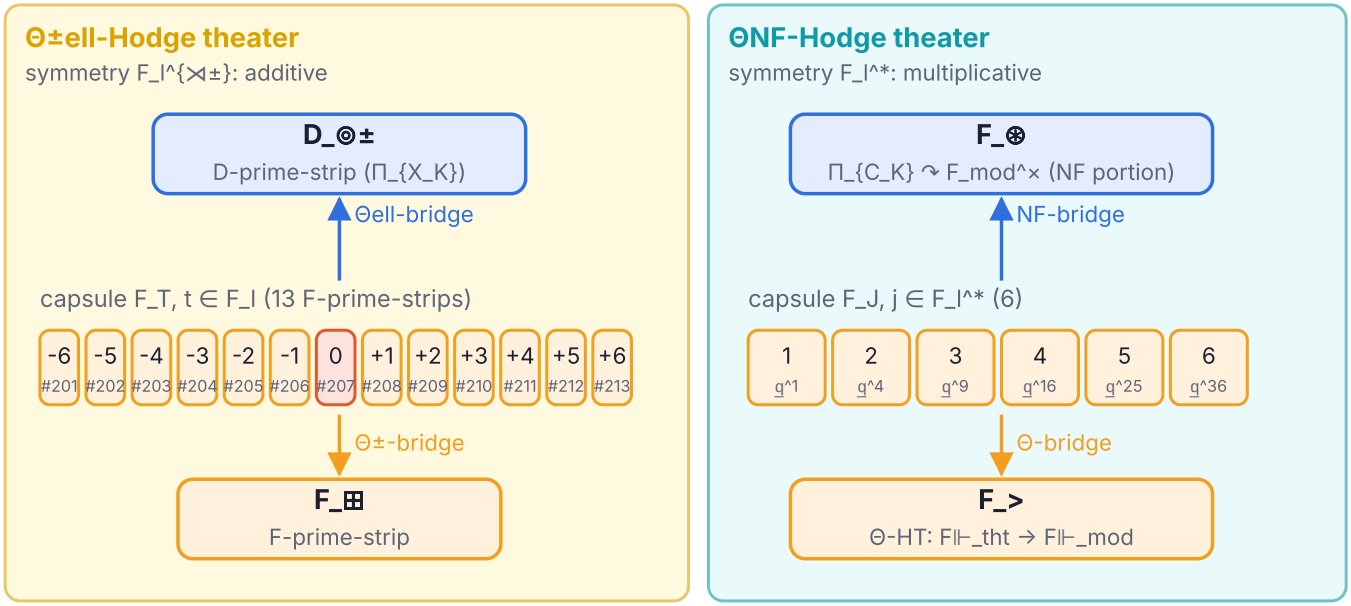
q-pilot {q_v}: −|log q| = -0.931 Θ-pilot {q_v^{j²}}: -14.118 (naive, ×l(l+1)/12)

Ⅱ⊢-strips: FⅡ⊢_Δ (q-pilot) · FⅡ⊢_env · FⅡ⊢_gau · FⅡ⊢_LGP (Θ-pilot) · FⅡ⊢_lgp

carries log-shells (codomain of a log-link)

stage 3 of 5 — the capsule F_J (j ∈ F_l^*) — **[IUT I]** Def 6.4 p. 162, Def 5.5 p. 151, Def 6.13 p. 182; **[Alien]** Fig 3.8 p. 67

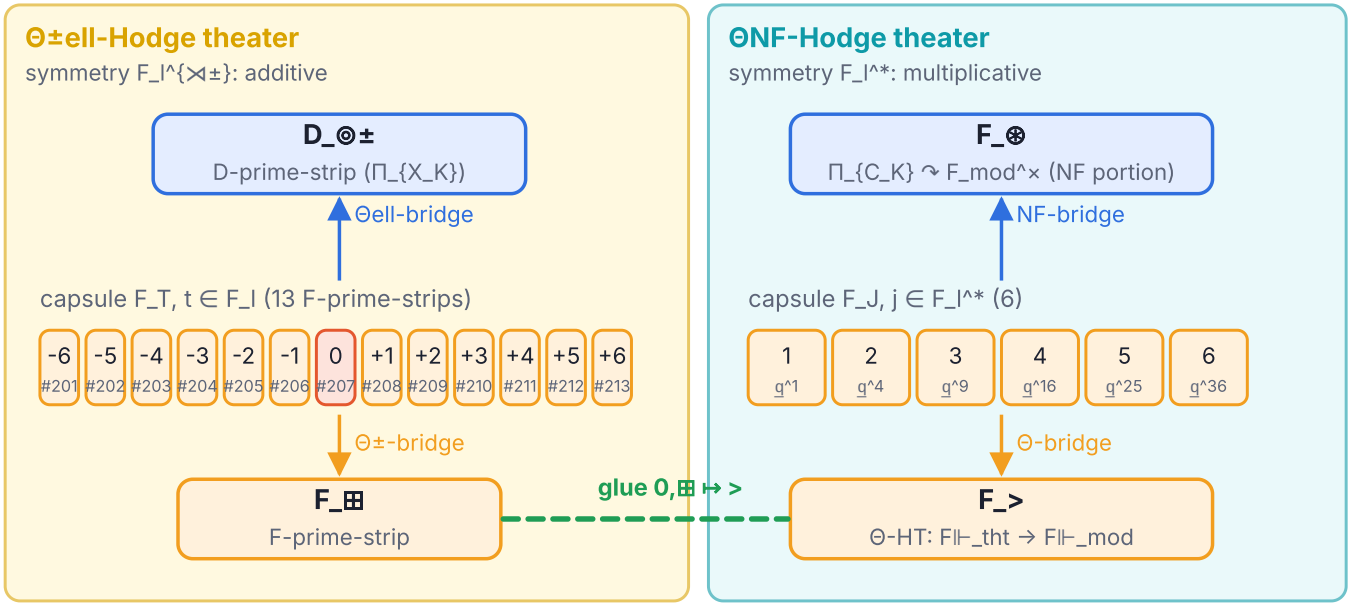
†HT, universe #1 [IUTchI] Def 6.13; [Alien] Fig 3.8



q-pilot {q_v}: −|log q| = -0.931 Θ-pilot {q_v^{j²}}: -14.118 (naive, ×l(l+1)/12)
IH-strips: FIH_Δ (q-pilot) · FIH_env · FIH_gau · FIH_LGP (Θ-pilot) · FIH_lgp
carries log-shells (codomain of a log-link)

stage 4 of 5 — the Θ-bridge F_J → F_> and the NF-bridge F_J → F_⊛ — **[IUT I]** Def 6.4 p. 162, Def 5.5 p. 151, Def 6.13 p. 182; **[Alien]** Fig 3.8 p. 67

†HT, universe #1 [IUTchI] Def 6.13; [Alien] Fig 3.8

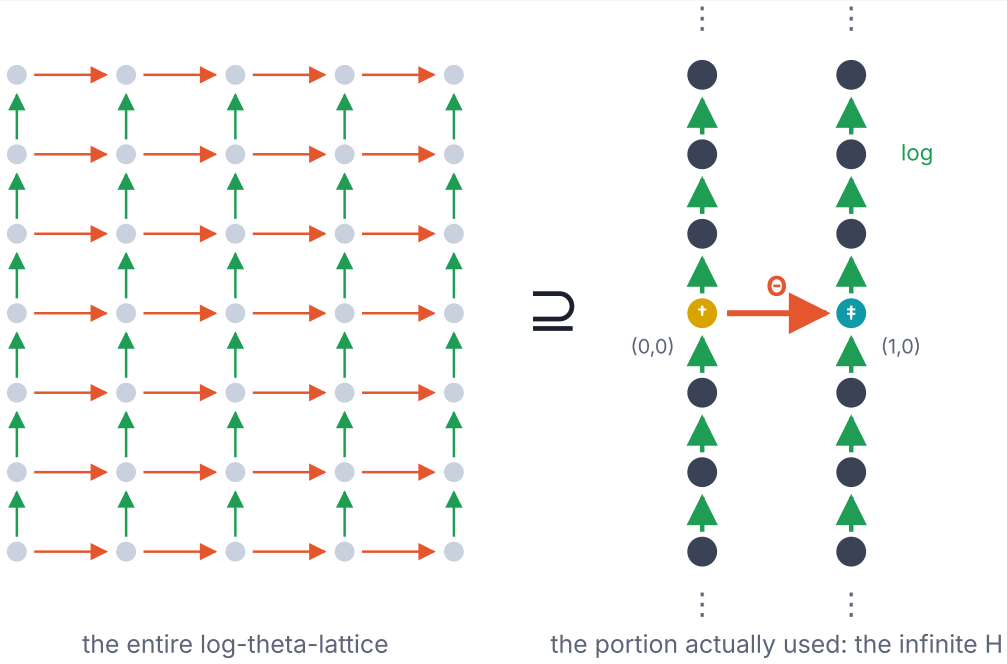


q-pilot {q_v}: −|log q| = -0.931 Θ-pilot {q_v^{j²}}: -14.118 (naive, ×l(l+1)/12)
IH-strips: FIH_Δ (q-pilot) · FIH_env · FIH_gau · FIH_LGP (Θ-pilot) · FIH_lgp
carries log-shells (codomain of a log-link)

stage 5 of 5 — the gluing 0, ⊞ ↦ > along F_l ⊇ F_l^× ↠ F_l^* — **[IUT I]** Def 6.4 p. 162, Def 5.5 p. 151, Def 6.13 p. 182; **[Alien]** Fig 3.8 p. 67

15 · Walk the log-theta-lattice

`iut walk --rows 2 --example 2` — the infinite H on columns 0 and 1 with the Θ×μ_LGP-link (0,0) → (1,0), the Hodge theater at the focus, the alien copies across the Θ-link, the log-shells and the multiradial log-volume estimate.



the entire log-theta-lattice ⊇ the infinite H actually used — **[IUT III]** Def 1.4 p. 45; **[Alien]** Fig 3.6 p. 60; **[EssLgc]** §3.3 p. 86

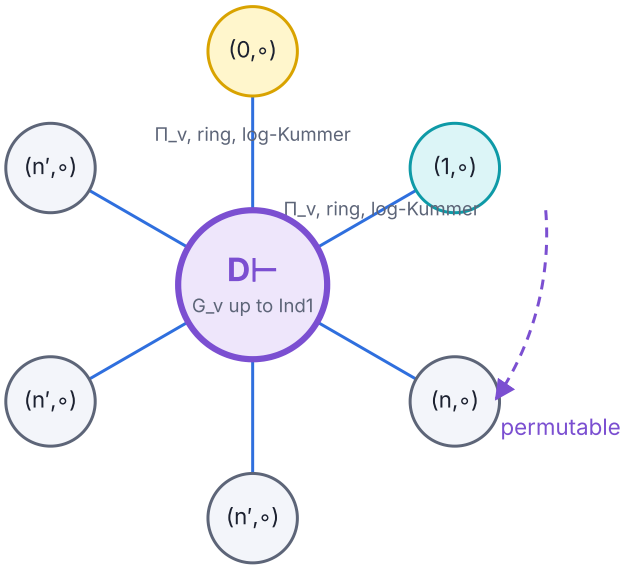
Θ-values acting on tensor packets of log-shells over processions [Alien] Fig 3.16–3.18; [IUTchIII] Thm 3.11



Each row is a tensor packet of mono-analytic log-shells (zero-mass containers) acted on by the theta value q^{j^2} and a copy of F_mod^×, up to (Ind1)–(Ind3). The input is only an abstract F⊢×μ-prime-strip (∧-Input, [EssLgc] §3.6), so the algorithm applies to †HT and ‡HT alike.

the multiradial representation of the Θ-pilot — **[IUT III]** Thm 3.11 p. 153; **[Alien]** Fig 3.16 –3.18 p. 103

étale-picture: every column (n,∘) is radial data over the one bi-coric core D⊢ [Alien] Fig 3.5, 3.12



Multiradial = the radial algorithm is compatible with arbitrary permutations of the spokes (parallel transport over the coric data); uniradial data, e.g. holomorphic structures, is not ([Alien] Fig 3.1).

the étale-picture: every column (n,∘) around the one bi-coric core D⊢ — **[IUT I]** Cor 6.10 p. 170; **[IUT III]** Cor 3.12 p. 173; **[Alien]** Fig 3.5 p. 57

```
Initial Θ-data  (F̃/F, X_F, l, C_K, V, V^bad_mod, ε_)      [IUTchI] Def 3.1
  name      : abc(1,2400,2401) l=13
  curve     : Frey y^2=x(x-1)(x+2400)
  fields    : F_mod ⊆ F ⊆ K = F(E[l]) ⊆ F̃     [F_mod:Q]=1  [F:Q]≤276480  [K:Q]≤7245987840
  l         : 13   (l* = 6, l± = 7)   ε_ = the cusp ±1
  log(q)    : 24.2023   |log(q)| = log(q)/2l = 0.9309   ht_E = log(q)/6 = 4.0337
  log(f)    : 4.6540 over V^bad_mod (the f of Thm 1.10); 5.3471 over every bad place   log(d^Ftpd) = 0.0000   h(j) ≈ 21.112
  log(d^K)  : ≈ 10.4770 (an estimate; ≤ 30.7839 by Thm 1.10, Step (ii))
```

```
  V ≅ V_mod : 6 valuations
    name   kind  p    e/f_m e/f_K reduction  ord(q)  V^bad_mod
    v2     non   2    1/1   ≥26/≥1 bad mult.  2       -
    v3     non   3    1/1   ≥26/≥1 bad mult.  2       yes
    v5     non   5    1/1   ≥52/≥1 bad mult.  4       yes
    v7     non   7    1/1   195/≥12 bad mult.  8       yes
    v13    non   13   1/1   ≥12/≥1 good       0       -
    vinf   arc   -    -     -     -          -       -
    v2: bad multiplicative but over 2, so it stays in V^good (Def 3.1 (b), (c); Thm 1.10 allows bad reduction over 2l)
  asserted  : F/F_mod Galois ✓ | V-1, E[6] ⊆ F ✓ | E[5] ⊆ F ✓ | SL2 ⊆ image ✓ | stable red. ✓ | C_K K-core ✓
  labels    : F_l = {-6..6}  |F_l| = {0..6}  F_l^* = {1..6}
  symmetries: F_l^{×±} (additive, 0±ell-HT)  acts on F_l;  F_l^* (multiplicative, 0NF-HT) acts on F_l^*

log-theta-lattice (0×μ_LGP-link horizontal; vertical = log-links)     [IUTchIII] Def 1.4, 3.8 (iii)
   m\n    +0        +1
   +2   •       •
          ↑         ↑
   +1   •       •
          ↑         ↑
   +0  [†•]—0→ ‡•
          ↑         ↑
   -1   •       •
          ↑         ↑
   -2   •       •

  • = 0±ell NF-Hodge theater (its own universe / label)   † = (0,0)   ‡ = (1,0)   [ ] = focus
  ↑ = log-link (vertically coric: D-prime-strips shared)   —0→ = 0×μ_LGP-link (horizontally coric: F⊢×μ shared)
  10 vertices, 9 links. The diagram does NOT commute ([IUTchIII] Rem 1.4.1).

0±ell NF-Hodge theater  †HT  at (n,m) = (0,0)   universe #1   log-iterate 2
  [ThetaPmEllNFHodgeTheater] = 0±ell-HT  ∪_{0,⊠ ↦ >}  0NF-HT                       [IUTchI] Def 6.13 (iii)

  0±ell-HT  (symmetry F_l^{×±}, additive / geometric, conjugate synchronization)
    capsule F_T, t ∈ F_l :  [-6] [-5] [-4] [-3] [-2] [-1] [+0] [+1] [+2] [+3] [+4] [+5] [+6]
    0±-bridge  F_T  → F_⊠   ✓        0ell-bridge  D_T → D_⊛± (Π_{X_K})   ✓
  0NF-HT    (symmetry F_l^*, multiplicative / arithmetic)
    capsule F_J, j ∈ F_l^*:  [1] [2] [3] [4] [5] [6]
    0-bridge   F_J  → F_>   ✓        NF-bridge    F_J → F_⊛ (Π_{C_K} ↷ F_mod^×)   ✓
  gluing 0,⊠ ↦ > : ✓    (F_l ⊇ F_l^× ↠ F_l^* ,  ±j ↦ j)

          0±ell-HT                                  0NF-HT
      D_⊛± (Π_{X_K})                            F_⊛ (Π_{C_K}, F_mod^×)
          ↑ 0ell-bridge                             ↑ NF-bridge
   [-l*]…[-1][0][+1]…[+l*]  = F_T    ↦ (0,⊠ ↦ >)    [1]…[l*] = F_J
          ↓ 0±-bridge                               ↓ 0-bridge
         F_⊠  ═══════════════ identified ═══════  F_>  ⊇  F⊩-_tht ← 0-Hodge theater → F⊩-_mod

  0-Hodge theater: 0__v evaluated at j ∈ F_l^*:  q_^1, q_^4, q_^9, q_^16, q_^25, q_^36
  q-pilot  {q__v}            log-vol -|log(q)| = -0.9309    (in F⊩-_Δ)
  0-pilot  {q__v^{j²}}_{j≤l*} naive log-vol  = -14.1180    (in F⊩-_gau / F⊩-_LGP; = l(l+1)/12 · (-|log q|))

  D-prime-strip  'CircDotPm'  [DPrimeStrip]  étale-like, holomorphic
    v2     Π_v#271002 G_v#271001
    v3     Π_v#271012 G_v#271011
    v5     Π_v#271022 G_v#271021
    v7     Π_v#271032 G_v#271031
    v13    Π_v#271042 G_v#271041
    vinf   Π_v#271052 G_v#271051
  F-prime-strip  'BoxPlus'  [FPrimeStrip]  Frobenius-like, holomorphic
    v2     Π_v#270002 G_v#270001 O× #170003 O▶{} ×-Kummer
    v3     Π_v#270012 G_v#270011 O× #170013 O▶{q_^1} ×-Kummer
    v5     Π_v#270022 G_v#270021 O× #170023 O▶{q_^1} ×-Kummer
    v7     Π_v#270032 G_v#270031 O× #170033 O▶{q_^1} ×-Kummer
    v13    Π_v#270042 G_v#270041 O× #170043 O▶{} ×-Kummer
    vinf   Π_v#270052 G_v#270051 O× #170053 O▶{} ×-Kummer
  F-prime-strip  'Gt'  [FPrimeStrip]  Frobenius-like, holomorphic
    v2     Π_v#270002 G_v#270001 O× #172003 O▶{} ×-Kummer
    v3     Π_v#270012 G_v#270011 O× #172013 O▶{q_^1} ×-Kummer
    v5     Π_v#270022 G_v#270021 O× #172023 O▶{q_^1} ×-Kummer
    v7     Π_v#270032 G_v#270031 O× #172033 O▶{q_^1} ×-Kummer
    v13    Π_v#270042 G_v#270041 O× #172043 O▶{} ×-Kummer
    vinf   Π_v#270052 G_v#270051 O× #172053 O▶{} ×-Kummer
  F-prime-strip  'CircAst'  [FPrimeStrip]  Frobenius-like, holomorphic
    v2     Π_v#273002 G_v#273001 O× #173003 O▶{} ×-Kummer
    v3     Π_v#273012 G_v#273011 O× #173013 O▶{q_^1} ×-Kummer
    v5     Π_v#273022 G_v#273021 O× #173023 O▶{q_^1} ×-Kummer
    v7     Π_v#273032 G_v#273031 O× #173033 O▶{q_^1} ×-Kummer
    v13    Π_v#273042 G_v#273041 O× #173043 O▶{} ×-Kummer
    vinf   Π_v#273052 G_v#273051 O× #173053 O▶{} ×-Kummer
  F⊩-prime-strip  'MonoGlobMod'  [FMonoGlobPrimeStrip]  Frobenius-like, mono-analytic, + C⊩ (global realified Frobenioid)
    pilot degree: -0.9309
    v2      (no Π)  G_v#274001 O× #174003 O▶{} split
    v3      (no Π)  G_v#274011 O× #174013 O▶{q_^1} split
    v5      (no Π)  G_v#274021 O× #174023 O▶{q_^1} split
    v7      (no Π)  G_v#274031 O× #174033 O▶{q_^1} split
    v13     (no Π)  G_v#274041 O× #174043 O▶{} split
    vinf    (no Π)  G_v#274051 O× #174053 O▶{} split
  F⊩-prime-strip  'MonoGlobTht'  [FMonoGlobPrimeStrip]  Frobenius-like, mono-analytic, + C⊩ (global realified Frobenioid)
    pilot degree: -14.1180
    v2      (no Π)  G_v#275001 O× #175003 O▶{} split
    v3      (no Π)  G_v#275011 O× #175013 O▶{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} split
    v5      (no Π)  G_v#275021 O× #175023 O▶{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} split
    v7      (no Π)  G_v#275031 O× #175033 O▶{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} split
    v13     (no Π)  G_v#275041 O× #175043 O▶{} split
    vinf    (no Π)  G_v#275051 O× #175053 O▶{} split
  F⊩▶×μ-prime-strip  'Triangle'  [FMonoGlobValUnitsModTorsPrimeStrip]  Frobenius-like, mono-analytic, + C⊩ (global realified Frobenioid)
    pilot degree: -0.9309
    v2      (no Π)  G_v#280001 O×μ#180003 O▶{} split ×μ-Kummer(Ism)
    v3      (no Π)  G_v#280011 O×μ#180013 O▶{q_^1} split ×μ-Kummer(Ism)
    v5      (no Π)  G_v#280021 O×μ#180023 O▶{q_^1} split ×μ-Kummer(Ism)
    v7      (no Π)  G_v#280031 O×μ#180033 O▶{q_^1} split ×μ-Kummer(Ism)
    v13     (no Π)  G_v#280041 O×μ#180043 O▶{} split ×μ-Kummer(Ism)
```

```
    vinf    (no Π)  G_v#280051 O×μ#180053 O▶{} split ×μ-Kummer(Ism)
  F⊩▶×μ-prime-strip  'LGPTorsor'  [FMonoGlobValUnitsModTorsPrimeStrip]  Frobenius-like, mono-analytic, + C⊩ (global realified Frobenioid)
    pilot degree: -14.1180
    v2      (no Π)  G_v#283001 O×μ#183003 O▶{} split ×μ-Kummer(Ism)
    v3      (no Π)  G_v#283011 O×μ#183013 O▶{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} split ×μ-Kummer(Ism)
    v5      (no Π)  G_v#283021 O×μ#183023 O▶{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} split ×μ-Kummer(Ism)
    v7      (no Π)  G_v#283031 O×μ#183033 O▶{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} split ×μ-Kummer(Ism)
    v13     (no Π)  G_v#283041 O×μ#183043 O▶{} split ×μ-Kummer(Ism)
    vinf    (no Π)  G_v#283051 O×μ#183053 O▶{} split ×μ-Kummer(Ism)
  log-shells I_v = (p*)^{-1}·log(O×_{K_v}) (codomain of log-link from (0,-1)):
    v2     p* = 4    I_v = p^{-25/26}·O_v   μ_log(I_v) = +1.3063   holomorphic Frobenius-like
    v3     p* = 3    I_v = p^{-6/13}·O_v   μ_log(I_v) = +1.0141   holomorphic Frobenius-like
    v5     p* = 5    I_v = p^{-19/26}·O_v   μ_log(I_v) = +1.5475   holomorphic Frobenius-like
    v7     p* = 7    I_v = p^{-54/65}·O_v   μ_log(I_v) = +1.9359   holomorphic Frobenius-like
    v13    p* = 13   I_v = p^{-5/6}·O_v   μ_log(I_v) = +2.1375   holomorphic Frobenius-like
    vinf   I_v = closed disc of radius π ⊆ C

0±ell NF-Hodge theater  HT  at (n,m) = (0,1)   universe #4   log-iterate 3
  [ThetaPmEllNFHodgeTheater] = 0±ell-HT  ω_{0,⊠ ↦ >}  0NF-HT                      [IUTchI] Def 6.13 (iii)

  0±ell-HT  (symmetry F_l^{×±}, additive / geometric, conjugate synchronization)
    capsule F_T, t ∈ F_l :  [-6] [-5] [-4] [-3] [-2] [-1] [+0] [+1] [+2] [+3] [+4] [+5] [+6]
    0±-bridge  F_T  → F_⊠   ✓        0ell-bridge  D_T → D_◎± (Π_{X_K})   ✓
  0NF-HT    (symmetry F_l^*, multiplicative / arithmetic)
    capsule F_J, j ∈ F_l^*:  [1] [2] [3] [4] [5] [6]
    0-bridge   F_J  → F_>   ✓        NF-bridge    F_J → F_⊛ (Π_{C_K} ↷ F_mod^×)   ✓
  gluing 0,⊠ ↦ > : ✓    (F_l ⊇ F_l^× ↠ F_l^* ,  ±j ↦ j)

          0±ell-HT                                  0NF-HT
      D_◎± (Π_{X_K})                              F_⊛ (Π_{C_K}, F_mod^×)
          ↑ 0ell-bridge                               ↑ NF-bridge
   [-l*]…[-1][0][+1]…[+l*]  = F_T    ↦ (0,⊠ ↦ >)    [1]…[l*] = F_J
          ↓ 0±-bridge                                 ↓ 0-bridge
        F_⊠  ═══════════════════ identified ═══════  F_>  ⊇  F⊩_tht ← 0-Hodge theater → F⊩_mod

  0-Hodge theater: 0_v evaluated at j ∈ F_l^*:  q_^1, q_^4, q_^9, q_^16, q_^25, q_^36
  q-pilot  {q_,v}           log-vol -|log(q)| = -0.9309    (in F⊩_Δ)
  0-pilot  {q_,v^{j²}}_{j≤l*} naive log-vol  = -14.1180    (in F⊩_gau / F⊩_LGP; = l(l+1)/12 · (-|log q|))

  D-prime-strip  'CircDotPm'  [DPrimeStrip]  étale-like, holomorphic
    v2     Π_v#271002 G_v#271001
    v3     Π_v#271012 G_v#271011
    v5     Π_v#271022 G_v#271021
    v7     Π_v#271032 G_v#271031
    v13    Π_v#271042 G_v#271041
    vinf   Π_v#271052 G_v#271051
  F-prime-strip  'BoxPlus'  [FPrimeStrip]  Frobenius-like, holomorphic
    v2     Π_v#270002 G_v#270001 O× #470003 O▶{} ×-Kummer
    v3     Π_v#270012 G_v#270011 O× #470013 O▶{q_^1} ×-Kummer
    v5     Π_v#270022 G_v#270021 O× #470023 O▶{q_^1} ×-Kummer
    v7     Π_v#270032 G_v#270031 O× #470033 O▶{q_^1} ×-Kummer
    v13    Π_v#270042 G_v#270041 O× #470043 O▶{} ×-Kummer
    vinf   Π_v#270052 G_v#270051 O× #470053 O▶{} ×-Kummer
  F-prime-strip  'Gt'  [FPrimeStrip]  Frobenius-like, holomorphic
    v2     Π_v#270002 G_v#270001 O× #472003 O▶{} ×-Kummer
    v3     Π_v#270012 G_v#270011 O× #472013 O▶{q_^1} ×-Kummer
    v5     Π_v#270022 G_v#270021 O× #472023 O▶{q_^1} ×-Kummer
    v7     Π_v#270032 G_v#270031 O× #472033 O▶{q_^1} ×-Kummer
    v13    Π_v#270042 G_v#270041 O× #472043 O▶{} ×-Kummer
    vinf   Π_v#270052 G_v#270051 O× #472053 O▶{} ×-Kummer
  F-prime-strip  'CircAst'  [FPrimeStrip]  Frobenius-like, holomorphic
    v2     Π_v#273002 G_v#273001 O× #473003 O▶{} ×-Kummer
    v3     Π_v#273012 G_v#273011 O× #473013 O▶{q_^1} ×-Kummer
    v5     Π_v#273022 G_v#273021 O× #473023 O▶{q_^1} ×-Kummer
    v7     Π_v#273032 G_v#273031 O× #473033 O▶{q_^1} ×-Kummer
    v13    Π_v#273042 G_v#273041 O× #473043 O▶{} ×-Kummer
    vinf   Π_v#273052 G_v#273051 O× #473053 O▶{} ×-Kummer
  F⊩-prime-strip  'MonoGlobMod'  [FMonoGlobPrimeStrip]  Frobenius-like, mono-analytic, + C⊩ (global realified Frobenioid)
    pilot degree: -0.9309
    v2      (no Π)  G_v#274001 O× #474003 O▶{} split
    v3      (no Π)  G_v#274011 O× #474013 O▶{q_^1} split
    v5      (no Π)  G_v#274021 O× #474023 O▶{q_^1} split
    v7      (no Π)  G_v#274031 O× #474033 O▶{q_^1} split
    v13     (no Π)  G_v#274041 O× #474043 O▶{} split
    vinf    (no Π)  G_v#274051 O× #474053 O▶{} split
  F⊩-prime-strip  'MonoGlobTht'  [FMonoGlobPrimeStrip]  Frobenius-like, mono-analytic, + C⊩ (global realified Frobenioid)
    pilot degree: -14.1180
    v2      (no Π)  G_v#275001 O× #475003 O▶{} split
    v3      (no Π)  G_v#275011 O× #475013 O▶{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} split
    v5      (no Π)  G_v#275021 O× #475023 O▶{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} split
    v7      (no Π)  G_v#275031 O× #475033 O▶{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} split
    v13     (no Π)  G_v#275041 O× #475043 O▶{} split
    vinf    (no Π)  G_v#275051 O× #475053 O▶{} split
  F⊩▶×μ-prime-strip  'Triangle'  [FMonoGlobValUnitsModTorsPrimeStrip]  Frobenius-like, mono-analytic, + C⊩ (global realified Frobenioid)
    pilot degree: -0.9309
    v2      (no Π)  G_v#280001 O×μ#480003 O▶{} split ×μ-Kummer(Ism)
    v3      (no Π)  G_v#280011 O×μ#480013 O▶{q_^1} split ×μ-Kummer(Ism)
    v5      (no Π)  G_v#280021 O×μ#480023 O▶{q_^1} split ×μ-Kummer(Ism)
    v7      (no Π)  G_v#280031 O×μ#480033 O▶{q_^1} split ×μ-Kummer(Ism)
    v13     (no Π)  G_v#280041 O×μ#480043 O▶{} split ×μ-Kummer(Ism)
    vinf    (no Π)  G_v#280051 O×μ#480053 O▶{} split ×μ-Kummer(Ism)
  F⊩▶×μ-prime-strip  'LGPTorsor'  [FMonoGlobValUnitsModTorsPrimeStrip]  Frobenius-like, mono-analytic, + C⊩ (global realified Frobenioid)
    pilot degree: -14.1180
    v2      (no Π)  G_v#283001 O×μ#483003 O▶{} split ×μ-Kummer(Ism)
    v3      (no Π)  G_v#283011 O×μ#483013 O▶{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} split ×μ-Kummer(Ism)
    v5      (no Π)  G_v#283021 O×μ#483023 O▶{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} split ×μ-Kummer(Ism)
    v7      (no Π)  G_v#283031 O×μ#483033 O▶{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} split ×μ-Kummer(Ism)
    v13     (no Π)  G_v#283041 O×μ#483043 O▶{} split ×μ-Kummer(Ism)
    vinf    (no Π)  G_v#283051 O×μ#483053 O▶{} split ×μ-Kummer(Ism)
  log-shells I_v = (p*)^{-1}·log(O×_{K_v}) (codomain of log-link from (0,0)):
    v2     p* = 4    I_v = p^{-25/26}·O_v   μ_log(I_v) = +1.3063   holomorphic Frobenius-like
    v3     p* = 3    I_v = p^{-6/13}·O_v   μ_log(I_v) = +1.0141   holomorphic Frobenius-like
```

```
    v5     p* = 5    I_v = p^{-19/26}·O_v   μ_log(I_v) = +1.5475   holomorphic Frobenius-like
    v7     p* = 7    I_v = p^{-54/65}·O_v   μ_log(I_v) = +1.9359   holomorphic Frobenius-like
    v13    p* = 13   I_v = p^{-5/6}·O_v   μ_log(I_v) = +2.1375   holomorphic Frobenius-like
    vinf   I_v = closed disc of radius π ⊆ C

Mutually alien copies across the 0×μ_LGP-link  (0,0) → (1,0)        [Alien] §1, §3.6; [IUTchII] Cor 4.10
  †HT = Dag (universe #1)        ‡HT = Ddag (universe #8)

  datum          † (domain)  F⊩-_LGP       ‡ (codomain) F⊩-_Δ       status
  -------------- ---------------------------- ---------------------------- ------------------------------
  v2             Π_v#270002                   Π_v#670002                   ALIEN: ring/holomorphic structure not shared
                 G_v#283001                   G_v#283001                   shared (up to Aut(G_v): Ind1)
                 O×μ#183003                   O×μ#183003                 shared (up to Ism: Ind2)
                 O► (good)                  O► (good)                  -
  v3             Π_v#270012                   Π_v#670012                   ALIEN: ring/holomorphic structure not shared
                 G_v#283011                   G_v#283011                   shared (up to Aut(G_v): Ind1)
                 O×μ#183013                   O×μ#183013                 shared (up to Ism: Ind2)
                 q_^{1,4,9,16,25,36}        q_^{1}                     DILATED: {q_^{j²}} ↦ {q_}
  v5             Π_v#270022                   Π_v#670022                   ALIEN: ring/holomorphic structure not shared
                 G_v#283021                   G_v#283021                   shared (up to Aut(G_v): Ind1)
                 O×μ#183023                   O×μ#183023                 shared (up to Ism: Ind2)
                 q_^{1,4,9,16,25,36}        q_^{1}                     DILATED: {q_^{j²}} ↦ {q_}
  v7             Π_v#270032                   Π_v#670032                   ALIEN: ring/holomorphic structure not shared
                 G_v#283031                   G_v#283031                   shared (up to Aut(G_v): Ind1)
                 O×μ#183033                   O×μ#183033                 shared (up to Ism: Ind2)
                 q_^{1,4,9,16,25,36}        q_^{1}                     DILATED: {q_^{j²}} ↦ {q_}
  v13            Π_v#270042                   Π_v#670042                   ALIEN: ring/holomorphic structure not shared
                 G_v#283041                   G_v#283041                   shared (up to Aut(G_v): Ind1)
                 O×μ#183043                   O×μ#183043                 shared (up to Ism: Ind2)
                 O► (good)                  O► (good)                  -
  vinf           Π_v#270052                   Π_v#670052                   ALIEN: ring/holomorphic structure not shared
                 G_v#283051                   G_v#283051                   shared (up to Aut(G_v): Ind1)
                 O×μ#183053                   O×μ#183053                 shared (up to Ism: Ind2)
                 O► (good)                  O► (good)                  -

  gluing data of the 0×μ_LGP-link ([IUTchII] Cor 4.10; [Alien] §3.6):
    (a0) G_v ↷ O×μ  of †F⊩-_LGP   ≅   G_v ↷ O×μ  of ‡F⊩-_Δ          — horizontally coric ✓
    (b0) {q_,v^{j²}}_{j=1..l*}   ↦   {q_,v}                            — value groups dilated by l(l+1)/12 = 15.167
    (c0) global realified Frobenioid  †C⊩  ≅  ‡C⊩                    — arithmetic degrees: † -14.1180  vs  ‡ -0.9309
    Π_v  NOT identified: ✓; ring structures NOT identified: ✓; full poly-isomorphism: ✓

  Kummer-detachment: to compare † and ‡ one passes to the étale-like D⊢-prime-strips (bi-coric) and
  reconstructs the Frobenius-like data up to (Ind1),(Ind2),(Ind3) — the multiradial representation.

Multiradial representation R^LGP_{0,◦} of the 0-pilot       [IUTchIII] Thm 3.11, Cor 3.12; [IUTchIV] Thm 1.10
  (a) tensor packets of log-shells over the processions S^±_{j+1} = {0..j}
  (b) splitting monoids of the LGP-monoids: {q_^{j²}}
  (c) copies (F_mod^×)_j of the global number field
  indeterminacies: (Ind1) perms of S^±_{j+1}, Aut(G_v)  (Ind2) Ism on O×μ  (Ind3) upper semi-compatibility

  j    |S±_{j+1}| (j+1)!     hull  (j+1)·log d^K   0-shift -(j²/2l)·log q
  0    1          1          10.4770                -0.0000
  1    2          2          20.9540                -0.9309
  2    3          6          31.4310                -3.7234
  3    4          24         41.9080                -8.3777
  4    5          120        52.3850                -14.8937
  5    6          720        62.8620                -23.2714
  6    7          5040       73.3390                -33.5108

  log(q) = 24.2023   |log(q)| = log(q)/2l = 0.9309   -|log(q)| = -0.9309
  Σ_{j=1}^{l*} [(j+1) log d^K - (j²/2l) log q] = 198.1713   ≈ (l²/48)(6 log d^K - log q) = 136.1146      [Alien] §3.7 (iv)
  procession-normalized (÷ l*), leading terms only:  hull 47.1465,  with the theta shift 33.0285   (an estimate, not a bound)
  -|log(θ)| ≤ B = 1530.5931: the bound of [IUTchIV] Thm 1.10, Steps (v)-(vii), with every term, summed over v_Q
  Cor 3.12 (an input, not re-proved here) gives B ≥ -|log(q)|:  1530.5931 ≥ -0.9309  → consistent;   C_0 = B/|log q| = 1644.29
  C_0 of the statement of Thm 1.10 (e*_mod = 2^12·3^3·5·e_mod, n_prm = 5·10^20 [ExpEst]) = 1.87999e+22  →  C_0 ≥ -1 ✓
  hence (1/6) log(q) ≤ (1 + 20 d_mod/l)(log d^Ftpd + log f^Ftpd) + 20(e*_mod·l + n_prm):  4.0337 ≤ 1e+22
  overall: the consequences of Cor 3.12 are consistent with this data

Étale-picture: the bi-coric D⊢-prime-strip at the centre, Hodge theaters as spokes   [IUTchIII] Cor 3.12; [IUTchI] Cor 6.10

                    (0,◦) column                      (1,◦) column
           R^LGP_{0,◦}  …  ↖        ↗  …  R^LGP_{1,◦}
                             \      /

                          ┌──────────────┐
                          │ D⊢-prime-strip │    {G_v}_{v∈V}  +  mono-analytic log-shells I⊢_v
                          │  (bi-coric)  │    shared by ALL 10 Hodge theaters built so far
                          └──────────────┘

                             /      \
           {n,m}D-HT  …  ↙          ↘  …  {n',m'}D-HT

  Every spoke is a D-0±ell NF-Hodge theater (étale-like, vertically coric); the Frobenius-like
  data of each vertex is recovered from the core only up to (Ind1),(Ind2),(Ind3).
  0-link mutations: Π_v (holomorphic) is NOT part of the core; G_v and O×μ are (up to Ind1, Ind2).

Validation report: 0 error(s), 0 warning(s), 12 checks
  [  ok ] LPrime                l = 13 prime, l* = 6, l± = 7   ([IUTchI] Def 3.1 (c))
  [  ok ] Fmod                  F_mod ⊆ F ⊆ K with [F_mod:Q]=1, [F:Q]≤276480, [K:Q]≤7245987840   ([IUTchI] Def 3.1)
  [  ok ] VBadMod               |V^bad_mod| = 3 odd primes of bad multiplicative reduction (2 is bad but excluded from V^bad_mod)   ([IUTchI] Def 3.1 (b))
  [info ] FBarOverF             asserted (not machine-checkable here): F Galois over F_mod of degree prime to l   ([IUTchI] Def 3.1 (b))
  [info ] FBarOverF             asserted (not machine-checkable here): √-1 ∈ F, and the 2·3-torsion points of E_F are rational over F   ([IUTchI] Def 3.1 (a), (b))
  [info ] KField                asserted (not machine-checkable here): Im(Gal(K/F) → GL2(F_l)) ⊇ SL2(F_l)   ([IUTchI] Def 3.1 (c))
  [info ] XF                    asserted (not machine-checkable here): E_F has stable reduction at all nonarchimedean valuations of F   ([IUTchI] Def 3.1 (b))
  [info ] CK                    asserted (not machine-checkable here): C_K is a K-core (no Shimura-curve coverings)   ([IUTchI] Def 3.1 (d); [CanLift] Rem 2.1.1)
  [  ok ] EpsilonUnder          ε_ = the cusp ±1 of C_K (nonzero; the canonical generator at every v ∈ V^bad)   ([IUTchI] Def 3.1 (f))
  [info ] InfiniteH             10 Hodge theaters, 8 log-links, 1 0-links (0×μ_LGP-link); only the row m=0 carries a 0-link: this is the 'infinite H' used in the proof
of Cor 3.12   ([EssLgc] §3.10; [Alien] §3.11)
  [info ] LogThetaLattice       the diagram is highly non-commutative: no square of links is required to commute   ([IUTchIII] Rem 1.4.1)
  [  ok ] LogThetaLattice       LGP-Gaussian log-theta-lattice well-formed   ([IUTchIII] Def 3.8 (iii))
```

16 · Species and mutations

`iut species` — the species table (which struct implements which species) and the mutation table (which function implements which mutation, and whether it is a core), with the mutation-history of the infinite H.

```
Species / mutations   ([IUTchIV] §3 Def 3.1, 3.3): the log-theta-lattice as a mutation-history

  species (C struct)                    notation                          ref
  InitialThetaData                      initial Θ-data                    [IUTchI] Def 3.1
  ThetaPmEllNFHodgeTheater              Θ±ell NF-Hodge theater            [IUTchI] Def 6.13
  DThetaPmEllNFHodgeTheater             D-Θ±ell NF-Hodge theater          [IUTchI] Def 6.13
  ThetaPmEllHodgeTheater                Θ±ell-Hodge theater               [IUTchI] Def 6.13
  ThetaNFHodgeTheater                   ΘNF-Hodge theater                 [IUTchI] Def 5.5
  ThetaHodgeTheater                     Θ-Hodge theater                   [IUTchI] Def 3.6
  DPrimeStrip                           D-prime-strip                     [IUTchI] Def 4.1
  DMonoPrimeStrip                       D⊢-prime-strip                    [IUTchI] Def 4.1
  FPrimeStrip                           F-prime-strip                     [IUTchI] Def 5.2
  FMonoPrimeStrip                       F⊢-prime-strip                    [IUTchI] Def 5.2
  FMonoUnitsModTorsPrimeStrip           F⊢×μ-prime-strip                  [IUTchII] Def 4.9
  FMonoValUnitsModTorsPrimeStrip        F⊢▶×μ-prime-strip                 [IUTchII] Def 4.9
  FMonoGlobPrimeStrip                   F⊩-prime-strip                    [IUTchI] Def 5.2
  FMonoGlobValUnitsModTorsPrimeStrip    F⊩▶×μ-prime-strip                 [IUTchII] Def 4.9
  LogShell                              log-shell                         [IUTchIII] Def 1.1
  TensorPacket                          tensor packet of log-shells       [IUTchIII] Prop 3.1
  QPilot                                q-pilot object                    [IUTchIII] Def 3.8
  ThetaPilot                            Θ-pilot object                    [IUTchIII] Def 3.8
  MultiradialRepresentation             multiradial representation R^LGP  [IUTchIII] Thm 3.11

  mutation (C function)          notation            domain                       → codomain                      core?
  LogLink                        log-link            ThetaPmEllNFHodgeTheater     → ThetaPmEllNFHodgeTheater      no
  ThetaUnitsModTorsLGPTorsorLink Θ×μ_LGP-link      ThetaPmEllNFHodgeTheater     → ThetaPmEllNFHodgeTheater      no
  ThetaUnitsModTorsLgpIdealLink  Θ×μ_lgp-link      ThetaPmEllNFHodgeTheater     → ThetaPmEllNFHodgeTheater      no
  ThetaUnitsModTorsGauLink       Θ×μ_gau-link      ThetaPmEllNFHodgeTheater     → ThetaPmEllNFHodgeTheater      no
  ThetaLink                      Θ-link              ThetaHodgeTheater            → ThetaHodgeTheater             no
  ForgetToD                      HT ↦ D-HT         ThetaPmEllNFHodgeTheater     → DThetaPmEllNFHodgeTheater     yes
  ForgetToDMono                  F ↦ D⊢          FPrimeStrip                  → DMonoPrimeStrip               yes
  ForgetToUnitsModTors           F⊩ ↦ F⊢×μ    FMonoGlobPrimeStrip          → FMonoUnitsModTorsPrimeStrip   yes
  LogShellConstruction           F ↦ I             FPrimeStrip                  → LogShell                      no
  EtaleLogShellConstruction      D ↦ I(D)          DPrimeStrip                  → LogShell                      yes
  TensorPacketConstruction       {I_t} ↦ ⊗       LogShell                     → TensorPacket                  no
  KummerDetachment               Frobenius-like ↦ étale-like FMonoGlobPrimeStrip          → DThetaPmEllNFHodgeTheater     no
  MultiradialAlgorithm           Thm 3.11            ThetaPilot                   → MultiradialRepresentation     no

  mutation-history Γ of the infinite H:   (0,m) →log (0,m+1),  (1,m) →log (1,m+1),  (0,0) →Θ (1,0)
  observable/core: D⊢-prime-strip (all arrows isomorphisms)  ⇒  the étale-picture.
```

17 · The whole pipeline

`iut pipeline --example 2` — the same Θ-data carried through every stage, from Def 3.1 to the height bound of [IUTchIV] Thm 1.10. Stage 5 has the bound of the proof of Thm 1.10 prime by prime; Cor 3.12 is an input, and the run reports whether the numbers are consistent with it. Under each stage are the placeholders that stand in for parts of the theory the program does not compute.

```
IUT pipeline   abc(1,2400,2401) l=13
  source: generated by the toolkit   curve: Frey y^2=x(x-1)(x+2400)   l = 13   rows ±2   horizontal: 0×μ_LGP-link

── stage 0 ── initial Θ-data ──────────────────────────────────────── [IUTchI] Def 3.1
Initial Θ-data  (F̄/F, X_F, l, C_K, V, V^bad_mod, ε_)      [IUTchI] Def 3.1
  name      : abc(1,2400,2401) l=13
  curve     : Frey y^2=x(x-1)(x+2400)
  fields    : F_mod ⊆ F ⊆ K = F(E[l]) ⊆ F̄     [F_mod:Q]=1  [F:Q]≤276480  [K:Q]≤7245987840
  l         : 13   (l* = 6, l± = 7)   ε_ = the cusp ±1
  log(q)    : 24.2023   |log(q)| = log(q)/2l = 0.9309   ht_E = log(q)/6 = 4.0337
  log(f)    : 4.6540 over V^bad_mod (the f of Thm 1.10); 5.3471 over every bad place   log(d^Ftpd) = 0.0000   h(j) ≈ 21.112
  log(d^K)  : ≈ 10.4770 (an estimate; ≤ 30.7839 by Thm 1.10, Step (ii))
  V ≅ V_mod : 6 valuations
    name   kind  p    e/f_m e/f_K reduction  ord(q)  V^bad_mod
    v2     non   2    1/1   ≥26/≥1 bad mult.  2       -
    v3     non   3    1/1   ≥26/≥1 bad mult.  2       yes
    v5     non   5    1/1   ≥52/≥1 bad mult.  4       yes
    v7     non   7    1/1   195/≥12 bad mult.  8       yes
    v13    non   13   1/1   ≥12/≥1 good       0       -
    vinf   arc   -    -     -     -          -       -
    v2: bad multiplicative but over 2, so it stays in V^good (Def 3.1 (b), (c); Thm 1.10 allows bad reduction over 2l)
  asserted  : F/F_mod Galois ✓ | V-1, E[6] ⊆ F ✓ | E[5] ⊆ F ✓ | SL2 ⊆ image ✓ | stable red. ✓ | C_K K-core ✓
  labels    : F_l = {-6..6}  |F_l| = {0..6}  F_l^* = {1..6}
  symmetries: F_l^{×±} (additive, 0±ell-HT)  acts on F_l;  F_l^* (multiplicative, 0NF-HT) acts on F_l^*
  ε_         : the cusp ±1 of C_K   ([IUTchI] Def 3.1 (f): from a nonzero element of Q ≅ Z/13Z)
  placeholders in this stage (2; `iut gaps` has the detail):
    ▒ approximate LocalFieldsOfK: The local fields K_v are known only through bounds
    ▒ open        GeneratorsFreyOnly: Initial Θ-data is generated only for Frey curves over Q

── stage 1 ── conditions (a)-(f) ──────────────────────────────────── [IUTchI] Def 3.1
  ✓ (a)  F is a number field with √-1 ∈ F; F̄ an algebraic closure of F
      asserted: F = F_mod(√-1, E[2·3·…]) contains √-1 by construction (flag h_F_contains_six_torsion)
  ✓ (b)  X_F once-punctured E_F with stable reduction; V^bad_mod ≠ ∅, odd, bad multiplicative; F/F_mod Galois of degree prime to l; E[2·3] ⊆ E(F)
      checked: V^bad_mod ≠ ∅, odd, multiplicative; the bound on [F:F_mod] is prime to l.  asserted: stable reduction, F/F_mod Galois, E[6] ⊆ E(F)
  ✓ (c)  l ≥ 5 prime with Im(G_F → GL_2(F_l)) ⊇ SL_2(F_l); K = F(E[l]); l prime to the residue characteristics of V^bad_mod and to the orders ord_v(q_v) on V^bad
      checked: l prime, l ∤ p_v and l ∤ ord_v(q_v) for every v ∈ V^bad_mod.  asserted: SL_2(F_l) ⊆ image (flag h_image_contains_SL2)
  ✓ (d)  C_K a hyperbolic orbicurve of type (1, l-tors)± with K-core C_K = C_F ×_F K; it determines X_K of type (1, l-tors)
      asserted: flag h_CK_is_K_core (fails only for the four exceptional j-invariants)
  ✓ (e)  V ⊆ V(K) is a section of V(K) ↠ V_mod (one v above each v_mod); at v ∈ V^bad, C_v of type (1, Z/lZ)±
      |V| = |V_mod| by construction of the data (one v per place listed)
  ✓ (f)  ε_ a cusp of C_K from a NONZERO element of Q ≅ Z/lZ; at v ∈ V^bad the cusp of the canonical generator ±1
      ε_ = the cusp ±1 (label 1 ∈ F_l^*)
  validation: 0 error(s), 0 warning(s), 9 checks
  placeholders in this stage (1; `iut gaps` has the detail):
    ▒ asserted    AssertedHypotheses: Five hypotheses of Def 3.1 and one of Thm 1.10 are flags

── stage 2 ── Hodge theaters ──────────────────────────────────────── [IUTchI] §3-§6, Def 6.13
  0±ell-Hodge theater   capsule F_T of 13 F-prime-strips, t ∈ F_13, symmetry F_l^{×±} of order 26 (orbit of t=0: 13)
    t:              -6    -5    -4    -3    -2    -1     0     1     2     3     4     5     6
    basepoint:     201   202   203   204   205   206   207   208   209   210   211   212   213
    0±-bridge F_T → F_⊠ ✓   0ell-bridge D_T → D_0± ✓
  0NF-Hodge theater     capsule F_J of 6 F-prime-strips, j ∈ F_l^*, symmetry F_l^* of order 6 (orbit of j=1: 6)
    j:              1     2     3     4     5     6
    0-bridge F_J → F_> ✓   NF-bridge F_J → F_⊛ ✓
  gluing 0,⊠ ↦ >        ✓   F_l^× ↠ F_l^*:  ±1 ↦ 1  ±2 ↦ 2  ±3 ↦ 3  ±4 ↦ 4  ±5 ↦ 5  ±6 ↦ 6   (0 ↦ >)
  0-Hodge theater at V^bad: q_v = q_v^{1/2l}, 0_v evaluated at j ∈ F_l^* is q_v^{j²}
    v3     ord(q_v) = 2   ord(q_v) = 2/26 = 1/13
    v5     ord(q_v) = 4   ord(q_v) = 4/26 = 2/13
    v7     ord(q_v) = 8   ord(q_v) = 8/26 = 4/13
  prime-strips at (0,0): F_⊠, D_0±, F_>, F_⊛, FH_mod, FH_tht, FH_Δ, FH_env, FH_gau, FH_LGP, FH_lgp  (each indexed by the 6 places of V)
  validation of the 0±ell NF-Hodge theater: 0 error(s) in 10 checks
  placeholders in this stage (2; `iut gaps` has the detail):
    ▒ open        HodgeTheatersAreBookkeeping: Hodge theaters are records of labels and copy numbers
    ▒ open        GlobalNumberFieldElements: No element of F_mod^× or κ-coric function is ever written down

── stage 3 ── Gaussian monoids (theta values) ──────────── [IUTchII] Cor 2.5, 3.5, 4.10
  Θ evaluated at the l = 13 evaluation points t ∈ F_l gives q_v^{t²} up to μ_26; it depends on |t| only.
  Ψ_gau at v ∈ V^bad = 0× · (q_v^{j²})_{j=1..6}.  ord_v(q_v^{j²}) = j²·ord(q_v)/26.
  v = v3 (p = 3, ord q_v = 2)
      j   j² ord(q_^{j²}) ord(q^{j²})
      1     1          1/13           2
      2     4          4/13           8
      3     9          9/13          18
      4    16         16/13          32
      5    25         25/13          50
      6    36         36/13          72
  v = v5 (p = 5, ord q_v = 4)
      j   j² ord(q_^{j²}) ord(q^{j²})
      1     1          2/13           4
      2     4          8/13          16
      3     9         18/13          36
      4    16         32/13          64
      5    25         50/13         100
      6    36         72/13         144
  v = v7 (p = 7, ord q_v = 8)
      j   j² ord(q_^{j²}) ord(q^{j²})
      1     1          4/13           8
      2     4         16/13          32
      3     9         36/13          72
      4    16         64/13         128
      5    25        100/13         200
      6    36        144/13         288
  log-volumes (normalized, [IUTchIII] Prop 3.9):  -(j²/2l)·log q for j = 1..6: -0.9309 -3.7234 -8.3777 -14.8937 -23.2714 -33.5108
```

```
  q-pilot {q__v}: -|log q| = -0.930856      0-pilot {q__v^{j²}}: -avg(j²)·|log q| = -(l(l+1)/12)·|log q| = -14.117983
  placeholders in this stage (1; `iut gaps` has the detail):
    ▒ approximate ThetaValuesOrdersOnly: Theta values are orders and units of q^{j²}, not values of the theta function

── stage 4 ── log-theta-lattice ──────────────────────────── [IUTchIII] Def 1.1, 1.4, 3.8
log-theta-lattice (0×μ_LGP-link horizontal; vertical = log-links)     [IUTchIII] Def 1.4, 3.8 (iii)
   m\n    +0        +1
   +2   •       •
          ↑         ↑
   +1   •       •
          ↑         ↑
   +0  [†•]—0→ ‡•
          ↑         ↑
   -1   •       •
          ↑         ↑
   -2   •       •

  • = 0±ell NF-Hodge theater (its own universe / label)   † = (0,0)   ‡ = (1,0)   [ ] = focus
  ↑ = log-link (vertically coric: D-prime-strips shared)   —0→ = 0×μ_LGP-link (horizontally coric: F⊢×μ shared)
  10 vertices, 9 links. The diagram does NOT commute ([IUTchIII] Rem 1.4.1).

  0×μ_LGP-link (0,0) → (1,0): glues F⊩►×μ-prime-strips, full poly-isomorphism ✓
    shares G_v ✓, 0×μ ✓; Π_v NOT shared (mutually alien), ring structure NOT kept; value-group dilation l(l+1)/12 = 15.1667
    v3     left (0,0) F⊩_LGP O►{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} ↦ right (1,0) F⊩_Δ O►{q_^1};  0×μ #183013 = #183013 ✓;  Π_v #270012 vs #670012
    v5     left (0,0) F⊩_LGP O►{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} ↦ right (1,0) F⊩_Δ O►{q_^1};  0×μ #183023 = #183023 ✓;  Π_v #270022 vs #670022
    v7     left (0,0) F⊩_LGP O►{q_^1,q_^4,q_^9,q_^16,q_^25,q_^36} ↦ right (1,0) F⊩_Δ O►{q_^1};  0×μ #183033 = #183033 ✓;  Π_v #270032 vs #670032
  log-links: 8 (columns n = 0, 1; rows -2..2). Each one identifies Π_v (vertical coricity) and builds the log-shells of its codomain:
    (I_v ∋ the module shown.  "exact": K_v is written down (p ł 30l, iut_local_field_K) and μ_log(I_v) is its true value;
     "model": e and m are the lower bounds forced by the roots of unity and by q__v ∈ K_v)
    v        p*    I_v ∋    μ_log(I_v)         K_v/Q_p: e, f, m, d    Π_v at (0,0) = at (0,1)?
    v2        4 p^{-25/26}·O      +1.306316  model  ≥26, ≥1, ≥2, ?         ✓
    v3        3 p^{-6/13}·O      +1.014104  model  ≥26, ≥1, ≥1, ?         ✓
    v5        5 p^{-19/26}·O      +1.547536  model  ≥52, ≥1, ≥1, ?         ✓
    v7        7 p^{-54/65}·O      +1.935931  exact  195, ≥12, 0, 0.9949    ✓
    v13      13 p^{-5/6}·O      +2.137458  model  ≥12, ≥1, ≥1, ?         ✓
    vinf      -    π-disc           0        -                        ✓
  validation of the lattice: 0 error(s) in 3 checks
    [info] 10 Hodge theaters, 8 log-links, 1 0-links (0×μ_LGP-link); only the row m=0 carries a 0-link: this is the 'infinite H' used in the proof of Cor 3.12
    [info] the diagram is highly non-commutative: no square of links is required to commute
  placeholders in this stage (2; `iut gaps` has the detail):
    ▒ approximate LogShellFromBounds: The log-shell I_v is given by an inner radius and a model log-volume
    ▒ open        LogLinkMovesNoElements: The log-link relabels copies; no element goes through a logarithm

── stage 5 ── multiradial representation, Cor 3.12 ──────── [IUTchIII] Thm 3.11, Cor 3.12
  procession S^±_{j+1} = {0..j} of tensor packets of log-shells, j = 0..l* = 6:
    (leading terms use the estimate log d^K ≈ 10.4770; the last column is the bound of Thm 1.10, Step (v)-(vii), before the average over j)
      j   |S|     (j+1)!   (j+1)·log d^K -(j²/2l)·log q      leading sum    full bound at j
      0     1         1        10.477006       -0.000000        10.477006 (j = 0: not averaged)
      1     2         2        20.954012       -0.930856        20.023156         690.002991
      2     3         6        31.431019       -3.723424        27.707595        1028.721317
      3     4        24        41.908025       -8.377704        33.530320        1365.577932
      4     5       120        52.385031      -14.893697        37.491334        1700.572834
      5     6       720        62.862037      -23.271401        39.590636        2033.706025
      6     7      5040        73.339044      -33.510818        39.828226        2364.977503
  indeterminacies: (Ind1) |Aut(Prc)| = Π_j (j+1)! = 125411328000, (Ind2) Ism on 7 direct summands, (Ind3) upper semi-compatibility ✓
  (a) 7 tensor packets, (b) 18 splitting monoids, (c) 6 copies of F_mod^×;  log-Kummer non-interference ✓;  compatible with the 0×μ_LGP-link ✓
  (0) |log q| = log q / 2l: log-volume of the q-pilot at (1,0), i.e. -|log q|                                    -0.930856   [IUTchIII] Cor 3.12; Def 3.8 (i)
  (1) the 0-pilot at (0,0): naive log-volume -avg(j²)·|log q|                                                  -14.118   [IUTchIII] Def 3.8 (i)
  (2) the 0×μ_LGP-link (0,0)→(1,0) identifies the 0-pilot with the q-pilot (value groups), units up to Ism       -0.930856   [IUTchIII] Def 3.8 (ii), Thm 3.11 (iii)
  (3) multiradial representation of the 0-pilot in terms of the mono-analytic data of (1,0): Ind1 count              5040   [IUTchIII] Thm 3.11 (i)
  (4) holomorphic hull, leading terms only: ((l+5)/4)·log d^K (an estimate, not a bound)                        47.1465   [Alien] §3.7 (iv)
  (5) leading terms with the theta shift: ((l+5)/4)·log d^K - ((l+1)/24)·log q (not a bound)                     33.0285   [Alien] §3.7 (iv); [IUTchIV] Rem 1.10.1
  (6) -|log 0| ≤ B: the bound of the proof of Thm 1.10 with every term, summed over v_Q                          1530.59   [IUTchIV] Thm 1.10, Steps (v)-(vii)
  (7) Cor 3.12 (an input, not re-proved here) gives B ≥ -|log q|, i.e. C_0 ≥ -1 for C_0 = B/|log q|          1644.29   [IUTchIII] Cor 3.12
  the hull bound, place by place ([IUTchIV] Thm 1.10, proof): l*_mod = log(e*_mod·l) = 15.7880
    v_Q    class                  log d^K_vQ ≤   log q_vQ   log s^Q ι          bound   source of the log d^K bound
    2      p | 2·3·5                    8.3178     0.0000    0.6931  1       322.3069   Step (ii): 11 log 2, + log 2 (tame, bad over 2)
    3      p | 2·3·5, Supp(q)           4.3944     2.1972    1.0986  1       303.7757   Step (ii): 3 log 3 + log f
    5      p | 2·3·5, Supp(q)           4.8283     6.4378    1.6094  1       303.7653   Step (ii): 2 log 5 + log f
    7      Supp(q)                      1.9359    15.5673    1.9459  1       285.7605   exact: (1 - 1/e)·log p, K_v tame
    13     p = l                        5.1299     0.0000    2.5649  1       309.8333   Step (ii): 2 log l
    ∞      archimedean                       -          -         -  -         5.1513   Step (vii): ((l+5)/4)·log π
    distinguished v_Q: ((l+5)/4)·log d^K_vQ - ((l+1)/24)·log q_vQ + log s^Q_vQ + (l+5)·l*_mod·ι     (Step (v))
  -|log 0| ≤ B = 1530.593100;   C_0 := B/|log q| = 1644.29, against the C_0 of the statement of Thm 1.10 = 1.87999e+22  ✓
  Cor 3.12 is an input here, not something this program verifies.  Its consequence B ≥ -|log q|: 1530.593100 ≥ -0.930856  consistent ✓
  placeholders in this stage (3; `iut gaps` has the detail):
    ▒ asserted    CorThreeTwelveIsAnInput: Cor 3.12 is used, not verified
    ▒ open        IndeterminaciesAreCounted: (Ind1)-(Ind3) are counted and named, their effect on a region is not computed
    ▒ open        DifferentSupport: The support of the different of F_tpd is not in the data model

── stage 6 ── the height bound ──────────────────────────── [IUTchIV] Thm 1.10, Cor 2.2
  log q = 24.202257   |log q| = 0.930856   log d^Ftpd = 0.000000   log f^Ftpd = 4.653960   e*_mod = 552960   n_prm = 5e+20
  hypotheses of Thm 1.10 beyond Def 3.1:  l ≥ 7 ✓   good reduction away from 2l outside V^bad ✓   F = F_mod(V-1, E[2·3·5]) ✓ (asserted)
  C_0 = 1.87999e+22 (≥ -1 ✓)
  Thm 1.10:  (1/6)·log q = 4.033710  ≤  (1 + 20·d_mod/l)(log d^Ftpd + log f^Ftpd) + 20(e*_mod·l + n_prm) = 1e+22   ✓
  Cor 2.2:   log qϑ = 25.588552 (2 included); (C1) wants 5.059 ≤ l ≤ 4.799e+08: l = 13 in range;  ε_E = 3.734e+15;  (C2) 4.264759 ≤ 1e+22 ✓
  Thm 1.10 holds here only as a comparison with its constant: main term 11.813899, constant 20(e*_mod·l + n_prm) = 1e+22.
  Cor 2.2 asserts nothing about this curve: it is in the exceptional set Exc (ε_E > 1, (log qϑ)^{1/2} < ξ_prm + n_prm; (log qϑ)^{1/2} = 5.059, ξ_prm + n_prm ≥ 5e+20).
  the proof of Cor 2.2 would take a prime with (P1)-(P3); the smallest one ≥ 7 is l = 11.
  placeholders in this stage (3; `iut gaps` has the detail):
    ▒ asserted    InexplicitConstants: n_prm, ξ_prm, H_unif, C_K, H_K are not given by the papers
    ▒ open        ExplicitEstimatesNotImplemented: The explicit abc inequality of [ExpEst] is not in the toolkit
    ▒ open        ReductionToGeneralPosition: From compactly bounded subsets to all curves: not modelled

pipeline complete: 0 problem(s)  (Def 3.1: 0, Hodge theater: 0, lattice: 0, consequence of Cor 3.12: consistent, hull bound vs Thm 1.10: below, cross-check: 0)
18 placeholder(s) stand in for parts of the theory that are not computed (`iut gaps`): "complete" means every stage ran, not that the theory was verified.
```

18 · Gaps and placeholders

`iut gaps` — the registry of everything the papers specify and the program does not do the way the papers do it (`src/iut_gaps.c`, `docs/GAPS.md`): **open** is not computed, **approximate** is computed from bounds, **asserted** is an input that is not checked, **fixed** was a gap and is closed.

```
Gaps and placeholders: 25 entries (12 open, 3 approximate, 3 asserted, 7 fixed)

── stage 0, initial Θ-data
  [approximate] LocalFieldsOfK — The local fields K_v are known only through bounds
      papers : K = F(E[l]) and, for each v ∈ V, the local field K_v with its ramification index e_v, residue degree f_v, different exponent d_v and p-power roots of
unity μ_{p^m}; they enter log(O×_{K_v}), the log-shell I_v and every log-volume.   ([IUTchI] Def 3.1 (c), (e); [IUTchIV] Prop 1.2–1.4, Thm 1.10 Step (iii))
      code   : At a place with p ∤ 30l (F_mod = Q, F = F_mod(√-1, E[2·3·5])) iut_local_field_K writes K_v down: at a Tate-curve place K_v = Q_p(ζ_60l, q_v^{1/30l}) up
to an unramified quadratic twist, so e = 30l/gcd(30l, ord q_v), d = 1 - 1/e, m = 0 unless (p - 1) | e, and f from the order of the unit of q_v (JSON data); at a good
place e = 1. At p | 30l, where K_v is wildly ramified, and for any other F, e_K and m_K are still the lower bounds forced by q_v^{1/2l} ∈ K_v and by √-1, μ_3, μ_5, μ_l
⊆ K, and f_K = 1. [F:Q] and [K:Q] are upper bounds.
      where  : iut_core.c: iut_local_field_K, iut_data_finish, iut_log_shell_volume
      fix    : Partly done: the tame places are exact (each printed value says = or ≥). Left: the wild places p | 30l, which need the Néron model at p (ordinary or
supersingular reduction, and for p = l the image of inertia in GL_2(F_l)); the residue degree at good places, which needs the order of Frobenius on E[60l]; and whether
the reduction at a Tate-curve place is split, which decides a factor 2 in f when f is odd.
  [open] GeneratorsFreyOnly — Initial Θ-data is generated only for Frey curves over Q
      papers : Initial Θ-data for any elliptic curve over a number field F with F_mod = Q(j_E), in particular the points of the λ-line of degree ≤ d of Cor 2.2.
([IUTchI] Def 3.1; [IUTchIV] Cor 2.2)
      code   : `--abc` builds the Frey curve of an abc triple with F_mod = Q; `--theta-json` reads four worked files (also F_mod = Q); `--random` with an odd seed
invents a number field by its degree only.
      where  : iut_core.c: iut_gen_from_abc, iut_gen_random; iut_input.c
      fix    : Open. Needs a number-field layer (minimal polynomial of j, places of F_mod with e, f). The JSON schema already has Fmod.degree and per-place data, so the
reader is the place to extend.
  [fixed] ChoiceOfLFixed — l could not be chosen the way the proof chooses it
      papers : A prime l with h^{1/2} ≤ l ≤ 10δ·h^{1/2}·log(2δh), dividing no local height ord_v(q_v), and with ord_v(q_v) < h^{1/2} if l = p_v.   ([IUTchIV] Cor 2.2,
proof, (P1)–(P3))
      code   : Was: l always given by hand. Now `--l auto` picks the smallest such prime ≥ 7 within IUT_MAX_L (iut_choose_l).
      where  : iut_ops_iv.c: iut_choose_l; iut_main.c: iut_options_make_data
      fix    : Done in this change.
  [fixed] BadPlaceOverL — A bad place over l was forced into V^bad_mod and the data rejected
      papers : V^bad_mod is a chosen nonempty set of odd bad places prime to l; E may be bad at places of V^good, and Thm 1.10 asks for good reduction only away from
2l.   ([IUTchI] Def 3.1 (b), (c); [IUTchIV] Thm 1.10)
      code   : Was: every odd bad multiplicative place was put in V^bad_mod, so abc(1,2400,2401) with l = 7 failed with an error. Now a bad place over l stays in V^good
with a note, and log q, the theta values and f^F leave it out.
      where  : iut_core.c: iut_data_finish, iut_validate_data
      fix    : Done in this change.

── stage 1, Def 3.1
  [asserted] AssertedHypotheses — Five hypotheses of Def 3.1 and one of Thm 1.10 are flags
      papers : √-1 ∈ F; F/F_mod Galois; E[2·3] ⊆ E(F); stable reduction; Im(G_F → GL_2(F_l)) ⊇ SL_2(F_l); C_K a K-core; and for Thm 1.10, F = F_mod(√-1, E[2·3·5]).
([IUTchI] Def 3.1 (a)–(d); [IUTchIV] Thm 1.10)
      code   : Each is a boolean the generator sets to true, or the JSON file records with its proof. The C code does not check any of them.
      where  : iut_core.c: iut_validate_data; iut_i.c: iut_def_three_one_check; iut_ops_iv.c: iut_thm110_compute, iut_gap_serre_SL2
      fix    : Prepared: iut_gap_serre_SL2 is the placeholder for Serre's criterion (three Frobenius elements), which scripts/infographic/initialtheta-data.js already
runs for the four examples; port it to C. K-core: test j ∉ {the four exceptional j-invariants} and no CM. Stable reduction: Raynaud's criterion from E[3], E[5] ⊆ E(F).
  [fixed] DefThreeOneLettering — The clauses of Def 3.1 were lettered differently from the paper
      papers : (a) √-1 ∈ F; (b) X_F, stable reduction, V^bad_mod, F/F_mod, E[2·3]; (c) l, SL_2, K, l prime to V^bad_mod and ord q; (d) C_‗K, K-core; (e) V a section;
(f) ε_.   ([IUTchI] Def 3.1 (a)–(f))
      code   : Was: the checker put SL_2 and the K-core under (b), the section V under (d) and V^bad_mod under (e), and the validator cited (a) and (e) for conditions
of (b). Now both follow the paper, and a clause that is part computed and part asserted says which part is which.
      where  : iut_i.c: iut_def_three_one_check; iut_core.c: iut_validate_data
      fix    : Done in this change.
  [fixed] DefThreeOneNotCounted — A failed clause (c) did not count as a problem
      papers : l is prime to the orders of the q-parameters at V^bad.   ([IUTchI] Def 3.1 (c))
      code   : Was: l | ord_v(q_v) was a warning in the validator, so `iut pipeline --abc 1 2187 2188 --l 7` printed × at (c) and still ended with 0 problems. Now it is
an error and the pipeline counts it.
      where  : iut_core.c: iut_validate_data
      fix    : Done in this change.

── stage 2, Hodge theaters
  [open] HodgeTheatersAreBookkeeping — Hodge theaters are records of labels and copy numbers
      papers : Categories: tempered Frobenioids F_‗v, their base categories D_v = B^temp(Π_v)^0, global Frobenioids F^⊛_mod, poly-isomorphisms as sets of isomorphisms,
bridges as poly-morphisms of capsules.   ([IUTchI] §3–§6; [EtTh]; [FrdI], [FrdII]; [AbsTopIII])
      code   : A prime-strip is, per v, a tuple of copy numbers (Π_v, G_v, O×) and a list of valuations; a bridge is a flag; a poly-isomorphism is a flag
`full_poly_isomorphism` and, in iut_i.c, a count.
      where  : iut_core.c: iut_prime_strip_build, iut_hodge_theater_new; iut_i.c
      fix    : Open, and by design of the data model (docs/DATA_MODEL.md §8). A first real step: represent Π_v at v ∈ V^bad by generators of the tempered fundamental
group of the Tate curve and G_v by its (p, e, f) invariants, so that an isomorphism of prime-strips is an actual map.
  [open] GlobalNumberFieldElements — No element of F_mod^× or κ-coric function is ever written down
      papers : The monoid M^⊛_mod = F_mod^×, the κ-coric rational functions on C_K and their values at the strictly critical points, which give the NF-bridge its
content.   ([IUTchI] Ex 5.1, Def 5.2; Rem 3.1.7)
      code   : Pseudo-monoids are described by name and by Kummer-ready/Kummer-blind flags; the κ-coric check takes a caller-supplied description of a function.
      where  : iut_i.c: iut_pseudo_monoid_get, iut_kappa_coric_check
      fix    : Open. For F_mod = Q a rational-function type with exact coefficients would do; the global Kummer theory of Prop 3.10 then becomes a statement about real
elements.

── stage 3, Gaussian monoids
  [approximate] ThetaValuesOrdersOnly — Theta values are orders and units of q^{j²}, not values of the theta function
      papers : The étale theta function Θ_‗v evaluated at the l-torsion points labelled j gives q_‗v^{j²} = (q_v^{1/2l})^{j²} up to μ_2l, an element of K_v.
([IUTchII] Cor 2.5, 2.6, 3.5; [EtTh] Prop 1.4)
      code   : The order j²·ord(q_v)/2l is exact, and the unit of q_v^{j²} is computed mod p^8 from the Tate parameter. The 2l-th root q_‗v and the theta series itself
are not computed.
      where  : iut_pipeline.c: stage 3; iut_input.c: iut_tate_unit_pow; iut_ops_iv.c: iut_gap_theta_value
      fix    : Prepared: iut_gap_theta_value is the placeholder. Extracting the 2l-th root needs K_v (gap LocalFieldsOfK); once K_v = F_v(ζ_l, q^{1/l}) is represented,
q_ is a uniformizer power times a unit by Hensel, and the series Σ (-1)^n q^{(n²+n)/2}(U^{2n+1}) can be summed to the same p-adic precision.

── stage 4, log-theta-lattice
  [approximate] LogShellFromBounds — The log-shell I_v is given by an inner radius and a model log-volume
      papers : I_v = (p⋆_v)^{-1}·log(O×_{K_v}) ⊆ K_v, a specific compact Z_p-module, with μ_log(I_v) = log p⋆ - (1/e + m/(ef)) log p; at an archimedean v, the closed
disc of radius π.   ([IUTchIII] Def 1.1, Prop 1.2; [AbsTopIII] Def 5.4 (iii); [IUTchIV] Prop 1.2, 1.4 (ii))
      code   : The radius printed is that of the largest p^λ·O_v the series argument puts inside I_v. Where K_v is written down (p ∤ 30l, see LocalFieldsOfK) μ_log(I_v)
= (1 - 1/e)·log p is exact and marked so; at p | 30l it uses the lower bounds for e and m and is marked a model. The archimedean log-volume is set to 0.
      where  : iut_lattice.c: iut_apply_log_link; iut_ops_iii.c: iut_log_shell_make; iut_core.c: iut_log_shell_volume
      fix    : The wild places follow LocalFieldsOfK. The archimedean term log π per factor is carried in the hull bound (Step (vii)) and should be shown on the shell
as well.
  [open] LogLinkMovesNoElements — The log-link relabels copies; no element goes through a logarithm
```

```
      papers : log_v: O×_{†v} → †K_v, whose image becomes the additive module of the next copy ‡; the log-Kummer correspondence follows elements up the column and is
only upper semi-compatible.   ([IUTchIII] Def 1.1 (iii), Prop 1.2 (iii), (iv), Prop 3.5)
      code   : The link copies Π_v and G_v, gives the units a new copy number and attaches the log-shells. Upper semi-compatibility is a table of three cases.
      where  : iut_lattice.c: iut_apply_log_link; iut_ops_iii.c: iut_log_link_strip, iut_upper_semi_compatibility
      fix    : Open. With p-adic units to precision p^8 (already used for the Tate unit) the p-adic logarithm of a unit is a few lines; the first use would be to show
log(q-unit) inside I_v at each v ∈ V^bad.

── stage 5, multiradial representation
  [fixed] HullBoundWasLeadingTerm — The upper bound for -|log Θ| was the two leading terms only
      papers : For each v_Q the procession-normalized bound of Step (v) (distinguished), (vi) (unramified: 0) or (vii) (archimedean: ((l+5)/4) log π), summed over v_Q.
([IUTchIV] Thm 1.10, proof, Steps (iv)-(viii))
      code   : Was: ((l+5)/4)·log d^K - ((l+1)/24)·log q with an estimated log d^K, called an upper bound and used for the verdict. Now iut_hull_bound_compute follows
Steps (v)-(vii) place by place with every term, checks that the result is below the C_0 of the statement, and the verdict rests on it.
      where  : iut_ops_iv.c: iut_hull_bound_compute; iut_lattice.c: iut_multiradial_build
      fix    : Done in this change. What remains is in the two entries below.
  [asserted] CorThreeTwelveIsAnInput — Cor 3.12 is used, not verified
      papers : -|log Θ| ≥ -|log q|: the hull of all possible images of the Θ-pilot, in the multiradial representation and subject to (Ind1)-(Ind3), contains a region of
the log-volume of the q-pilot.   ([IUTchIII] Thm 3.11, Cor 3.12)
      code   : The toolkit computes an upper bound B for -|log Θ| and reports whether the consequence B ≥ -|log q| is consistent with the data. It cannot test the
inequality itself, and it used to print it with a ✓ as if it had.
      where  : iut_lattice.c: iut_multiradial_build; iut_ops_iii.c: iut_cor312_steps; iut_pipeline.c: stage 5
      fix    : The wording is fixed in this change. The statement itself is disputed in the literature (Scholze-Stix 2018; the 2026 Lean work), and no computation here
bears on that.
  [open] IndeterminaciesAreCounted — (Ind1)-(Ind3) are counted and named, their effect on a region is not computed
      papers : (Ind1) automorphisms of the procession of D▷-prime-strips, (Ind2) Ism acting on each direct summand of each tensor packet, (Ind3) upper semi-
compatibility; the hull is that of the union of the images under all of them.   ([IUTchIII] Thm 3.11 (i), (ii); Prop 3.2, 3.5; [IUTchIV] Prop 1.2, 1.5)
      code   : (Ind1) is the number Π(j+1)!, (Ind2) the number of direct summands, (Ind3) a flag. The union and its hull are replaced by the container of Prop 1.2 (ii)
/ 1.4 (iii) as in the proof.
      where  : iut_ops_iii.c: iut_multiradial_full_build; iut_ops_iv.c: iut_hull_bound_compute, iut_gap_hull_exact
      fix    : Prepared: iut_gap_hull_exact is the placeholder for the exact log-volume. It needs the tensor packets as actual Z_p-lattices; with LocalFieldsOfK closed,
a packet at an unramified p is a free Z_p-module with an explicit basis and the action of Ism is a group of matrices.
  [open] DifferentSupport — The support of the different of F_tpd is not in the data model
      papers : V^dst_Q contains every prime under Supp(d^{F_tpd}); each such prime contributes log p + (l+5)·l*_mod to the bound.   ([IUTchIV] Thm 1.10, proof, Step
(iii), (D4)-(D7))
      code   : Only the total log d^{F_tpd} is stored. When it is not 0 the hull bound adds ((l+5)/4)·log d^{F_tpd} without those per-prime terms and marks itself
incomplete. For the Frey curves F_tpd = Q and nothing is missing.
      where  : iut_ops_iv.c: iut_hull_bound_compute
      fix    : Open; goes with GeneratorsFreyOnly: store the ramified primes of F_tpd with their exponents next to dmod.

── stage 6, height bound
  [fixed] ThmOneTenHypotheses — The hypotheses of Thm 1.10 were not looked at, and its f^F counted the wrong places
      papers : E has good reduction at every v ∈ V(F)^good not dividing 2l; F = F_mod(V-1, E[2·3·5]), hence l ≥ 7; f^F has the support of the q-divisor, V^bad_mod.
([IUTchIV] Thm 1.10)
      code   : Was: the inequality was evaluated for any data, and log f^{F_tpd} included a bad place over 2 (so `--abc 1 8 9` used log 6 where the theorem has log 3).
Now the three hypotheses are reported, the verdict line says when the theorem does not apply, and f^F is summed over V^bad_mod.
      where  : iut_ops_iv.c: iut_thm110_compute, iut_cor22_compute
      fix    : Done in this change.
  [asserted] InexplicitConstants — n_prm, ξ_prm, H_unif, C_K, H_K are not given by the papers
      papers : [IUTchIV] only says that n_prm and ξ_prm ≥ 5 exist, and that H_unif, C_K, H_K exist for a compactly bounded subset; the exceptional set Exc_d is finite
but not listed.   ([IUTchIV] Prop 1.6, Prop 2.1 (ii), Cor 2.2 (ii); [ExpEst] §1)
      code   : n_prm = 5·10^20 is borrowed from [ExpEst] (where it serves a sharper form of the prime number theorem); ξ_prm is used only through ξ_prm ≥ 5; Exc is
tested by the two conditions the proof makes explicit; C_K is the constant of Thm 1.10 with e_mod = 1.
      where  : iut_ops.h: IUT_ETA_PRM_VALUE; iut_ops_iv.c: iut_cor22_compute
      fix    : Open. For the 4/3 form of Prop 1.6 a far smaller n_prm is admissible (Rosser-Schoenfeld), which would shrink the constant of Thm 1.10 by many orders of
magnitude; it needs a citation checked before it goes in.
  [open] ExplicitEstimatesNotImplemented — The explicit abc inequality of [ExpEst] is not in the toolkit
      papers : An explicit ε-version of abc and its consequences (e.g. the Fermat exponent bound), with the modified constants of [ExpEst] §1 and the case p = 2 handled
by μ_6-initial Θ-data.   ([ExpEst] Thm A, Thm B, Cor C)
      code   : Nothing beyond the constant n_prm. The pipeline ends at [IUTchIV] Thm 1.10 and the conditions of Cor 2.2.
      where  : iut_ops_iv.c: iut_gap_explicit_abc
      fix    : Prepared: iut_gap_explicit_abc is the placeholder. The estimates of [ExpEst] §5 are elementary once the hull bound exists per place, which it now does.
  [open] ReductionToGeneralPosition — From compactly bounded subsets to all curves: not modelled
      papers : Cor 2.3 obtains Vojta/abc for all hyperbolic curves from Cor 2.2 through [GenEll] (Belyi maps, compactly bounded subsets, bounded discrepancy classes).
([IUTchIV] Cor 2.2 (i), Cor 2.3; [GenEll] Thm 2.1)
      code   : iut_cor23_bound evaluates the slack of one inequality; there is no notion of a compactly bounded subset or of ≈.
      where  : iut_ops_iv.c: iut_cor23_bound
      fix    : Open. Not needed to run one curve through the theory; needed to state what the run proves.

── stage 7, cross-check
  [open] CrossCheckIsShallow — The cross-check compares four numbers, for JSON data only
      papers : —   (—)
      code   : log q, |log q|, log f and ht are compared with the values stored in the JSON file. Generated data has no independent check, and nothing else in the
pipeline is recomputed a second way.
      where  : iut_pipeline.c: stage 7
      fix    : Prepared: the hull bound now gives a second check (C_0 place by place must lie below the C_0 of the statement), which the pipeline counts. Next:
recompute ord_v(q_v) from the j-invariant in C for `--abc` data.

── the toolkit as a whole
  [open] ManualUsesAJavaScriptPort — The manual is built from a second implementation in JavaScript
      papers : —   (scripts/manual/lib/iut.js, ops.js)
      code   : docs/IUT_MANUAL.pdf is produced by scripts/manual, which carries its own JavaScript copy of the data model and of the operations for its diagrams and
progressions. The fixes of November 2026 (V^bad_mod, Def 3.1 lettering, e_K and m_K bounds, tame K_v, f^F over V^bad_mod, Thm 1.10 hypotheses, the hull bound, Cor 3.12
as an input) are ported and tested (scripts/manual/lib/test.js; the hull bound agrees with the C value), and the manual now prints `iut pipeline` and `iut gaps` from
the C program. It is still a second implementation that the next C change will not reach.
      where  : scripts/manual/lib/iut.js, scripts/manual/lib/ops.js
      fix    : Open: make the diagrams read `iut … --json` (as the guides and articles do) and delete the port.
  [fixed] ViewsDoNotShowGaps — The GUI and the TUI did not show the gap list or the hull table
      papers : —   (—)
      code   : Was: only `iut pipeline` and the web app showed them. Now iut-gui has the hull table on the stage 5 page, the placeholders under every stage and in the
object tree; the TUI has a gaps screen (key g) and lists the placeholders of stages 5 and 6 under the multiradial screen, which shows B but not the table prime by
prime.
      where  : src/gui/gui_draw.c: draw_mr, draw_gaps; src/gui/gui_tree.c; src/iut_tui.c
      fix    : Done.
  [open] LeanNotInSync — The Lean mirror of the data types does not have the new fields
      papers : —   (lean/)
      code   : lean/ mirrors the structs of iut.h as of an earlier state: no m_K, no h_F_contains_five_torsion, the old rule for V^bad_mod.
      where  : lean/
      fix    : Open.
  [open] PrerequisitePapers — The prerequisite papers have no operations layer
      papers : The étale theta function and its rigidities ([EtTh]), mono-anabelian reconstruction ([AbsTopIII]), Frobenioids ([FrdI], [FrdII]), general position
([GenEll]).   (pdf/: [EtTh], [AbsTopIII], [FrdI], [FrdII], [GenEll], [SemiAnbd], …)
      code   : Their results appear as named facts and flags (cyclotomic rigidity kinds, reconstruction steps of Cor 1.2, tempered facts); none of their constructions
```

```
is run.
      where  : iut_i.c, iut_ops_ii.c; iut_ops_iv.c: iut_gap_reconstruct
      fix    : Prepared: iut_gap_reconstruct is the placeholder for a reconstruction algorithm run on a concrete profinite/tempered group. The two guides
(anabelian_guide.pdf, multiradial_guide.pdf) describe the algorithms step by step and are the specification to code against.
```

[m487]

Appendix

A · C structs and the identifiers they implement

struct / enum	header	identifiers
`iut_initial_theta_data`	iut.h	InitialThetaData, FBarOverF, Fmod, KField, EpsilonUnder
`iut_valuation`	iut.h	VSet, VNon, VArc, VBadMod, VGoodMod, VBad, VGood, PV
`iut_prime_strip`	iut.h	BasicPrimeStrip, DPrimeStrip, DMonoPrimeStrip, FPrimeStrip, FMonoPrimeStrip, FMonoUnitsPrimeStrip, FMonoUnitsModTorsPrimeStrip, FMonoValUnitsModTorsPrimeStrip, FMonoGlobPrimeStrip, FMonoGlobValUnitsModTorsPrimeStrip, FBoxPlus, DCircDotPm, FGt, FCircAst, FMonoGlobTriangle, FMonoGlobEnv, FMonoGlobGau, FMonoGlobLGPTorsor, FMonoGlobLgpIdeal, FMonoGlobMod, FMonoGlobTht, DGt, DMonoGt, DMonoTriangle
`iut_local_datum`	iut.h	BasicPrimeStrip, PiV, GV, OValV, OUnitsV, OUnitsModTorsV, OTorsV, OValgroupV, UnitsModTorsKummer, UnitsKummer
`iut_pilot`	iut.h	QPilot, ThetaPilot
`iut_log_shell`	iut.h	LogShell, HolLogShell, MonoLogShell, FrobLogShell, EtLogShell, PStar
`iut_capsule_member`	iut.h	Capsule, FT, FJ, LabelT, LabelJ
`iut_theta_hodge_theater`	iut.h	ThetaHodgeTheater
`iut_theta_pm_ell_hodge_theater`	iut.h	ThetaPmEllHodgeTheater
`iut_theta_nf_hodge_theater`	iut.h	ThetaNFHodgeTheater
`iut_hodge_theater`	iut.h	ThetaPmEllNFHodgeTheater, NMHodgeTheater, DagHodgeTheater, DdagHodgeTheater, EtaleHodgeTheater, Bullet
`iut_link`	iut.h	ThetaLink, ThetaUnitsModTorsLink, ThetaUnitsModTorsGauLink, ThetaUnitsModTorsLGPTorsorLink, ThetaUnitsModTorsLgpIdealLink, LogLink, DThetaPmEllNFLink, ThetaDilation, AlienCopy
`iut_indeterminacies`	iut.h	IndOne, IndTwo, IndThree
`iut_procession_step`	iut.h	Procession, SPmJPlusOne, SJPlusOne
`iut_multiradial_rep`	iut.h	MultiradialRepresentation, CTheta, AbsLogQ, AbsLogTheta, HolomorphicHull
`iut_log_theta_lattice`	iut.h	LogThetaLattice, LGPTorsorGaussianLogThetaLattice, LgpIdealGaussianLogThetaLattice, InfiniteH, NCircColumn
`iut_species`	iut.h	ZeroSpecies, OneSpecies, SpeciesObject, SpeciesMorphism
`iut_mutation`	iut.h	Mutation, MorphismOfMutations, MutationHistory, Observable, Core
`iut_ident`	iut.h	Dag
`iut_pseudo_monoid`	iut_i.h	PseudoMonoid, MCircAst, MCircAstMod, MCircAstSol, MCircAstKappa, MCircAstInfKappa, MCircAstInfKappaUnits, MCircAstKappaSol, MV, MKappaV, MInfKappaV, MInfKappaUnitsV
`iut_poly_morphism`	iut_i.h	Capsule, PolyMorphism, PolyIsomorphism, FullPolyIsomorphism, PlusFullPolyIsomorphism, CapsuleFullPolyMorphism, CapsuleFullPolyIsomorphism, PolyAutomorphism, Isomorph
`iut_anabelioid`	iut_i.h	BPi, BX, BXZero, BTemp, Temperoid
`iut_field_kind`	iut_i.h	PV, NF, MLF, CAF
`iut_orbicurve`	iut_i.h	XF, COrbicurve, XUnderK, CK, XArrow, CArrow, XUnderUnderK, CUnderUnderV, CUnderV, OrbicurveType
`iut_cusps_section_one`	iut_i.h	CuspEpsilon, CuspEpsilonPrime, CuspEpsilonDoublePrime, CuspTwoEpsilon, CuspZero, InertiaEpsilonPrime, DeltaEpsilon, DeltaE, JX, JC, SectionSigma, IotaInvolution
`iut_cor_one_two`	iut_i.h	CorOneTwo, HSubgroup, PiXArrow, PiCArrow, PiXUnder, PiCUnder
`iut_tempered_fact`	iut_i.h	PiTpG, PiGProfinite, TemperedConjugates, CommensurablyTerminal, TemperedSubcover, TemperedVerticialEdge, DecompositionSynchronized, DiscreteAnalogue, GaloisCountable
`iut_tempered_at_v`	iut_i.h	PiTpG, PiGProfinite, TemperedConjugates, CommensurablyTerminal, TemperedSubcover, TemperedVerticialEdge, DecompositionSynchronized, DiscreteAnalogue, GaloisCountable
`iut_def_three_one_item`	iut_i.h	DefThreeOne
`iut_theta_approach`	iut_i.h	ThetaApproach, MXCyclotome, LDeltaTheta
`iut_s_mod`	iut_i.h	SMod
`iut_kappa_function`	iut_i.h	CFmod, CriticalPoint, StrictlyCriticalPoint, KappaCoric, InfKappaCoric, InfKappaUnitsCoric, KappaSolvable, FMuL, AutKappaSol
`iut_kappa_verdict`	iut_i.h	CFmod, CriticalPoint, StrictlyCriticalPoint, KappaCoric, InfKappaCoric, InfKappaUnitsCoric, KappaSolvable, FMuL, AutKappaSol
`iut_kappa_sol`	iut_i.h	CFmod, CriticalPoint, StrictlyCriticalPoint, KappaCoric, InfKappaCoric, InfKappaUnitsCoric, KappaSolvable, FMuL, AutKappaSol
`iut_local_frobenioid`	iut_i.h	DV, DMonoV, DThetaV, FUnderV, FBiratV, YDdotV, CV, CMonoV, CThetaV, PhiCMonoV, TauMonoV, TauThetaV, FMonoV, FThetaV, MuTwoL, LogPhi, LogPV, LogTheta, TMCategory, AutHolOrbispace, ABox, KappaKummerV
`iut_global_realified`	iut_i.h	ThetaDilation, CMonoGlobMod, CMonoGlobTht, DMonoGlobMod, LogModPV, RhoThetaV, RhoDV, FMonoGlobD, FrobeniusElement, DefThreeSix, CorThreeSeven
`iut_f_monoglob`	iut_i.h	ThetaDilation, CMonoGlobMod, CMonoGlobTht, DMonoGlobMod, LogModPV, RhoThetaV, RhoDV, FMonoGlobD, FrobeniusElement, DefThreeSix, CorThreeSeven
`iut_theta_hodge_theater_def`	iut_i.h	ThetaDilation, CMonoGlobMod, CMonoGlobTht, DMonoGlobMod, LogModPV, RhoThetaV, RhoDV, FMonoGlobD, FrobeniusElement, DefThreeSix, CorThreeSeven
`iut_labcusp_torsor`	iut_i.h	LabCusp, EtaV, DCircAstGlobal, VDCircAst, AutEpsilon, AutSL, BorelSubgroup, SemiUnipotentSubgroup, VPmUn, VBor, VK, PhiNFBulletV, ArithLocalAnalyticSection
`iut_global_base`	iut_i.h	LabCusp, EtaV, DCircAstGlobal, VDCircAst, AutEpsilon, AutSL, BorelSubgroup, SemiUnipotentSubgroup, VPmUn, VBor, VK, PhiNFBulletV, ArithLocalAnalyticSection
`iut_valuation_orbit`	iut_i.h	LabCusp, EtaV, DCircAstGlobal, VDCircAst, AutEpsilon, AutSL, BorelSubgroup, SemiUnipotentSubgroup, VPmUn, VBor, VK, PhiNFBulletV, ArithLocalAnalyticSection
`iut_eval_points`	iut_i.h	MuMinus, EvaluationPoint, EvaluationSection, PhiThetaVJ, PhiLC, ChiJ, ZetaLabel, DNFBridge, DThetaBridge, DThetaNFHodgeTheater, DThetaNFLink, SynchronizedLabels
`iut_theta_bridge_piece`	iut_i.h	MuMinus, EvaluationPoint, EvaluationSection, PhiThetaVJ, PhiLC, ChiJ, ZetaLabel, DNFBridge, DThetaBridge, DThetaNFHodgeTheater, DThetaNFLink, SynchronizedLabels
`iut_label_transport`	iut_i.h	MuMinus, EvaluationPoint, EvaluationSection, PhiThetaVJ, PhiLC, ChiJ, ZetaLabel, DNFBridge, DThetaBridge, DThetaNFHodgeTheater, DThetaNFLink, SynchronizedLabels
`iut_iso_torsor`	iut_i.h	MuMinus, EvaluationPoint, EvaluationSection, PhiThetaVJ, PhiLC, ChiJ, ZetaLabel, DNFBridge, DThetaBridge, DThetaNFHodgeTheater, DThetaNFLink, SynchronizedLabels

struct / enum	header	identifiers
`iut_forgetful_counts`	iut_i.h	MuMinus, EvaluationPoint, EvaluationSection, PhiThetaVJ, PhiLC, ChiJ, ZetaLabel, DNFBridge, DThetaBridge, DThetaNFHodgeTheater, DThetaNFLink, SynchronizedLabels
`iut_global_frobenioid`	iut_i.h	FCircAstGlobal, FCircAstPrime, FCircAstMod, FCircAstJ, FCircAstAngleJ, FCircAstJCapsule, PhiCircAst, PiRat, DCircAstMod, DistributionVJ, KummerReady, KummerBlind, DeltaValuation, PsiNFBridge, PsiThetaBridge, PivotalDistribution, FDCircAst, StripLifting
`iut_distribution`	iut_i.h	FCircAstGlobal, FCircAstPrime, FCircAstMod, FCircAstJ, FCircAstAngleJ, FCircAstJCapsule, PhiCircAst, PiRat, DCircAstMod, DistributionVJ, KummerReady, KummerBlind, DeltaValuation, PsiNFBridge, PsiThetaBridge, PivotalDistribution, FDCircAst, StripLifting
`iut_strip_functor`	iut_i.h	FCircAstGlobal, FCircAstPrime, FCircAstMod, FCircAstJ, FCircAstAngleJ, FCircAstJCapsule, PhiCircAst, PiRat, DCircAstMod, DistributionVJ, KummerReady, KummerBlind, DeltaValuation, PsiNFBridge, PsiThetaBridge, PivotalDistribution, FDCircAst, StripLifting
`iut_fl_semi_pm_set`	iut_i.h	FlSemiPmGroup, FlSemiPmTorsor, PositiveElement, DPmV, LabCuspPm, EtaZeroV, EtaPmV, AutPlus, AutMinus, AutAlpha, SignVector, DCircDotPmGlobal, AutPm, AutCsp, VPm, MinusOneFl, AlphaThetaPm, PhiThetaEllBulletV, GammaPm, TAbs, TStar, DThetaPmBridge, DThetaEllBridge, DThetaPmEllHodgeTheater, ZetaPm, XiSync, PropSixSeven, DThetaPmEllLink, DagHTDThetaPmEllNF, HalfPlaneDictionary, GeometricBasepoint, FigSixSix
`iut_labcusp_pm_torsor`	iut_i.h	FlSemiPmGroup, FlSemiPmTorsor, PositiveElement, DPmV, LabCuspPm, EtaZeroV, EtaPmV, AutPlus, AutMinus, AutAlpha, SignVector, DCircDotPmGlobal, AutPm, AutCsp, VPm, MinusOneFl, AlphaThetaPm, PhiThetaEllBulletV, GammaPm, TAbs, TStar, DThetaPmBridge, DThetaEllBridge, DThetaPmEllHodgeTheater, ZetaPm, XiSync, PropSixSeven, DThetaPmEllLink, DagHTDThetaPmEllNF, HalfPlaneDictionary, GeometricBasepoint, FigSixSix
`iut_sign_vector`	iut_i.h	FlSemiPmGroup, FlSemiPmTorsor, PositiveElement, DPmV, LabCuspPm, EtaZeroV, EtaPmV, AutPlus, AutMinus, AutAlpha, SignVector, DCircDotPmGlobal, AutPm, AutCsp, VPm, MinusOneFl, AlphaThetaPm, PhiThetaEllBulletV, GammaPm, TAbs, TStar, DThetaPmBridge, DThetaEllBridge, DThetaPmEllHodgeTheater, ZetaPm, XiSync, PropSixSeven, DThetaPmEllLink, DagHTDThetaPmEllNF, HalfPlaneDictionary, GeometricBasepoint, FigSixSix
`iut_prop_six_seven`	iut_i.h	FlSemiPmGroup, FlSemiPmTorsor, PositiveElement, DPmV, LabCuspPm, EtaZeroV, EtaPmV, AutPlus, AutMinus, AutAlpha, SignVector, DCircDotPmGlobal, AutPm, AutCsp, VPm, MinusOneFl, AlphaThetaPm, PhiThetaEllBulletV, GammaPm, TAbs, TStar, DThetaPmBridge, DThetaEllBridge, DThetaPmEllHodgeTheater, ZetaPm, XiSync, PropSixSeven, DThetaPmEllLink, DagHTDThetaPmEllNF, HalfPlaneDictionary, GeometricBasepoint, FigSixSix

B · Index of identifiers

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